

UPPER LIMB

Outline

- ◆ Introduction
- ◆ Bones of the upper limb
- ◆ Superficial structures of upper limb
- ◆ Axioappendicular Muscles
- ◆ Axilla
- ◆ Brachial plexus
- ◆ Arm
- ◆ Cubital fossa
- ◆ Forearm
- ◆ Hand
- ◆ Surface anatomy

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Introduction

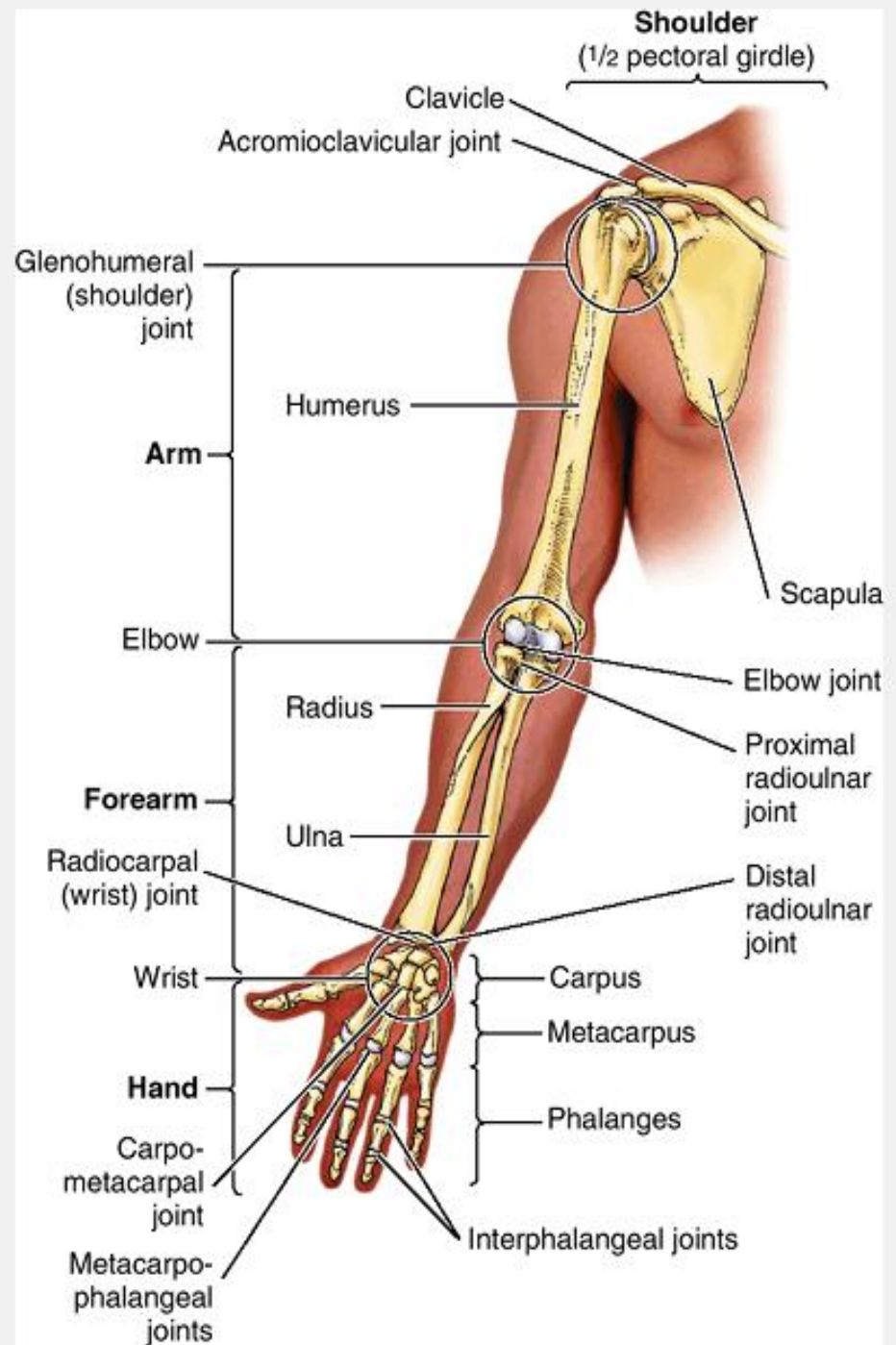
**Upper limb
consists of 4
segments**

**Shoulder:
pectoral girdle**

Arm: humerus

**Forearm: ulna
and radius**

**Hand: carpus,
metacarpus
and phalanges**

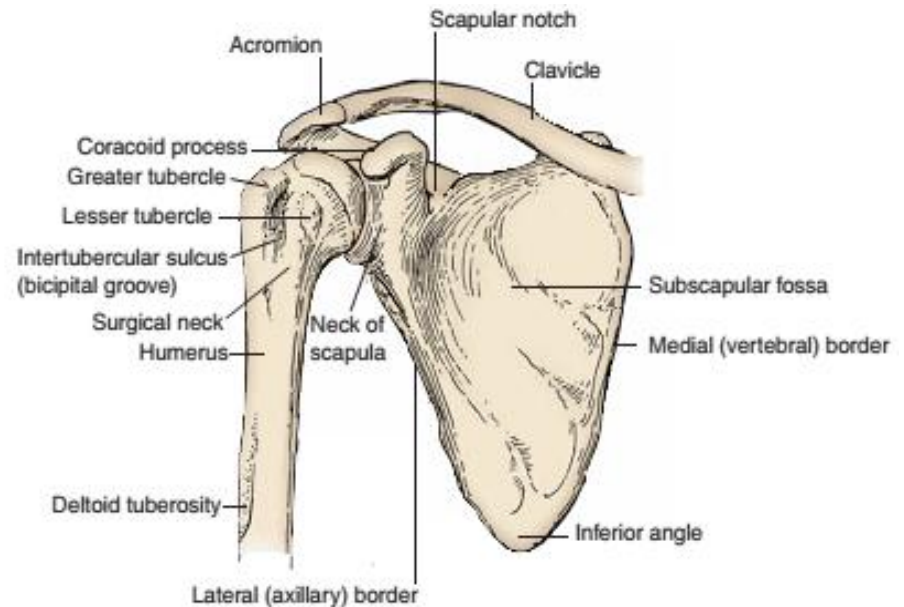


Bones of the upper limb

- The superior appendicular skeleton composed of
 - ✚ Pectoral girdle
 - ✚ Bones of the free part of the upper limb
- Articulates with the axial skeleton only at the **sternoclavicular joint**
 - Allows great mobility

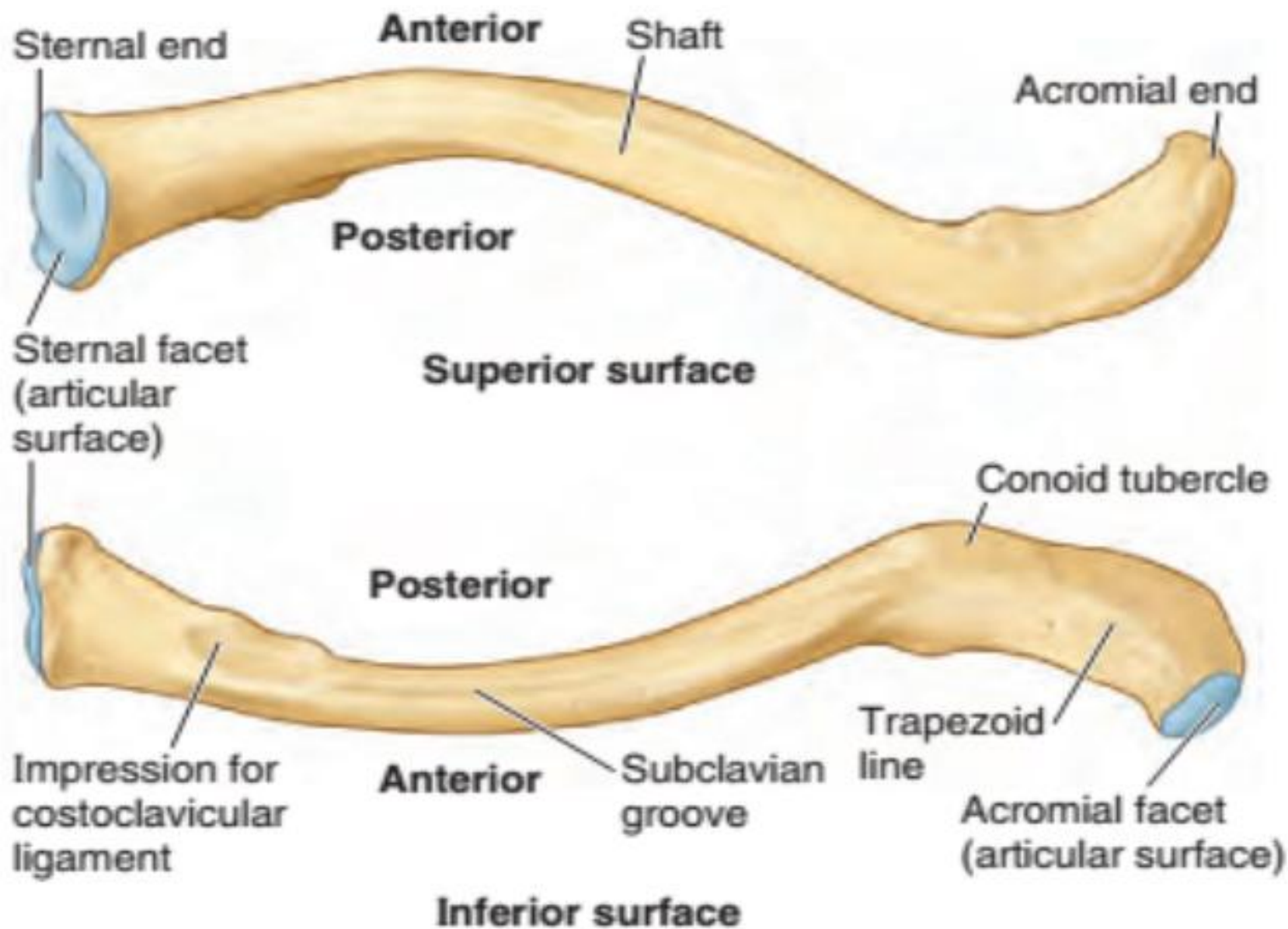
Pectoral (Shoulder) Girdle

- ◆ Consists of two bones:
clavicle and scapula
- ◆ Loosely attached, held in place largely by musculature
- ◆ Frees girdle to move over the thorax



Clavicle (collar bone)

- ➡ Double curved bones extending along the superior thorax
- ➡ The medial two thirds of the shaft of the clavicle are convex anteriorly, whereas the lateral third is flattened and concave anteriorly
 - curvatures increase the resilience of the clavicle
- ➡ **Sternal end:** articulates with manubrium of the sternum at the sternoclavicular (SC) joint
- ➡ **Acromial end:** articulates with acromion of scapula at the acromioclavicular (AC) joint
- ➡ It has two surface superior & inferior surface

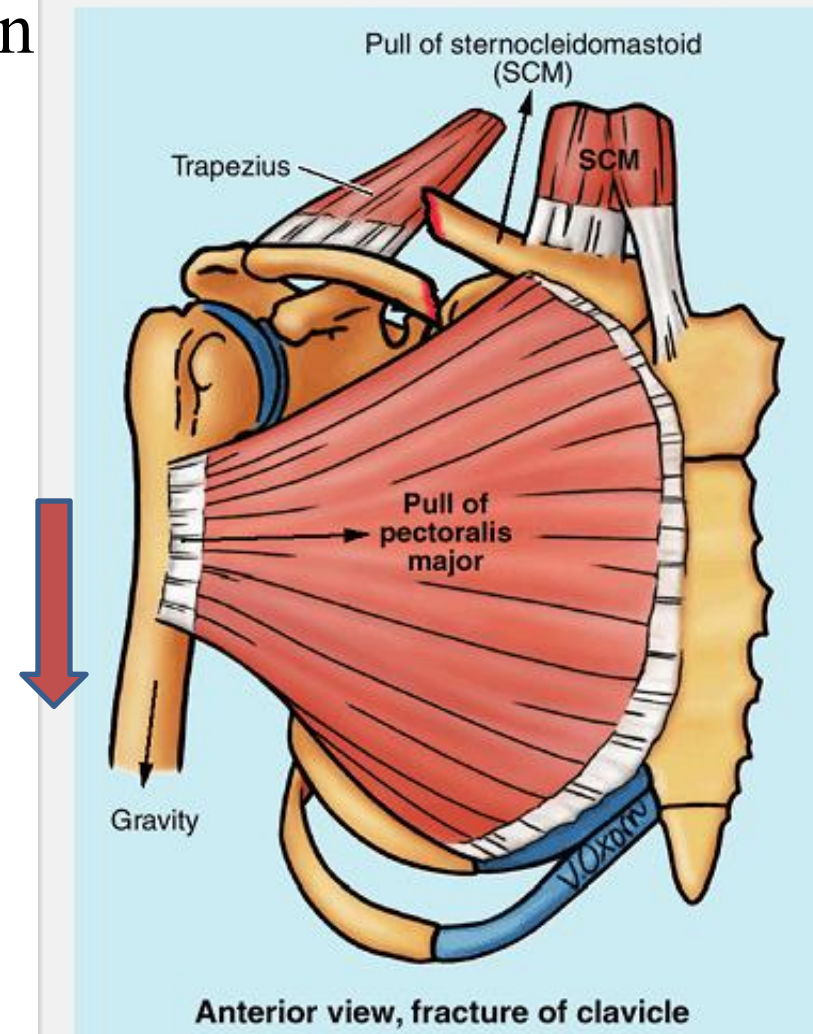


Clavicle: function

- ✚ Connects the upper limb to the trunk
- ✚ Attachment site for muscles of the thorax and shoulder
- ✚ Position scapula away from thorax so that the arm has maximum freedom of motion
- ✚ Afford protection to the neurovascular bundle supplying the upper limb
- ✚ Transmits shocks (traumatic impacts) and mechanical weight from the upper limb to the axial skeleton

Fracture of clavicle

- The weakest part is at the junction of its middle and lateral thirds
- **After fracture**
 - Sternocleidomastoid (SCM) muscle elevates the medial fragment of bone
 - Trapezius muscle is unable to hold up the lateral fragment owing to the weight of the upper limb
 - Results the **shoulder drops**



Scapula (shoulder blade)

- Triangular flat bone
- Lies on the posterolateral aspect of the thorax, overlying the **2nd-7th** ribs
- It has two surface

Posterior surface

- Convex
- Divided by spine into supraspinous and infraspinous fossa
- Spine continues laterally as flat expanded **acromion** which articulates with the acromial end of the clavicle

Scapula (shoulder blade)...

✚ Anterior surface

- Forms subscapular fossa

✚ Has a glenoid cavity laterally

- Shallow, concave, oval fossa
- Directed anterolaterally and slightly superiorly
- Articulates with the head of the humerus - glenohumeral (shoulder) joint
- Considerably smaller than the head of the humerus

✚ *Scapula has three processes :spinous , acromial ,and coracoid processes*

Scapula (shoulder blade)...

✚ **Borders:** medial, lateral and superior

✚ **Angles:** superior, lateral, and inferior

✚ **Lateral border**

- the thickest part, includes head of the scapula where the **glenoid cavity** is located

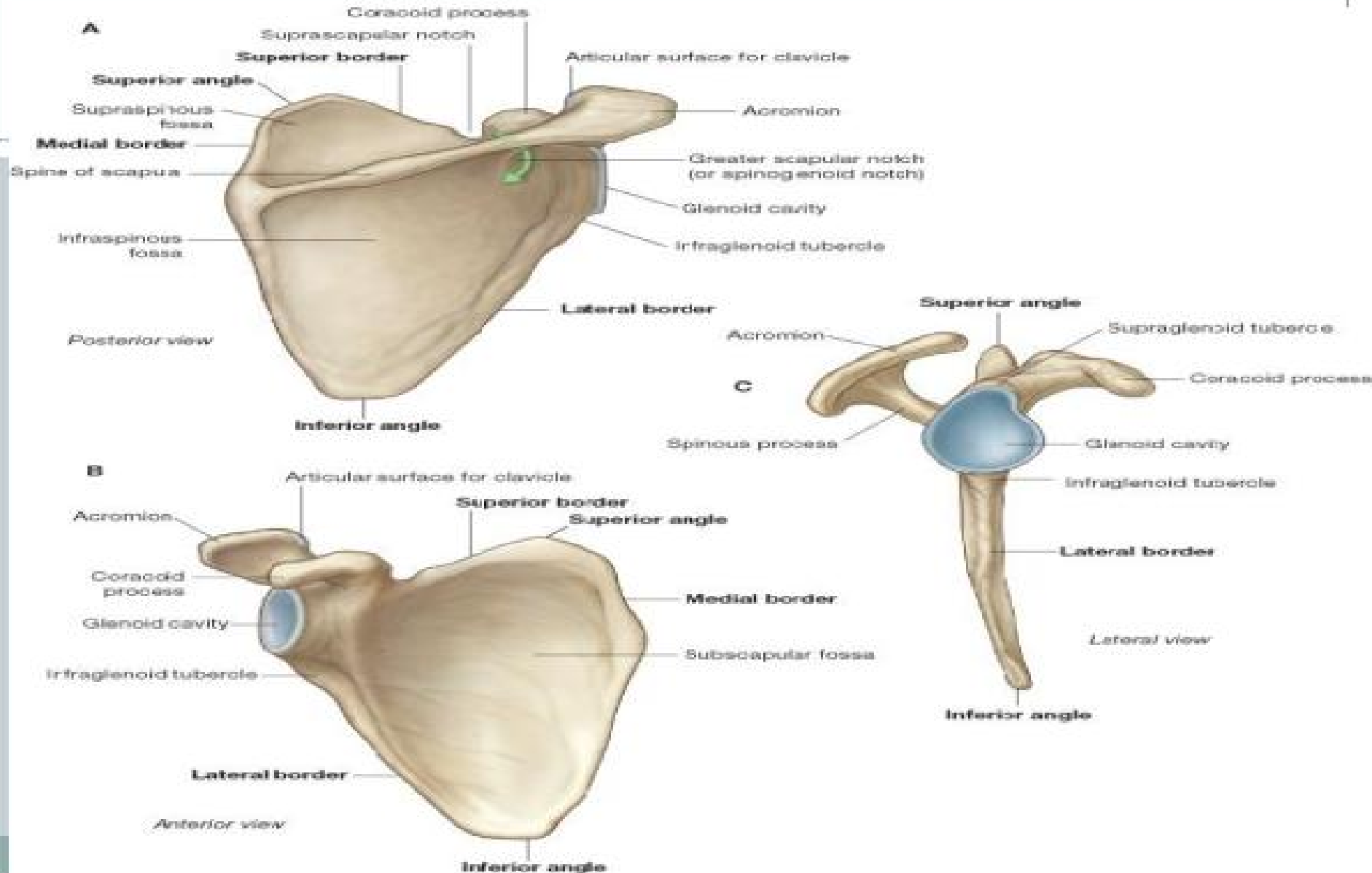
✚ **Neck**

- just inferior to the head

✚ **Superior border**

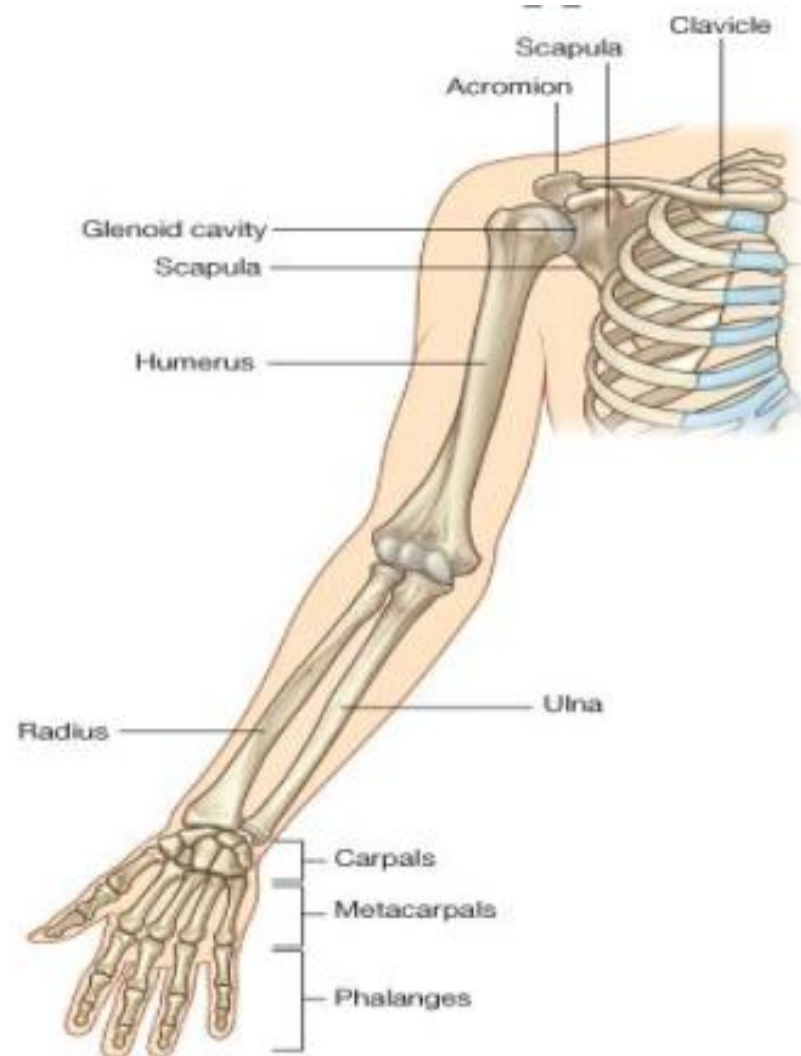
- marked near the junction of its medial two thirds and lateral third by the *suprascapular notch and laterally spinoglenoid notch*

Scapula: anterior & posterior view



The free upper limb

- **Arm**
 - Humerus
- **Forearm**
 - Ulna
 - Radius
- **Hand**
 - Wrist (8 carpal bones)
 - Palm (5 metacarpal bones)
 - Fingers (14 phalanges)

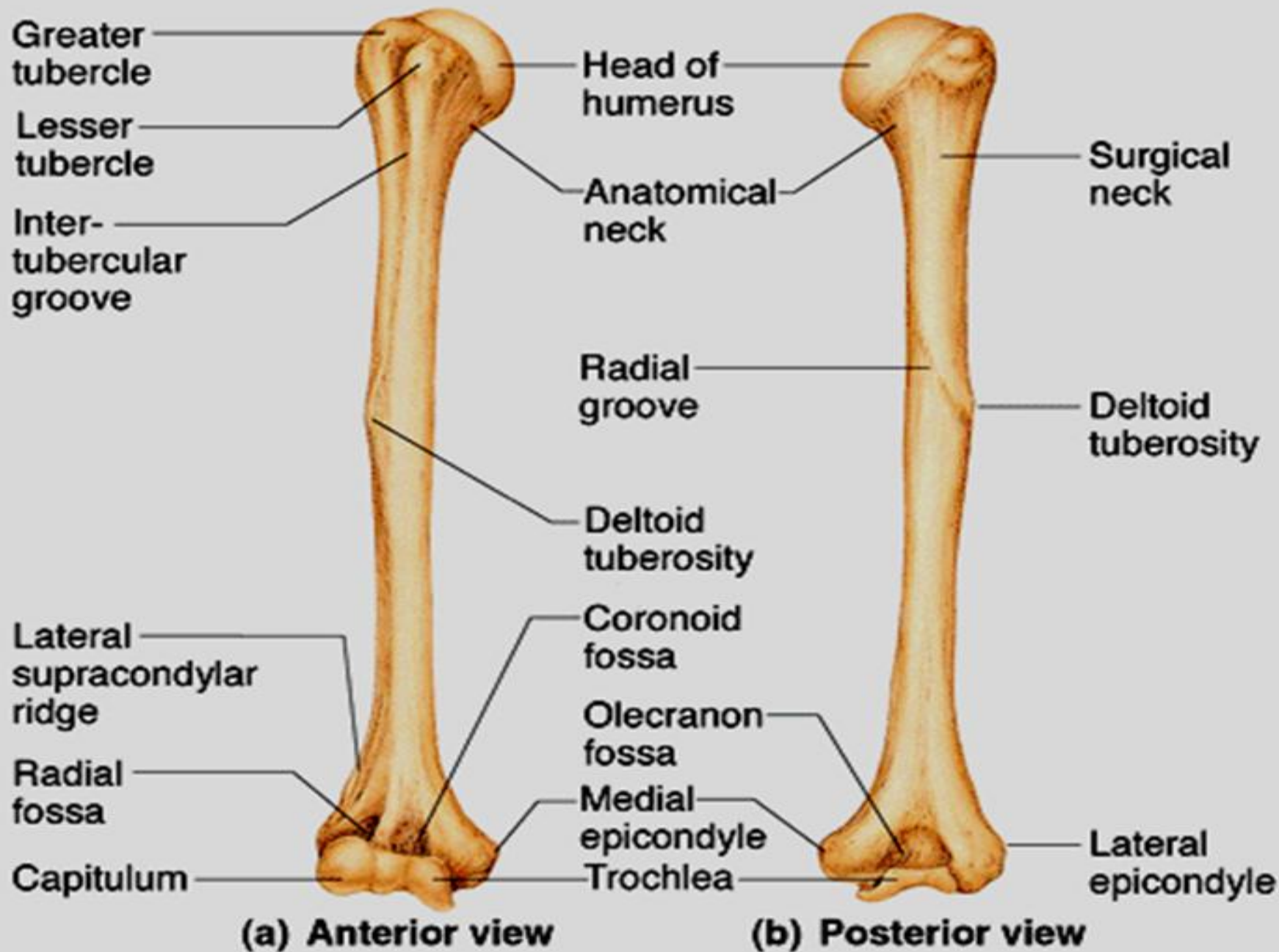


Humerus

- The longest bone of upper extremity
- Consist of head, shaft (body), anatomical & surgical necks, tubercles (greater & lesser)
- **Head**
 - ball-shaped
 - articulate with the **glenoid cavity** of the scapula
- **Tubercles**
 - lesser and greater tubercle
 - **intertubercular groove** (bicipital groove) separates the two

Humerus...

- ✚ **Anatomical neck:-** just distal to the humeral head
 - separates the head from the tubercles
- ✚ **Surgical neck :-** narrow part just distal to the tubercles
- ✚ **Shaft (body) :-** two prominent features:
 - **deltoid tuberosity** laterally for attachment of deltoid muscle
 - **radial (spiral) groove** posteriorly for **radial nerve and deep artery of the arm**
- ✚ **Inferior end**
 - widens as sharp medial and lateral supracondylar ridges
 - ends distally as medial and lateral epicondyle



Humerus...

❖ Condyles of the humerus

- distal end of the humerus
- include trochlea, capitulum, olecranon, coronoid, and radial fossae
- has two articular surfaces
 - **capitulum**: for articulation with the head of the radius
 - **trochlea**: for articulation with the trochlear notch of the ulna
- **Coronoid fossa**
 - superior to the trochlea anteriorly
 - receives the coronoid process of the ulna during **full flexion** of the elbow

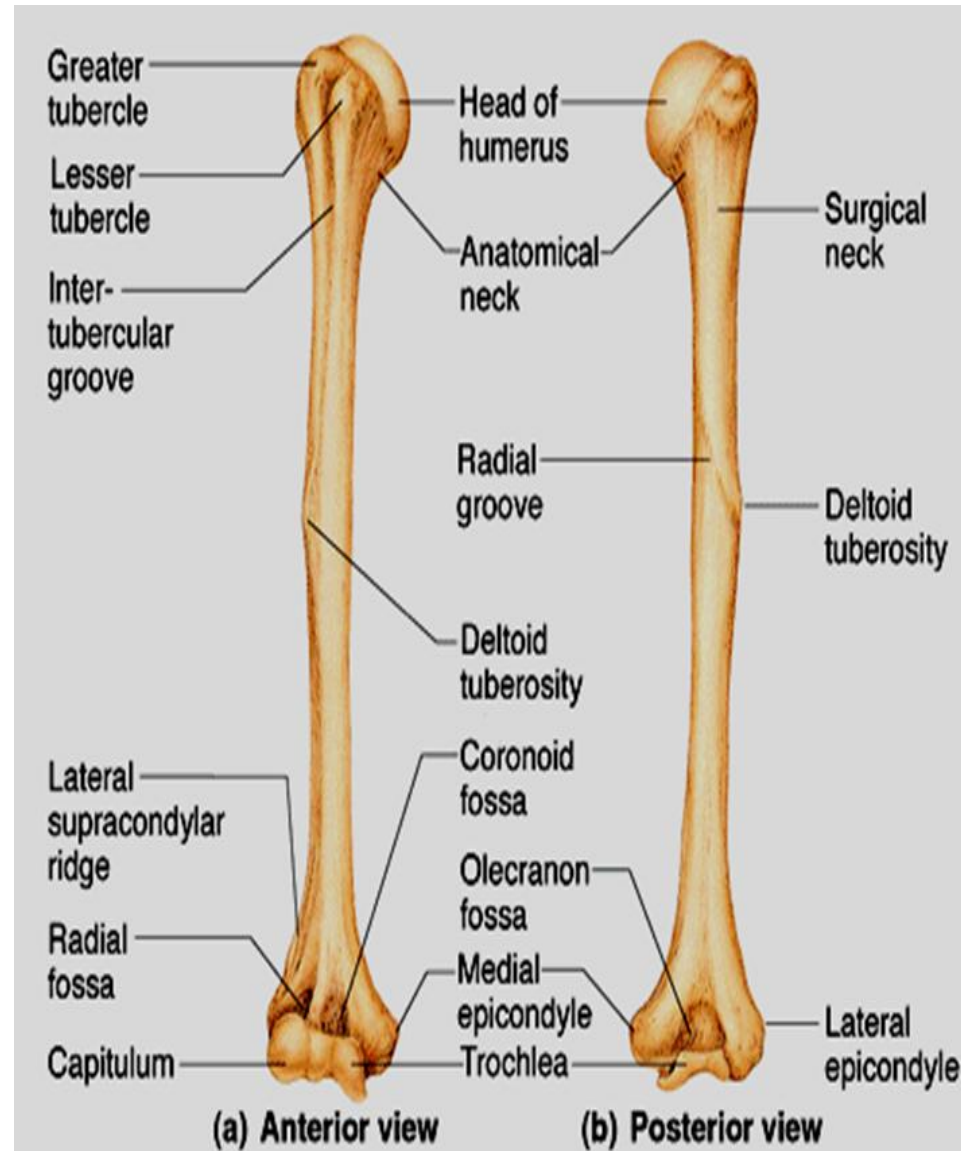
Humerus...

○ Olecranon fossa

- posteriorly
- accommodates the olecranon of the ulna during **extension** of the elbow

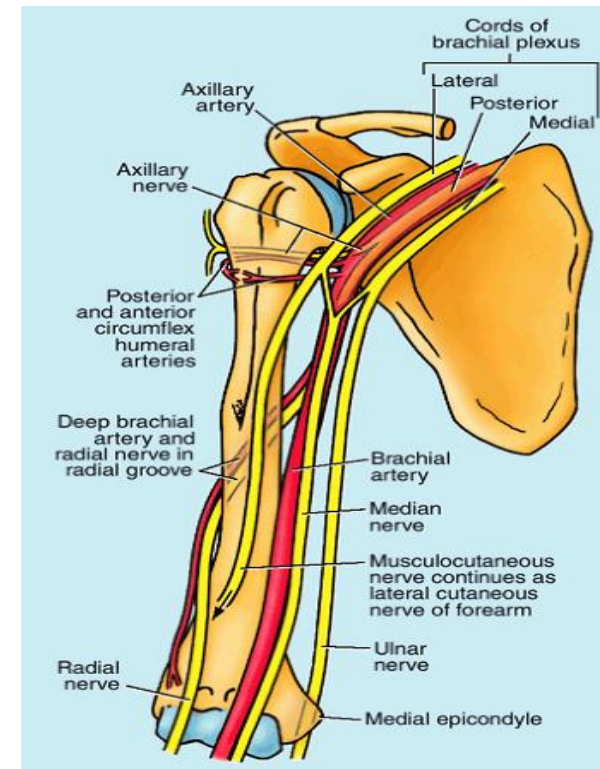
○ Radial fossa

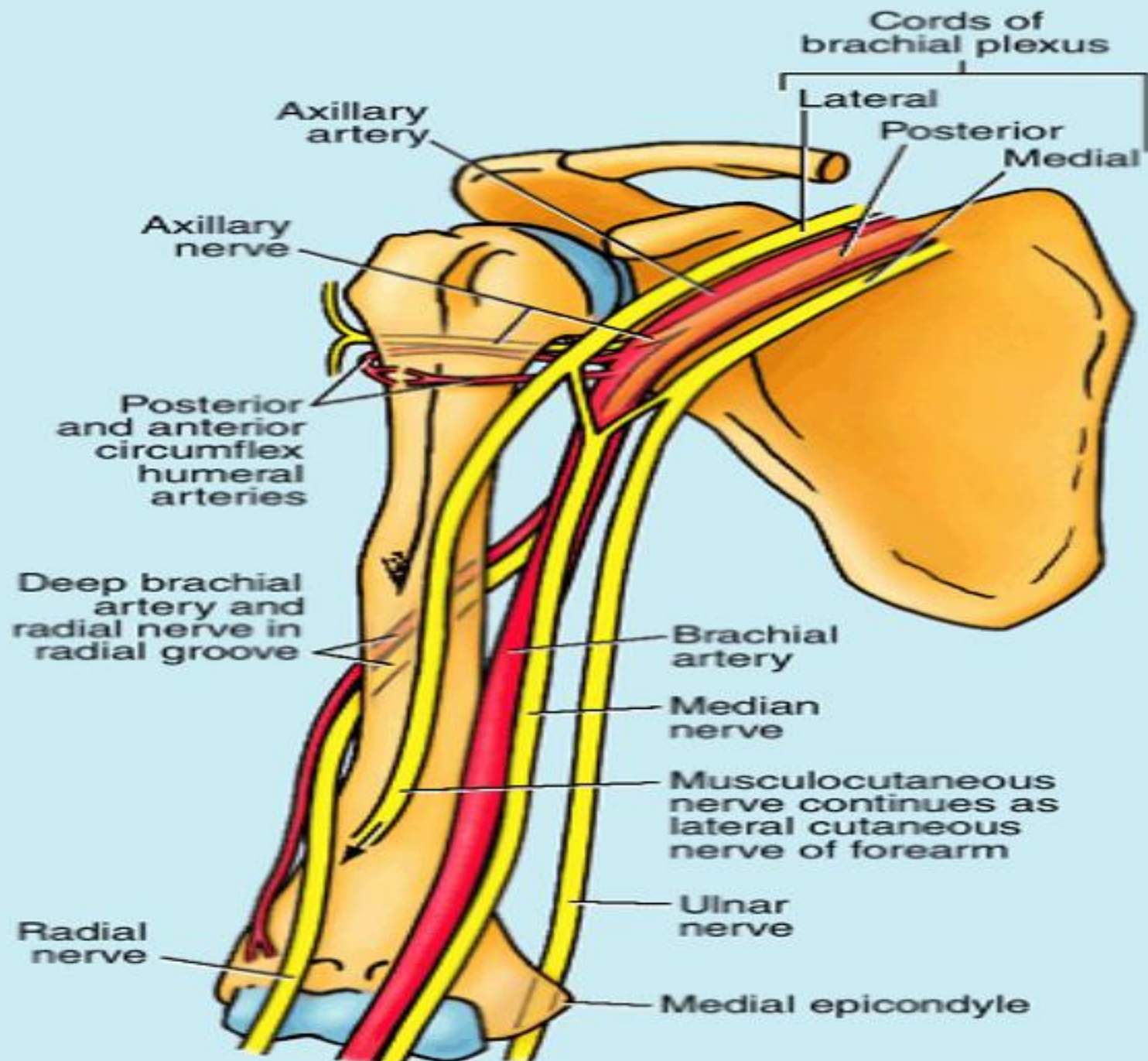
- superior to the capitulum anteriorly
- accommodates the edge of the head of the radius when the elbow is **fully flexed**



Fracture of the humerus

- Fractures of the **surgical neck** of the humerus
 - common in elderly people
 - usually result from a fall on the hand
- Because nerves are in contact with the humerus, they may be injured when the associated part of the humerus is fractured
 - **Surgical neck** – axillary nerve
 - **Radial groove** – radial nerve
 - **Distal humerus** – median nerve
 - **Medial epicondyle** – ulnar nerve

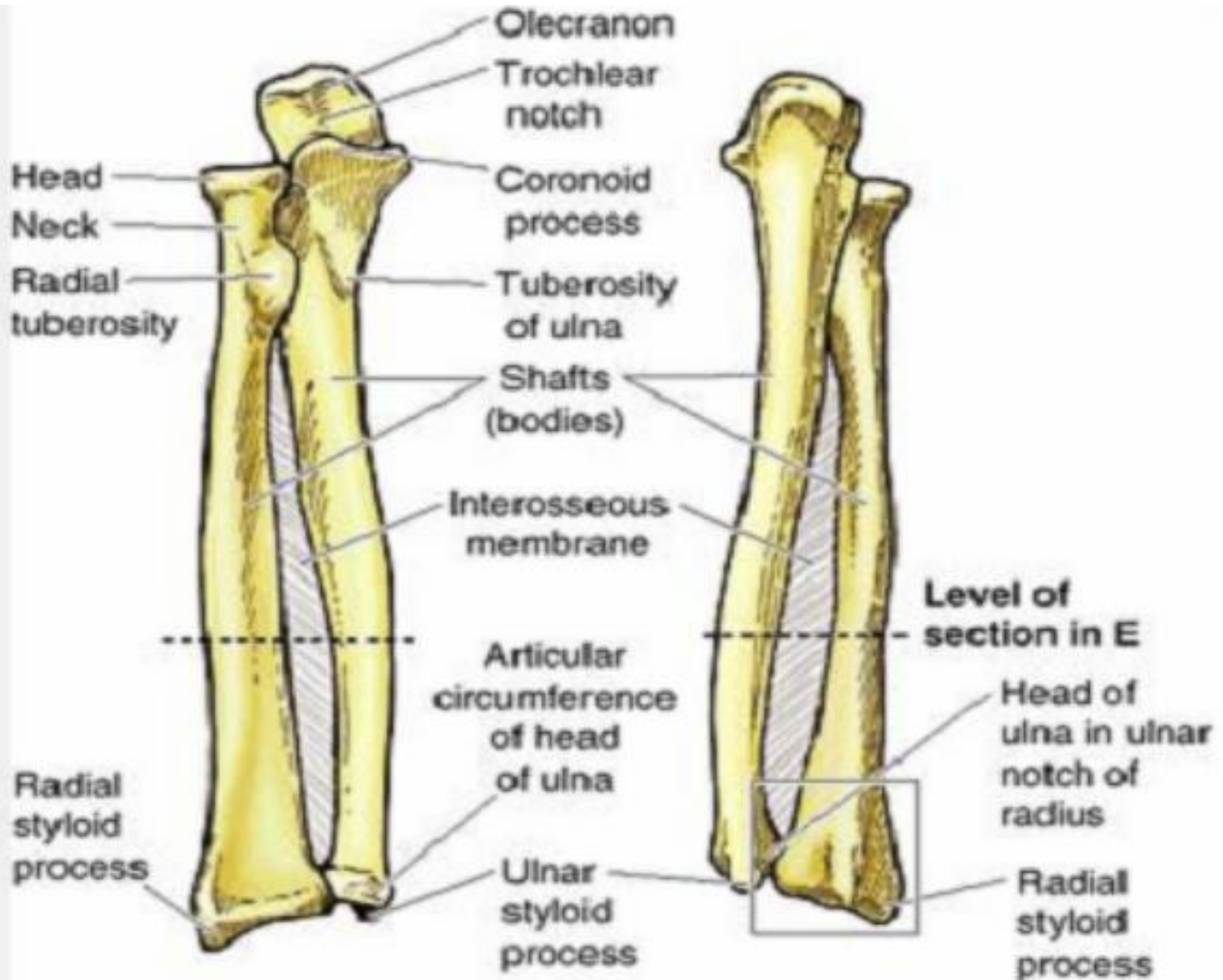




ulna

- ➡ The medial and longer of the two forearm bones
- ➡ Its proximal end has two prominent projections
 - ✚ Olecranon posteriorly
 - ✚ Coronoid process anteriorly
 - The two form the walls of the trochlear notch
- ➡ Trochlear notch
 - Articulates with the trochlea of the humerus
- ➡ Radial notch
 - On the lateral side of the coronoid process
 - Articulates with the head of radius

Fore arm bone



ulna

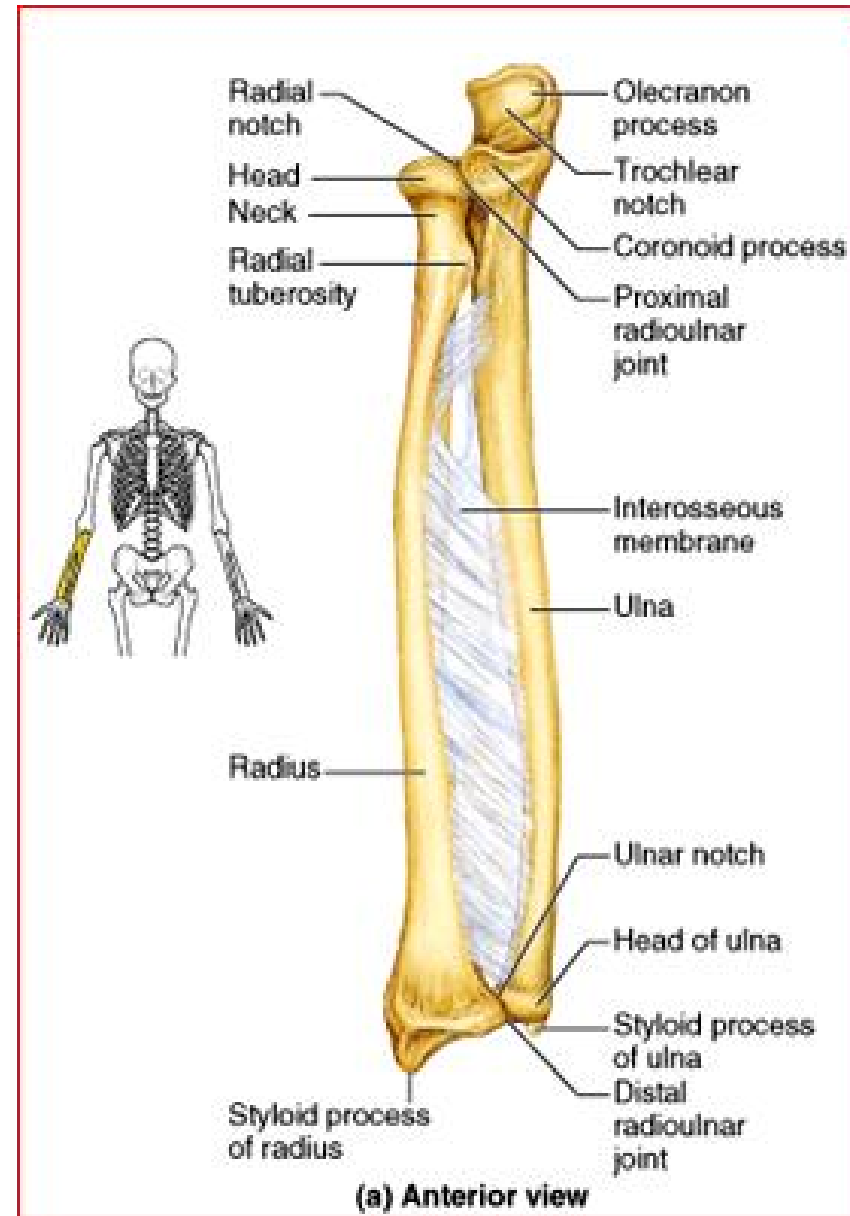
- **Shaft (body)**
 - proximally it is thick, but it tapers distally
- **Head**
 - narrow distal end
 - rounded with small, conical ulnar styloid process
- The ulna does not reach, and therefore does not participate in the **wrist (radiocarpal) joint**

Radius

- ✿ The lateral and shorter of the two forearm bones
- ✿ Its proximal end consists of a cylindrical head, a short neck, a projection from the medial surface (radial tuberosity)
- ✿ Head
 - superior surface is concave for articulation with capitulum of humerus
 - medially articulate with radial notch of ulna

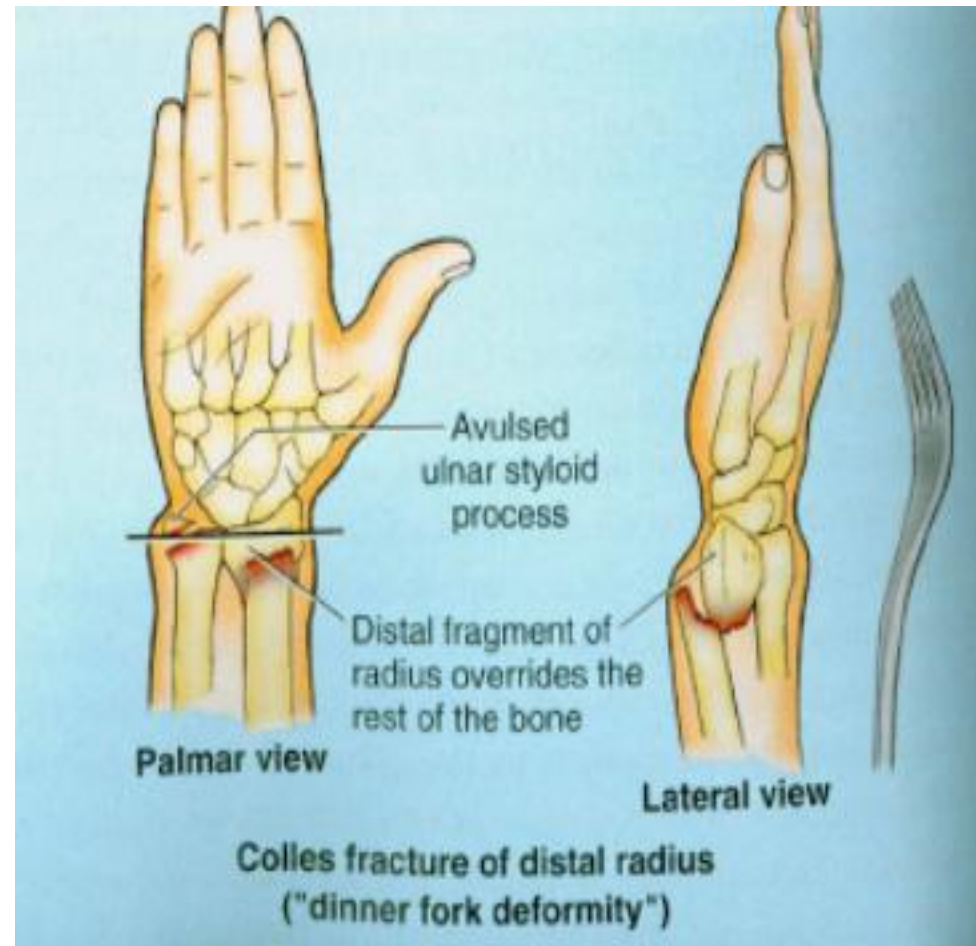
Radius...

- **Neck**
 - narrow part between head and radial tuberosity
- **Shaft (body)**
 - gradually enlarges distally
- **Distal end**
 - medially - ulnar notch for head of ulna
 - laterally - radial styloid process



Fracture of ulna and radius

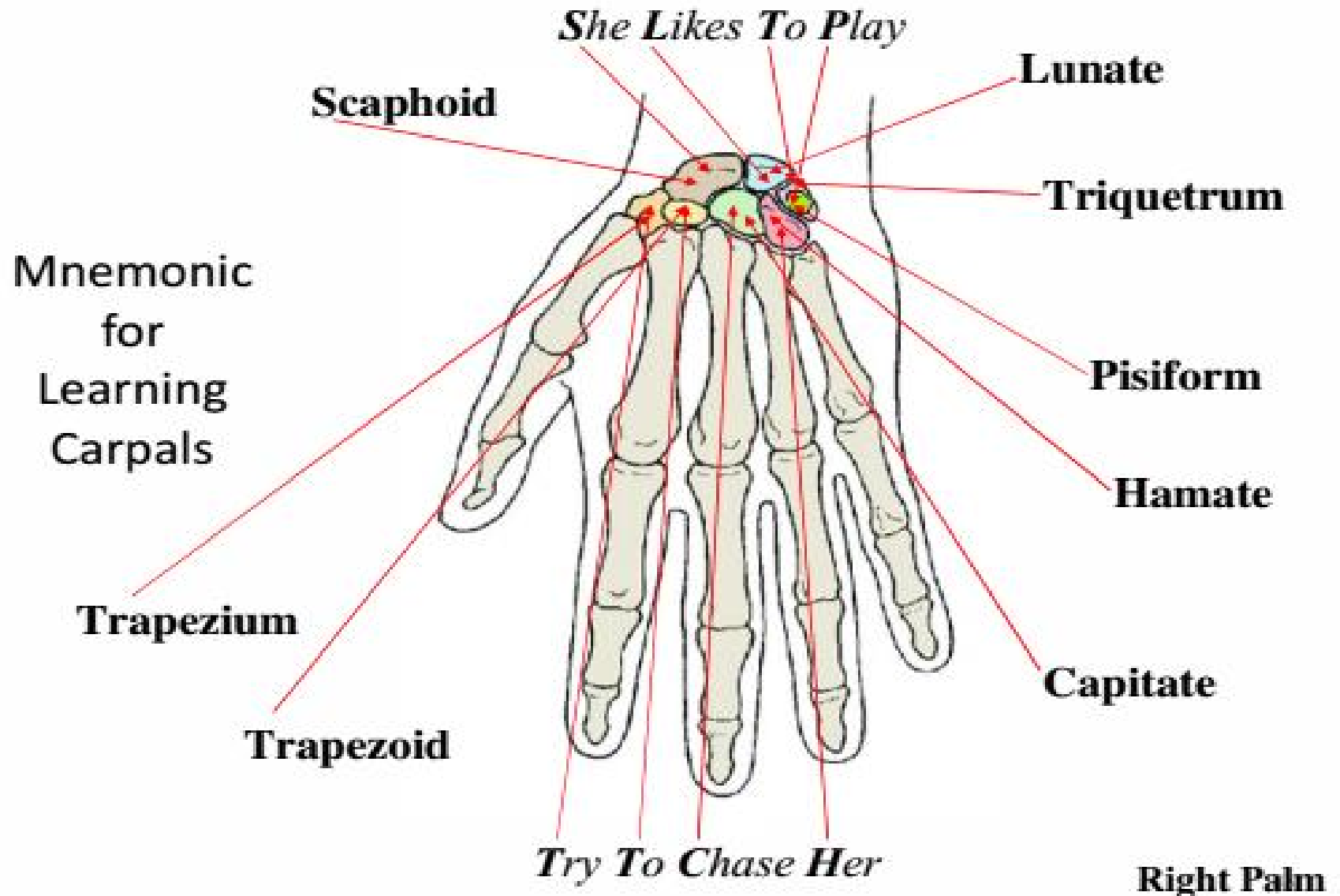
- **Colles fracture**
 - A complete fracture of the distal end of the **radius**
 - Results from forced dorsiflexion of the hand
 - The distal fragment of the radius is displaced dorsally and often broken into pieces



Carpal Bones

- 8 bones, form the wrist (carpus)
- Arranged in two transverse rows of 4 bones
- Carpus is convex from side to side posteriorly and concave anteriorly
- **Proximal row**
 - articulate with radius and articular disc of wrist joint proximally and distal row of carpals distally
 - **Proximal row from lateral to medial: “She Like To Play”**
 - **Scaphoid**: boat-shaped with scaphoid tubercle, largest bone in proximal row
 - **Lunate**: moon-shaped
 - **Triquetrum**: pyramidal
 - **Pisiform**: pea-shaped; lies on palmar surface of triquetrum₂₇

Carpal Bones...



Carpal Bones...

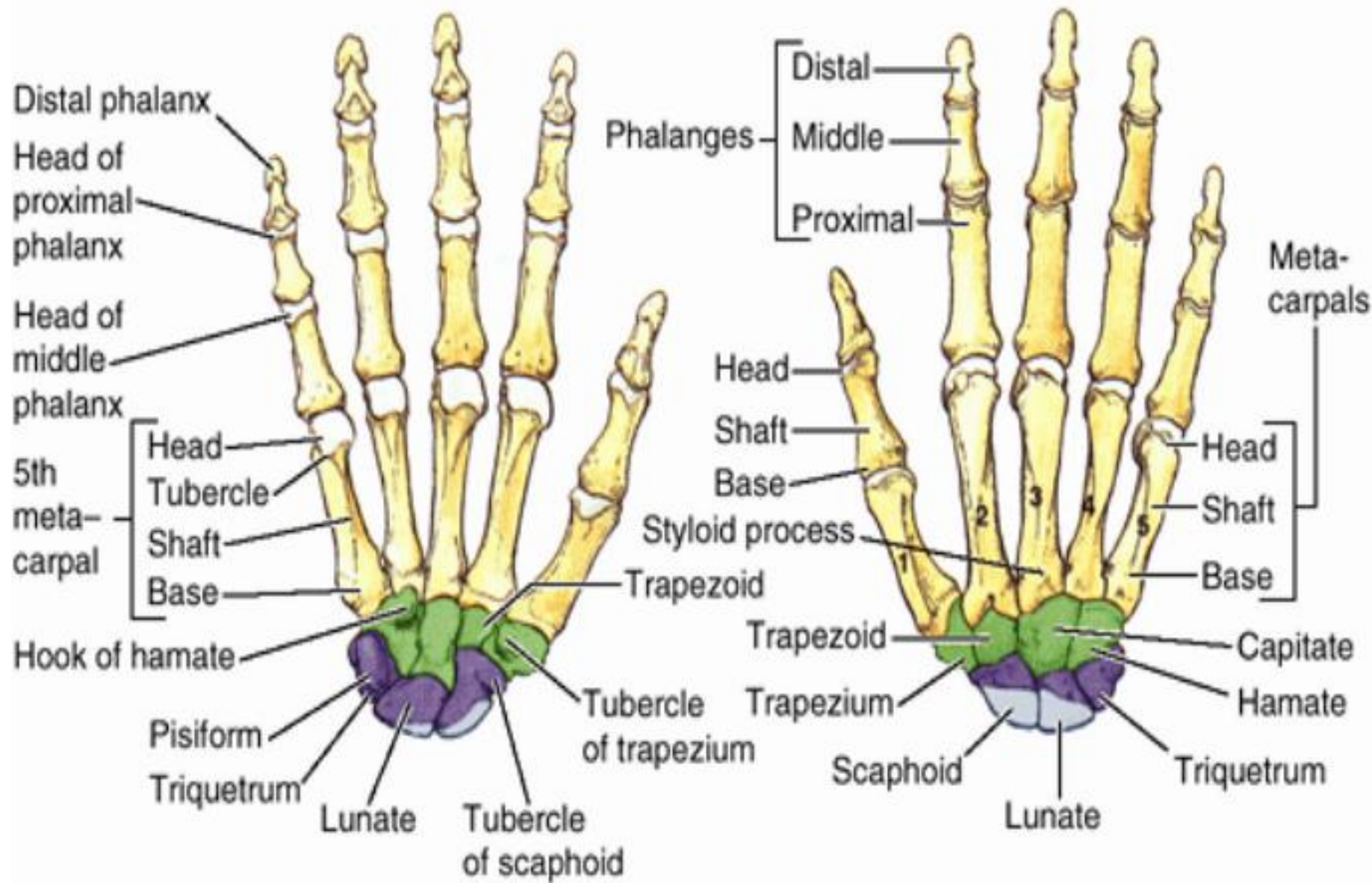
- **Distal row**
 - Articulate with proximal row of carpals proximally and metacarpals distally
 - Distal row from lateral to medial: “**Try To Chase Her**”
 - **Trapezium**: four sided
 - **Trapezoid**: wedge-shaped
 - **Capitate**: head like; largest in distal row
 - **Hamate**: wedge-shaped, has hooked process (hook of hamate)

Metacarpals

- Forms the skeleton of the palm
- Composed of **five** metacarpal bones
- They count from lateral to medial
- Each metacarpal consists of a **base, shaft, and head**
- The proximal bases of the metacarpals articulate with the carpal bones
- The distal heads of the metacarpals articulate with the proximal phalanges and form the knuckles

Phalanges

- **14** bones; form the fingers
- Each digit has three phalanges (proximal, middle, and distal) *except for the first (thumb)*, which has only two (proximal and distal)
- Each phalanx has a **base** proximally, a **shaft (body)**, and a **head** distally
- The distal phalanges are flattened and expanded at their distal ends, which underlie the nail beds



(A) Anterior (palmar) view

(B) Posterior (dorsal) view

Joints of the Upper Limb

1. Joints of the Shoulder Girdle

- i. sternoclavicular joint
- ii. acromioclavicular joint
- iii. glenohumeral joint

2. The elbow joint

3. Radioulnar Joints

4. The wrist (radiocarpal) joint

5. Intercarpal Joints

6. Carpometacarpal joints

7. Metacarpophalangeal joints

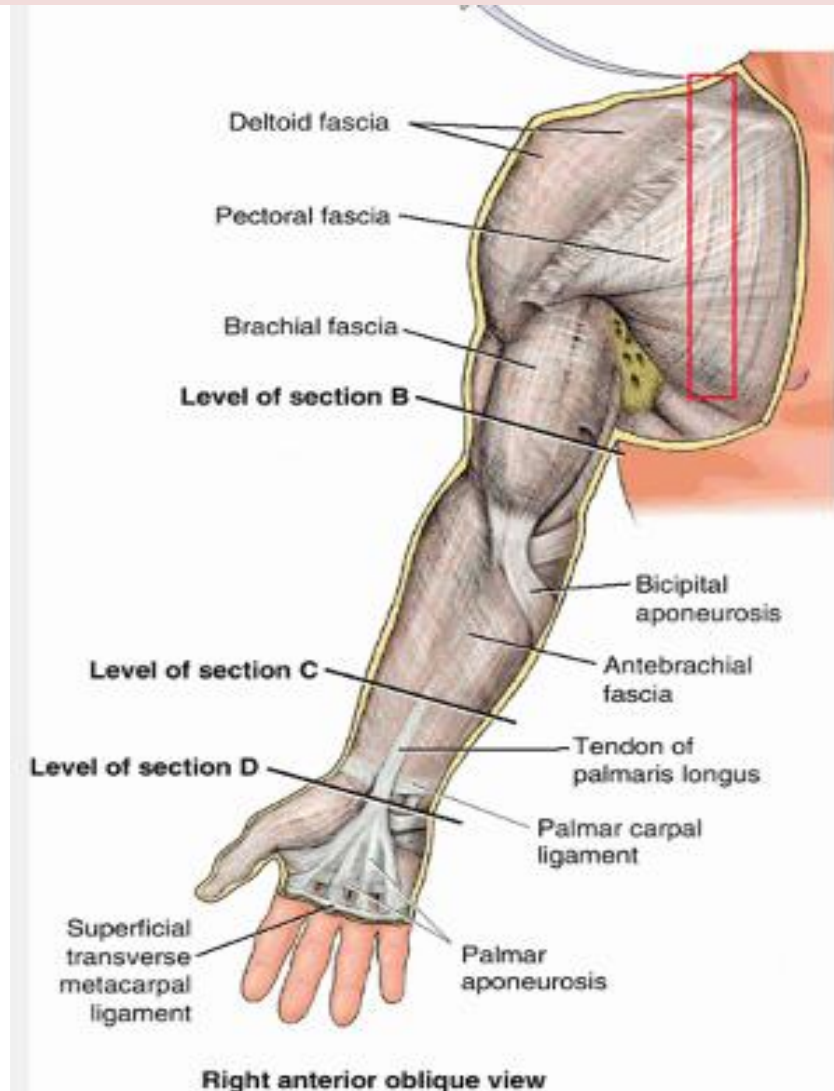
8. Interphalangeal joints

Superficial structures of upper limb

Fascia of upper limb

■ Pectoral fascia

- ▶ Invests the pectoralis major
- ▶ Continuous inferiorly with fascia of anterior abdominal wall
- ▶ Leaves lateral border of pectoralis major and becomes axillary fascia



■ Clavipectoral fascia

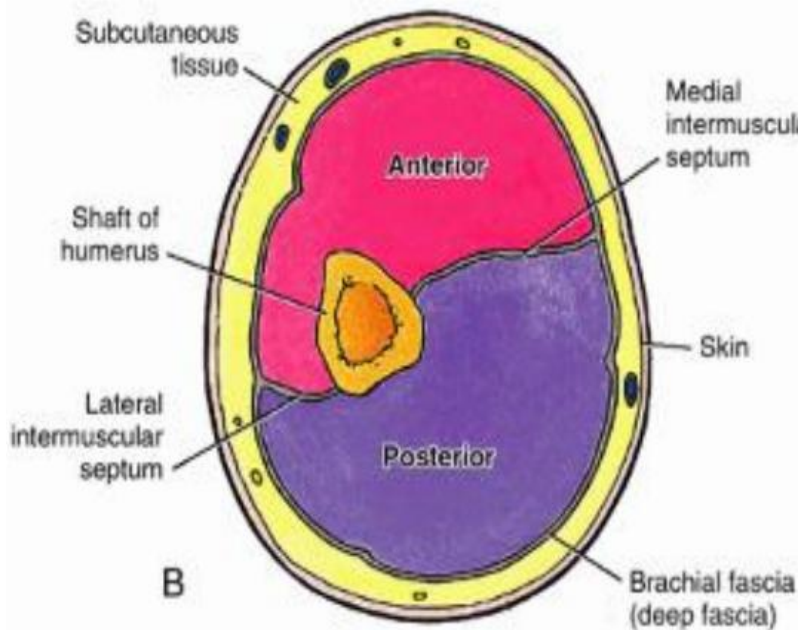
- Deep to pectoralis major
- Descends from the clavicle, enclosing subclavius and then the pectoralis minor
- Becoming continuous inferiorly with the axillary fascia

■ Deltoid fascia

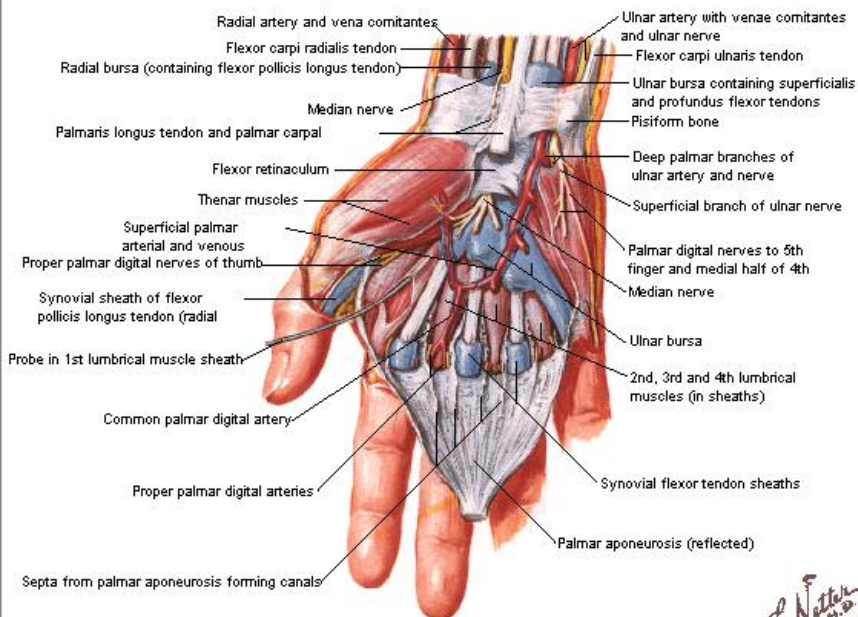
- Invests deltoid muscle

■ Brachial fascia

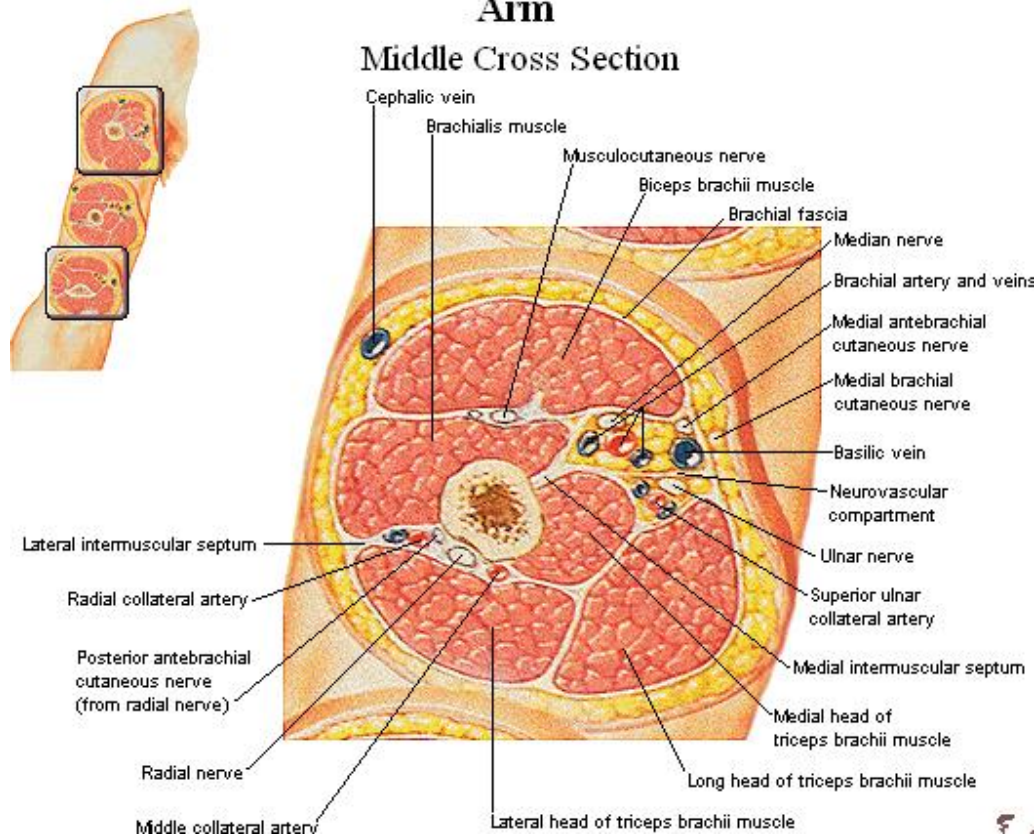
- ✚ encloses the arm
- ✚ continuous with deltoid, pectoral, axillary, infrapinnous and antebrachial fasciae
- ✚ form two intermuscular septa
 - medial and lateral
 - extend from the deep surface of the brachial fascia and attach to the humerus
 - divide the arm into anterior (flexor) and posterior (extensor) fascial compartments



Wrist and Hand Deeper Palmar Dissections

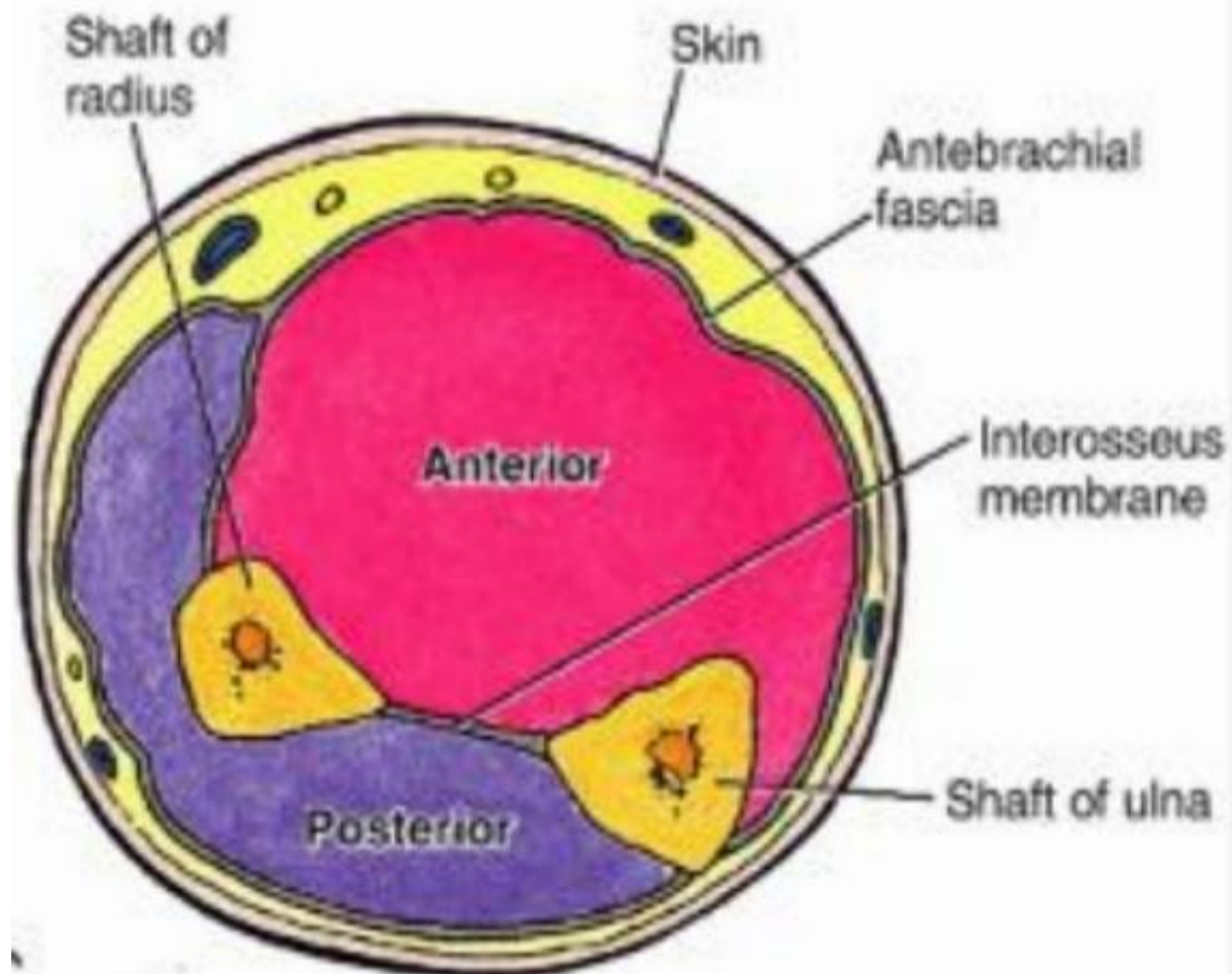


Arm Middle Cross Section



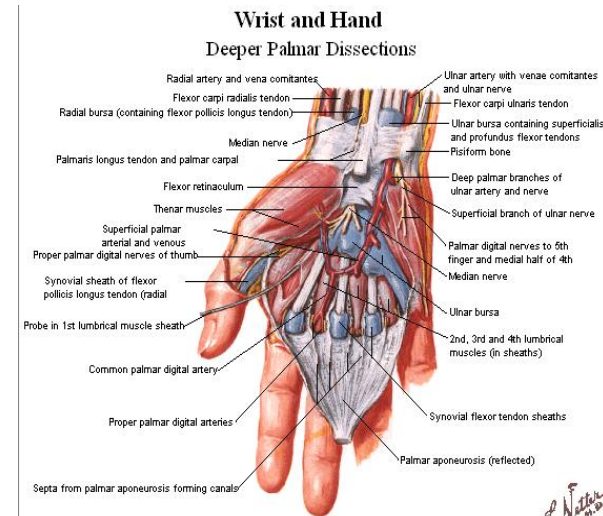
■ Antebrachial fascia

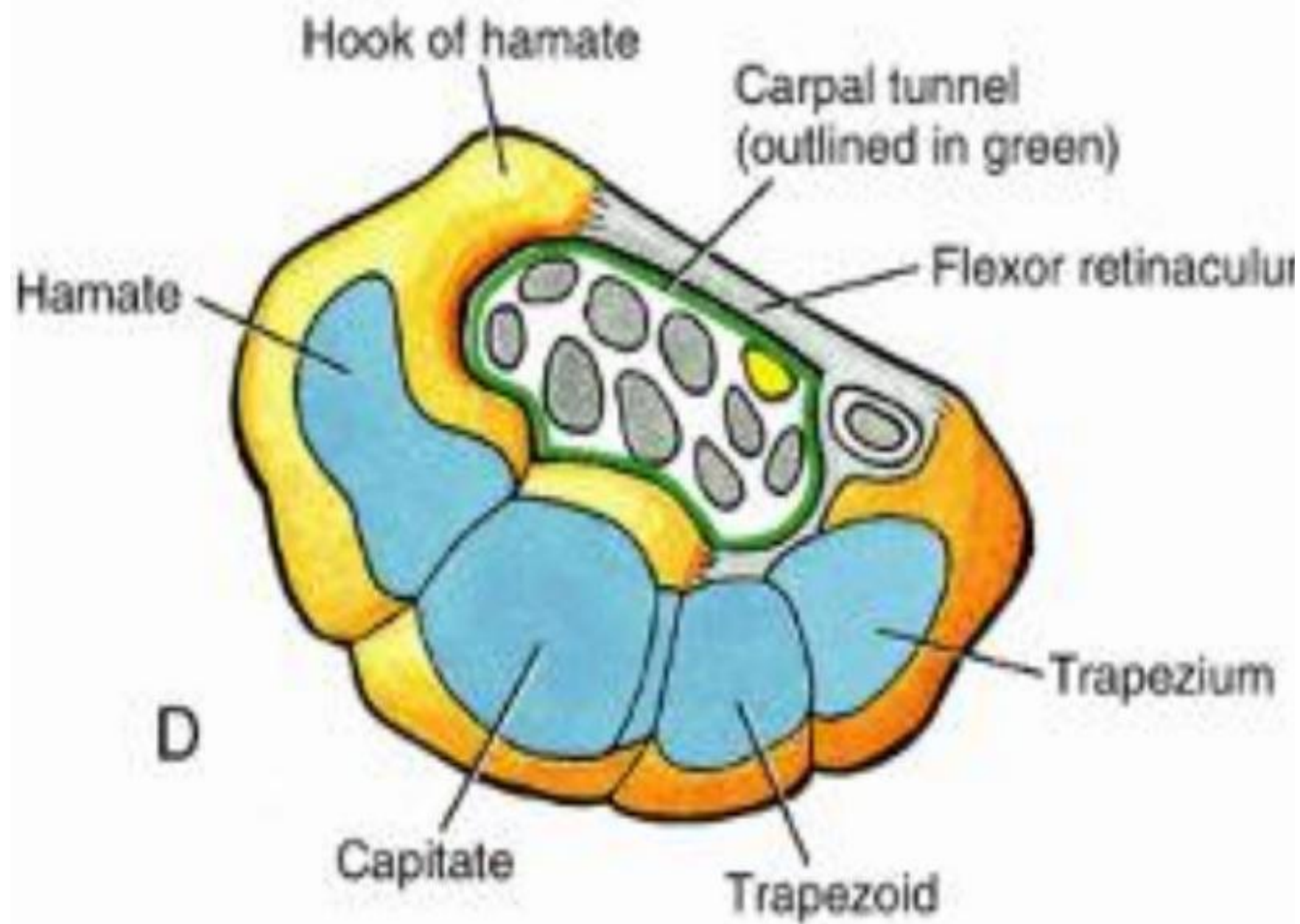
- ✚ In the forearm
- ✚ Surrounds fascial compartments separated by interosseous membrane
- ✚ Thickens posteriorly over the distal end to form a transverse band, **extensor retinaculum**, which holds the extensor tendons in position
- ✚ Anteriorly at distal end forms an anterior thickenings, palmar carpal ligament and **flexor retinaculum**



■ Palmar fascia

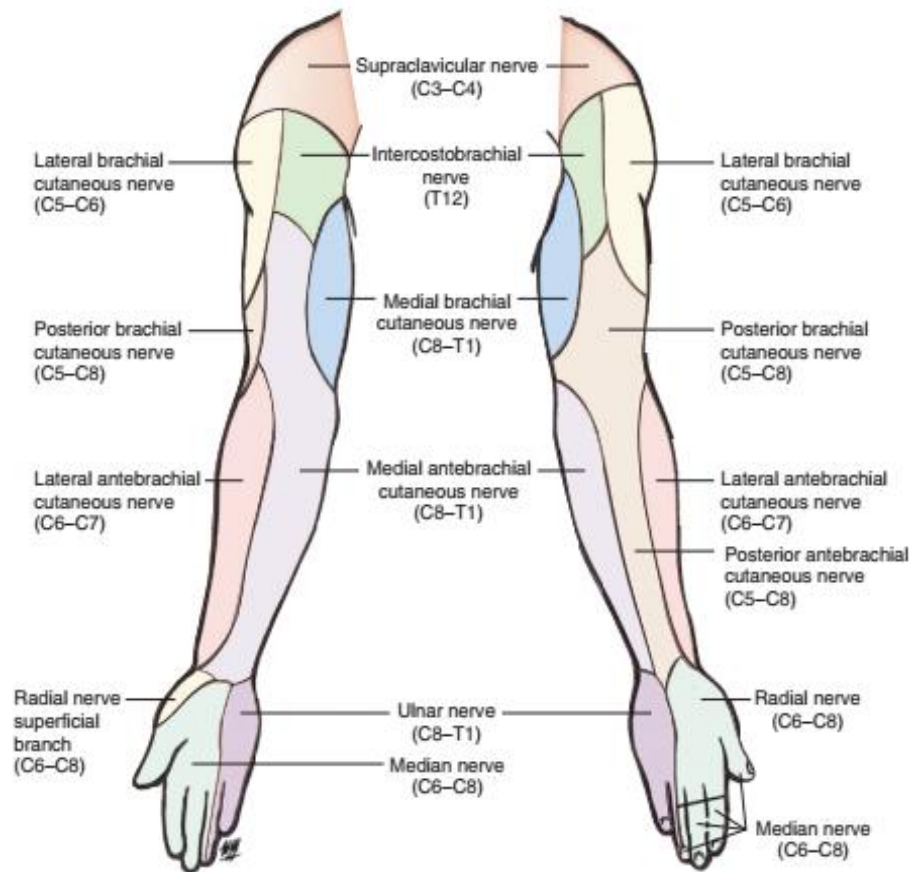
- ✚ Extension of deep fascia beyond retinacula
- ✚ Palmar aponeurosis
 - Central part, thick, triangular
 - Forms four thickenings that radiate to the base of fingers
 - Superficial transverse metacarpal ligament: transverse bands distally and form the base of palmar aponeuroses
 - Ligaments extend from aponeurosis to skin, hold the skin closer



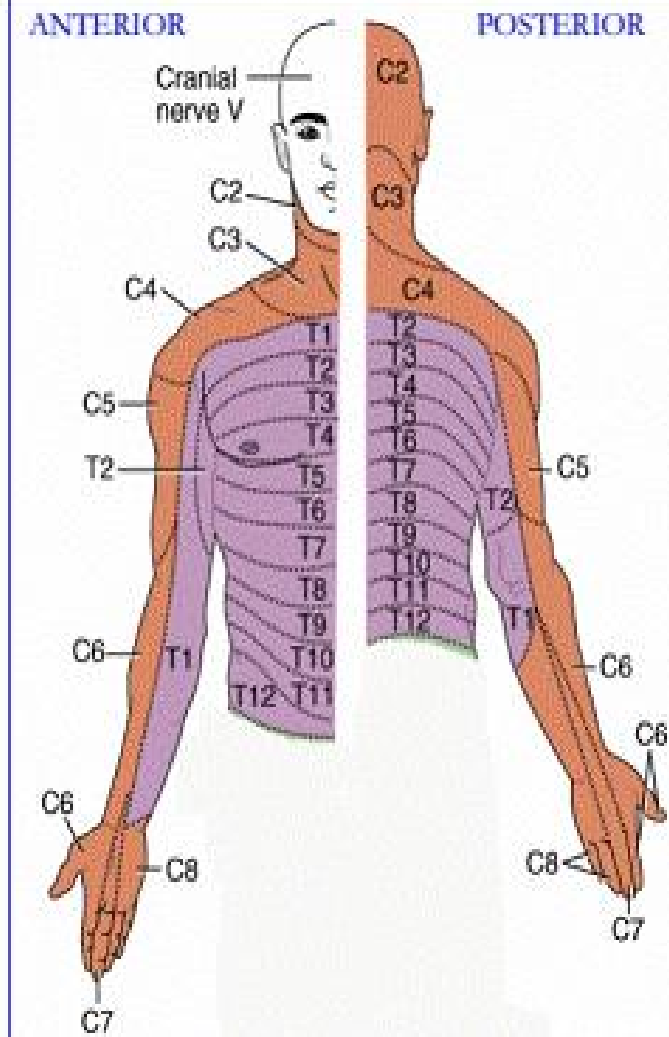


Cutaneous nerves of upper limb

- ✚ Most cutaneous nerves of upper limb are derived from brachial plexus
- ✚ Cutaneous nerves to shoulder are derived from cervical plexus
- ✚ C4: base of the neck, shoulder
- ✚ C5: arm laterally
- ✚ C6: forearm laterally and thumb
- ✚ C7: middle three fingers and middle of posterior surface
- ✚ C8: little finger, the medial side
- ✚ T1: middle of forearm to axilla
- ✚ T2: small part of arm and axilla



DERMATOMES



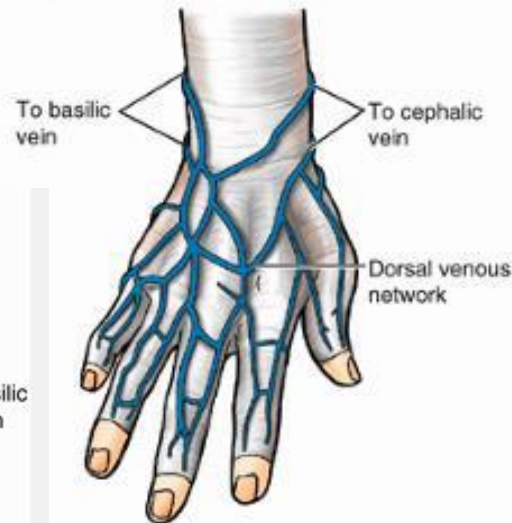
Superficial vessels of upper limb

■ Dorsal venous network

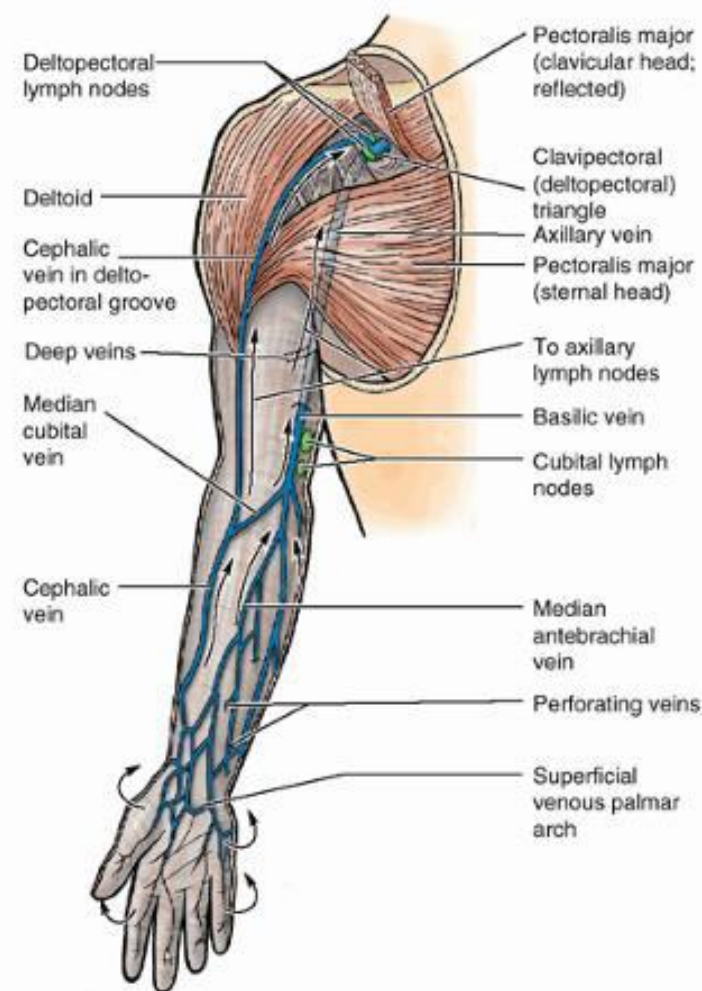
- On dorsum of the hand
- Give rise to cephalic and basilic veins

■ Cephalic vein

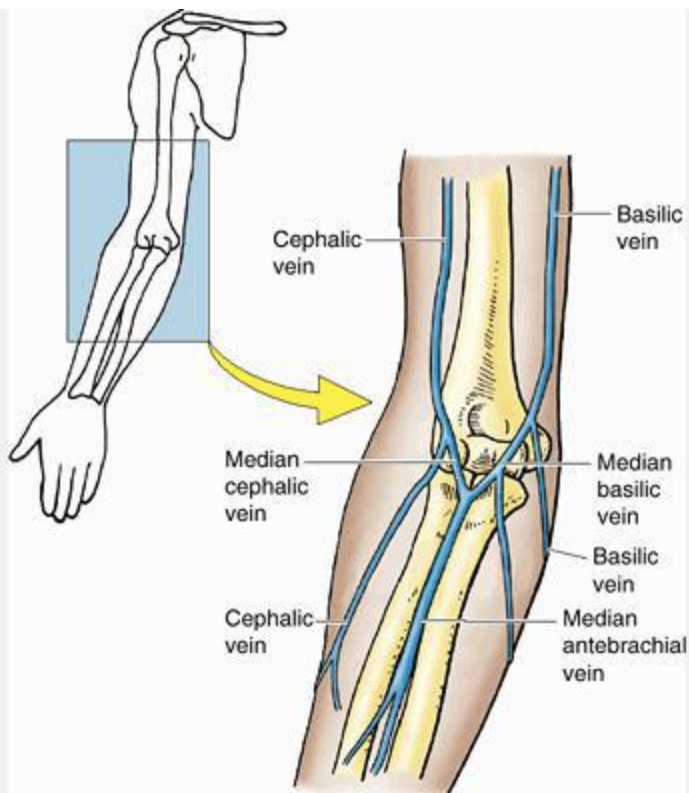
- Ascends from **lateral part** of dorsal venous network
- Proceed along anterolateral surface of forearm and arm
- Anterior to elbow communicate with **median cubital vein**
- Superiorly passes between deltoid and pectoralis major muscles and enters clavipectoral triangle
- Pierces clavipectoral fascia and joins **axillary vein**



Posterior (dorsal) views



(B) Anterior (palmar) views

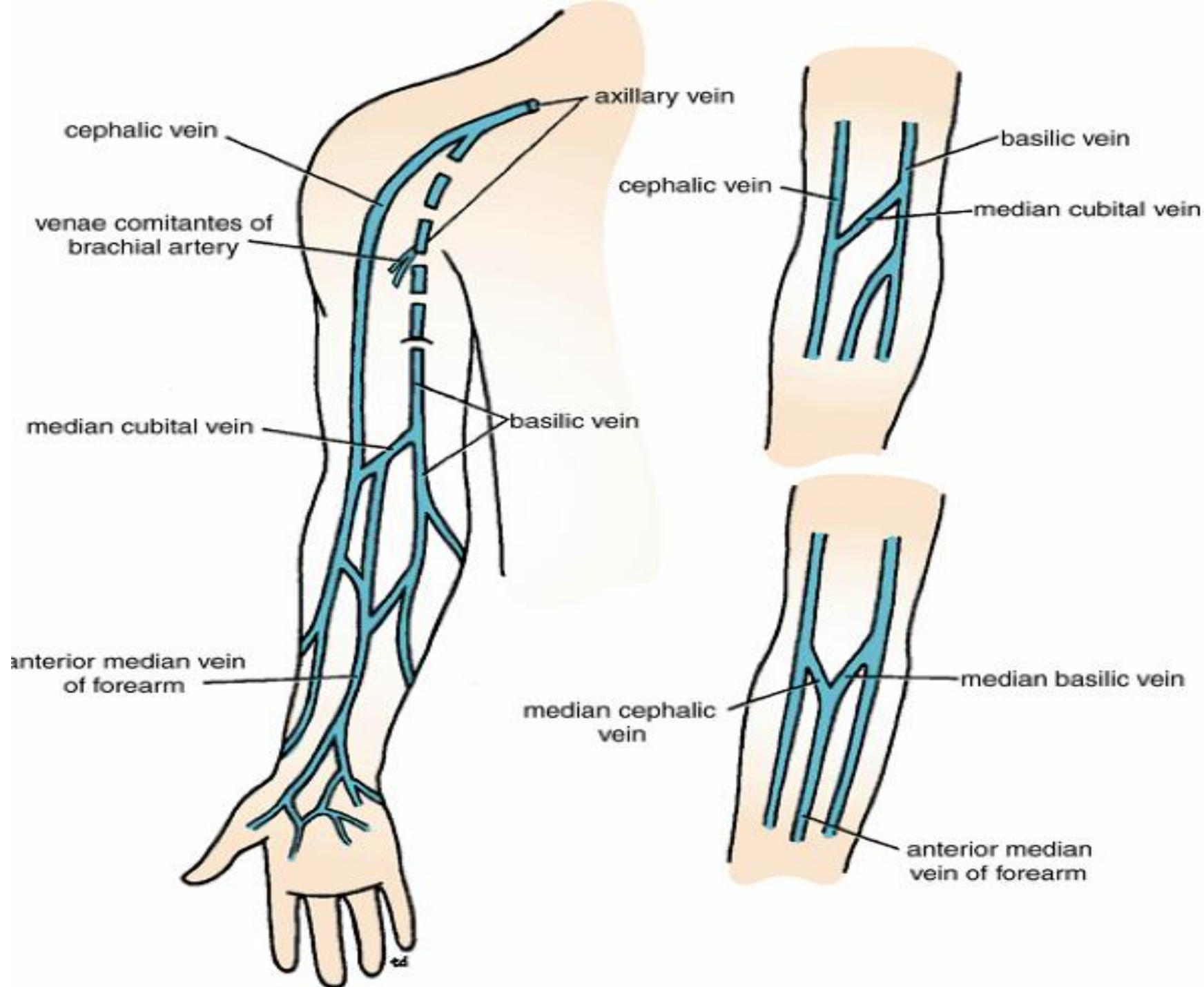


■ **Basilic vein**

- ✚ Ascends from **medial part** of dorsal venous network
- ✚ Proceed along medial side of forearm and inferior part of arm
- ✚ Passes deeply near the junction of the middle and inferior thirds of the arm, piercing the brachial fascia and running superiorly parallel to the brachial artery

■ **Median antebrachial vein** (median vein of forearm)

- ✚ Ascends in the middle of anterior aspect of forearm between cephalic and basilic veins



Venipuncture

- Puncture of a vein to draw blood or inject a solution
- The **median cubital** vein is commonly used for venipuncture
- The veins forming dorsal venous network and the cephalic and basilic veins arising from it are commonly used for long-term introduction of fluids

Lymphatic drainage of the upper limb

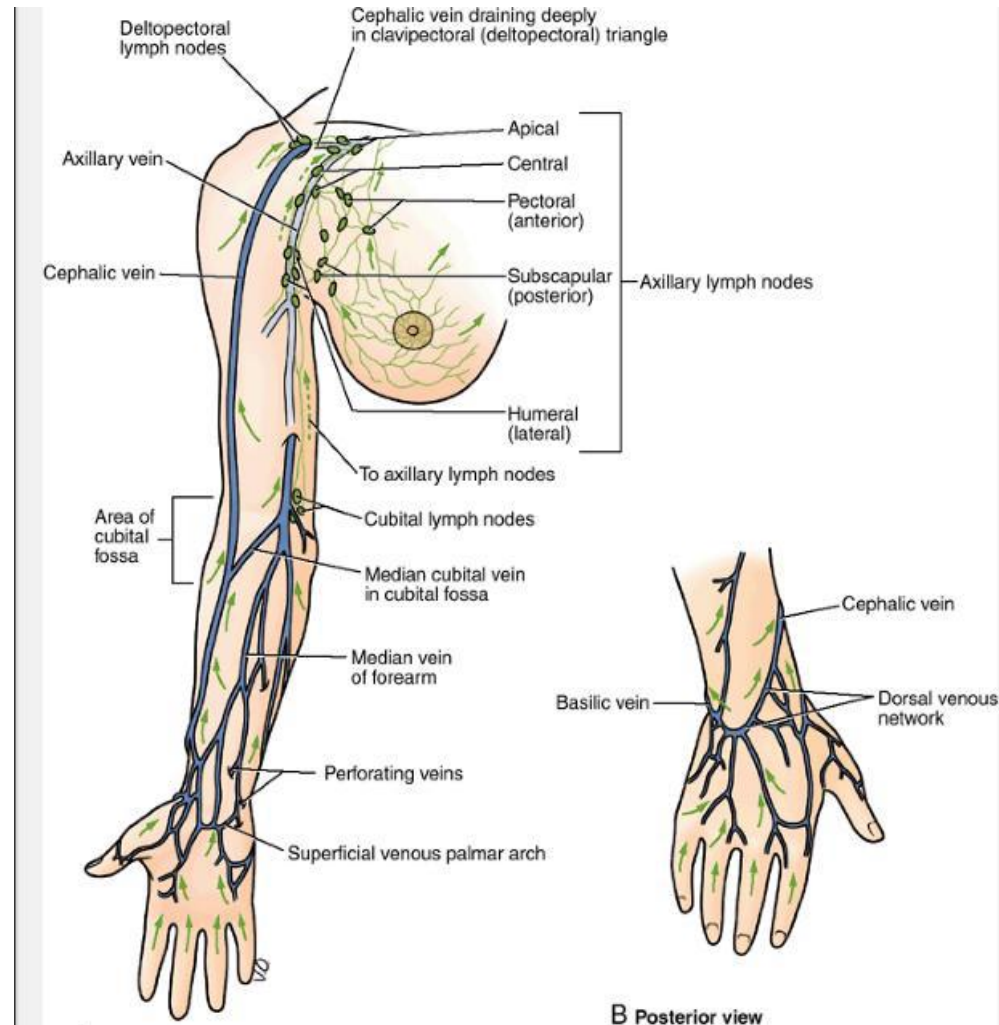
1. Superficial:- are more significant

Accompany superficial veins

- Most accompany cephalic veins & terminate in the apical axillary LN (most)

2. Deep:- less significant & course with deep veins

- Drains most of joints, bones nerves and muscles
- Terminate in the lateral axillary lymph nodes



Axioappendicular Muscles

Anterior Axioappendicular Muscles

- Four muscles that move the pectoral girdle
 - **Pectoralis major**
 - **Pectoralis minor**
 - **Subclavius**
 - **Serratus anterior**

Pectoralis major

- ✚ Fan-shaped
- ✚ Covers superior part of thorax
- ✚ Lateral border forms anterior wall of axilla (anterior axillary fold)
- ✚ With deltoid form deltopectoral groove
- ✚ Superiorly separate with deltoid and with clavicle form **deltopectoral triangle**
- ✚ Origin
 - **Clavicular head:** medial half of clavicle
 - **Sternocostal head:** sternum, superior six costal cartilages, aponeurosis of external oblique muscle

Insertion

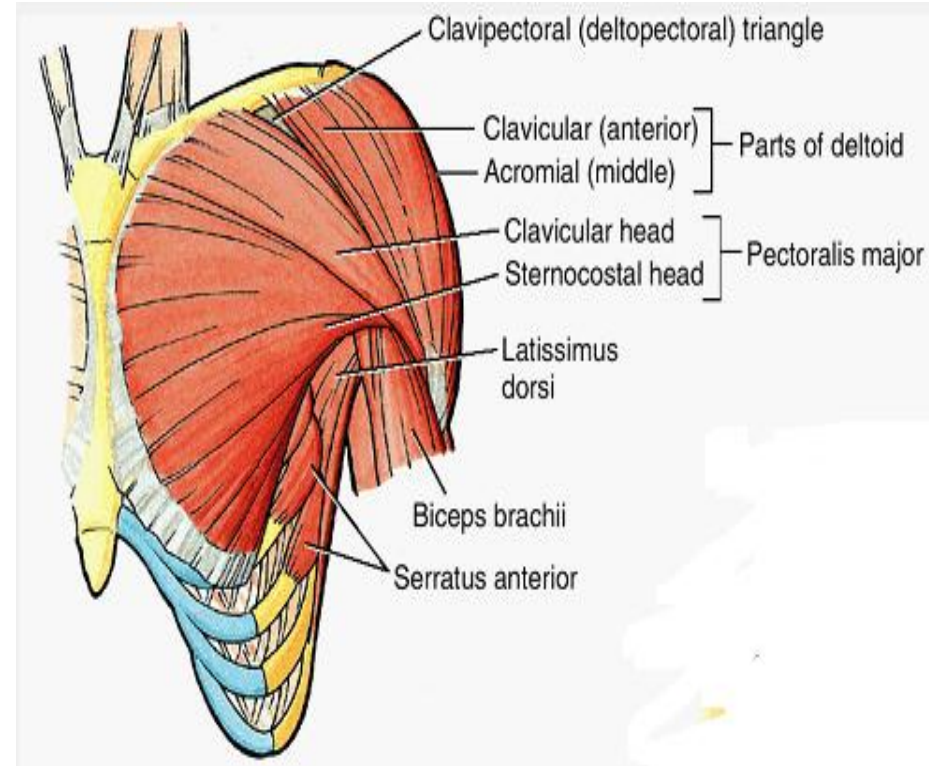
- Lateral lip of Intertubercular groove of humerus

Innervation

- Lateral and medial pectoral nerves

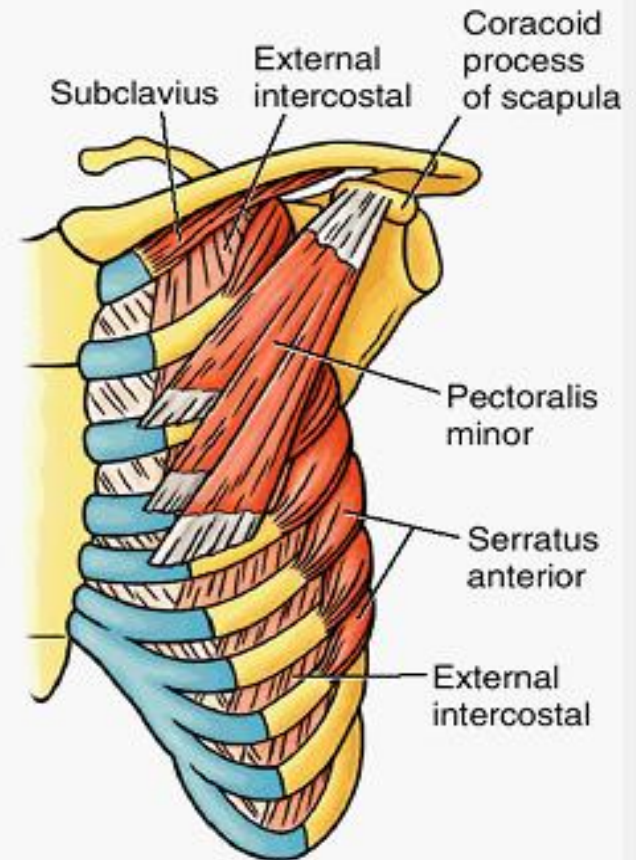
Action

- Adducts and medially rotates humerus; draws scapula anteriorly and inferiorly



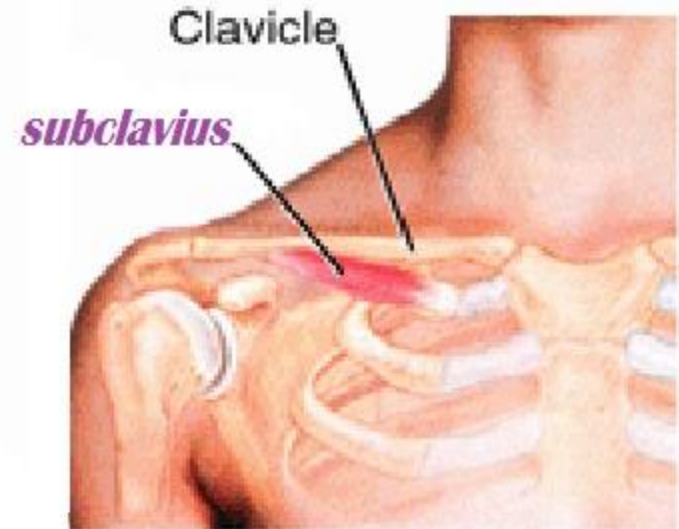
Pectoralis minor

- ✚ Triangular
- ✚ Lies in anterior wall of axilla
- ✚ Covered by pectoralis major
- ✚ Origin: 3rd to 5th ribs near their costal cartilages
- ✚ Insertion: coracoid process of scapula
- ✚ Innervation: **Medial pectoral nerve**
- ✚ Action:
 - Stabilizes scapula by drawing it inferiorly and anteriorly against thoracic wall
 - Elevates ribs in forced inspiration



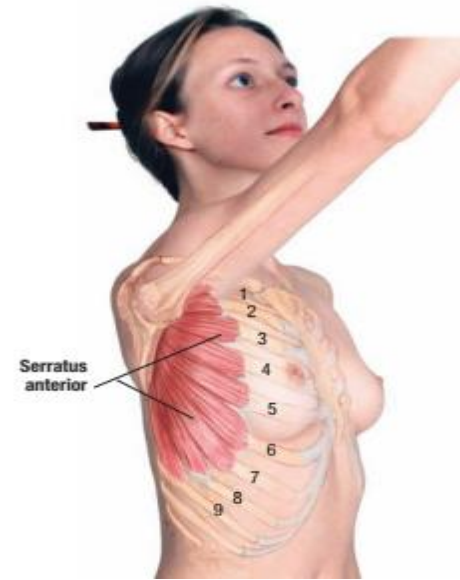
Subclavius

- ✦ Lies horizontally inferior to clavicle
- ✦ Small, round
- ✦ Origin: Junction of 1st rib and its costal cartilage
- ✦ Insertion: Inferior surface of middle third of clavicle
- ✦ **Innervation: Nerve to subclavius**
- ✦ **Action:** Anchors and depresses clavicle



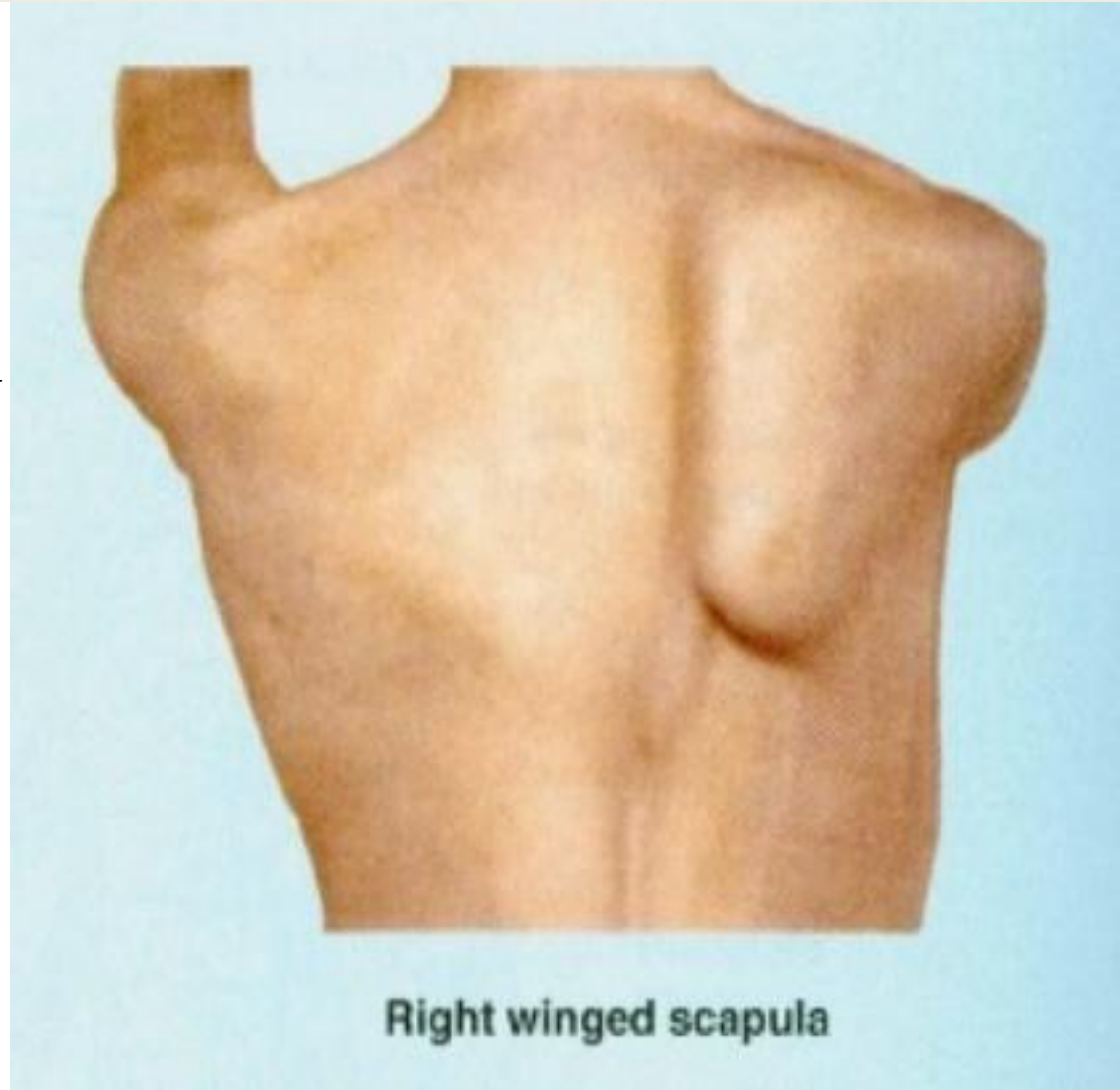
Serrates anterior

- ▶ Overlies lateral part of thorax
- ▶ Forms medial wall of axilla
- ▶ Anchors scapula, enabling other muscles to use it as a fixed bone for movement of humerus
- ▶ Origin: External surfaces of lateral parts of 1st to 8th ribs
- ▶ Insertion: Anterior surface of medial border of scapula
- ▶ **Innervation:** Long thoracic nerve
- ▶ Action: Protracts scapula and holds it against thoracic wall; rotates scapula



Paralysis of serratus anterior

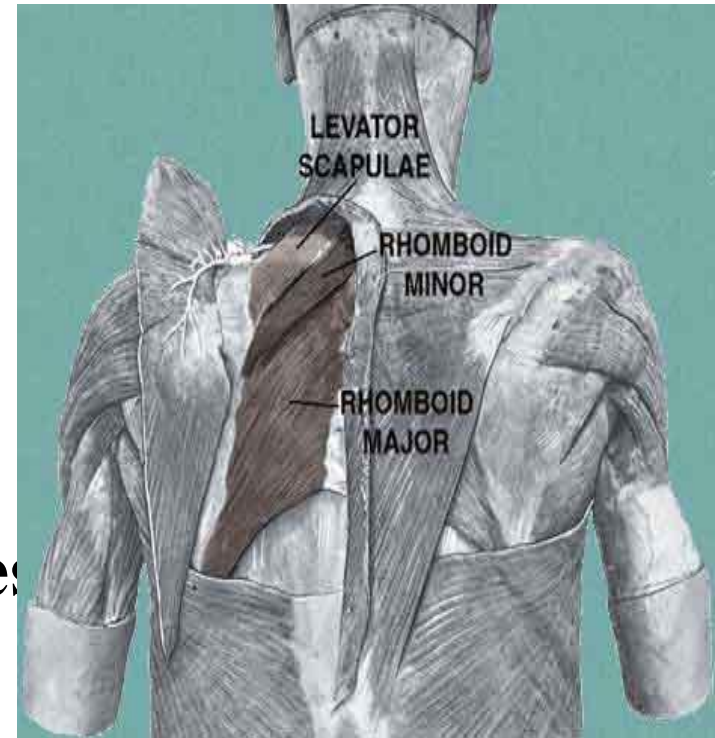
- **Due to injury of long thoracic nerve**
- Medial border of the scapula moves laterally and posteriorly away from thoracic wall
 - **winged scapula**
- Arm cannot be abducted above horizontal position



Right winged scapula

Posterior Axioappendicular Muscles

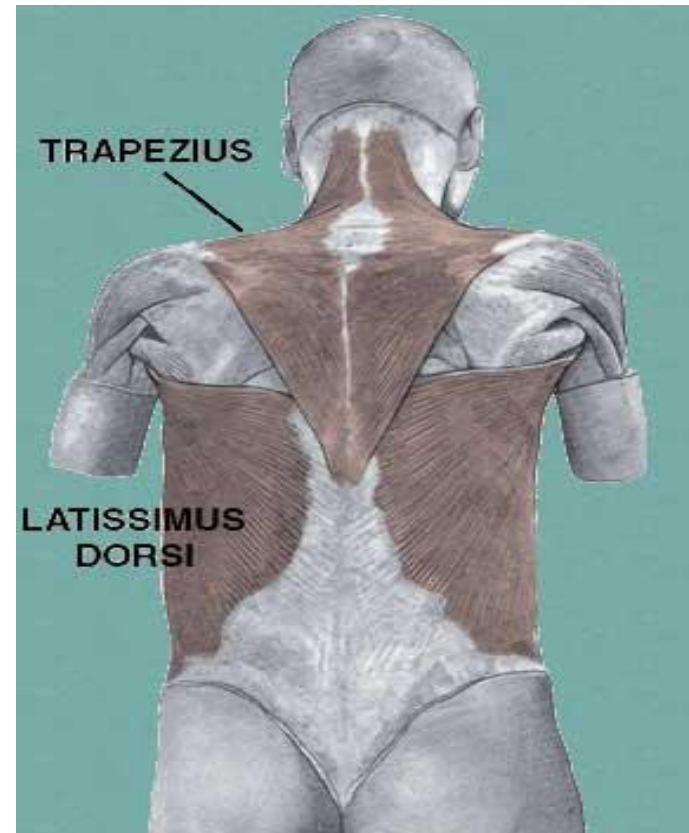
- ❖ Attach the superior appendicular skeleton to the axial skeleton
- ❖ **Divided into three groups**
 - **Superficial (extrinsic shoulder) muscles** : trapezius and latissimus dorsi
 - **Deep (extrinsic shoulder) muscles** : levator scapulae and rhomboids
 - **Scapulohumeral (intrinsic shoulder) muscles**: deltoid, teres major, and the four rotator cuff muscles (supraspinatus, infraspinatus, teres minor, and subscapularis)



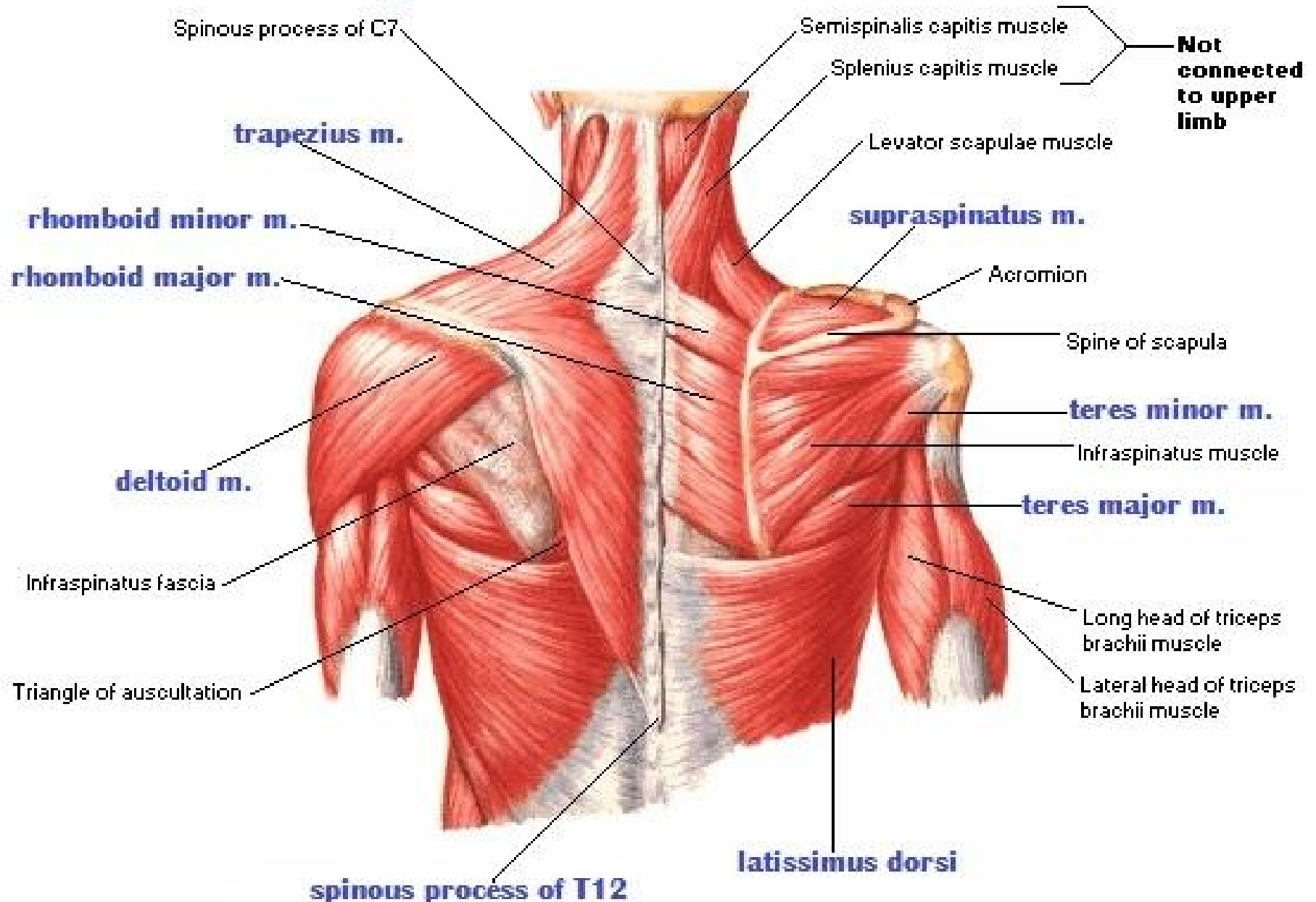
Superficial Muscles

Trapezius

- ✚ Provides direct attachment of pectoral girdle to trunk
- ✚ Covers posterior part of neck and superior half of trunk
- ✚ Attach pectoral girdle to cranium and vertebral column
- ✚ Fibers are divided into 3 parts
 - Superior part (descending fibers)
 - Middle part
 - Inferior part (ascending fibers)



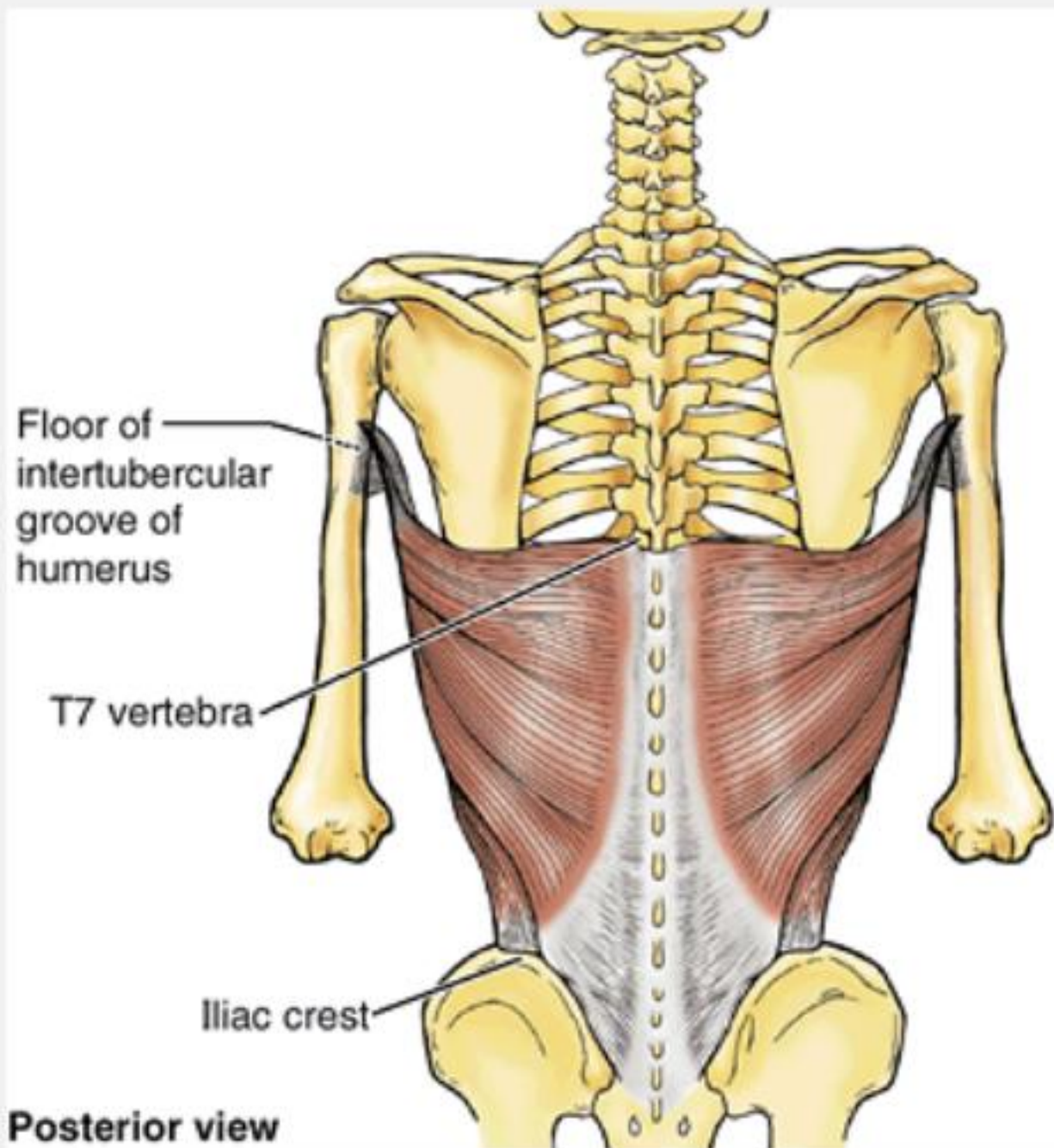
posterior View



- **Origin:** Medial third of **superior nuchal line**; **external occipital protuberance**; nuchal ligament; **spinous processes of C7 to T12 vertebrae**
- **Insertion:** Lateral third of clavicle; acromion and spine of scapula
- **Innervation:** Accessory nerve (CN XI) (motor fibers) and C3, C4 spinal nerves (pain and proprioceptive fibers)
- **Action:**
 - descending part elevates scapula
 - ascending part depresses scapula
 - middle part (or all parts together) retracts scapula
 - descending and ascending parts act together to rotate glenoid cavity superiorly

Latissimus dorsi

- Fan-shaped
- Covers wide area of back
- Passes from trunk to humerus
- Raises trunk to the arm
- Origin: Spinous processes of inferior 6 thoracic vertebrae, thoracolumbar fascia, iliac crest, and inferior 3 or 4 ribs
- **Insertion:** intertubercular groove of humerus
- **Innervation:** Thoracodorsal nerve
- Action: Extends, adducts, and medially rotates humerus; raises body toward arms during climbing



Deep Muscles

Levator scapulae

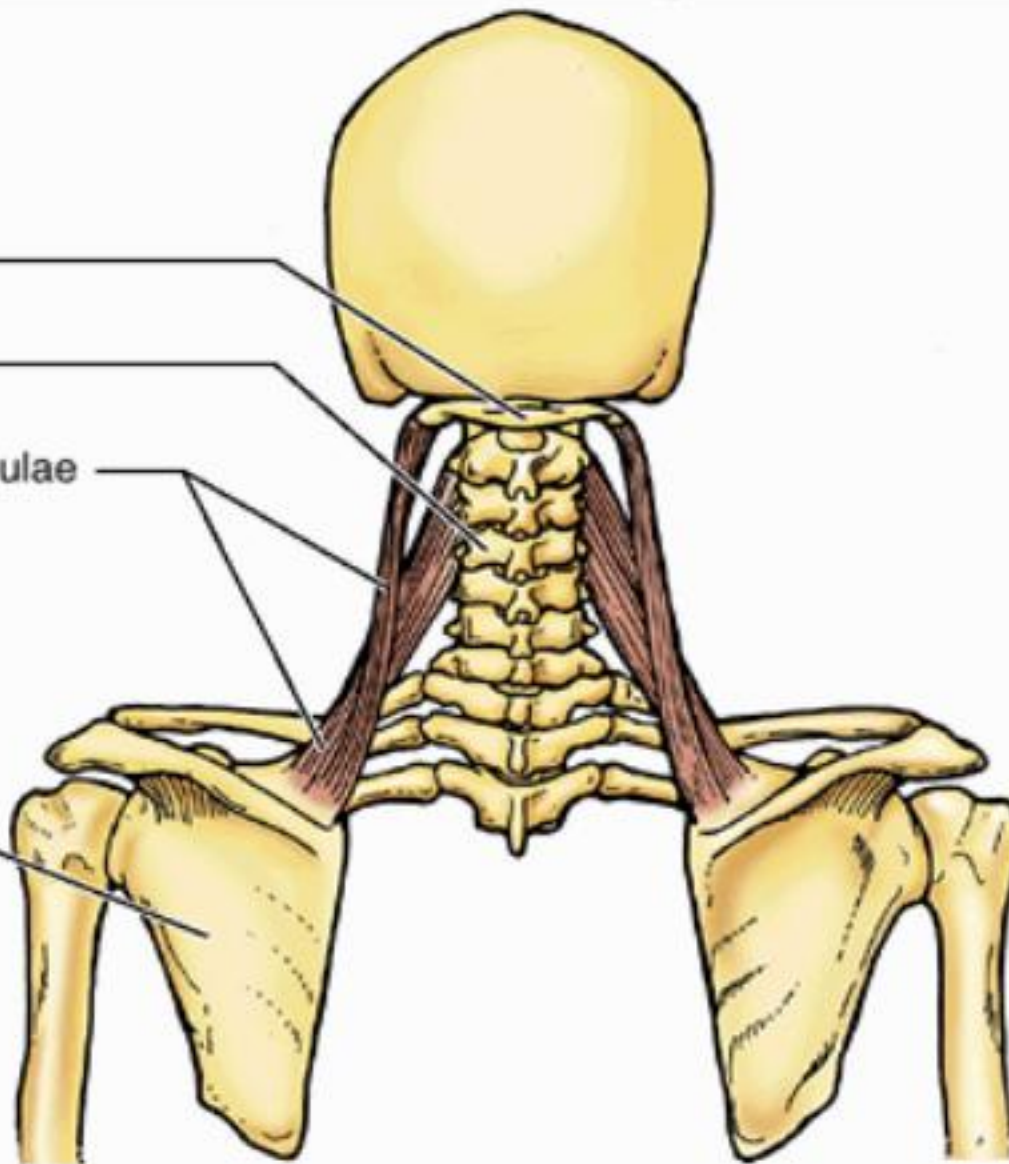
- Lies deep to **SCM** and **trapezius**
- **Origin:** Transverse processes of C1 to C4 vertebrae
- **Insertion:** Medial border of scapula superior to root of spine
- **Innervation:** Dorsal scapular (C5) and cervical (C3, C4) nerves
- **Action:**
 - Elevates scapula and tilts its glenoid cavity inferiorly by rotating scapula with the rhomboid and pectoralis minor
 - Acting **bilaterally**, extend the neck
 - Acting **unilaterally**, contribute to lateral flexion of the neck

C1 vertebra

C4 vertebra

Levator scapulae

Scapula

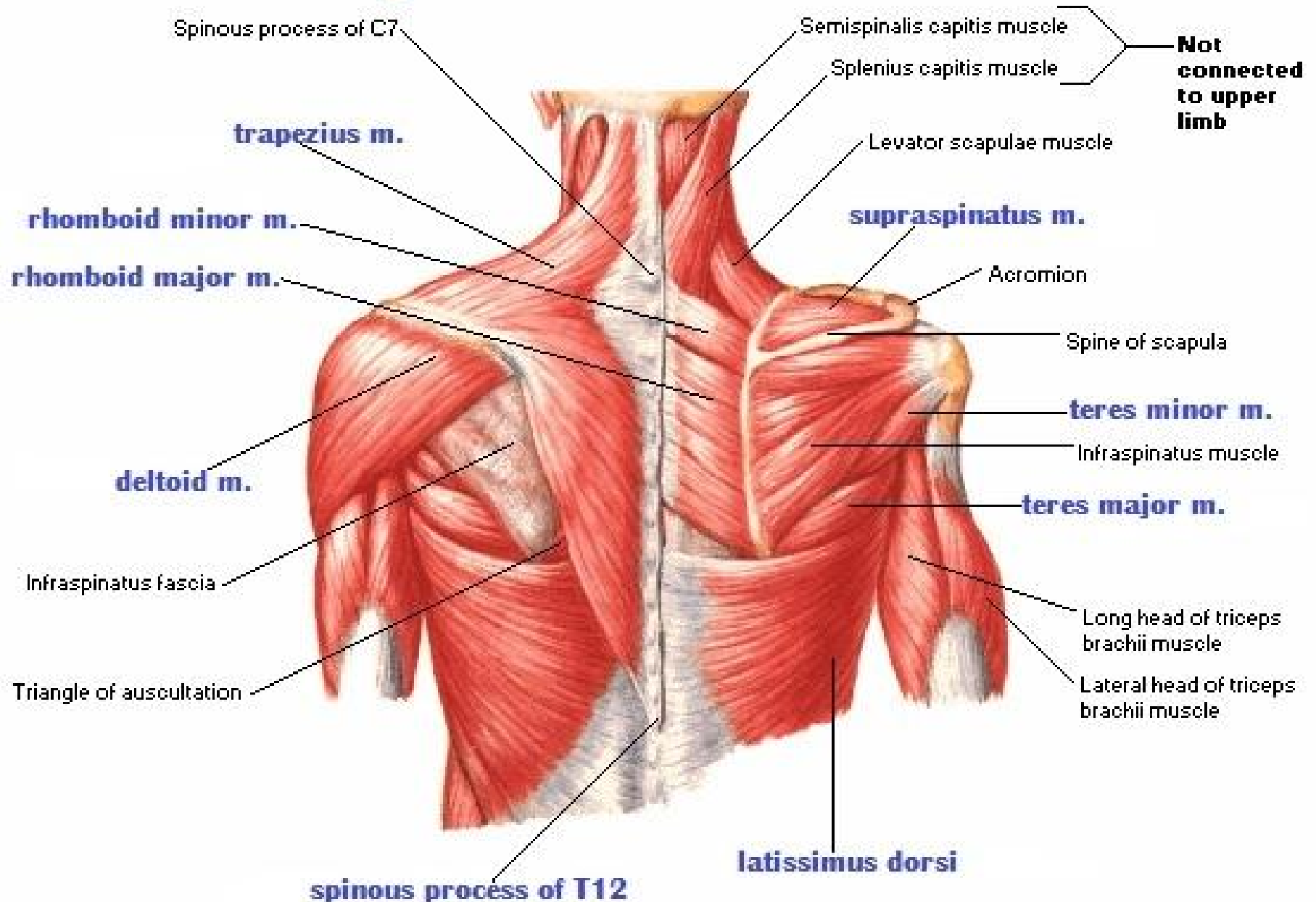


Posterior view

Rhomboids (major and minor)

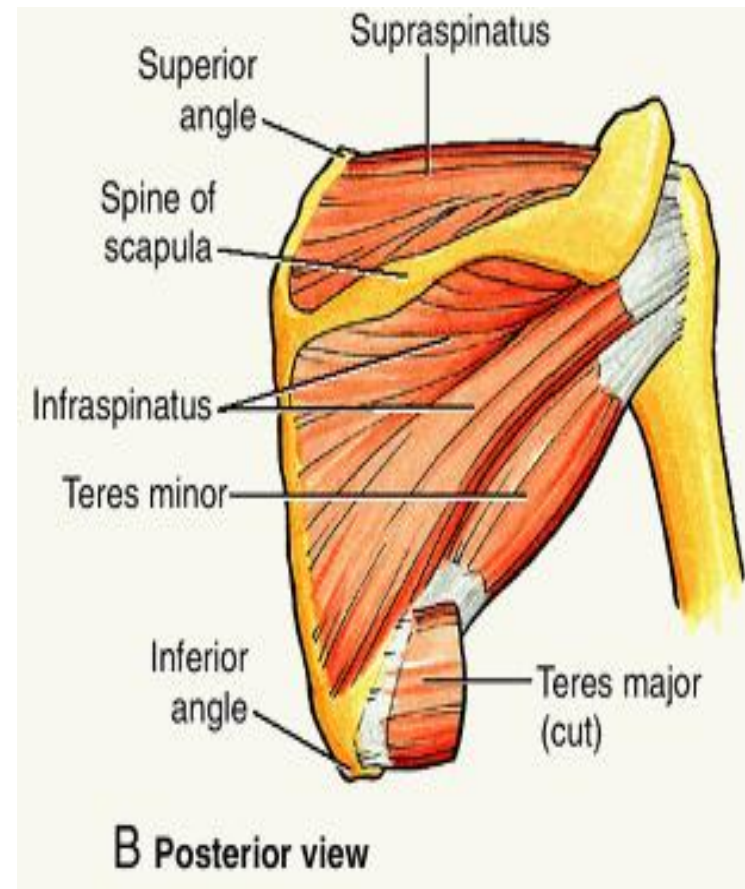
- Lie deep to trapezius
- Form parallel bands that pass inferolaterally from vertebrae to medial border of the scapula
- **Origin:**
 - Minor: nuchal ligament; spinous processes of C7 and T1 vertebrae
 - Major: spinous processes of T2 to T5 vertebrae
- **Insertion:**
 - Minor: medial end of scapular spine
 - Major: medial border of scapula from level of spine to inferior angle
- **Innervation: Dorsal scapular nerve**
- **Action:**
 - Retract scapula and rotate it to depress glenoid cavity
 - fix scapula to thoracic wall during movements of upper limb

posterior View



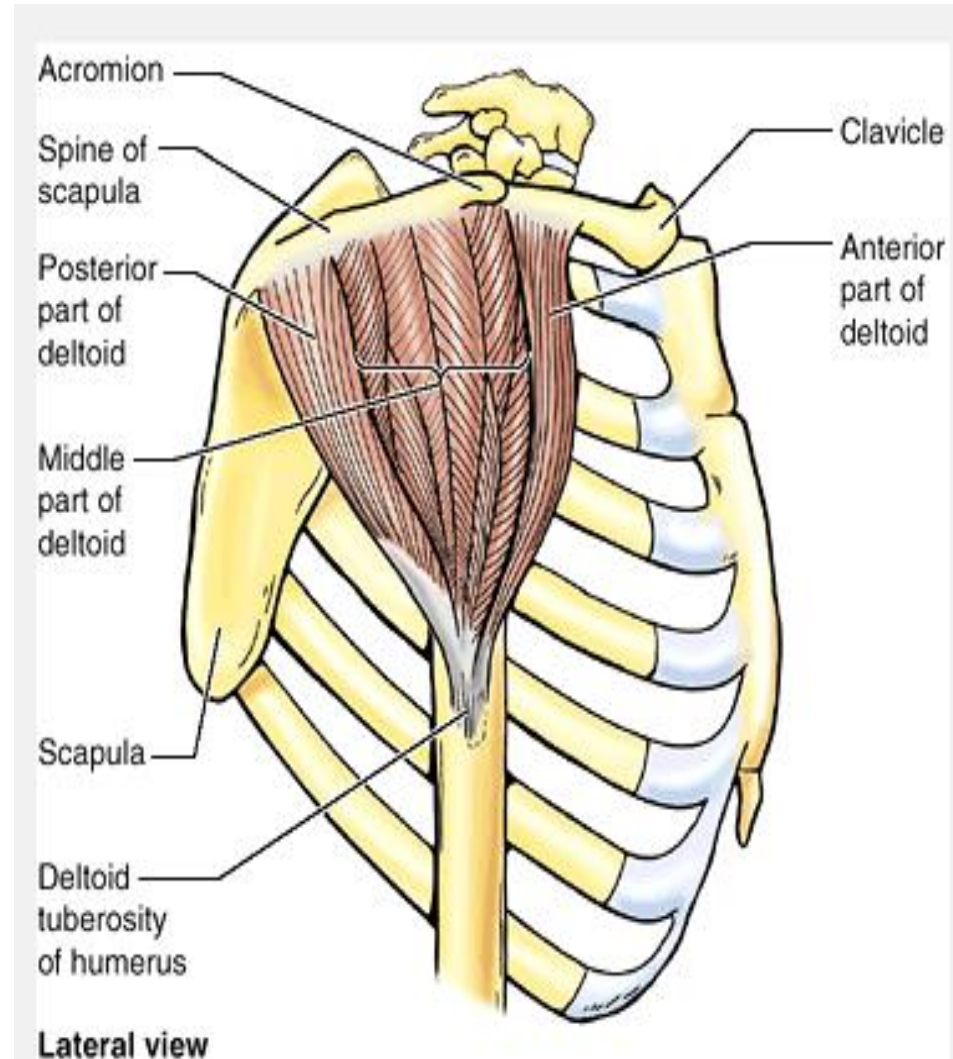
Scapulohumeral (intrinsic) Muscles

- Six muscles
 - **Deltoid**
 - **Teres major**
 - **Supraspinatus**
 - **Infraspinatus**
 - **Subscapularis**
 - **Teres minor**
- Short muscles that pass from scapula to humerus
- Act on **glenohumeral joint**



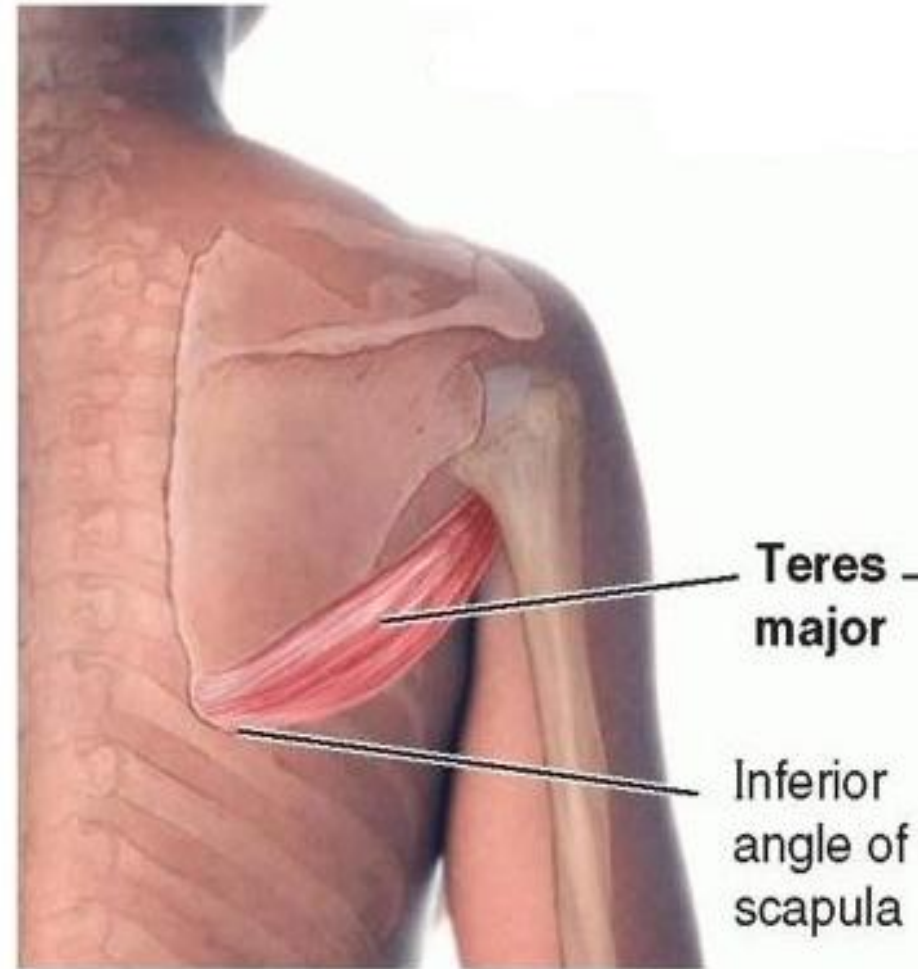
Deltoid muscle

- **Origin:** Lateral third of clavicle; acromion and spine of scapula
- **Insertion:** Deltoid tuberosity of humerus
- **Innervation:** *Axillary nerve*
- **Action:**
 - **Anterior part:** flexes and medially rotates arm
 - **Middle part:** abducts arm
 - **Posterior part:** extends and laterally rotates arm



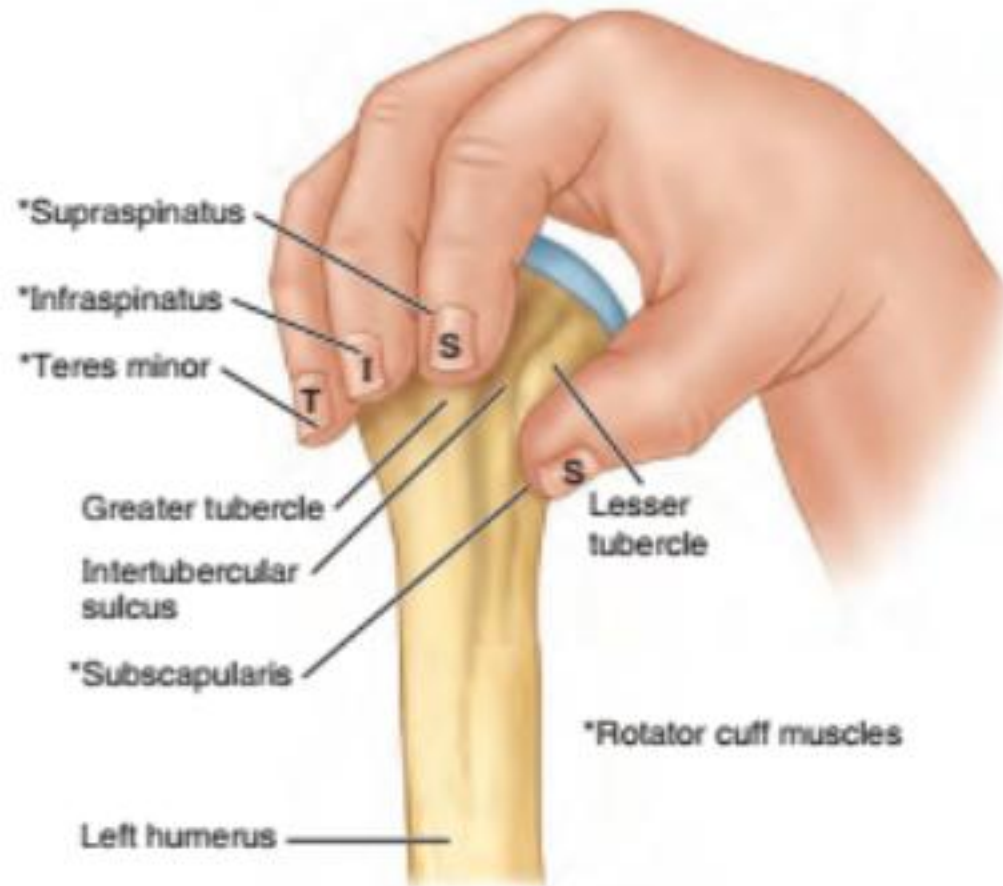
Teres major

- Thick rounded muscle that lies on the inferolateral third of the scapula
- **Origin:** Posterior surface of inferior angle of scapula
- **Insertion:** intertubercular groove of humerus
- **Innervation:** Lower subscapular nerve
- **Action:** Adducts and medially rotates arm



Rotator cuff muscles

- Four of the scapulohumeral muscles called rotator cuff because they form a musculotendinous cuff around glenohumeral joint
- All except the **supraspinatus** are rotators of the humerus
- The supraspinatus initiates and assists the deltoid in the abduction of the arm



Supraspinatus

- Origin: Supraspinous fossa of scapula
- Insertion: greater tubercle of humerus
- **Innervation:** Suprascapular nerve
- Action: Initiates and assists deltoid in abduction of arm and acts with rotator cuff muscles

Infraspinatus

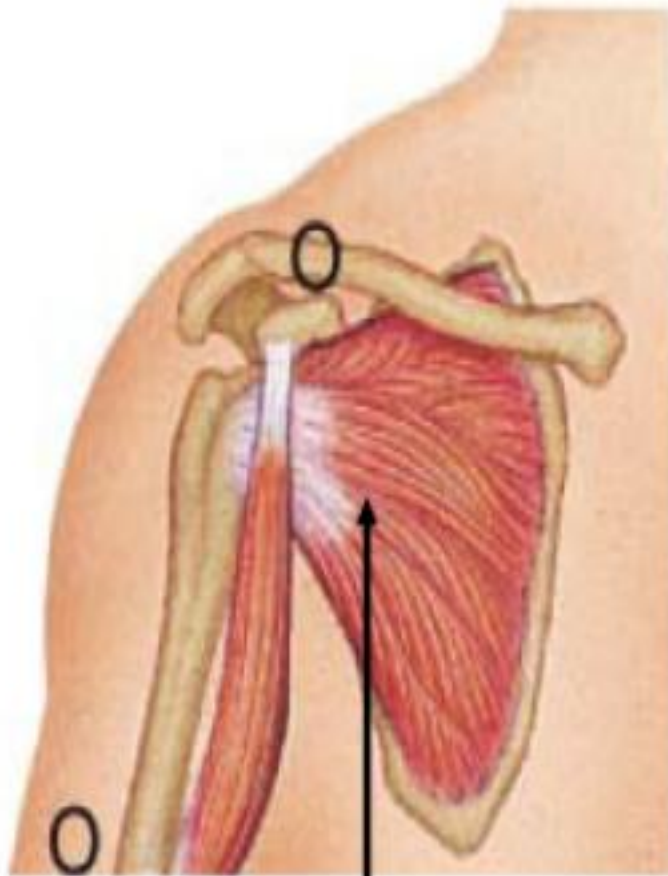
- Origin: Infraspinous fossa of scapula
- Insertion: greater tubercle of humerus
- **Innervation:** Suprascapular nerve
- Action: Laterally rotate arm; help hold humeral head in glenoid cavity of scapula

Teres minor

- Origin: Middle part of lateral border of scapula
- Insertion: greater tubercle of humerus
- **Innervation:** Axillary nerve
- Action: Laterally rotate arm; help hold humeral head in glenoid cavity of scapula

Subscapularis

- Origin: Subscapular fossa
- Insertion: Lesser tubercle of humerus
- **Innervation:** Upper and lower subscapular nerves
- Action: Medially rotates and adduct arm; helps hold humeral head in glenoid cavity



● Subscapularis



Abductor

Supraspinatus ●

Infraspinatus ●

Teres minor ●

Adductors

Injury to axillary nerve

- Occur when surgical neck of humerus is fractured
- Results in **atrophy of deltoid**
- Rounded contour of shoulder disappears
- To test deltoid (function of axillary nerve) the arm is abducted against resistance

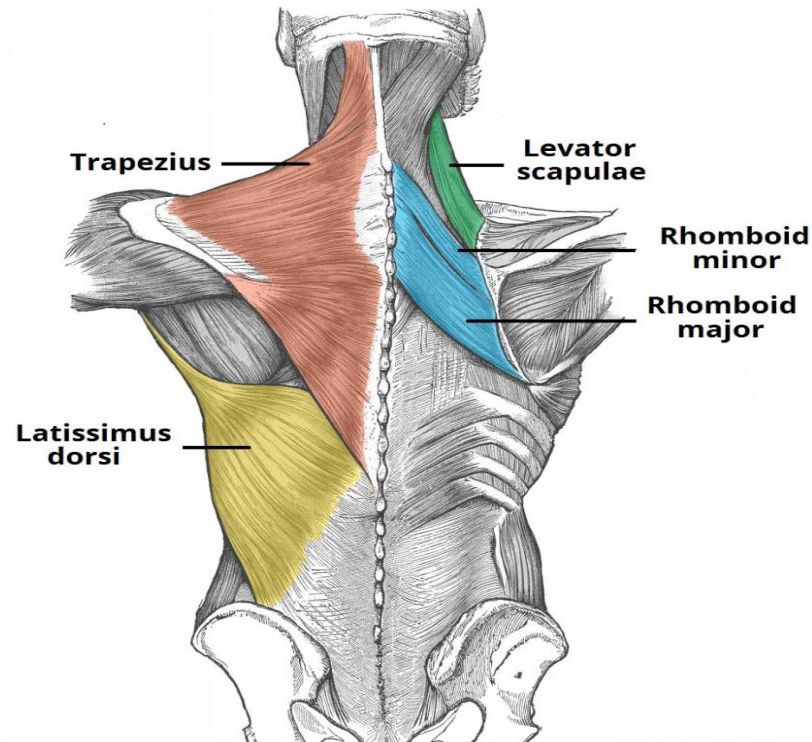
Rotator cuff injuries

- Produce instability of glenohumeral joint
- Rapture of supraspinous tendon is the **most common injury**

Surface anatomy

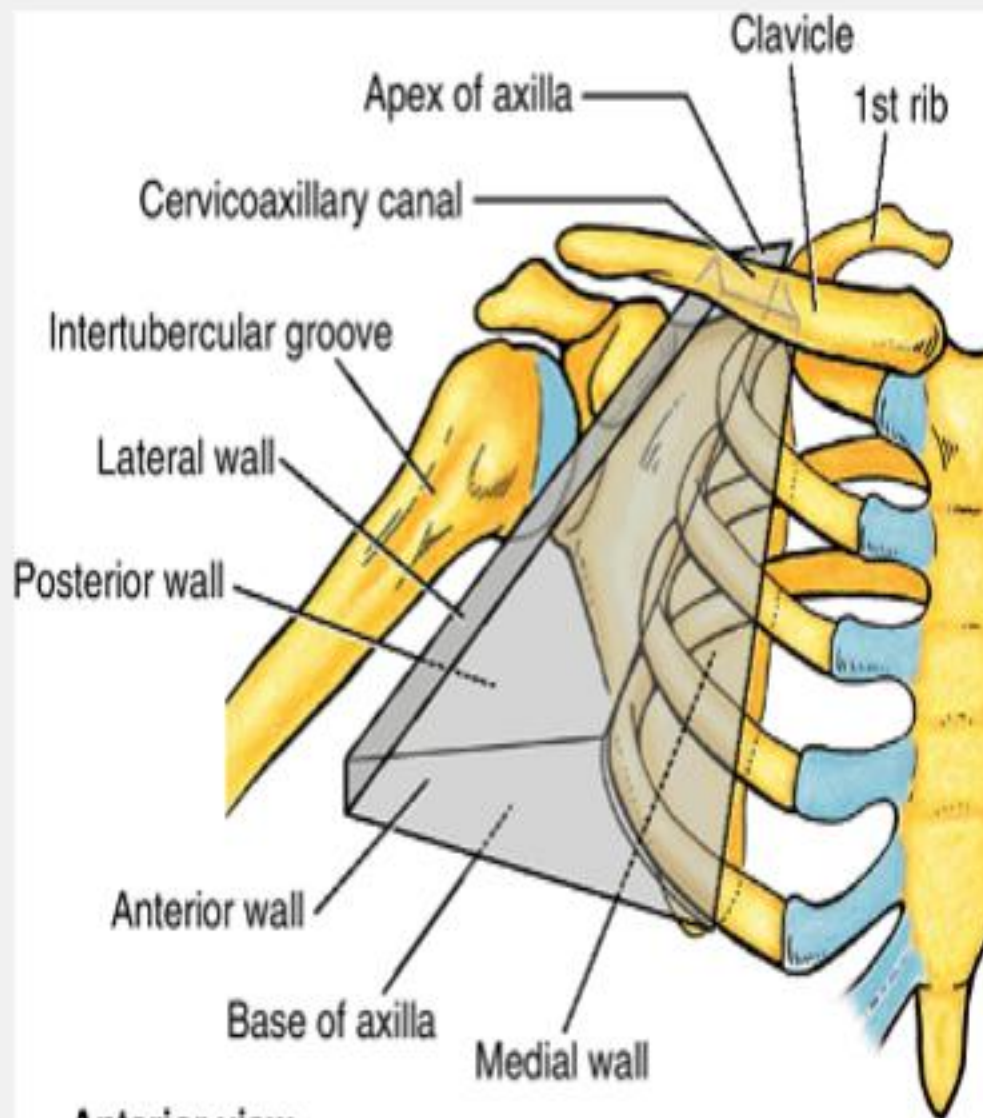
- **Triangle of auscultation**

- ✚ The area formed by the superior border of latissimus dorsi, the medial border of the scapula, and the inferolateral border of the trapezius
- ✚ This gap in the thick back musculature is a good place to examine posterior segments of the lungs with a **stethoscope**

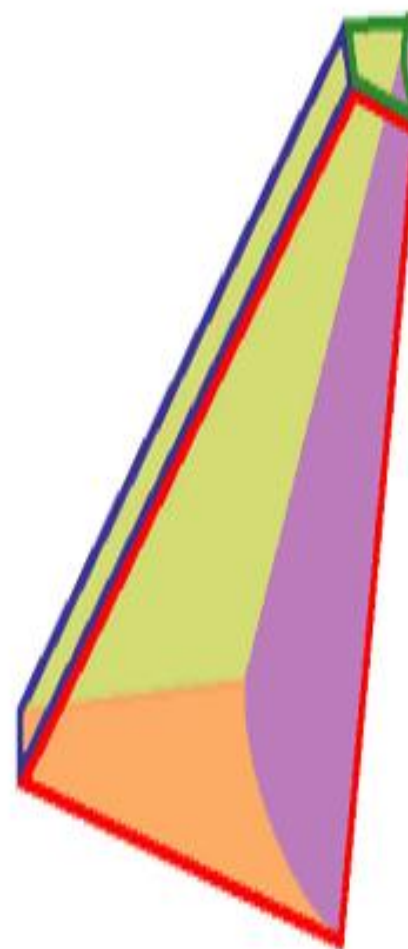


Axilla

- Pyramidal space inferior to glenohumeral joint and superior to skin and axillary fascia at junction of arm and thorax
- Provides a passageway for vessels and nerves to and from upper limb
- **Has apex, base and 4 walls**
- **Apex**
 - Is cervicoaxillary canal: passageway between neck and axilla
 - **Bounded** by 1st rib, clavicle and superior edge of scapula
- **Base**
 - Formed by concave skin, subcutaneous tissue, and axillary fascia
 - Axillary fossa (armpit)



Anterior view



Axillary boundaries

- Apex
- Base
- Anterior wall
- Lateral wall
- Medial wall
- Posterior wall

Axilla: walls

- **Anterior wall**

- Formed by pectoralis major and minor with their fascia
- Anterior axillary fold is inferior most part of this wall

- **Posterior wall**

- Formed by scapula and subscapularis and inferiorly by the teres major and latissimus dorsi
- Posterior axillary fold is the inferior most part of this wall

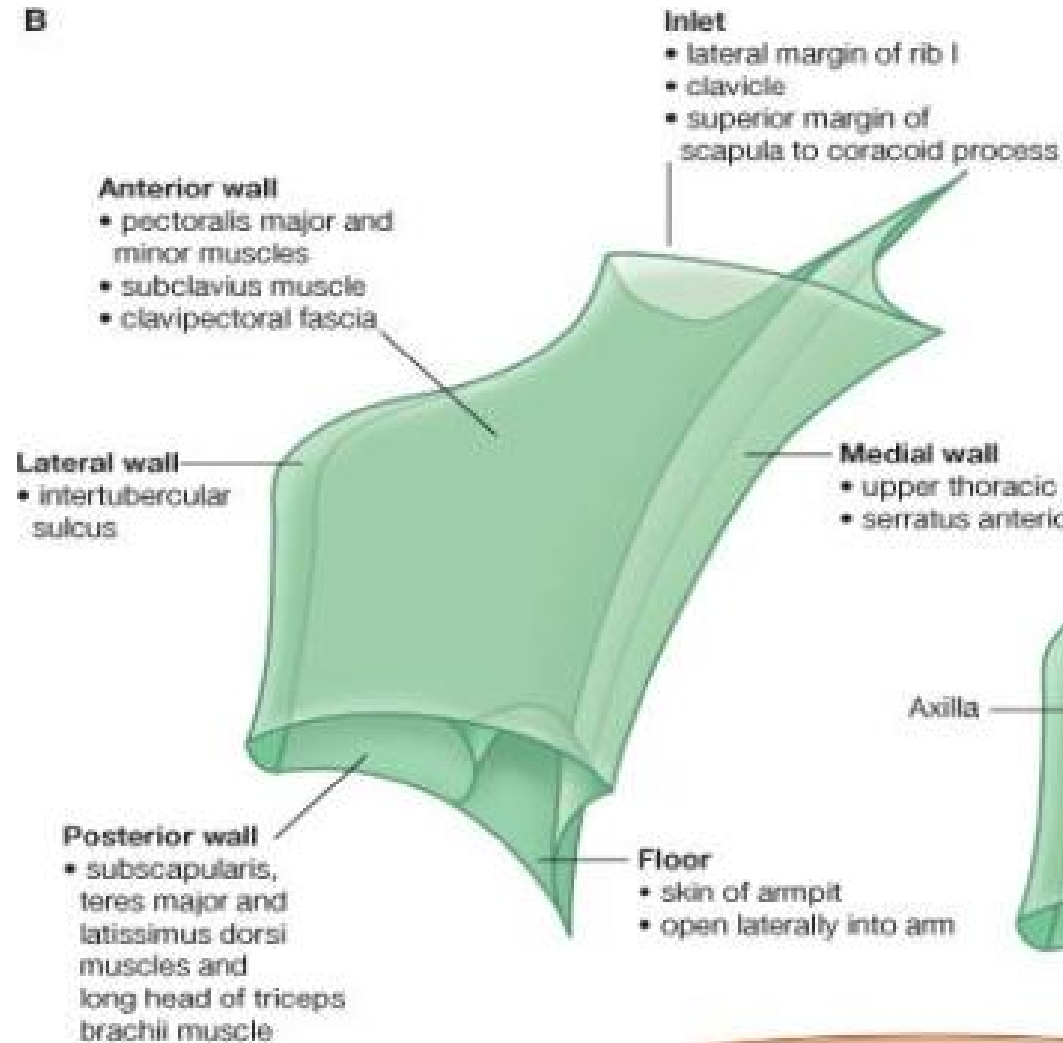
- **Medial wall**

- Formed by the thoracic wall (1st -4th ribs and intercostal muscles) and serratus anterior

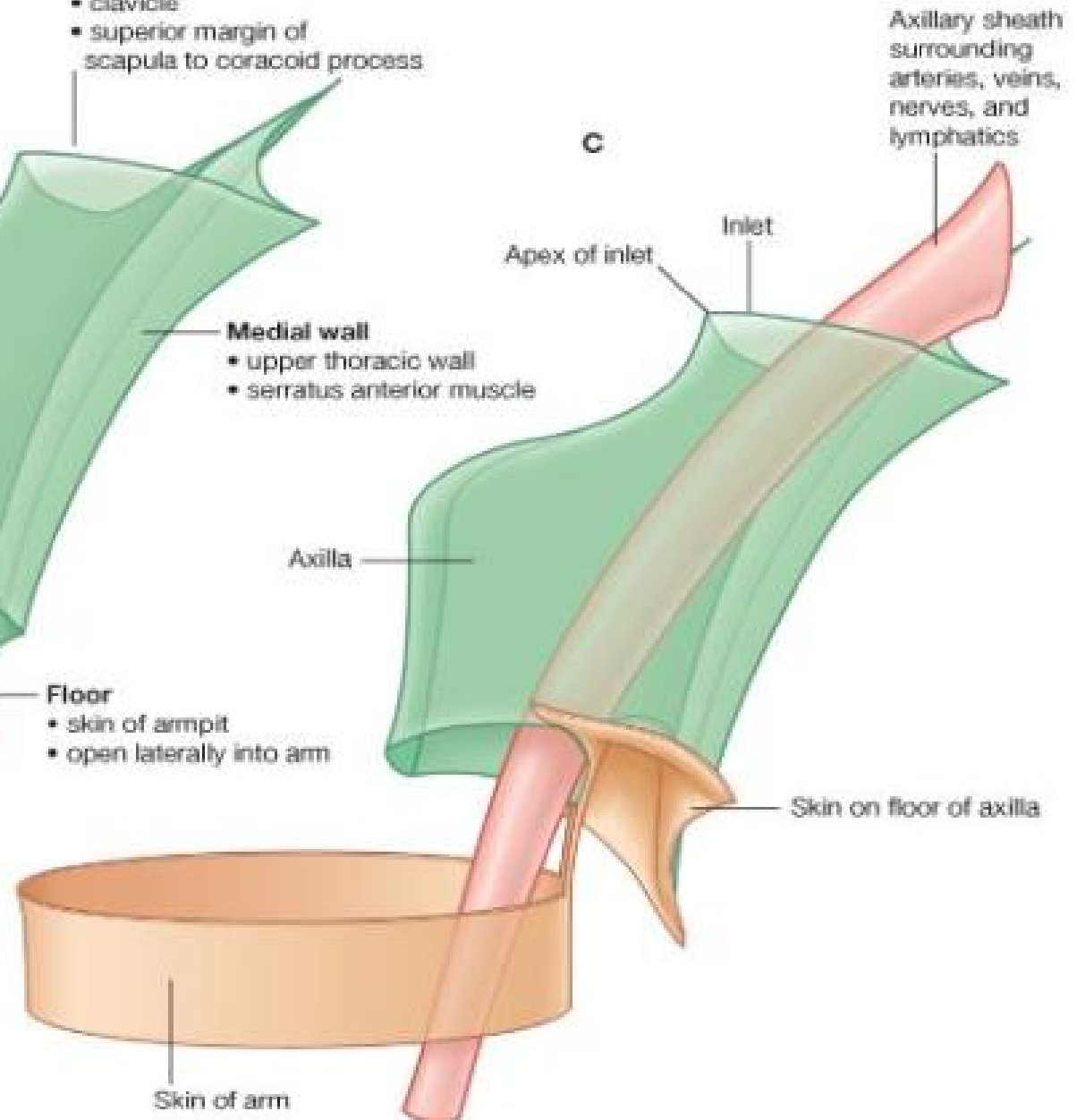
- **Lateral wall**

- Narrow bony wall, formed by humerus

B



C



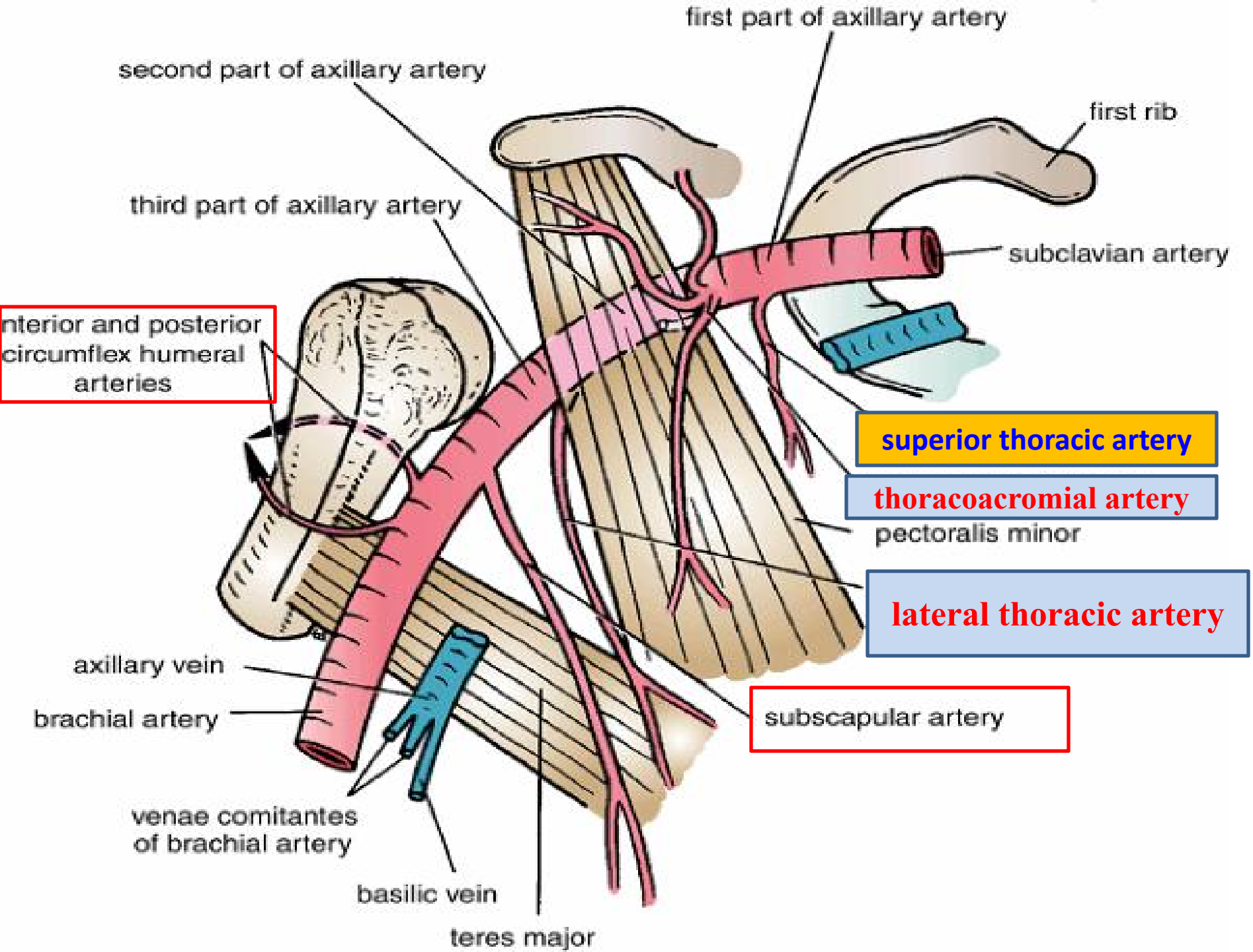
Axilla: content

The axilla contains:

- ✚ Axillary artery and its branches
- ✚ Axillary vein and its tributaries
- ✚ Nerves of the cords and branches of the brachial plexus
- ✚ Lymphatic vessels
- ✚ Groups of axillary lymph nodes

Axillary Artery

- Begins at the lateral border of 1st rib as the continuation of **subclavian artery**
- Ends at inferior border of **teres major**, becomes brachial artery
- Passes posterior to the pectoralis minor into the arm
- It is divided into three parts relative to **pectoralis minor**



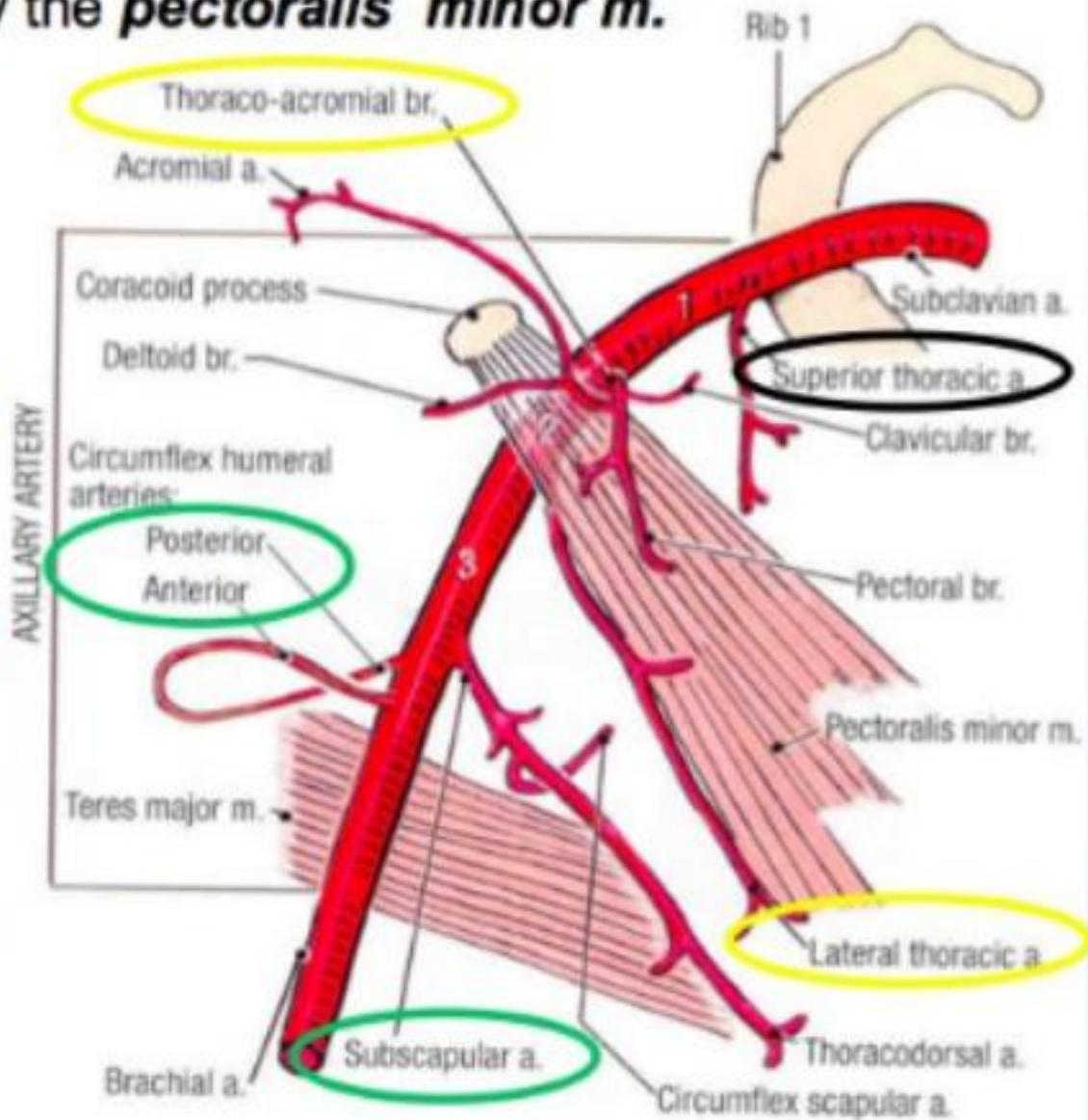
Axillary Artery: parts

- **First part**
 - located between lateral border of 1st rib and medial border of pectoralis minor
 - enclosed in axillary sheath
 - has one branch: **superior thoracic artery**
- **Second part**
 - lies posterior to pectoralis minor
 - has two branches: **thoracoacromial artery and lateral thoracic artery**
- **Third part**
 - extends from lateral border of pectoralis minor to the inferior border of teres major
 - has three branches: **subscapular artery, anterior circumflex humeral artery and posterior circumflex humeral artery**

Axilla: Axillary Artery

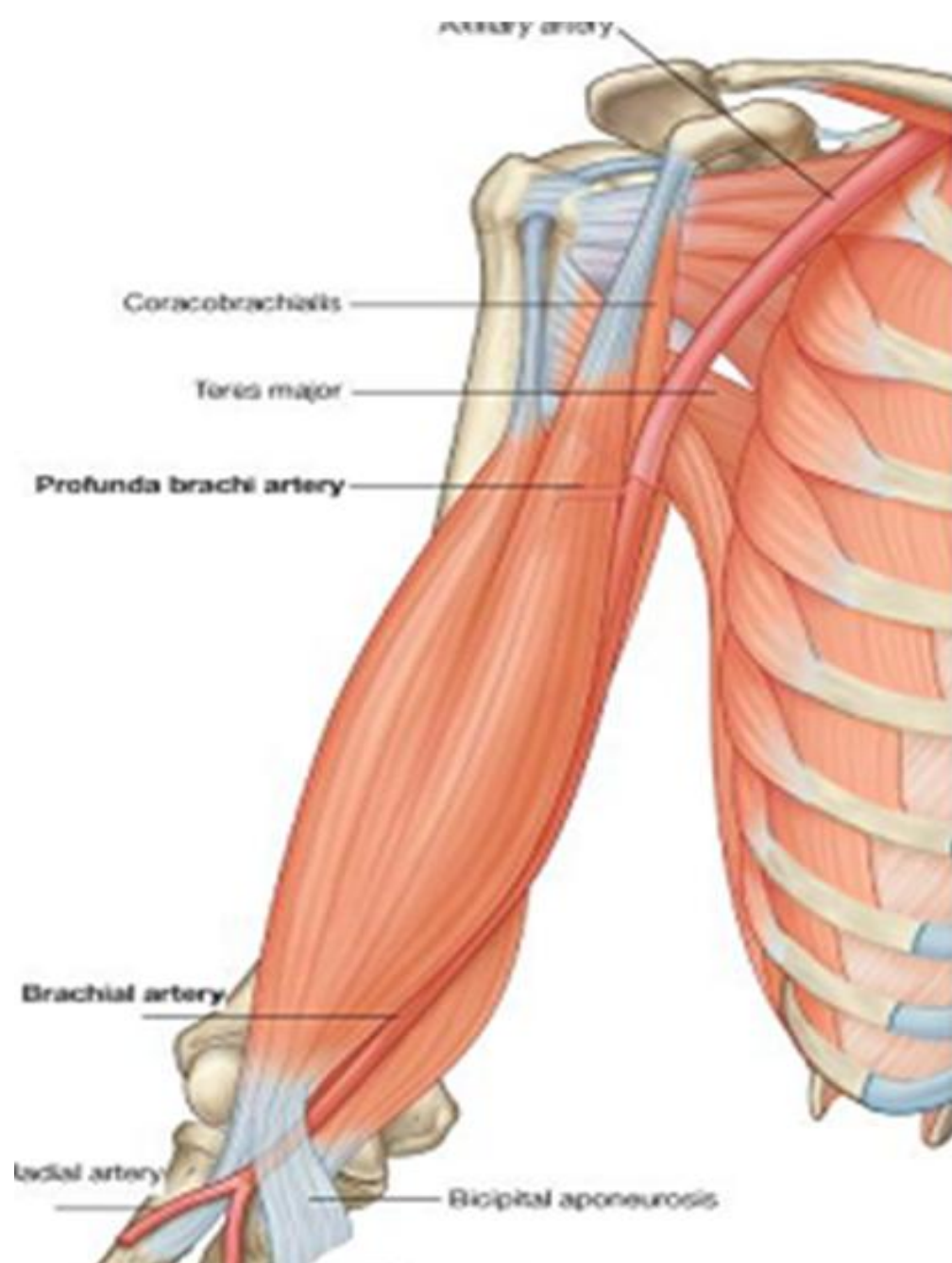
3 segments, defined by the *pectoralis minor m.*

- **1st segment** = 1 branch:
 - superior thoracic a.
(supplies 1st 2 intercostal spaces)
- **2nd segment** = 2 branches:
 - thoracoacromial a.
 - lateral thoracic a.
- **3rd segment** = 3 branches:
 - subscapular a.
 - circumflex scapular a
 - thoracodorsal a.
 - post. humeral circumflex a.
 - ant. humeral circumflex a.



Compression of axillary artery

- Compression of the third part of axillary artery against the humerus may be necessary when profuse bleeding occurs
- If compression is required at a more proximal site, the axillary artery can be compressed at its origin at the lateral border of the 1st rib by exerting downward pressure in the angle between the clavicle and the attachment of the SCM

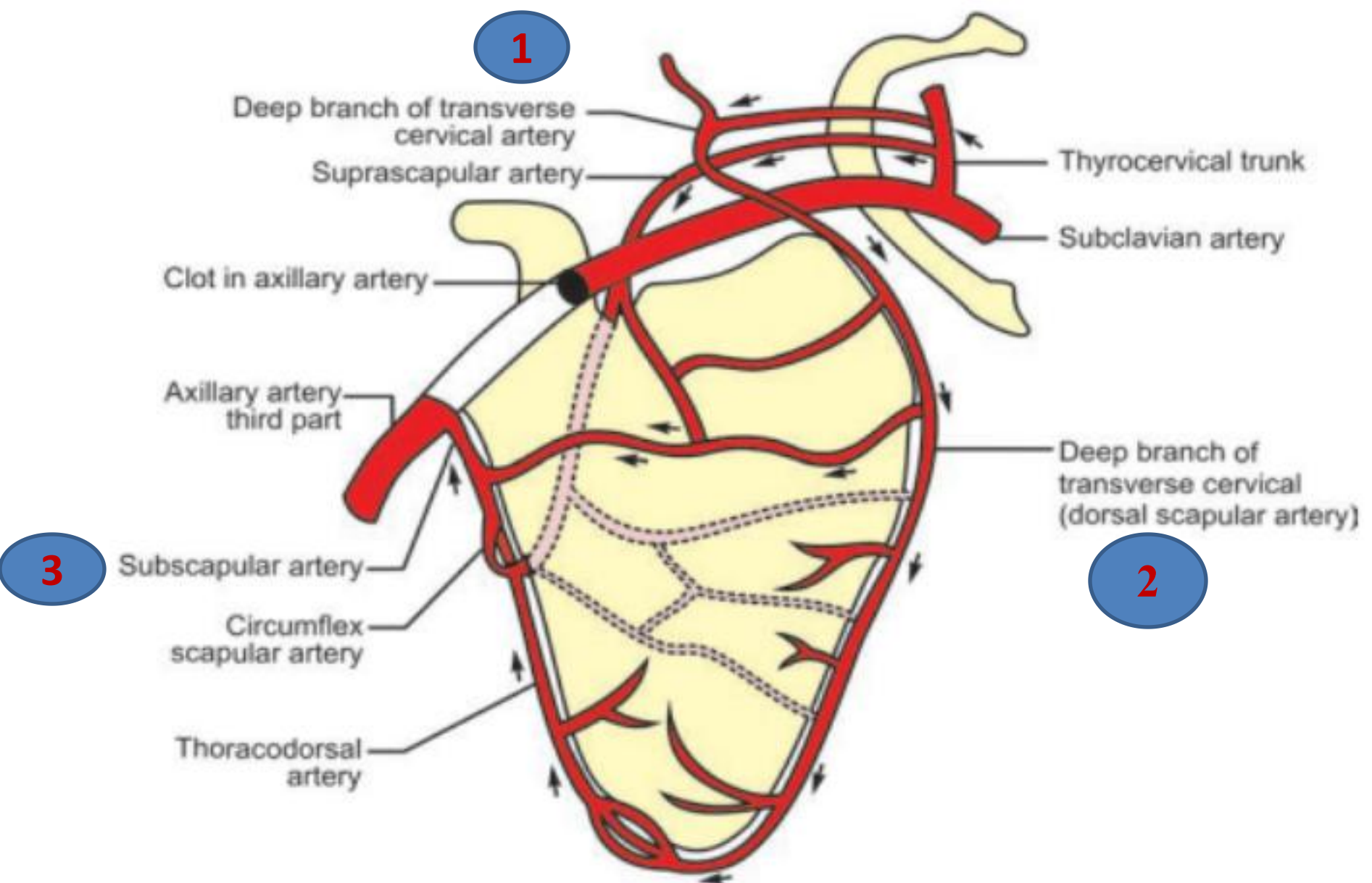


Arterial anastomoses around scapula

- Several arteries join to form networks on anterior and posterior surfaces of the scapula
 - Dorsal scapular artery
 - Suprascapular artery
 - Subscapular artery (via the posterior circumflex scapular branch)
- The collateral circulation made possible by these anastomoses is important when ligation of a lacerated subclavian or axillary artery is necessary or vascular stenosis (narrowing) of the axillary artery occurs
- In either case, the direction of blood flow in the subscapular artery is reversed, enabling blood to reach the third part of the axillary artery

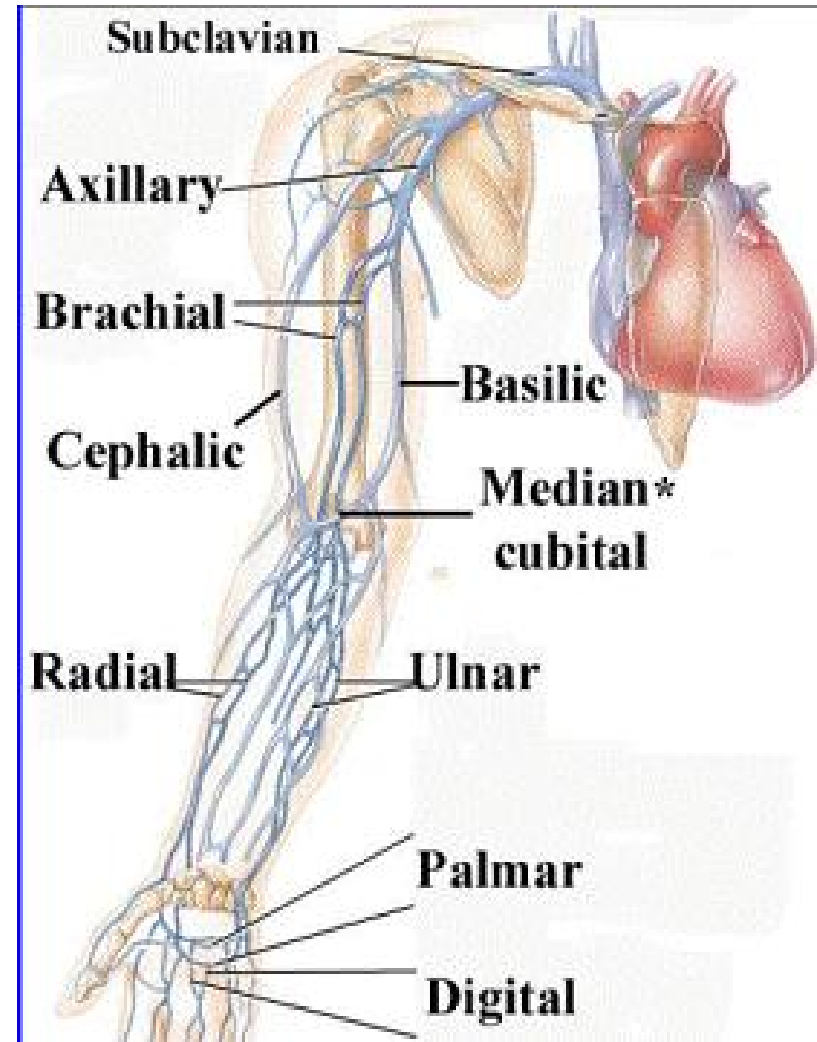
Arterial anastomoses around scapula

- **Subscapular artery** receives blood through several anastomoses with the **suprascapular artery**, **transverse cervical artery**, and **intercostal arteries**
- **Slow occlusion** of an artery (resulting from disease) often enables sufficient collateral circulation to develop, preventing ischemia (deficiency of blood)
- **Sudden occlusion** usually does not allow sufficient time for adequate collateral circulation to develop; as a result, ischemia of the upper limb may occur



Axillary vein

- ✚ Lies initially (distally) on the anteromedial side of the axillary artery
- ✚ Formed by the union of **brachial veins** and **basilic vein** at the inferior border of teres major
- ✚ Ends at lateral border of 1st rib where it becomes **subclavian vein**
- ✚ **Wounds in the axilla** often involve the axillary vein because of its large size and exposed position

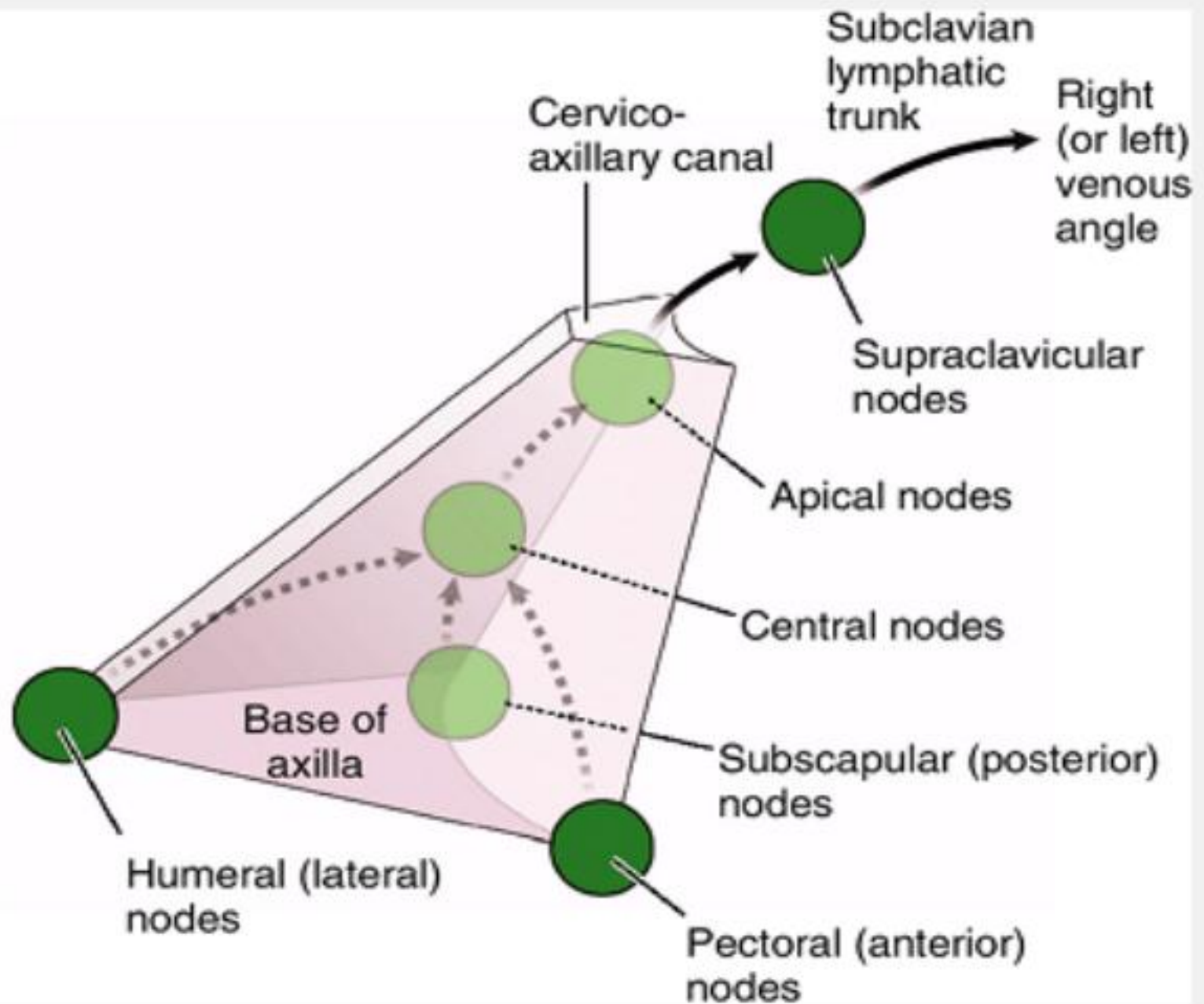


Axillary lymph nodes

- There are 5 principal groups of axillary lymph nodes: **pectoral**, **subscapular**, **humeral**, **central**, and **apical**

● **Pectoral (anterior) nodes**

- lie along the medial wall of the axilla, around the lateral thoracic vein and inferior border of the pectoralis minor
- receives lymph mainly from **anterior thoracic wall**, including most of the **breast**



● Subscapular (posterior) nodes

- lie along the posterior axillary fold and subscapular blood vessels
- receive lymph from posterior aspect of **thoracic wall** and **scapular region**

● Humeral (lateral) nodes

- lie along the lateral wall of the axilla, medial and posterior to the axillary vein
- receive nearly **all the lymph from the upper limb**, **except** that carried by lymphatic vessels accompanying the **cephalic vein** (drain to the **apical axillary** and **infraclavicular nodes**)

Central nodes

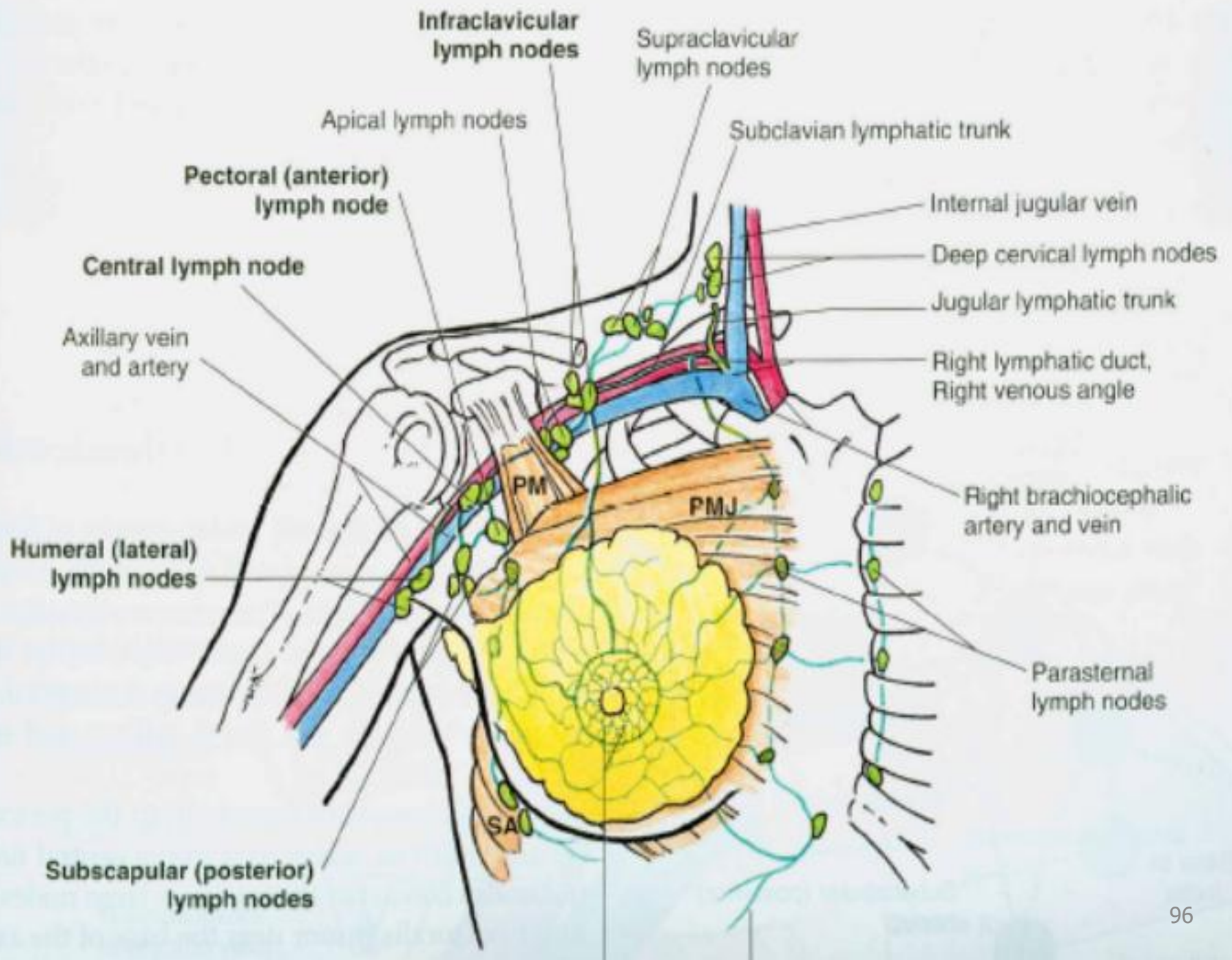
- situated deep to the pectoralis minor, in association with the second part of the axillary artery
- efferent lymphatic vessels from the pectoral, subscapular and humeral nodes pass to the central nodes
- efferent vessels from the central nodes pass to the apical nodes

● Apical nodes

- Located at apex of axilla along medial side axillary vein and first part of axillary artery
- Receive lymph from **all other groups of axillary nodes** and **from lymphatics** accompanying the cephalic vein
- Efferent vessels from these nodes unite to form **subclavian lymphatic trunk**

● Subclavian lymphatic trunk

- on right side
 - Joined by **jugular** and **bronchomediastinal** trunks to form **right lymphatic duct** or **enter right venous angle independently**
- on left side
 - Joins thoracic duct



Enlargement of axillary lymph nodes

- ✚ An infection in the upper limb can cause the axillary nodes to enlarge and become tender and inflamed, a condition called **lymphangitis**
- ✚ Infections in the pectoral region and breast, including the superior part of the abdomen, can also produce enlargement of the axillary nodes
- ✚ These nodes are also the most common site of metastases (spread) of cancer of the breast
- ✚ *Upper lateral quadrant* of breast most common site of spread of cancer

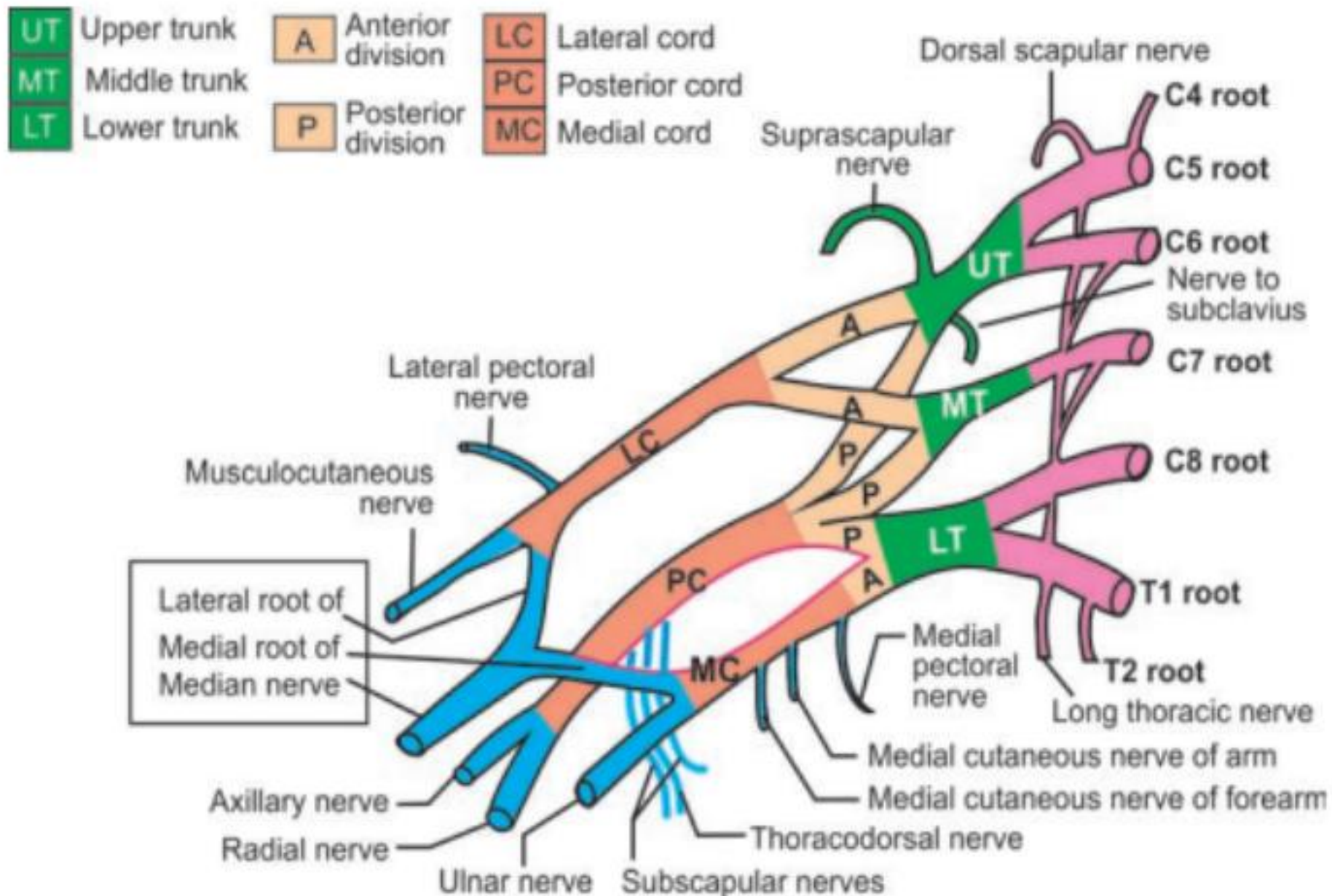
Brachial plexus

Formation

- Brachial plexus is a major network of nerves supplying the upper limb
- It begins in the lateral cervical region (posterior triangle) and extends into the axilla
- Formed by the union of the anterior rami of **C5 - T1** nerves, which constitute the roots of brachial plexus

Formation of Brachial plexus

- ✦ Terms used to describe the plexus from medial to lateral are: **Roots / Trunks / Divisions / Cords /branches**
- ✦ The five roots (rami C5-T1) lie deep to the SCM muscle
- ✦ At the lateral border of SCM, these nerves unite to form the **upper(C5&C6), middle(C7), and lower trunks (C8 &T1)**
- ✦ Each of the three trunks divides almost immediately to form **anterior and posterior divisions**
- ✦ The divisions give rise to three large fiber bundles called the **lateral, medial, and posterior cords**



Roots

- The roots usually pass through the gap between the anterior and middle scalene muscles with the subclavian artery
- In the inferior part of the neck, the roots unite to form three trunks

Trunks

- **Superior trunk**: union of C5 and C6
- **Middle trunk**: continuation of C7
- **Inferior trunk**: union of C8 and T1

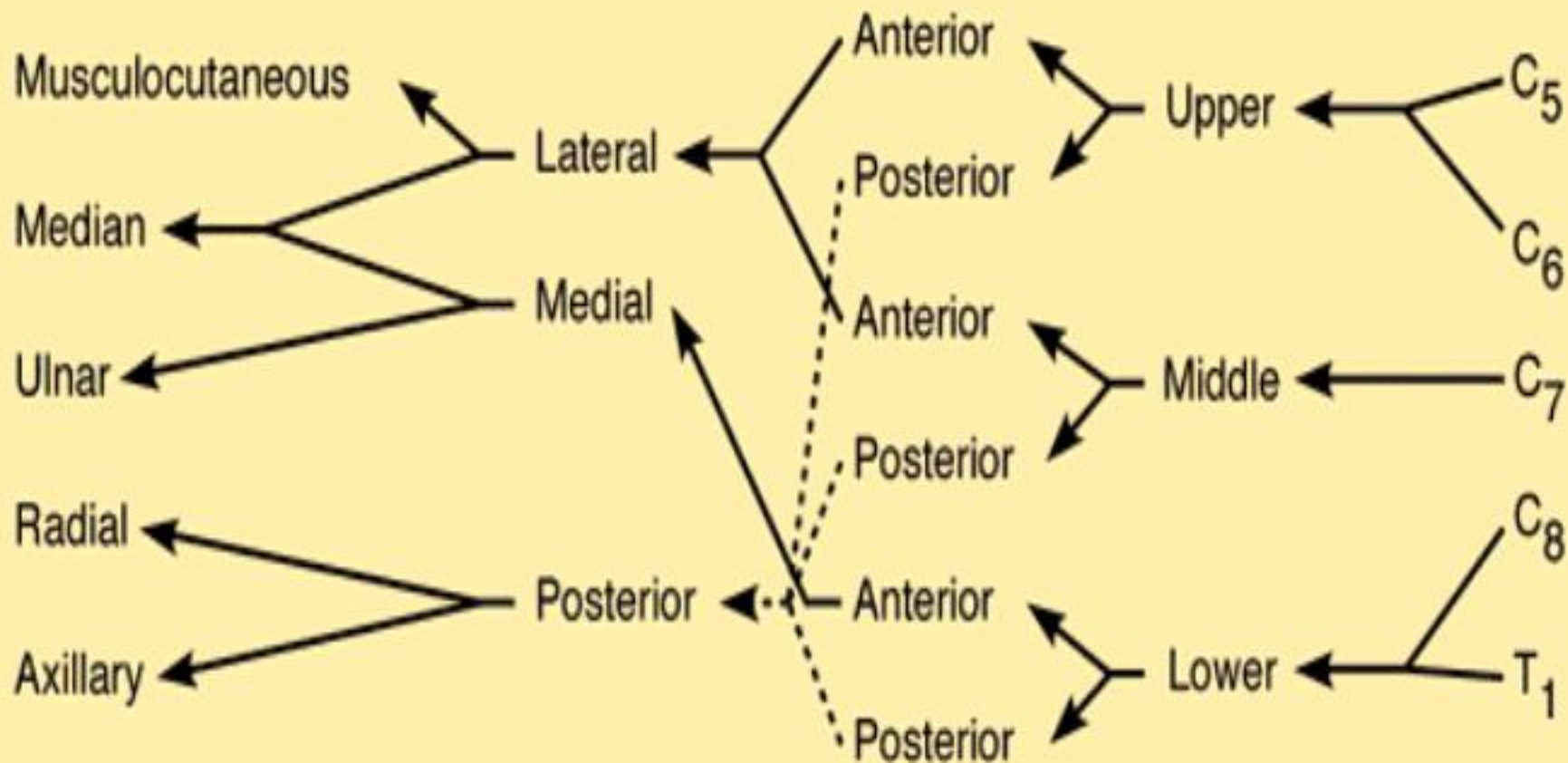
**Major terminal
branches
(peripheral nerves)**

Cords

Divisions

Trunks

**Roots
(ventral
rami)**

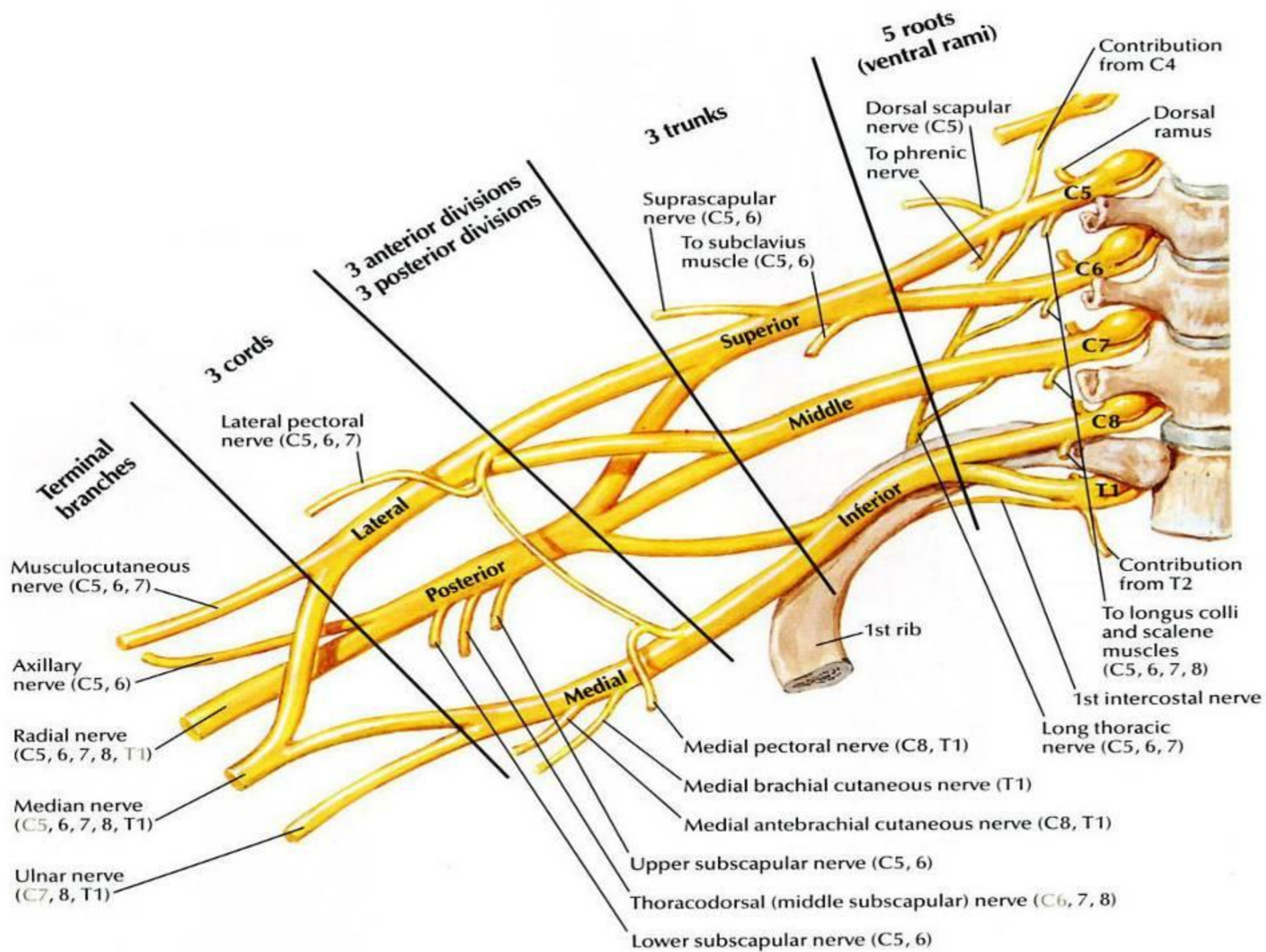


Divisions

- ✚ Each trunk divides into anterior and posterior divisions as the plexus passes through cervicoaxillary canal
- ✚ **Anterior division** supply anterior (flexor) compartments of upper limb
- ✚ **Posterior division** supply posterior (extensor) compartments of upper limb
- ✚ Anterior divisions of superior and middle trunks unite to form **lateral cords**
- ✚ Anterior divisions of lower trunk continues as **medial cord**
- ✚ Posterior divisions of all three trunks unite to form **posterior cord**

Terminal branches

- The cords continue distally to form the main nerves of the upper limb
 - The **lateral cord** continues as musculocutaneous nerve
 - The **medial cord**, as ulnar nerve
 - The **posterior cord**, as radial nerve and axillary nerve
 - A cross-communication between the lateral and medial cords forms **median nerve by forming “M” shape**



Summary “Really Tired Drink Coffee buddy”!!!

- The plexus is formed by Five roots
 - derived from the anterior rami of C5, 6, 7, 8 and T1
 - link up into:
- **Three trunks: formed by the union of**
 - C5 and 6 (upper)
 - C7 alone (middle)
 - C8 and T1 (lower)
 - Which split into:
- **Six divisions**
 - formed by each trunk dividing into anterior and posterior division
 - which link up again into:
- **Three cords**
 - Lateral: from anterior divisions of upper and middle trunks
 - Medial: from anterior division of lower trunk
 - Posterior: from the union of all three posterior divisions

Branches

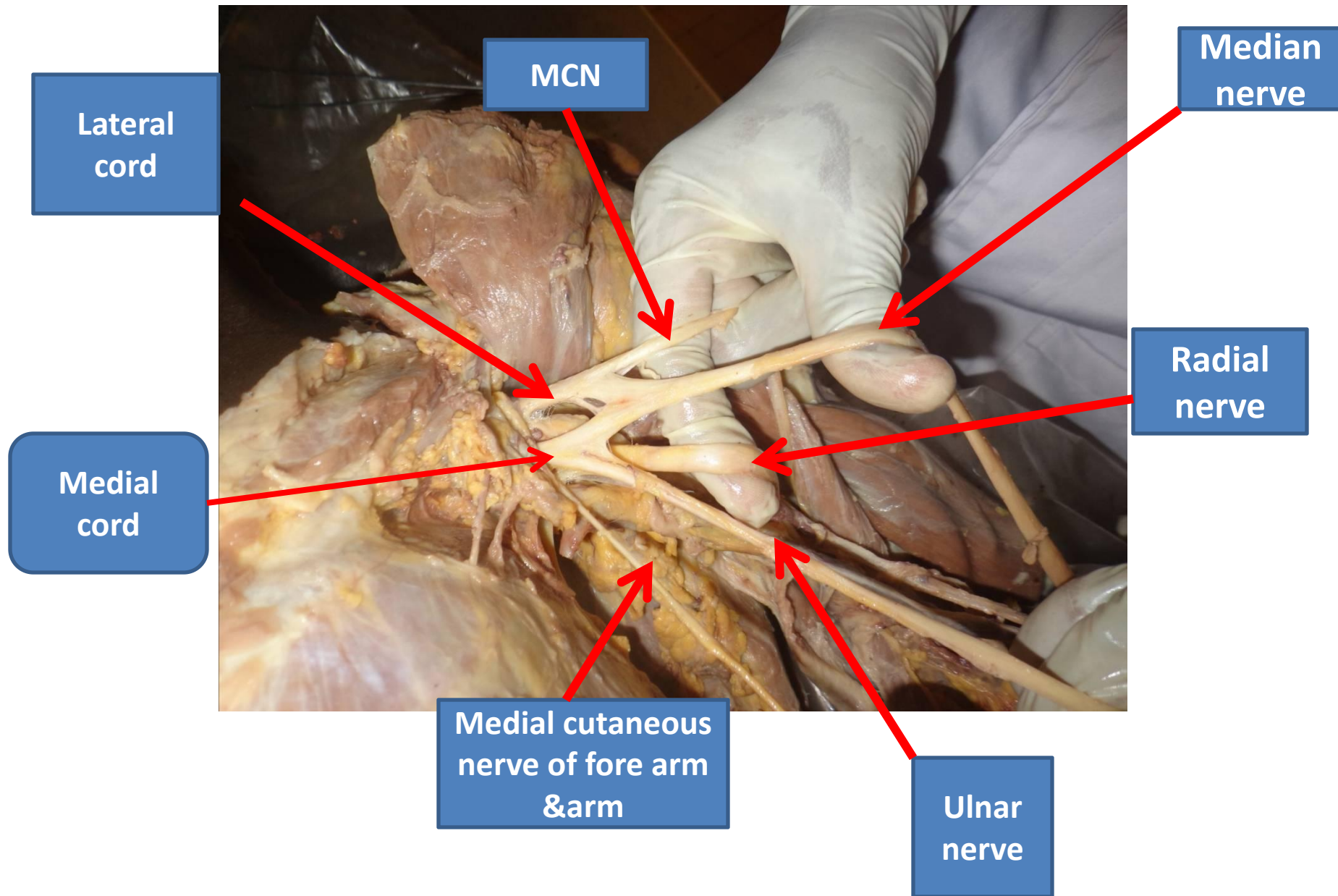
- **From the roots**
 - nerve to rhomboids (dorsal scapular nerve)
 - Phrenic nerve to diaphragm
 - nerve to serratus anterior = long thoracic nerve)
- **From the trunk**
 - nerve to subclavius
 - suprascapular nerve
- **From the lateral cord**
 - musculocutaneous nerve
 - lateral pectoral nerve
 - lateral root of **median nerve**

- **From the medial cord**
 - medial pectoral nerve
 - medial cutaneous nerves of arm and forearm
 - ulnar nerve
 - medial root of median nerve
- **From the posterior cord**
 - Upper subscapular nerves
 - Thoracodorsal nerve
 - Lower subscapular nerve
 - axillary nerve
 - radial nerve

- Dorsal scapular nerve: Rhomboids and levator scapulae
- Long thoracic nerve: serratus anterior
- Suprascapular nerve: Supraspinatus and infraspinatus muscles; glenohumeral (shoulder) joint
- Subclavian nerve (nerve to subclavius): Subclavius and sternoclavicular joint ; accessory phrenic root innervates diaphragm
- Lateral pectoral nerve: pectoralis major
- Medial pectoral nerve: Pectoralis minor and sternocostal part of pectoralis major

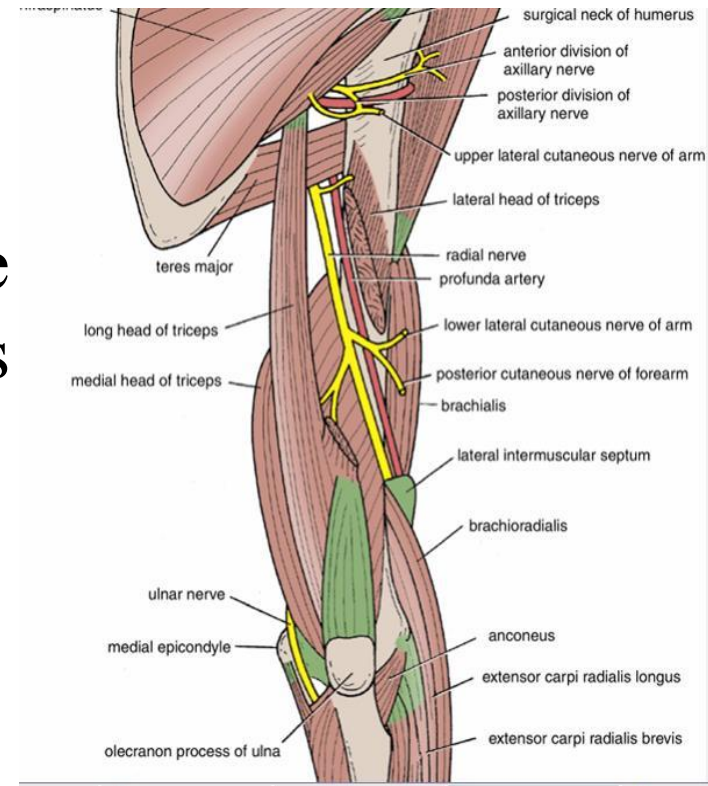
- Medial cutaneous nerve of arm: Skin of medial side of arm, as far distal as medial epicondyle of humerus and olecranon of ulna
- Median cutaneous nerve of forearm: Skin of medial side of forearm, as far distal as wrist
- Upper subscapular nerve: Superior portion of subscapularis
- Lower subscapular nerve: Inferior portion of subscapularis and teres major
- Thoracodorsal nerve: Latissimus dorsi

Brachial plexus



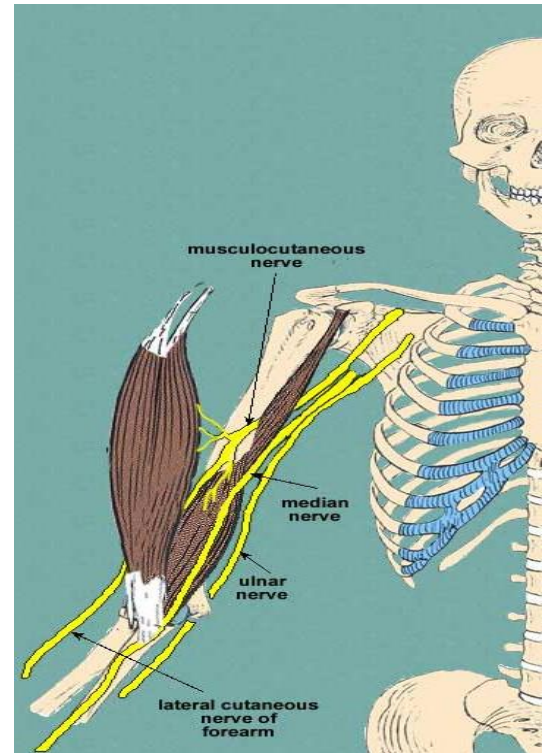
Axillary nerve

- **Origin:** Terminal branch of posterior cord
- **Course:** Exits axillary fossa posteriorly, pass with posterior circumflex humeral artery; gives rise to superior lateral brachial cutaneous nerve; then winds around surgical neck of humerus
- **Innervation:** Glenohumeral joint; teres minor and deltoid muscles; skin of superolateral arm



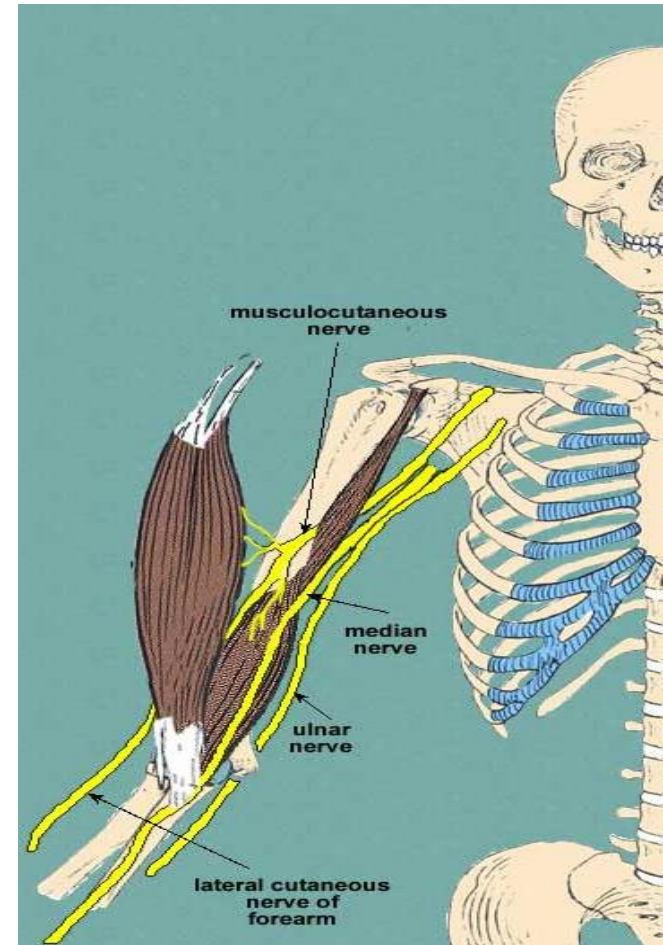
Radial nerve

- ✚ Origin: Larger terminal branch of posterior cord (C5-T1)
- ✚ Course: Exits axillary fossa posterior to axillary artery; passes posterior to humerus in radial groove with deep artery of arm, between lateral and medial heads of triceps; perforates lateral intermuscular septum; enters cubital fossa, dividing into superficial and deep braches
- ✚ **Innervation:** All muscles of posterior compartments of arm and forearm; skin of posterior and inferolateral arm, posterior forearm, and dorsum of hand lateral to axial line of digit 4



Ulnar nerve

- ✚ Origin: Larger terminal branch of medial cord (C7, C8 & T1)
- ✚ Course: Descends on medial arm; passes posterior to medial epicondyle of humerus; then descends on ulnar aspect of forearm to hand
- ✚ Innervation: Flexor carpi ulnaris and ulnar half of flexor digitorum profundus; most intrinsic muscles of hand; skin of hand medial to axial line of digit 4

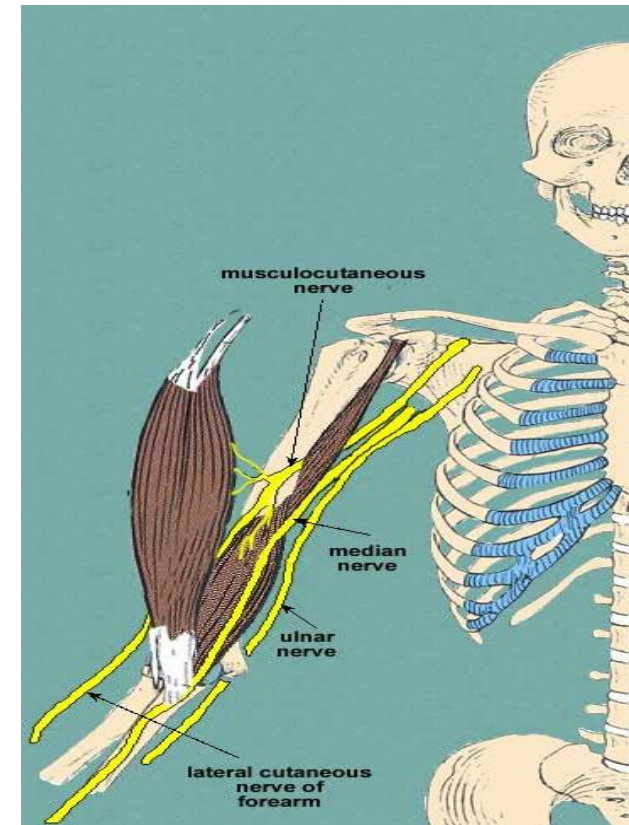


Median nerve

- ✚ Origin: lateral root of median nerve is a terminal branch of lateral cord (C6, C7); medial root of median nerve is a terminal branch of medial cord (C8, T1)
- ✚ Course: Lateral and medial roots merge to form median nerve lateral to axillary artery; descends through arm, cubital fossa, forearm to hand
- ✚ Innervation: Muscles of anterior forearm compartment (except for flexor carpi ulnaris and ulnar half of flexor digitorum profundus), five intrinsic muscles in thenar half of palm and palmar skin
- ✚ No muscular branch in the arm
- ✚ Vasomotor for brachial artery

Musculocutaneous nerve

- ✚ Origin: Terminal branch of lateral cord (C5-C7)
- ✚ Course: Exits axilla by *piercing coracobrachialis*; descends between biceps brachii and brachialis, supplying both; continues as lateral cutaneous nerve of forearm
- ✚ Innervation: Muscles of anterior compartment of arm (*coracobrachialis, biceps brachii and brachialis*); skin of lateral aspect of forearm



Brachial plexus injuries (clinical condition of UL)

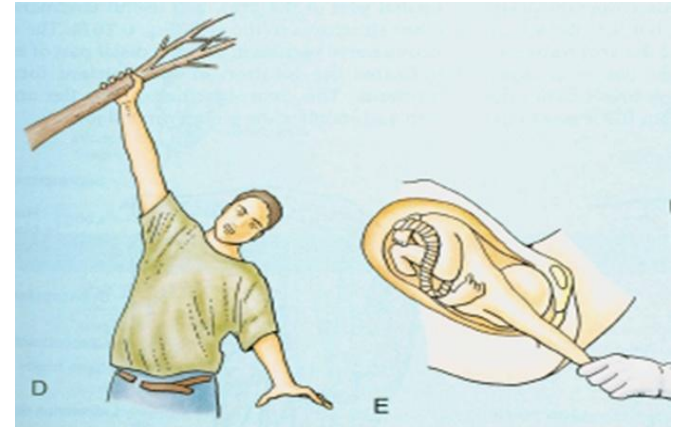
- ✚ Injuries to brachial plexus affect movements and cutaneous sensations in the upper limb
- ✚ Disease, stretching and wounds in posterior triangle of neck or in axilla may produce brachial plexus injuries

Brachial plexus injuries

- Injuries to **superior parts** of brachial plexus (C5 and C6) (**Erb-Duchenne palsy**)
 - result from an excessive increase in the angle between the neck and the shoulder
 - results in paralysis of the muscles of shoulder and arm supplied by C5 & C6
 - the characteristic position of the limb ("**waiter's tip position**")
 - an adducted shoulder, medially rotated arm, and extended elbow
 - lateral aspect of the upper limb also loses sensation



- Injuries to **inferior parts of brachial plexus (Klumpke paralysis)**
 - less common
 - occur when the upper limb is suddenly pulled superiorly
 - injure the inferior trunk (C8 & T1) and may tears the roots of spinal nerves
 - muscles of the hands are affected and results **in claw hand**

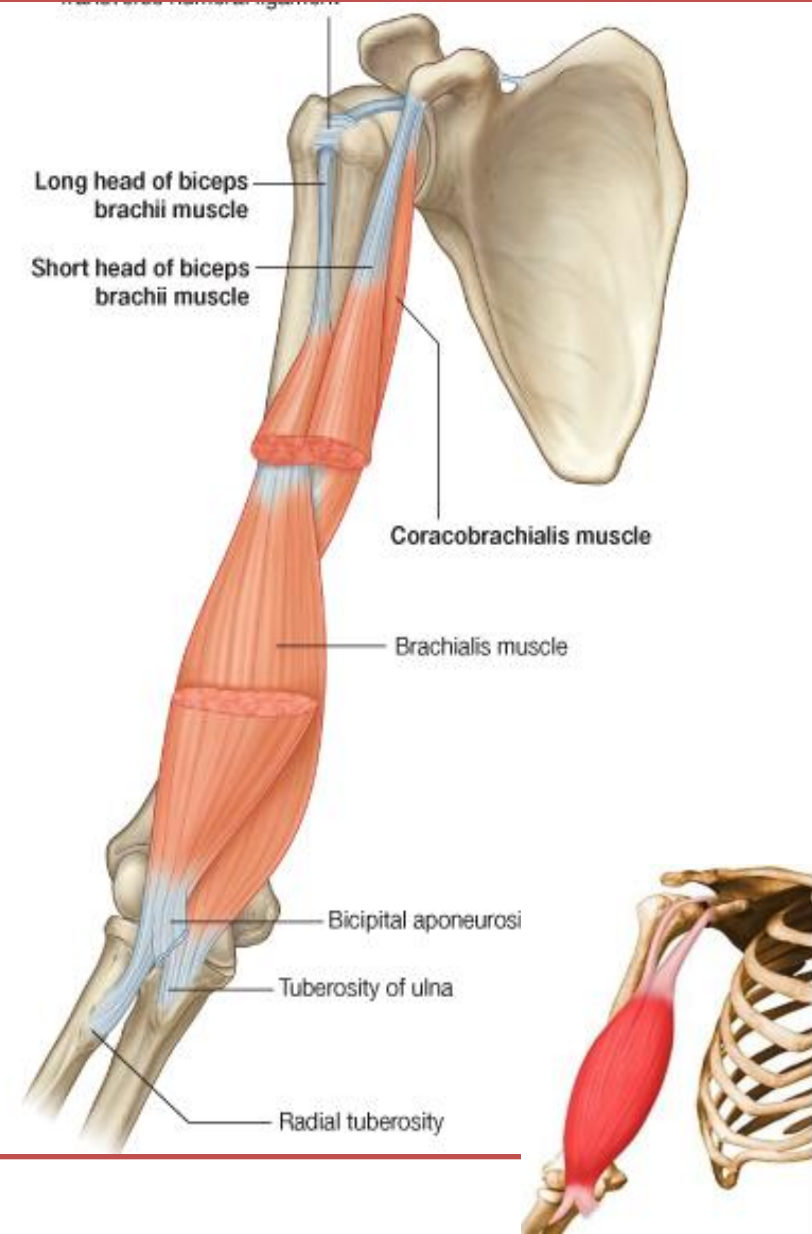


Muscles of arm

- Four arm muscles
- **Three flexors**
 - biceps brachii, brachialis, and coracobrachialis
 - in the anterior (flexor) compartment
 - supplied by musculocutaneous nerve
- **One extensor**
 - triceps brachii
 - in the posterior compartment
 - supplied by radial nerve
- **Anconeus**
 - a small triangular muscle on the posterior aspect of the elbow, covers the posterior aspect of the ulna proximally

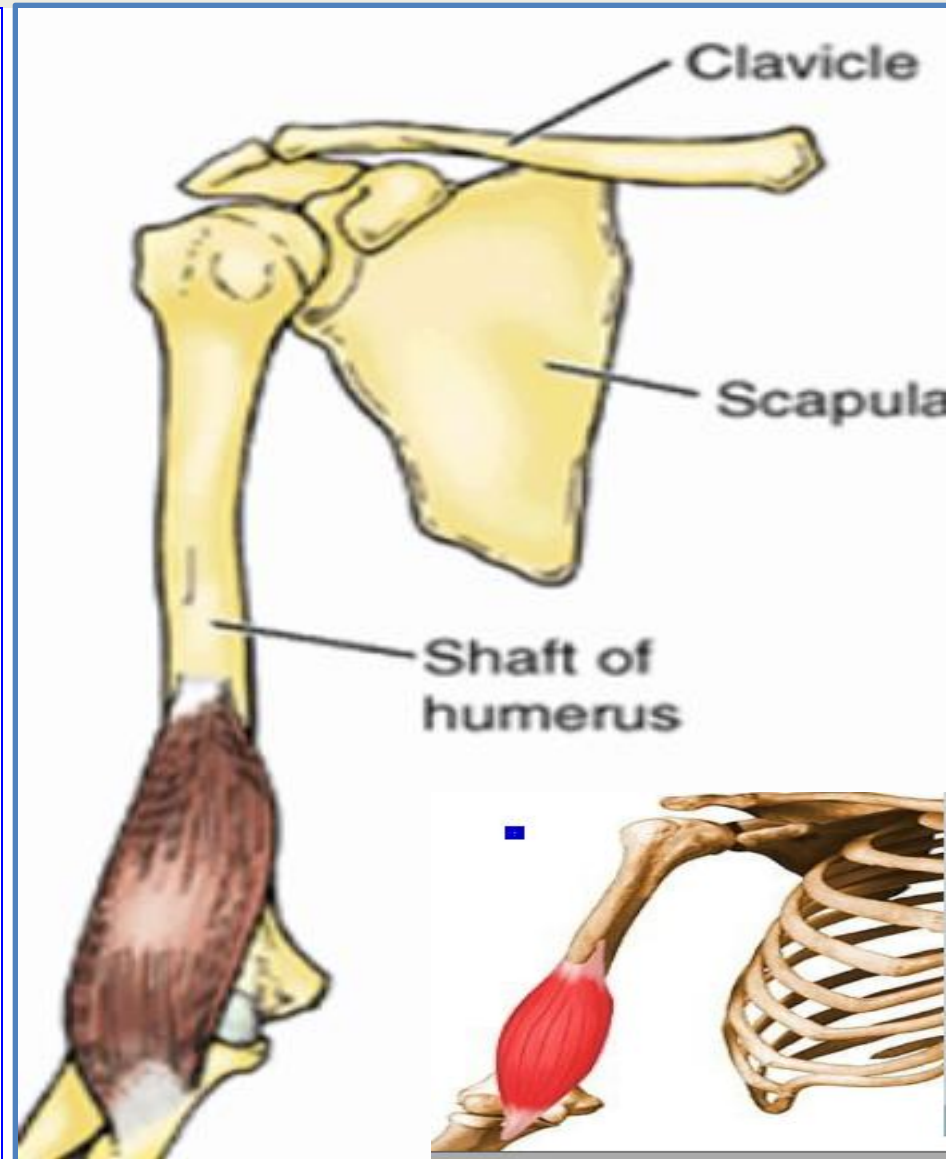
Biceps brachii

- Has two heads; long head and short head
- **Origin:**
 - **Short head:** tip of coracoid process of scapula
 - **Long head:** supraglenoid tubercle of scapula
- **Innervation:**
Musculocutaneous nerve (C5, C6)
- **Action:** Supinates forearm and, when it is supine, flexes forearm



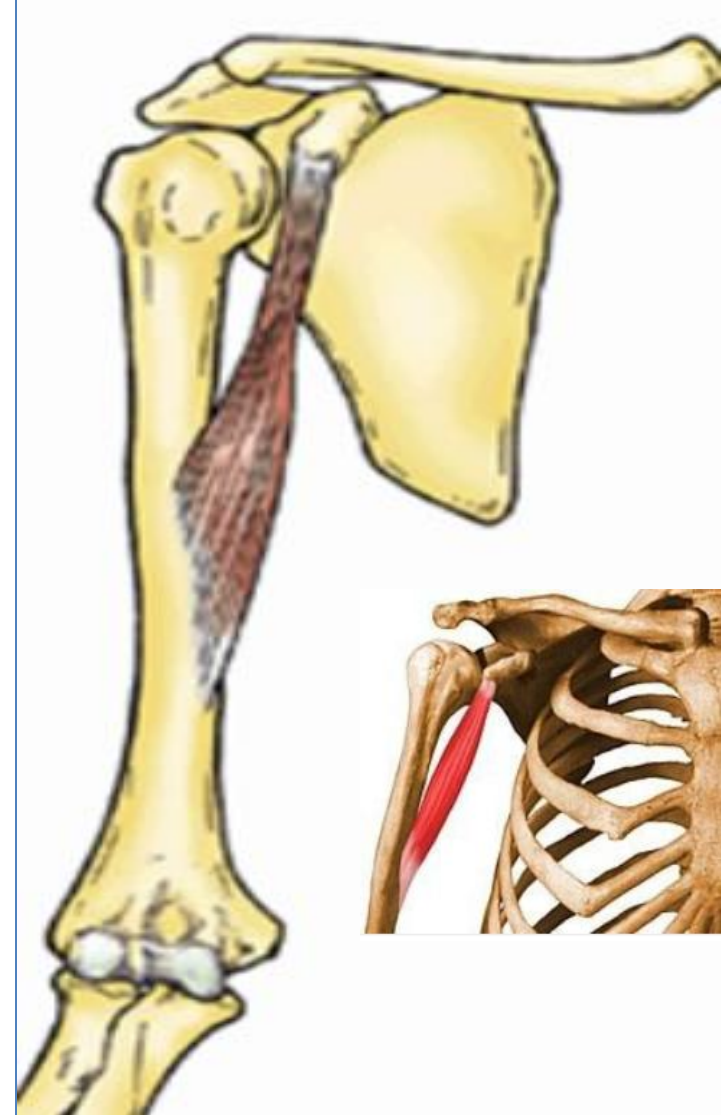
Brachialis

- Flattened fusiform muscle
- Lies posterior (deep) to the biceps
- It is the only **pure flexor**
- Innervation:
Musculocutaneous nerve (C5, C6)
- Action: Flexes forearm in all positions



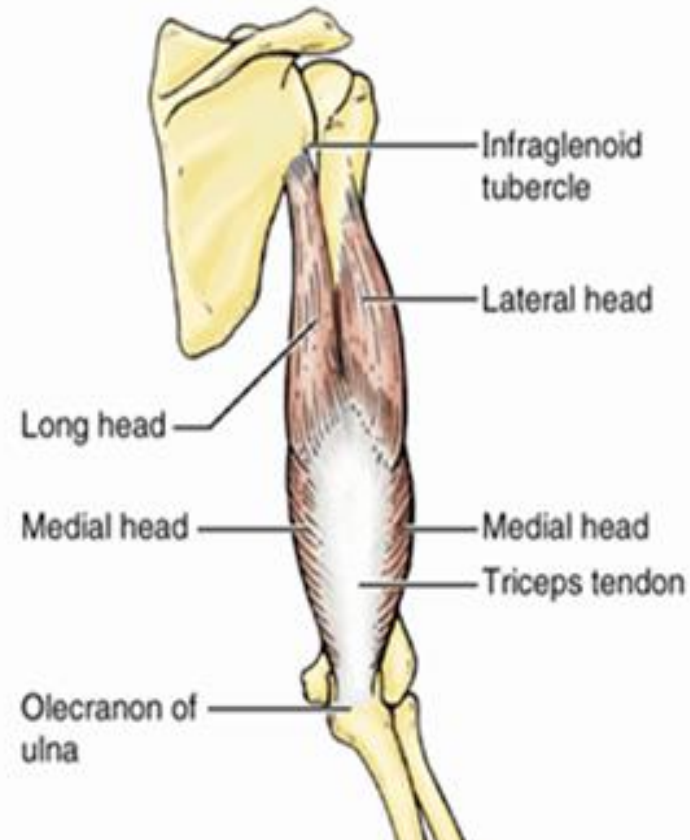
Coracobrachialis

- ✦ Elongated muscle in the superomedial part of the arm
- ✦ **Useful landmark** for locating other structures in the arm
- Musculocutaneous nerve pierces it
- Distal part of its attachment indicates the location of the nutrient foramen of the humerus
- Ulnar nerve pierce medial intramuscular septa
- Radial nerve pierce lateral intramuscular septa & median nn cross brachial artery
- ✦ Innervation: **Musculocutaneous nerve (C5, C6, C7)**
- ✦ Action: Helps flex and adduct arm



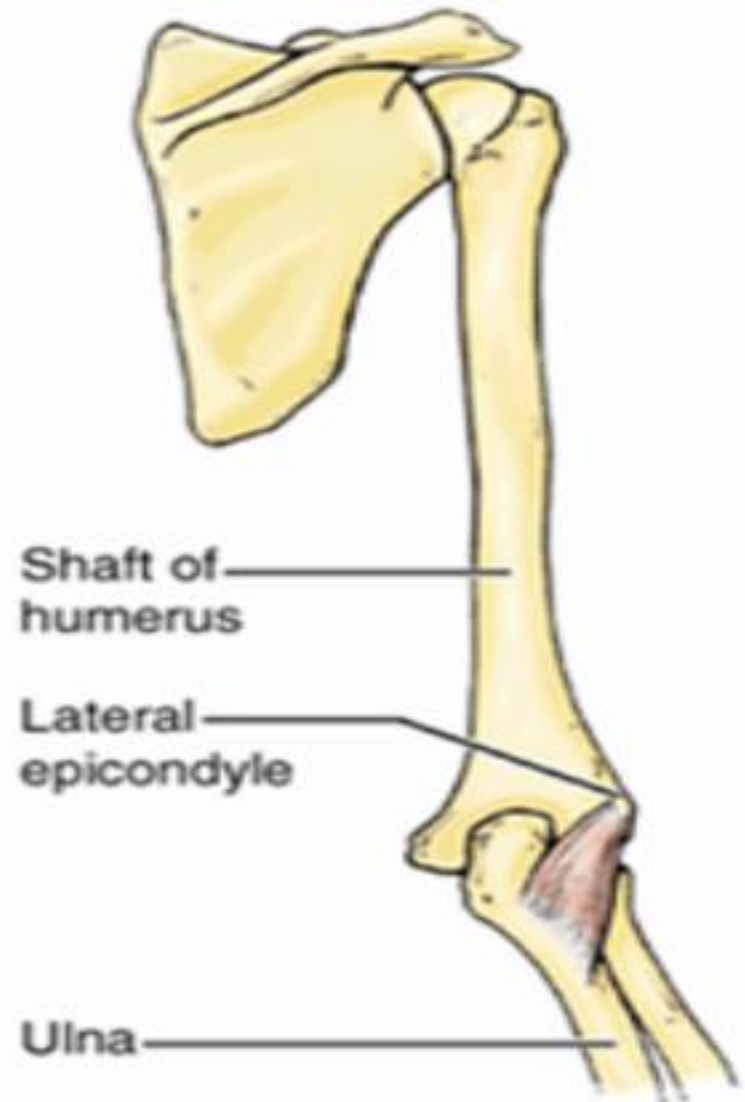
Triceps brachii

- Large fusiform muscle in posterior compartment
- Origin:
 - **Long head:** infra-glenoid tubercle of scapula
 - **Lateral head:** posterior surface of humerus, superior to radial groove
 - **Medial head:** posterior surface of humerus, inferior to radial groove
- Innervation: Radial nerve (C6, C7, C8)
- Action: Chief extensor of forearm; long head resists dislocation of humerus



Anconeus

- **Origin:** Lateral epicondyle of humerus
- **Insertion:** Lateral surface of olecranon and superior part of posterior surface of ulna
- **Innervation:** Radial nerve (C7, C8, T1)
- **Action:** Assists triceps in extending forearm; stabilizes elbow joint; may abduct ulna during pronation



Quadrangular space

It is the passageway for nerves and vessels passing the axilla to the posterior scapular region

its boundaries **are formed by:**

- **Superiorly** : *teres minor*;
- **Laterally** : *surgical neck of the humerus*;
- **Inferiorly** : *teres major*;
- **Medially** : *the long head of triceps brachii*.
- **Contents** : **axillary nerve and the posterior circumflex humeral artery and vein** pass through it.

Quadrangular Space

Supraspinatus

Infraspinatus

Teres minor

Deltoid

Triangular Interval

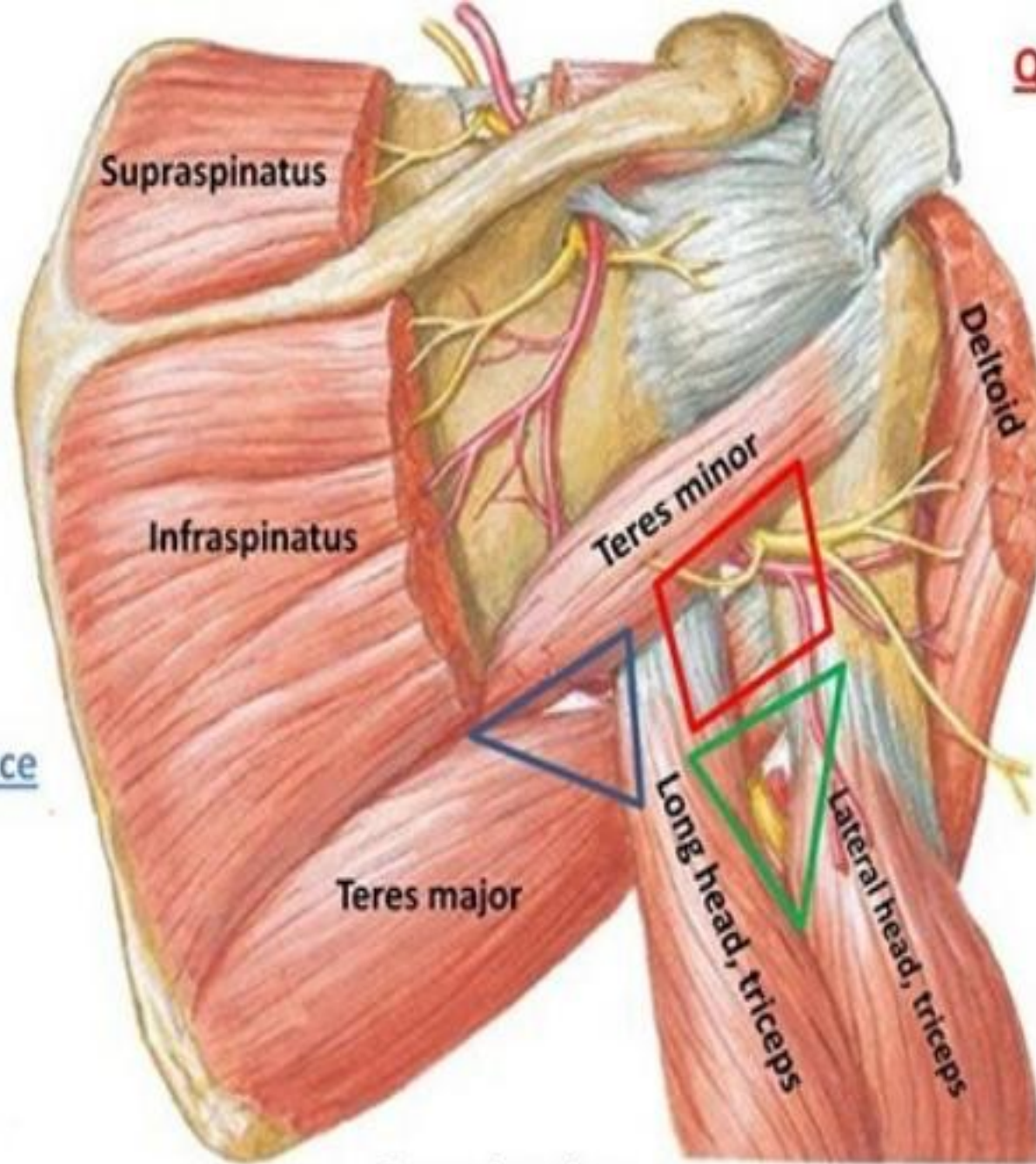
Triangular Space

Teres major

Long head, triceps

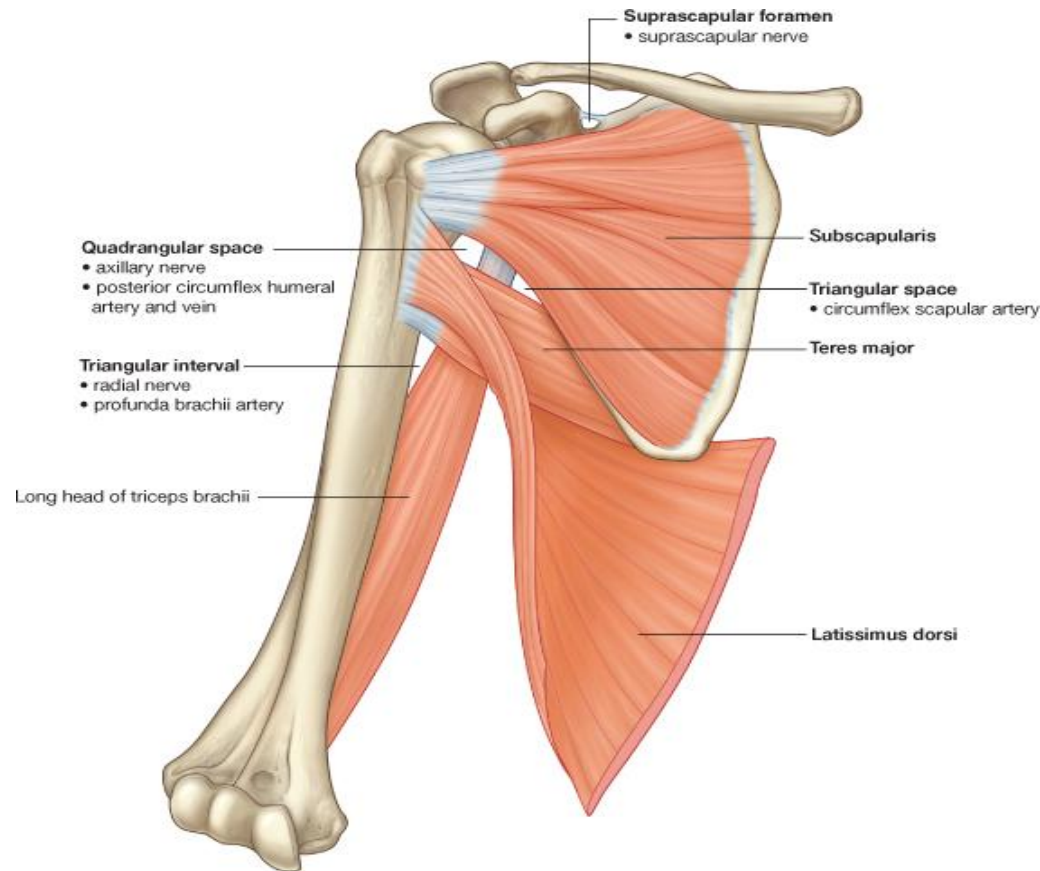
Lateral head, triceps

Posterior view



Triangular space

- the triangular space **is formed by:**
- **Laterally :** *long head of triceps brachii;*
- **Inferiorly :** *teres major;*
- **superiorly :** *teres minor.*
- Content : *circumflex scapular artery* and *vein* pass through this gap



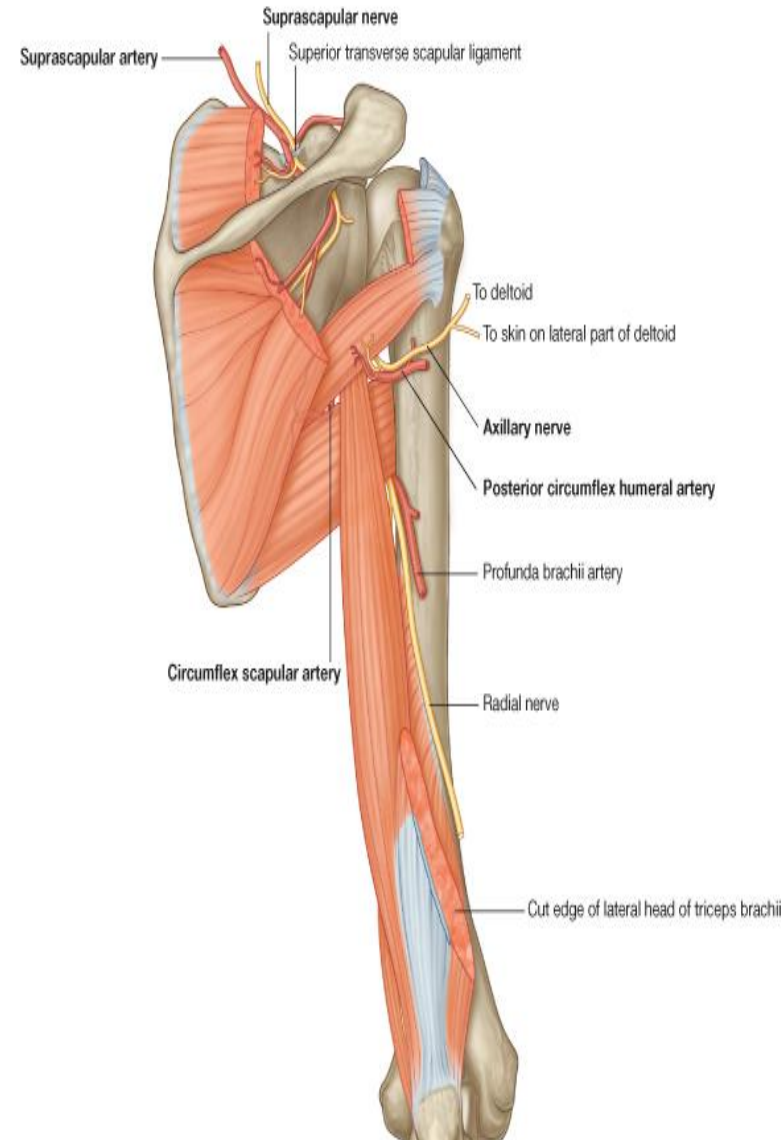
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Triangular interval

- The triangular interval is **formed by:**
- **Medially:** *lateral margin of the long head of triceps brachii;*
- **Laterally:** *the shaft of the humerus;*
- **Superiorly:** *inferior margin of teres major*

Contents :

radial nerve, the profunda brachii artery (deep artery of arm), and associated veins pass through it



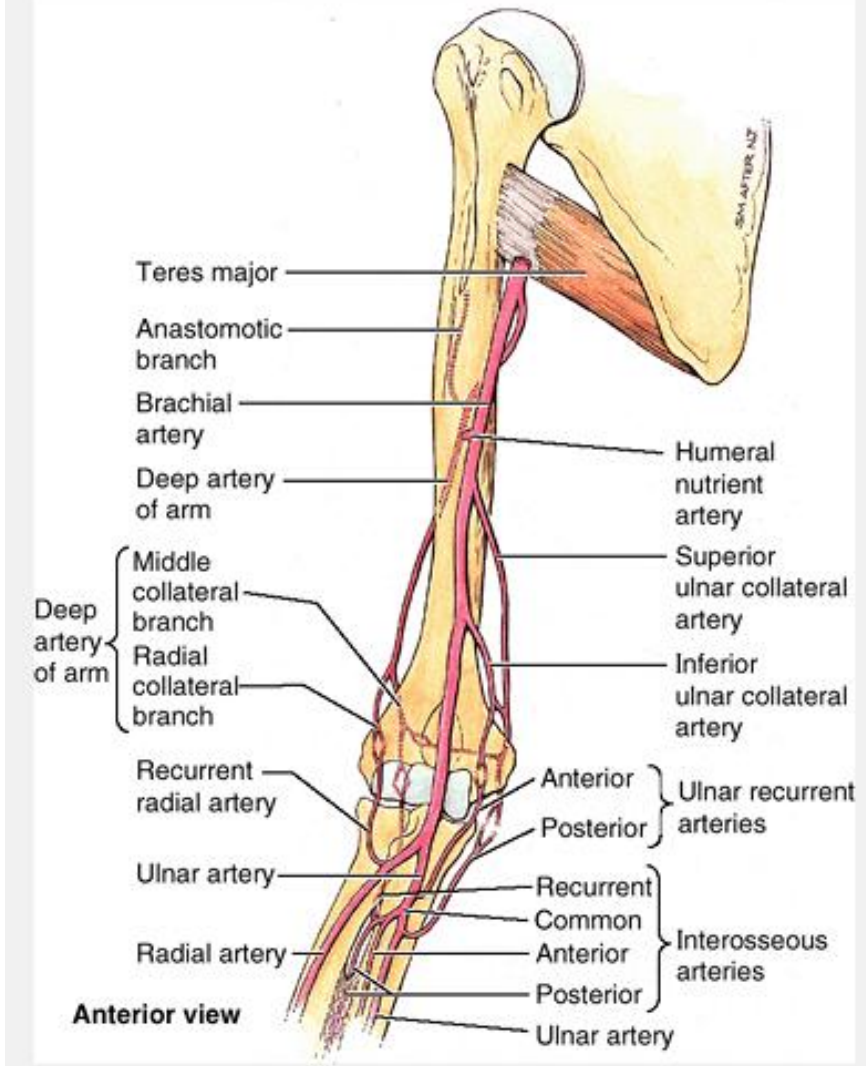
Arteries of Arm

Brachial artery

- Provides the main arterial supply to the arm
- Is the continuation of the axillary artery
- Begins at the inferior border of **teres major**
- Ends in the **cubital fossa opposite the neck of the radius** under cover of the bicipital aponeurosis, where it divides into the **radial and ulnar arteries**

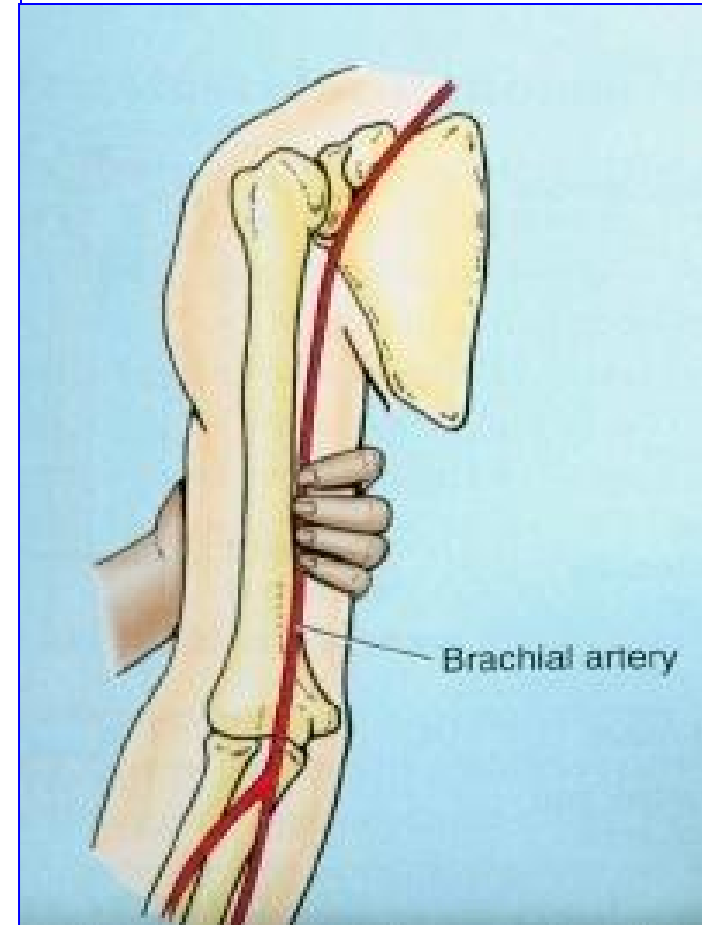
- **Branches**

- ✚ unnamed muscular branches
- ✚ humeral nutrient artery
- ✚ deep artery of arm
- ✚ superior and inferior ulnar collateral arteries



Compression of Brachial Artery

- The best place to compress the brachial artery to control **hemorrhage** is near the middle of the arm
- Because the arterial anastomoses around the elbow provide collateral circulation, the brachial artery may be clamped distal to the inferior ulnar collateral artery without producing tissue damage
- The ulnar and radial arteries still receive sufficient blood through the anastomoses
- Ischemia of the elbow and forearm results in clamping the brachial artery proximal to the deep artery of the arm for an extended period

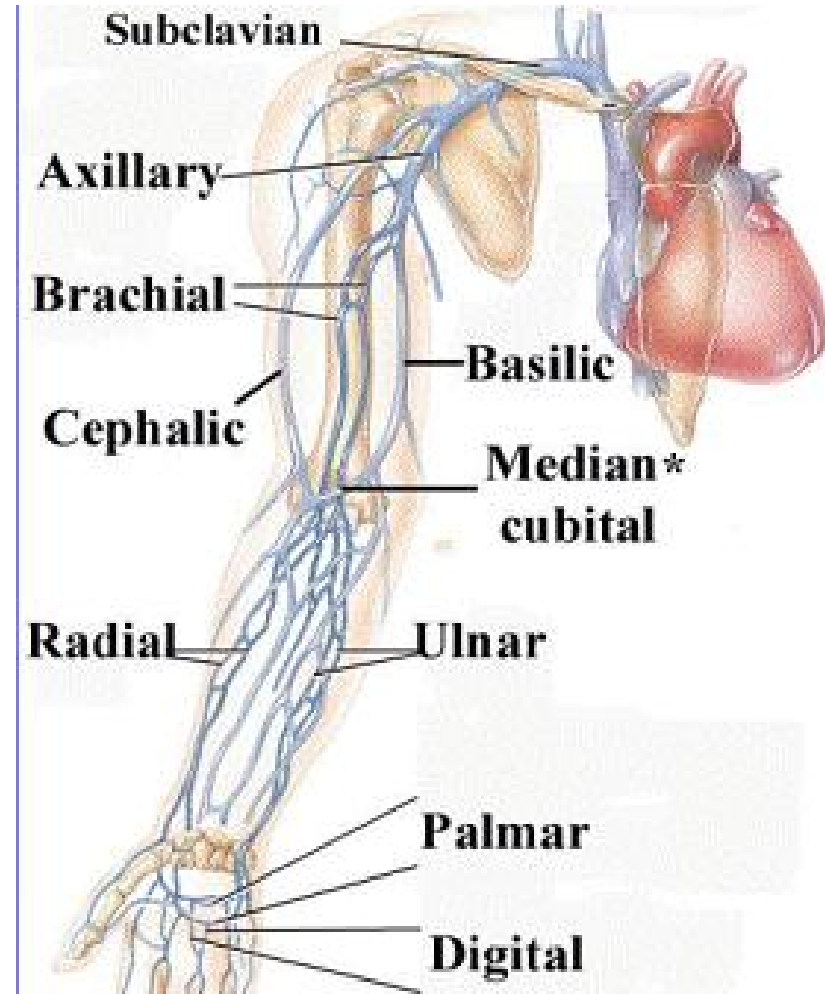


Veins of Arm

- Two sets of veins, superficial and deep, anastomose freely with each other

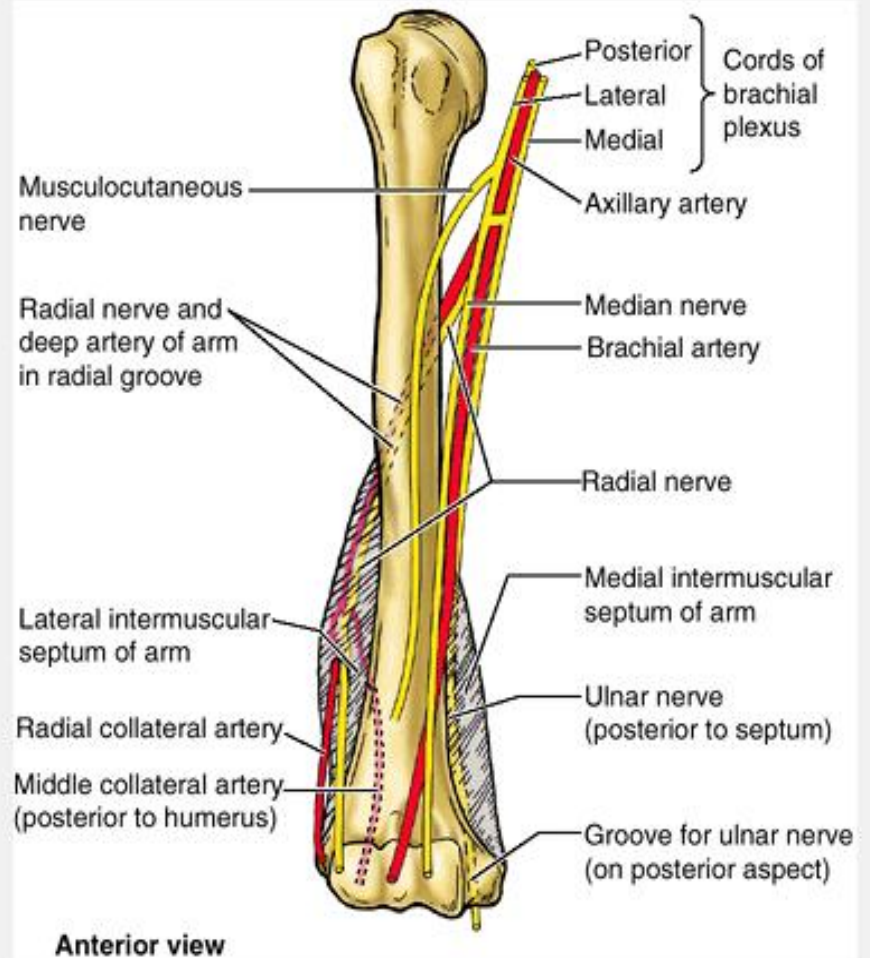
Brachial veins

- Paired deep veins, accompany the brachial artery
- Begins at the elbow by union of the accompanying veins of the ulnar and radial arteries
- Ends by merging with the basilic veins to form the **axillary vein**



Nerves of arm

- 4 main nerves
 - Median
 - Musculocutaneous
 - Ulnar
 - Radial



Median nerve

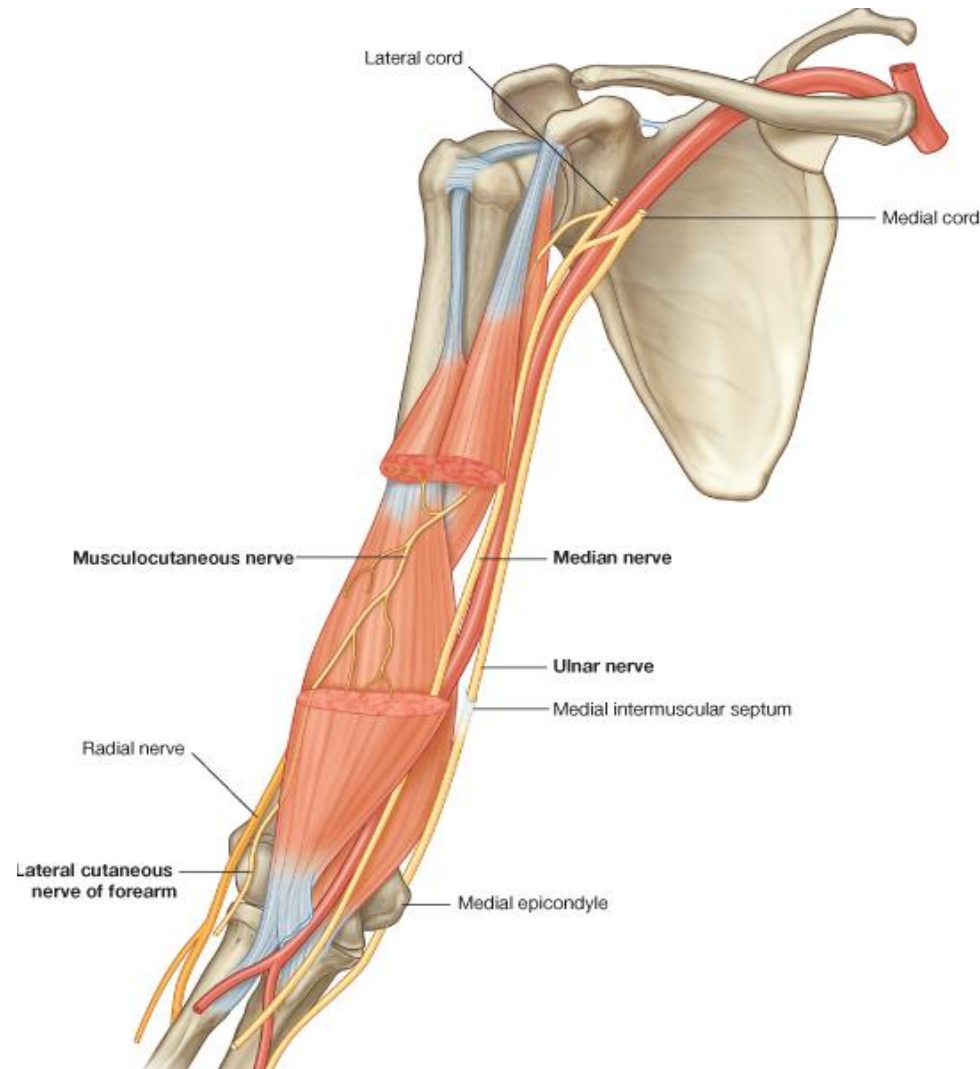
- Formed in the axilla by the union of a branch from medial and lateral cords, which unite anterior to the third part of axillary artery
- The nerve runs distally in the arm, initially on the lateral side of the brachial artery until it reaches the middle of the arm, where it crosses to the medial side and contacts the brachialis
- The median and ulnar nerves supply **no muscular branches to the arm**; however, they supply articular branches to the elbow joint

Ulnar nerve

- Arises from the medial cord of the brachial plexus
- It passes distally, anterior to the insertion of teres major and to the long head of triceps, on the medial side of the brachial artery
- Around the middle of the arm, it pierces the medial intermuscular septum with the superior ulnar collateral artery and descends between the septum and the medial head of triceps
- It then passes posterior to the medial epicondyle of the humerus to enter the forearm

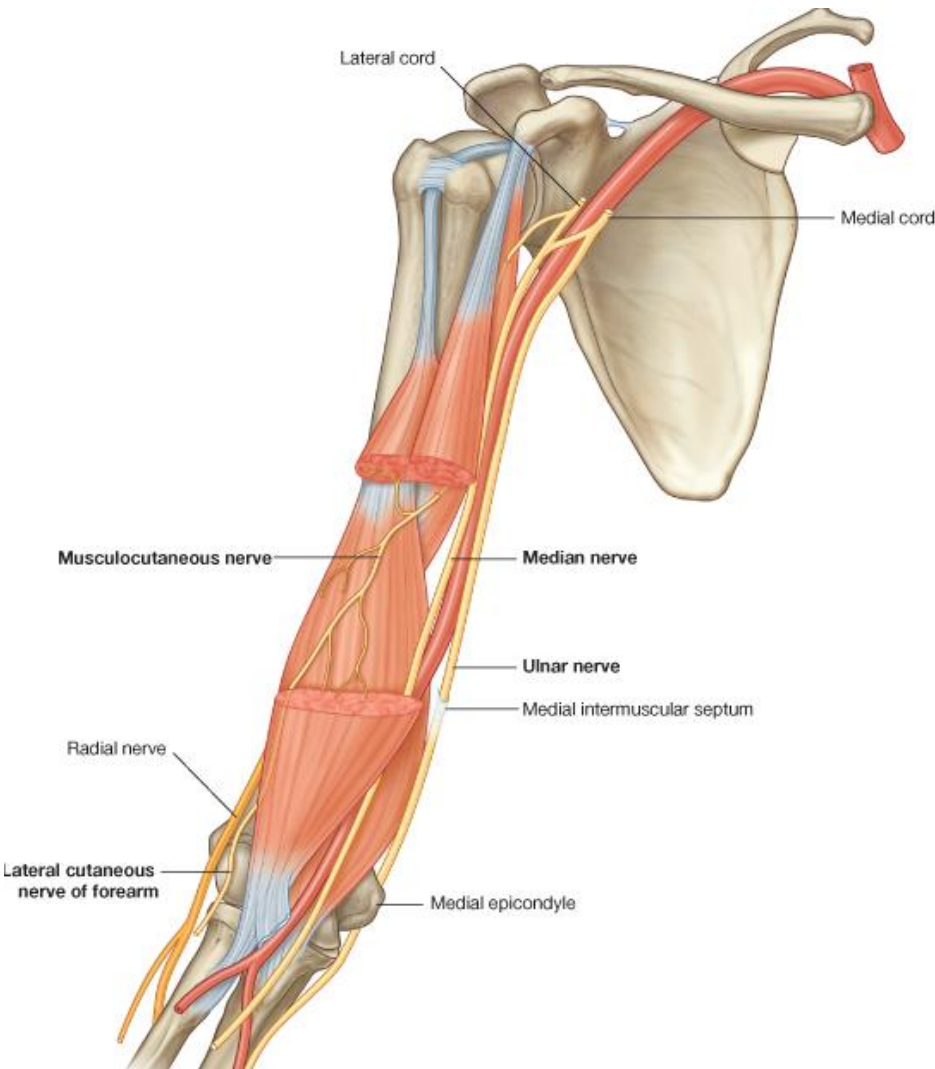
Musculocutaneous nerve

- Continues from the lateral cord of the plexus
- Pierces the **Coracobrachialis** and then continues distally between the brachialis and biceps
- It supply **all three muscles of the anterior** compartment of the arm & lateral cutaneous of the forearm



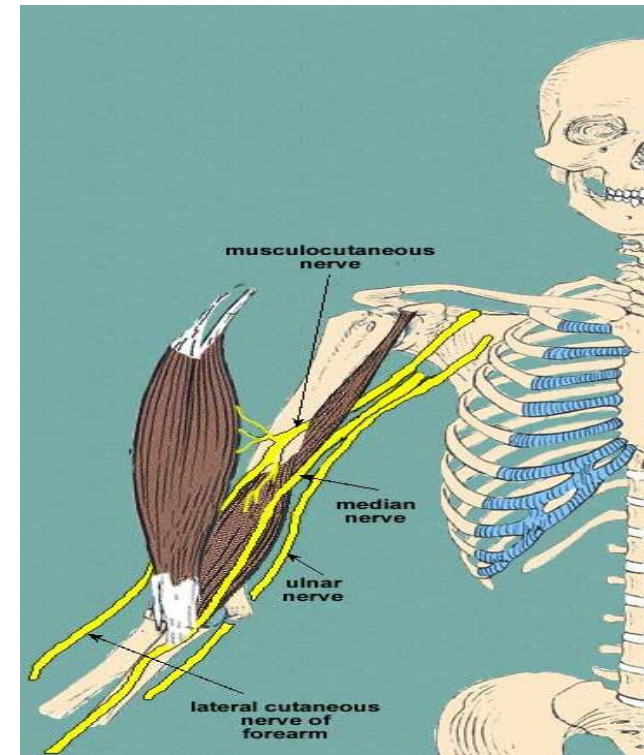
Radial nerve

- Enters the arm posterior to brachial artery, medial to the humerus, and anterior to long head of triceps
- In the cubital fossa, it divides into **deep** and **superficial** branches
- Supplies** the muscles in the posterior compartments of the arm and forearm and the overlying skin



Injury to Musculocutaneous Nerve

- ✚ Results in paralysis of the coracobrachialis, biceps, and brachialis
- ✚ Flexion of the elbow and supination of the forearm are greatly weakened
- ✚ Loss of sensation may occur on the lateral surface of the forearm supplied by the lateral cutaneous nerve of the forearm



Injury to Radial Nerve

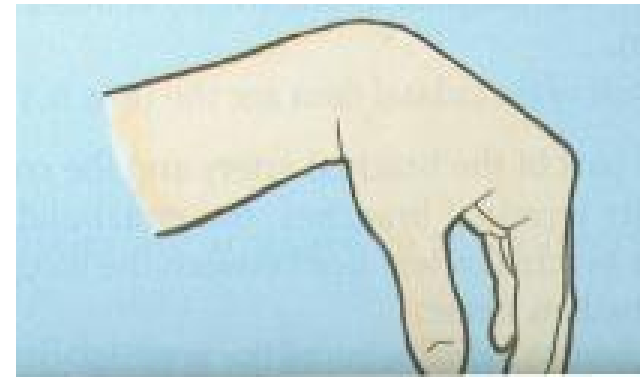
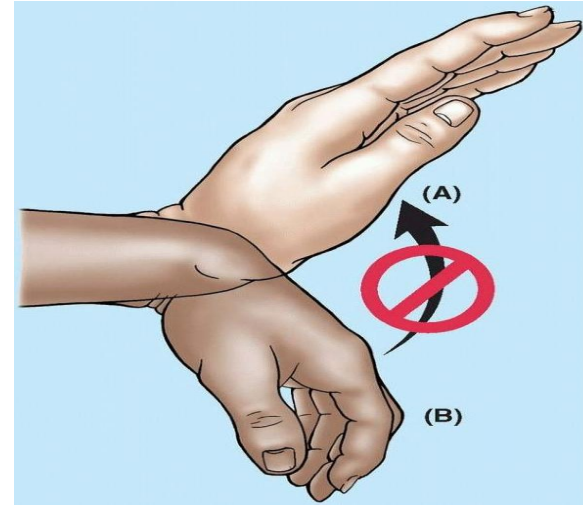
■ Injury to the radial nerve **superior to the origin** of its branches to the triceps brachii results in

- paralysis of triceps, brachioradialis, supinator, and extensor muscles of the wrist and fingers
- Loss of sensation occurs in areas of skin supplied by this nerve

■ **When the radial nerve is injured in radial groove**

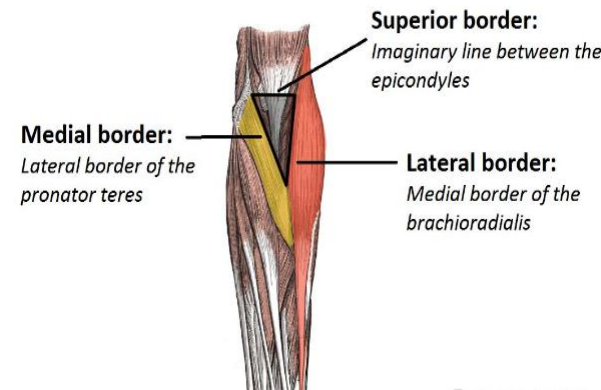
- the triceps is weakened
- the muscles in posterior compartment of the forearm are paralyzed

■ The clinical sign of radial nerve injury is **wrist drop**



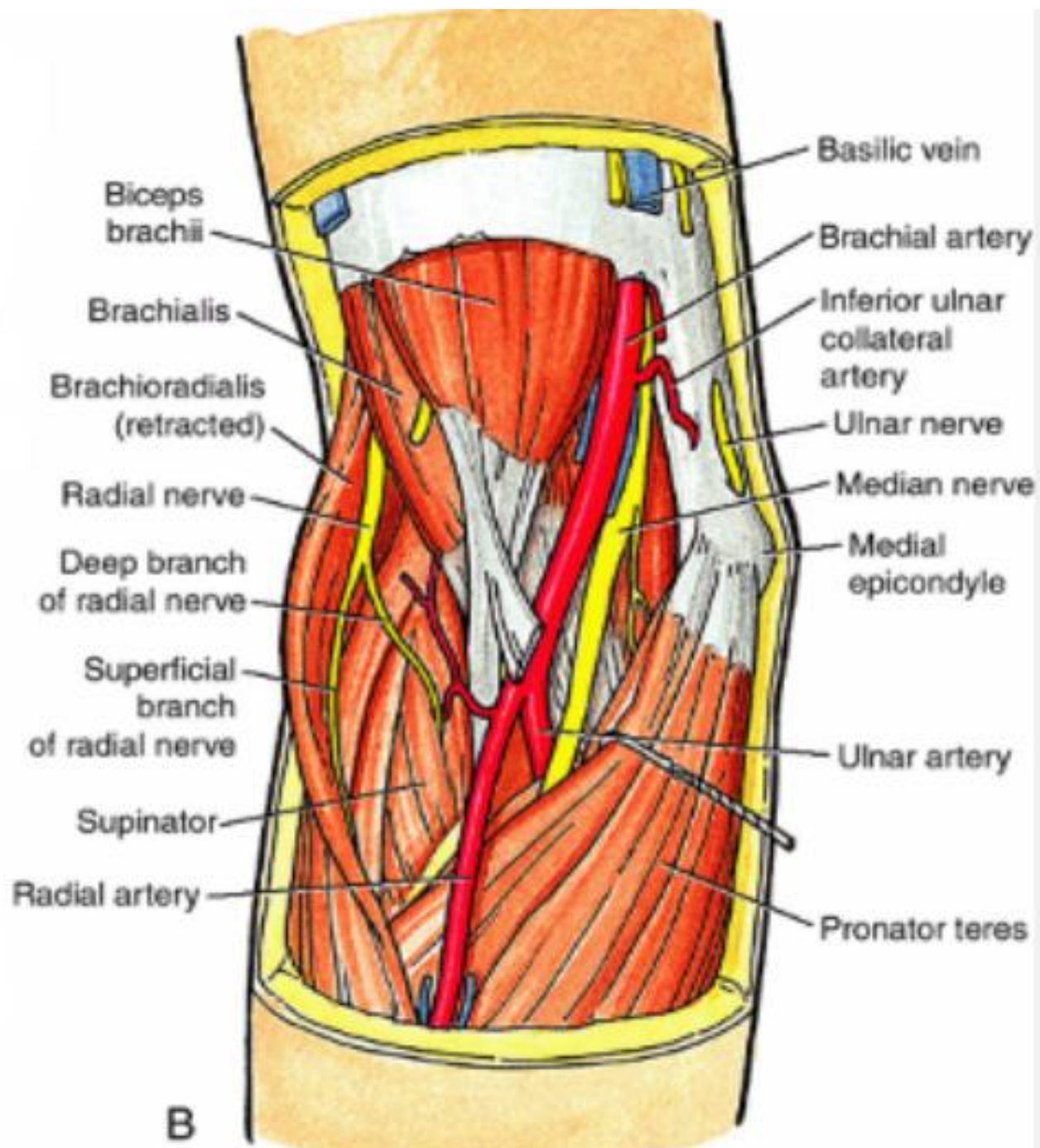
Cubital fossa

- Shallow triangular depression on anterior surface of the elbow
- **Boundaries**
 - ✚ **Superiorly**, an imaginary line connecting the medial and lateral epicondyles
 - ✚ **Medially**, **pronator teres**
 - ✚ **Laterally**, **brachioradialis**
- **Floor**
 - **Brachialis and supinator muscles**
- **Roof**
 - reinforced by the bicipital aponeurosis, subcutaneous tissue and skin



- **Contents**

- ✚ Brachial artery and branches, radial and ulnar arteries
- ✚ Accompanying veins of the arteries
- ✚ Biceps brachii tendon
- ✚ Median nerve
- ✚ Radial nerve, dividing into its superficial and deep branches
- ✚ Median cubital vein
- ✚ Medial and lateral cutaneous nerves of the forearm



Muscles of Forearm

- ✦ The tendons of the forearm muscles pass through the distal part of the forearm and continue in to the wrist, hand and fingers
- ✦ The **flexors** and **pronators** of the forearm are in the anterior compartment and are served **mainly** by the **median nerve**
- ✦ The **extensor** and **supinators** are in posterior compartment and are **all** innervated by the **radial nerve**

Flexor-pronator muscles of the forearm

- The flexor-pronator muscles are in the anterior compartment of the forearm
- The tendons of most flexor muscles pass across the anterior surface of the wrist and are held in place by **palmar carpal ligament** and **flexor retinaculum**
- The flexor-pronator muscles are arranged in three layers
 - **Superficial**
 - **Intermediate**
 - **Deep**

■ Superficial layer

- ✚ Four muscles: **Pronator teres, Flexor carpi radialis, Palmaris longus, Flexor carpi ulnaris**
- ✚ All are attached proximally by a common flexor tendon to **medial epicondyle of humerus** (common flexor origin)

■ Intermediate layer

- ✚ One muscle: **Flexor digitorum superficialis**

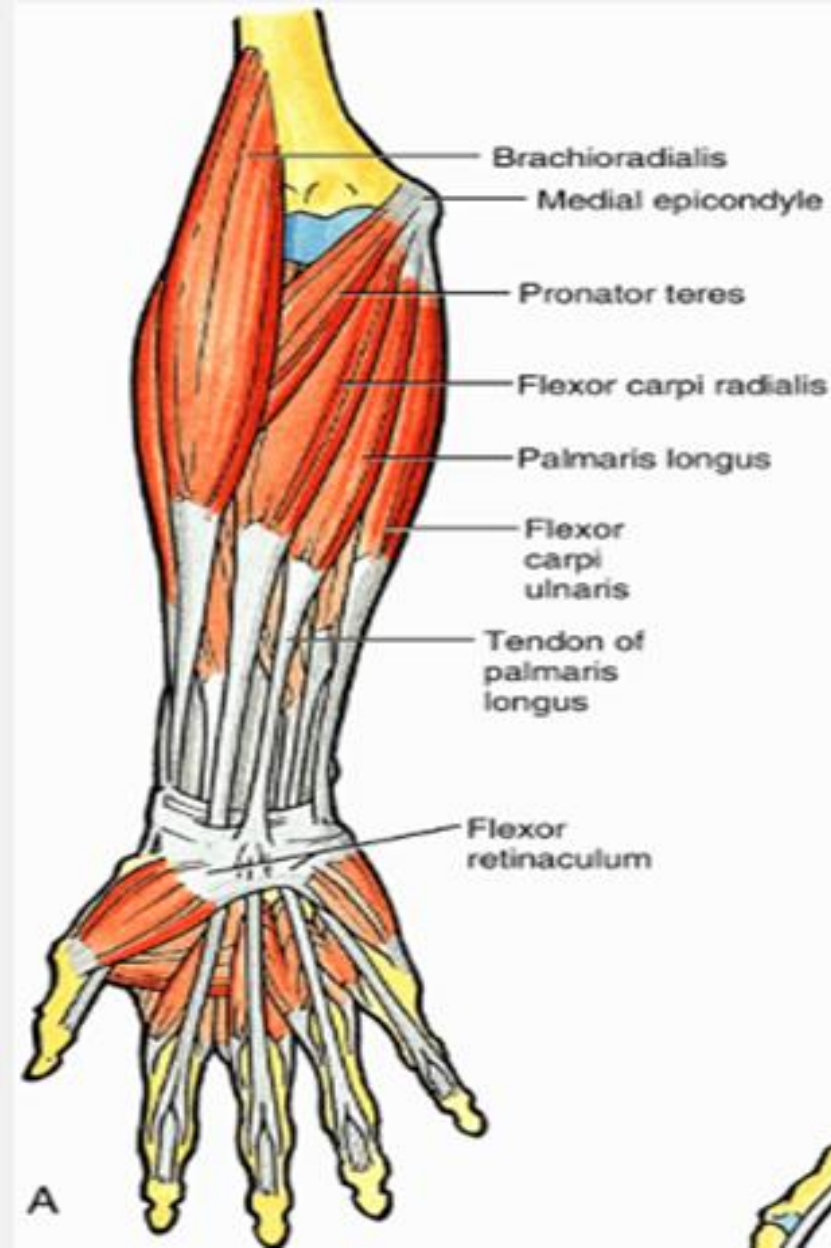
■ Deep layer

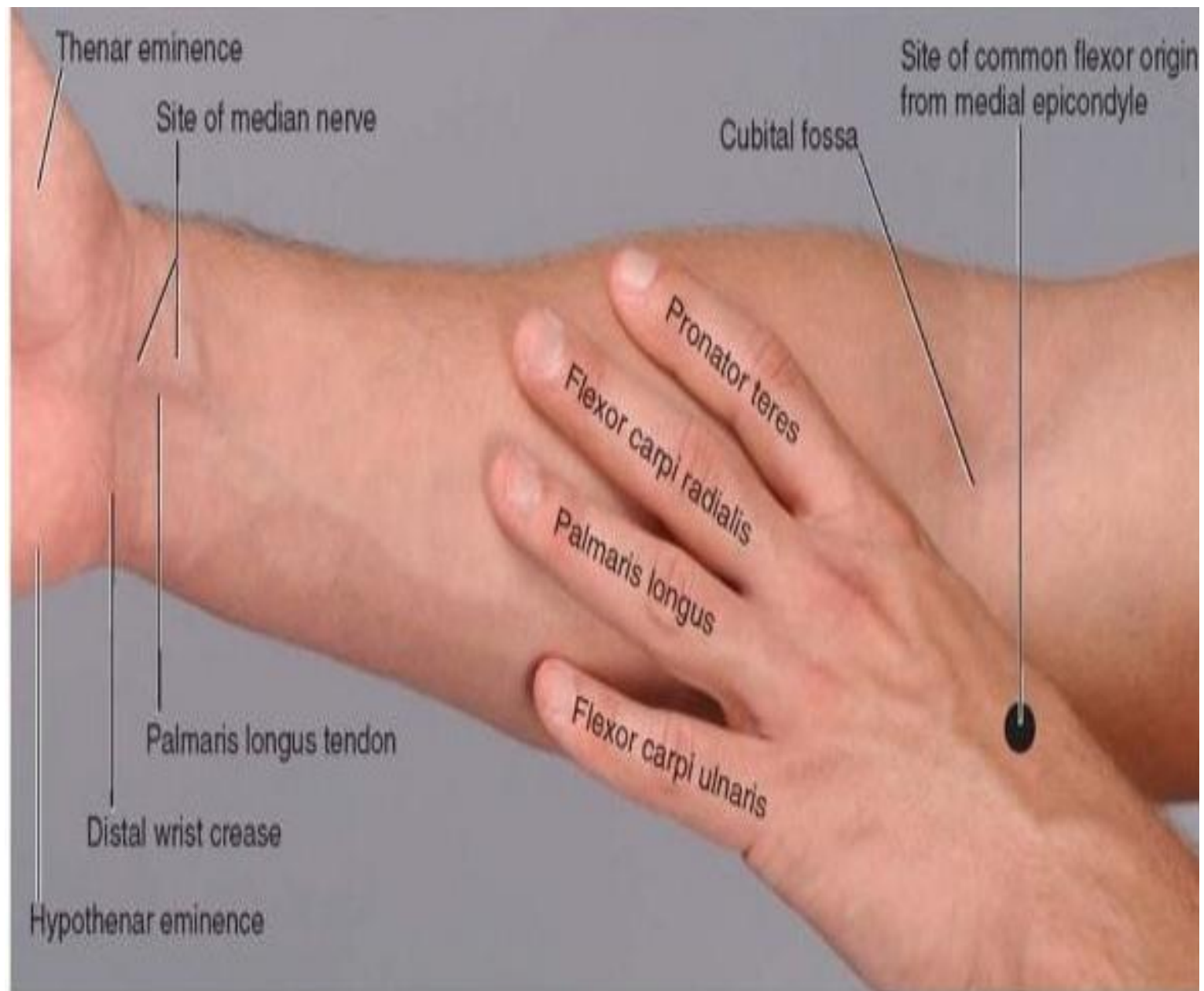
- ✚ Three muscles: **Flexor digitorum profundus, Flexor pollicis longus, Pronator quadratus**

- **Functionally**, the **brachioradialis** is a flexor of forearm, but it is located in posterior or extensor compartment and is thus supplied by **radial nerve**

Superficial (first) layer muscles

- **Pronator teres**
 - Innervation: median nerve (C6, C7)
 - Action: pronates and flexes forearm
- **Flexor carpi radialis**
 - Innervation: median nerve (C6, C7)
 - Action: Flexes and abducts hand (at wrist)





- **Palmaris longus**

- Innervation: median nerve (C7, C8)
- Action: Flexes hand (at wrist) and tenses palmar aponeurosis
- It may be absent(10-14 %)

- **Flexor carpi ulnaris**

- Origin
 - Humeral head: common flexor origin
 - Ulnar head: Olecranon and posterior border (via aponeurosis)
- Insertion: Pisiform, hook of hamate, 5th metacarpal
- ***Innervation: ulnar nerve (C7, C8)***
- Action: Flexes and adducts hand (at wrist)

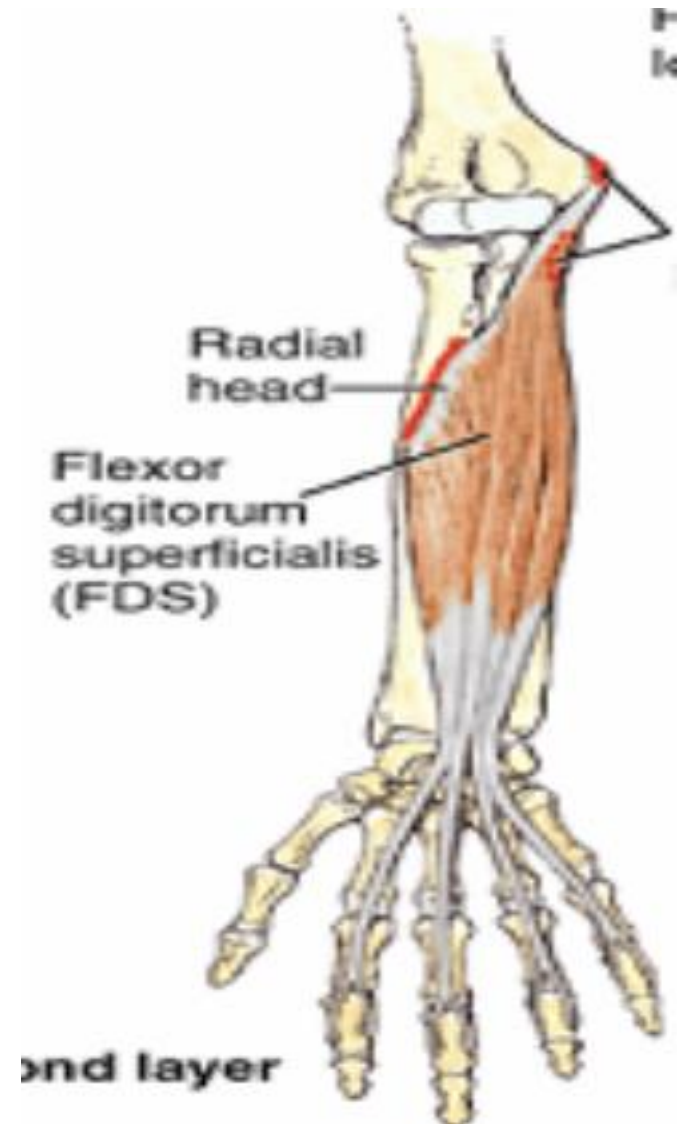
Intermediate (second) layer muscles

❖ Flexor digitorum superficialis (FDS)

✚ **Innervation:** Median nerve (C7, C8, T1)

✚ **Action:** Flexes middle phalanges at proximal interphalangeal joints of middle four fingers; acting more strongly,

✚ it also flexes proximal phalanges at metacarpophalangeal joints



Deep (third) layer muscles

❑ Flexor digitorum profundus (FDP)

– Innervation:

- Medial part: **Ulnar nerve** (C8, T1)
- Lateral part: **Anterior interosseous nerve**, from **median nerve** (C8, T1)

– Action:

- Medial part: Flexes **distal phalanges** 4 and 5 at distal interphalangeal joints
- Lateral part: Flexes **distal phalanges** 2 and 3 at distal interphalangeal joints



□ Flexor pollicis longus (FPL)

- Innervation: Anterior interosseous nerve, from **median nerve (C8, T1)**
- Action: Flexes phalanges of 1st digit (thumb)



□ Pronator quadratus

- Innervation: Anterior interosseous nerve, from **median nerve (C8, T1)**
- Action: Pronates forearm; deep fibers bind radius and ulna together



Extensor muscles of the forearm

- The extensor muscles are in the posterior (extensor-supinator) compartment of the forearm
- All are innervated by branches of the **radial nerve**

1. Three functional groups:

□ Muscles that **extend and abduct or adduct the hand at wrist joint**

- ✚ Extensor carpi radialis longus
- ✚ Extensor carpi radialis brevis
- ✚ Extensor carpi ulnaris

□ Muscles that extend the medial four digits

- ✚ extensor digitorum
- ✚ extensor indicis
- ✚ extensor digiti minimi

□ Muscles that extend or abduct the thumb

- ✚ abductor pollicis longus
- ✚ extensor pollicis brevis
- ✚ extensor pollicis longus

❖ Extensor retinaculum

- held extensor tendons in place in wrist region
- as tendons pass over the dorsum of the wrist, they are covered with synovial tendon sheaths to reduce friction

2. Two anatomical groups: superficial and deep layers

Superficial extensors

✚ attached proximally by a common extensor tendon to lateral epicondyle

▶ extensor carpi radialis brevis

▶ extensor digitorum

▶ extensor digiti minimi

▶ extensor carpi ulnaris

✚ attached proximally to the lateral supraepicondylar ridge and the adjacent lateral intermuscular septum

▶ brachioradialis

▶ extensor carpi radialis longus

Deep extensors

▪ Three act on the thumb

- deep to the superficial extensors and emerge ("crop out") from a furrow in the lateral part of the forearm = outcropping muscles

- Abductor pollicis longus

- Extensor pollicis brevis

- Extensor pollicis longus

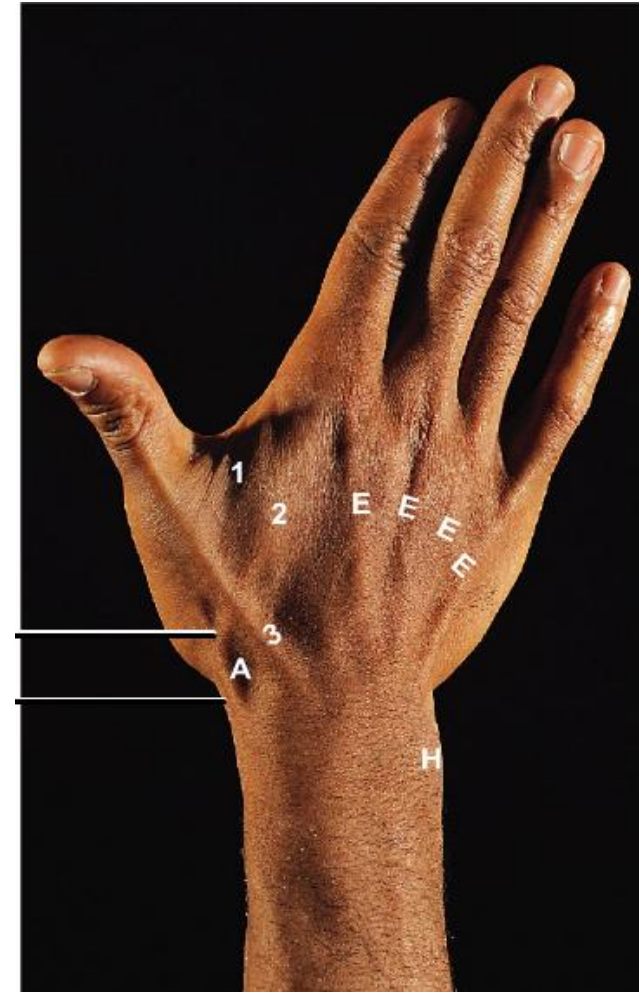
▪ One act on index finger

- Extensor indicis

- confers independence to the index finger

Anatomical snuff box

- A triangular shallow depression on the lateral aspect of the wrist when the thumb is extended fully
 - this draws the APL, EPB, and EPL tendons up and producing a concavity between them
- The tendons of the **APL** and **EPB** bound anatomical snuff box **laterally**, and tendon of **EPL** bounds it **medially**
- **Radial artery** lies on the floor of the snuffbox

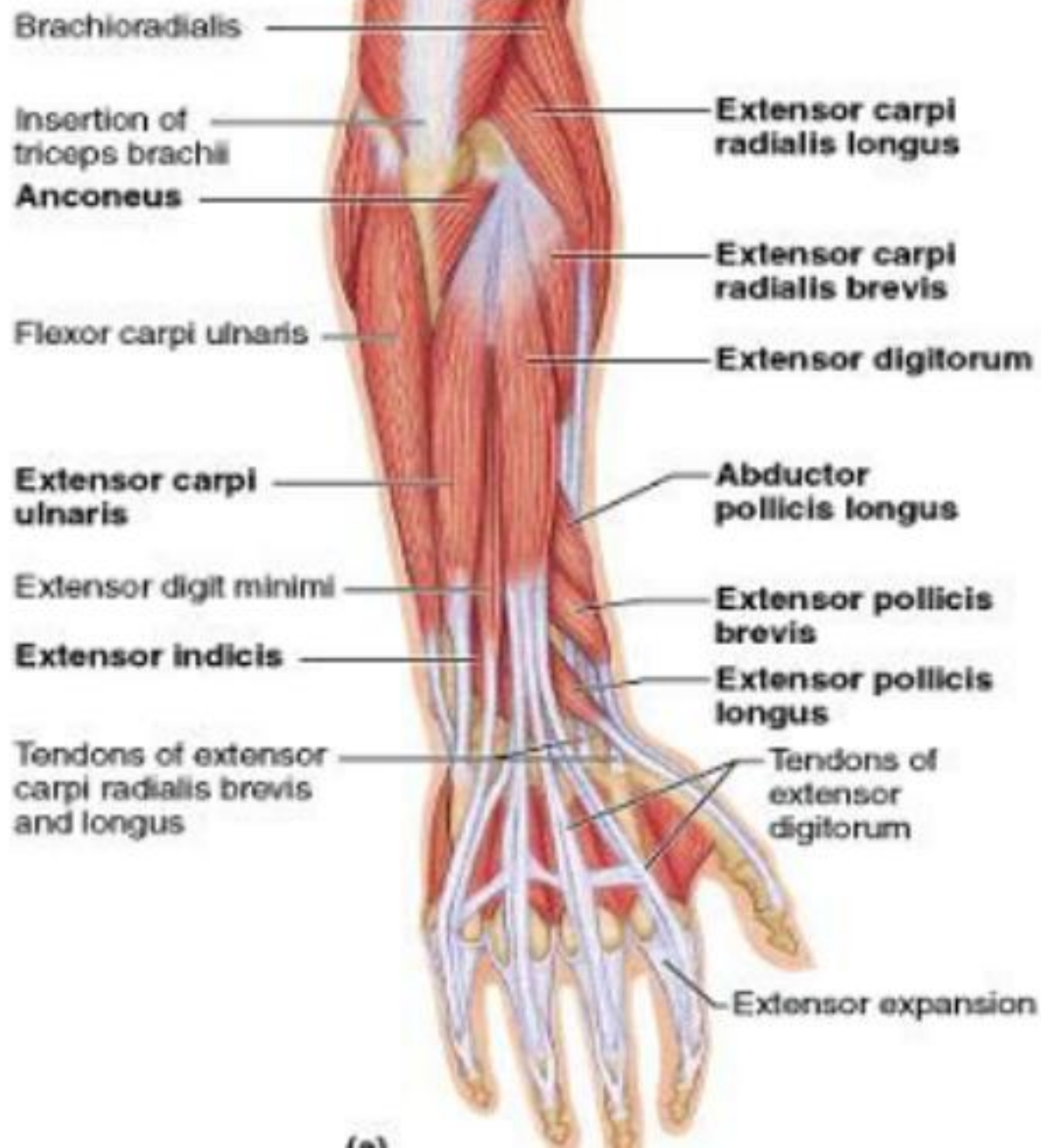


Muscles of the posterior compartment

Superficial layer

- **Brachioradialis**
 - Innervation: Radial nerve (C5,C6,C7)
 - Action: Relatively weak flexion of forearm
- **Extensor carpi radialis longus**
 - Innervation: Radial nerve (C6,C7)
 - Action: Extend and abduct hand at wrist joint
- **Extensor carpi radialis brevis**
 - Innervation: Deep branch of radial nerve (C7 & C8)
 - Action: Extend and abduct hand at wrist joints

- **Extensor digitorum**
 - Innervation: Posterior interosseous nerve (C7,C8)
 - Action: Extends medial 4 fingers at metacarpophalangeal and interphalangeal joints
- **Extensor digiti minimi**
 - Innervation: Posterior interosseous nerve (C7,C8)
 - Action: Extends 5th finger at metacarpophalangeal and interphalangeal joints
- **Extensor carpi ulnaris**
 - Innervation: Posterior interosseous nerve (C7,C8)
 - Action: Extend and abduct hand at wrist joint



Deep layer

❖ **Supinator**

- **Innervation:** Deep branch of radial nerve (C7, C8)
- **Action:** Supinates forearm; rotates radius to turn palm anteriorly or superiorly (if elbow is flexed)

❖ **Abductor pollicis longus (APL)**

- **Innervation:** Posterior interosseous nerve (C7, C8), continuation of deep branch of radial nerve
- **Action:** Abducts thumb and extends it at carpometacarpal joint

❖ **Extensor pollicis longus (EPL)**

Action: Extends distal phalanx of thumb at interphalangeal joint; extends metacarpophalangeal and carpometacarpal joints

❖ **Extensor pollicis brevis (EPB)**

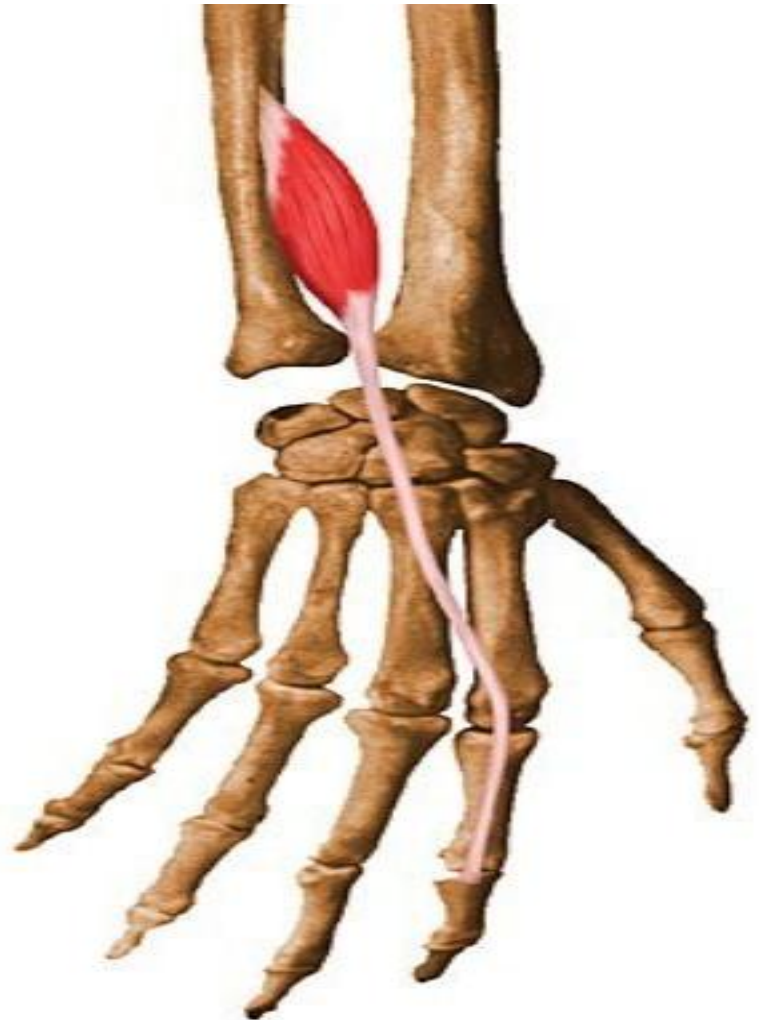
Action: Extends proximal phalanx of thumb at metacarpophalangeal joint; extends carpometacarpal joint

- **Innervation:** Posterior interosseous nerve (C7, C8), continuation of deep branch of radial nerve



❖ **Extensor indicis**

- **Innervation:** Posterior interosseous nerve (C7, C8), continuation of deep branch of radial nerve
- **Action:** Extends 2nd finger (enabling its independent extension); helps extend hand at wrist



Nerves of the forearm

- ✦ The major nerves of the forearm are the **median**, **ulnar**, and **radial**
- ✦ Although the radial nerve appears in the cubital region, it soon enters the posterior compartment of the forearm
- ✦ Besides the cutaneous branches, there are only two nerves of the anterior aspect of the forearm: the median and ulnar nerves

Median nerve

- ✦ Enters the forearm between the two **heads of pronator teres**
- ✦ Here the **nerve gives off its anterior interosseous branch**
 - supplies deep muscles of forearm (**flexor pollicis longus, flexor digitorum profundus (lateral half) and pronator quadratus**)
- ✦ Then lies on the deep aspect of flexor digitorum superficialis
- ✦ At the wrist, it becomes superficial on the ulnar side of flexor carpi radialis, exactly in the midline
 - Here it gives off a palmar cutaneous branch, which supplies the skin of the mid palm

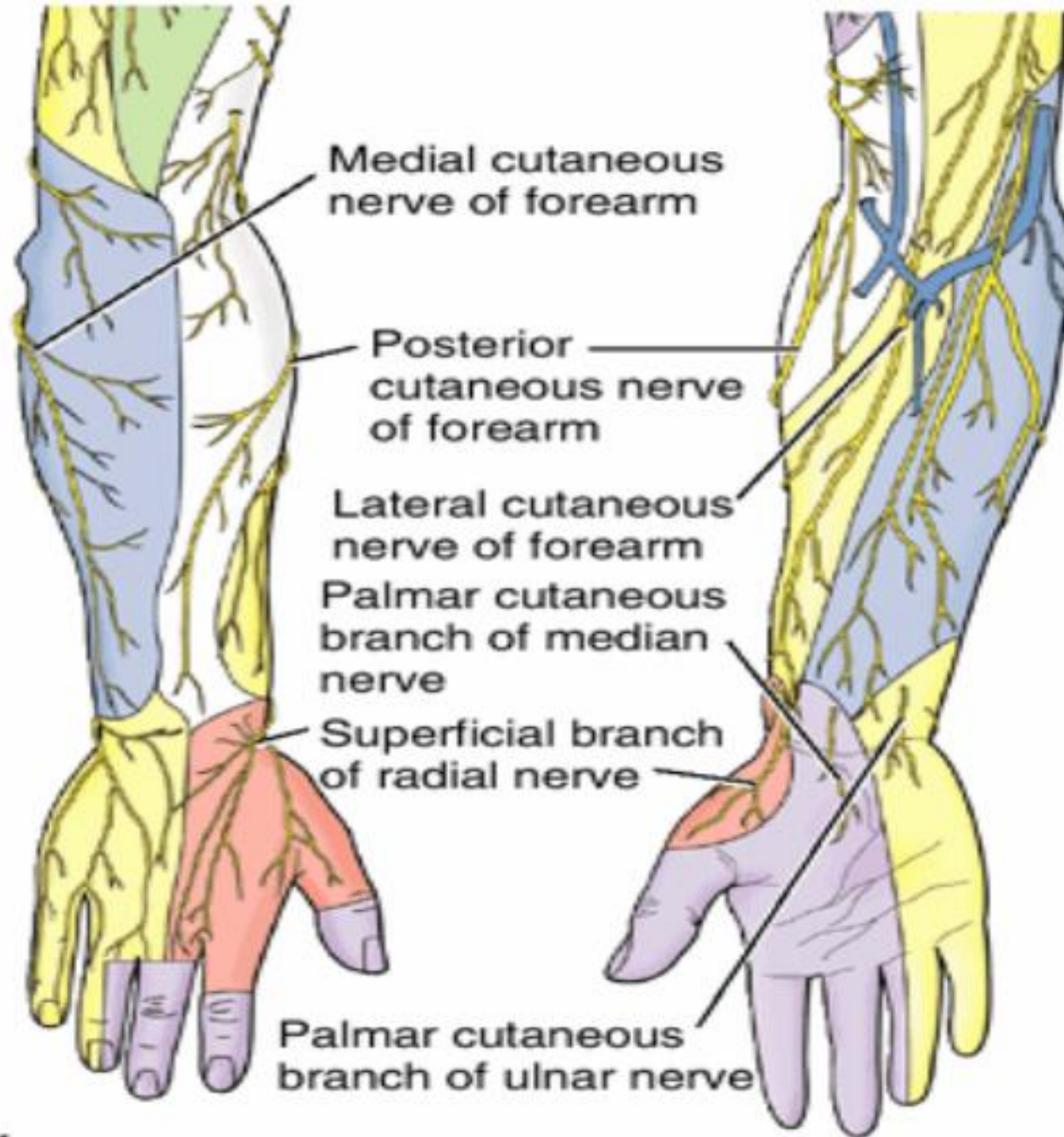
- It then passes deep to the **flexor retinaculum**, gives branches to
 - Thenar muscles, lateral two lumbricals and cutaneous branches
- **Branches of median nerve**

❖ **Muscular**

- all muscles of the flexor aspects of the forearm except (apart from flexor carpi ulnaris and the ulnar half of flexor digitorum profundus)
- thenar eminence muscles
- radial two lumbricals

❖ **Cutaneous**

- Lateral side of the palm
- palmar, and a variable degree of the dorsal, aspect of the lateral 3 & 1/2 digits



Cutaneous nerves

Ulnar nerve

- ✦ Passes behind the medial epicondyle to enter the forearm
- ✦ Descends beneath flexor carpi ulnaris until this muscle thins out into its tendon, leaving the nerve to lie superficially on its radial side
- ✦ In the distal two thirds of the forearm the nerve is accompanied by the **ulnar artery** which lies on the nerve's radial side
- ✦ About 5 cm above the wrist, a dorsal cutaneous branch passes deep to flexor carpi ulnaris to supply the dorsal aspects of the ulnar 1 & 1/2 fingers and the dorsal aspect or the ulnar side of the hand

- ✚ Crosses flexor retinaculum superficially to break up into
 - superficial terminal branch, supplying the ulnar 1 & 1/2 fingers
 - deep terminal branch which supplies the hypothenar muscles and the intrinsic muscles of the hand

✚ Branches

✚ Muscular

- flexor carpi ulnaris
- medial half of flexor digitorum profundus
- hypothenar muscles
- All Interossei
- 3rd and 4th lumbricals
- adductor pollicis

✚ Cutaneous

- ulnar side of both aspects of the hand
- both surfaces of the ulnar 1 & 1/2 fingers

Radial nerve

- ✚ At the level of the lateral epicondyle its posterior interosseous nerve is given off, which winds round the radius within the supinator muscle then sprays out to be distributed to the extensor muscles of the forearm
- ✚ The radial nerve itself continues as the superficial radial nerve, lying deep to brachioradialis
- ✚ Above the wrist, it emerges posteriorly from beneath this muscle to end by dividing into cutaneous nerves to the posterior aspects of the radial 3 & 1/2 digits

• Branches

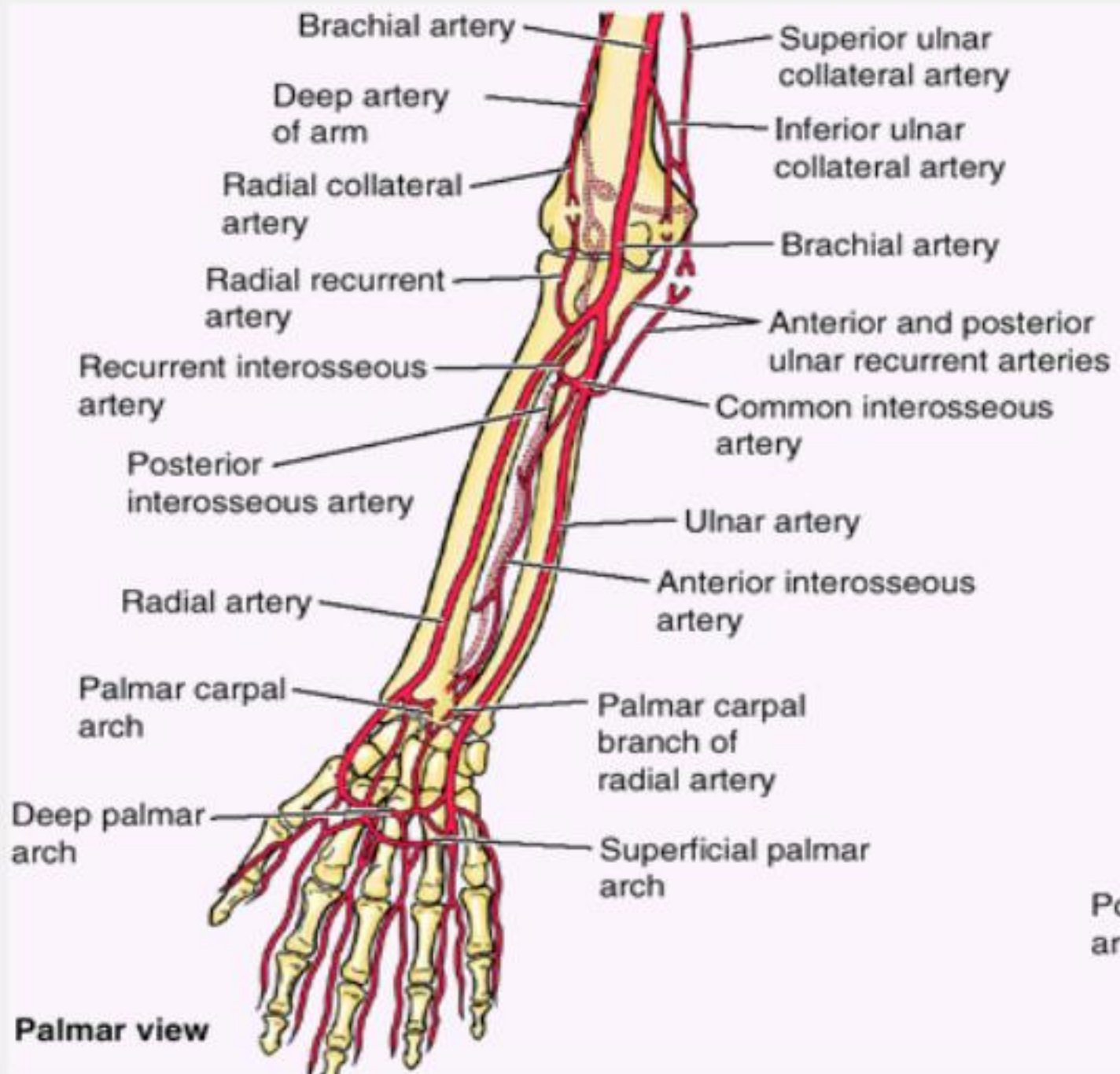
- ✚ The radial nerve is the nerve supply to the extensor aspect of the upper limb
- ✚ The **main trunk** itself innervates: triceps, anconeus, brachioradialis and extensor carpi radialis longus
- ✚ The **posterior interosseous** branch supplies all the remaining extensor muscles of the forearm together with the supinator and abductor pollicis longus
- ✚ Cutaneous branches are distributed to the back of the arm, forearm and radial side of the dorsum of the hand

Arteries and veins of forearm

- Brachial artery ends in distal part of cubital fossa opposite neck of radius by dividing into ulnar and radial arteries
 - The main arteries of the forearm
- There are superficial and deep veins in the forearm
 - Superficial veins lie in subcutaneous tissue
 - Deep veins accompany deep arteries

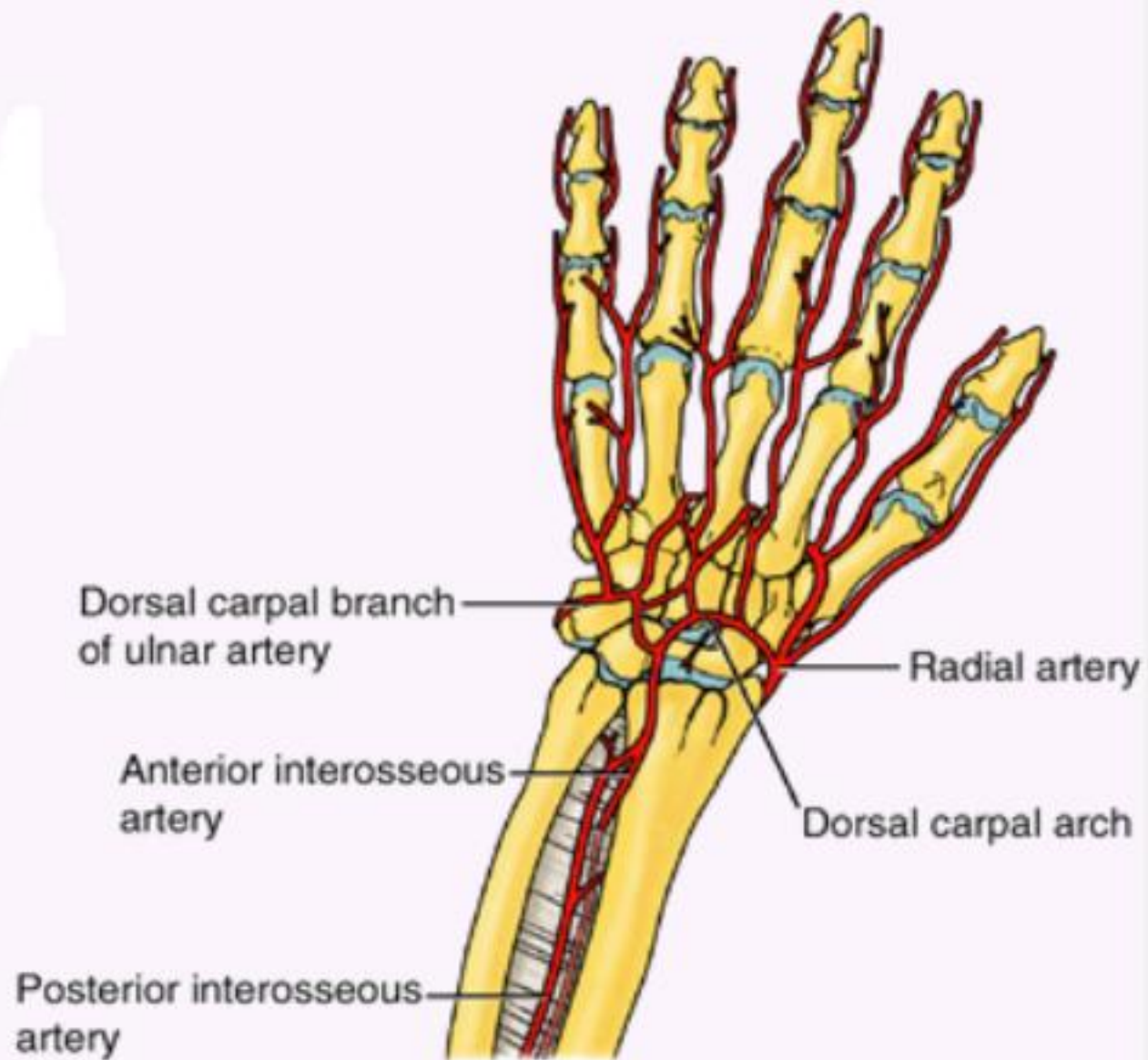
Ulnar artery

- **Origin:** As larger terminal branch of brachial artery in cubital fossa
- **Course:**
 - Descends inferiorly deep to pronator teres, palmaris longus and flexor digitorum superficialis to reach medial side of forearm
 - Passes superficial to flexor retinaculum at wrist in ulnar (Guyon) canal to enter hand
- **Branches**
 - ➡ Anterior ulnar recurrent artery
 - ➡ Posterior ulnar recurrent artery
 - ➡ Common interosseous artery
 - ➡ Anterior interosseous artery
 - ➡ Posterior Interosseous artery
 - ➡ Recurrent interosseous artery
 - ➡ Palmar carpal branch
 - ➡ Dorsal carpal branch



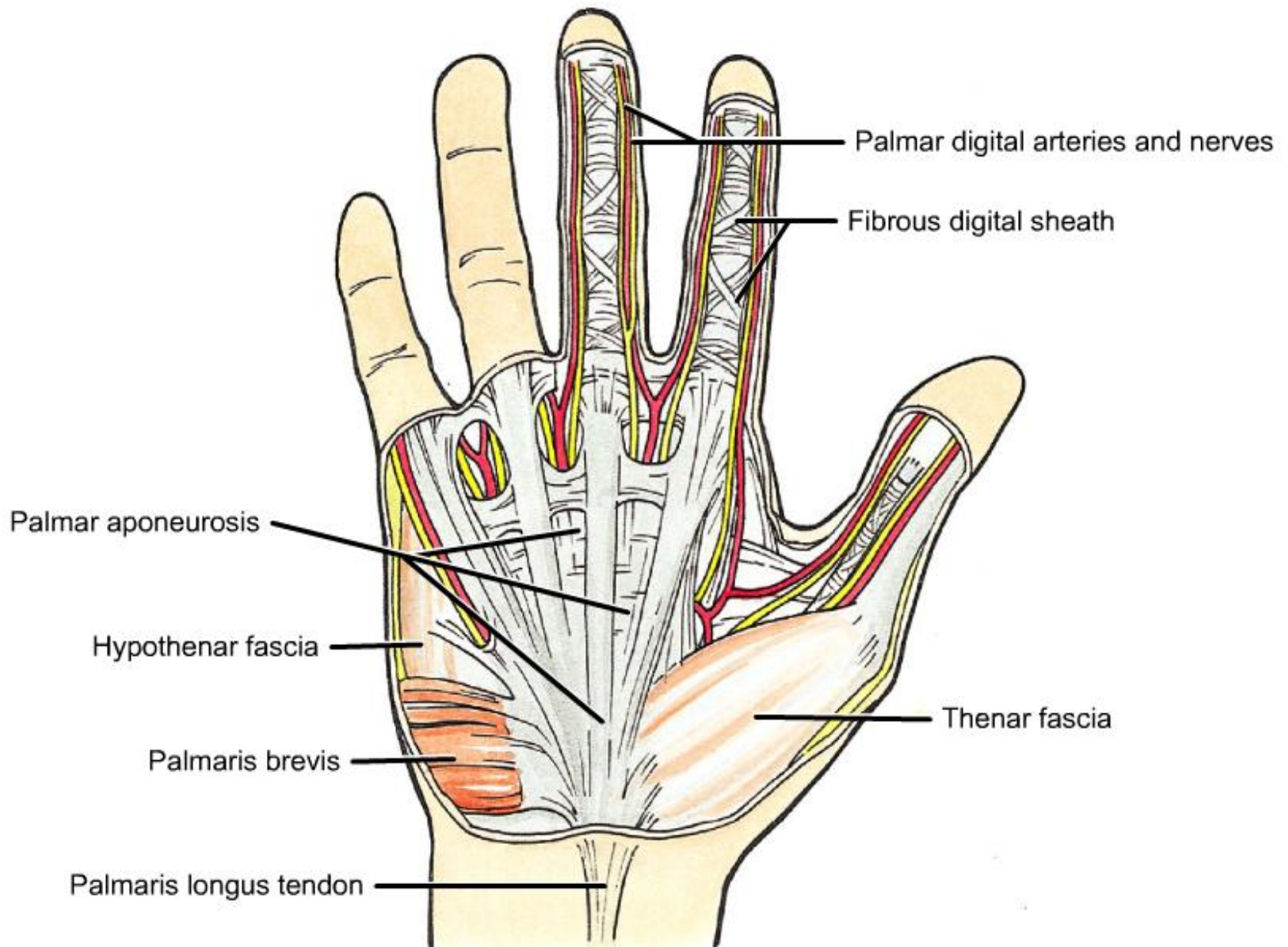
Radial artery

- Origin: As smaller terminal branch of brachial artery in cubital fossa
- Course: Runs inferolaterally under cover of brachioradialis; lies lateral to flexor carpi radialis tendon in distal forearm; winds around lateral aspect of radius and crosses floor of anatomical snuff box to pierce 1st dorsal interosseous muscle
- **Branches**
 - **Radial recurrent artery**
 - **Palmar carpal branch**
 - **Dorsal carpal branch**



Posterior view

The Hand



Fascia of palm

- Continuous with antebrachial fascia and the fascia of the dorsum of the hand
- Thin over thenar and hypothenar eminences, but it is thick centrally where it forms the **fibrous palmar aponeurosis** and in the fingers where it forms the **digital sheaths**

- **Palmar aponeurosis**

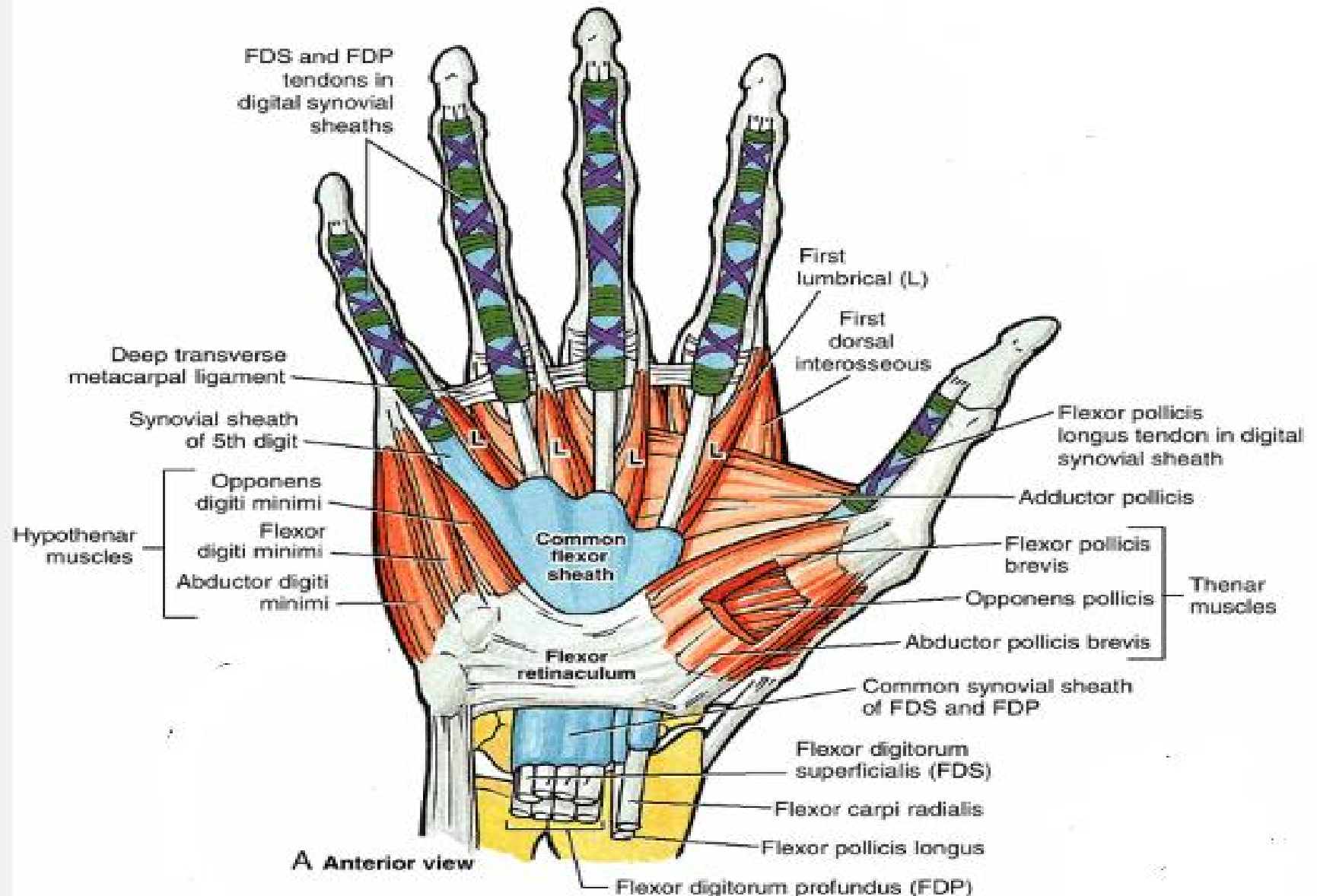
- strong, well defined part of the deep fascia of the palm
- covers the soft tissues and overlies the flexor tendons
- apex is continuous with the flexor retinaculum and palmaris longus tendon
- four longitudinal digital bands radiate from the apex and attach distally to the bases of the proximal phalanges

- **Fibrous digital sheaths**

- **ligamentous tubes** that enclose the flexor tendon and the synovial sheaths that surround them as they pass along the palmar aspect of their respective digit

Fibrous digital sheath

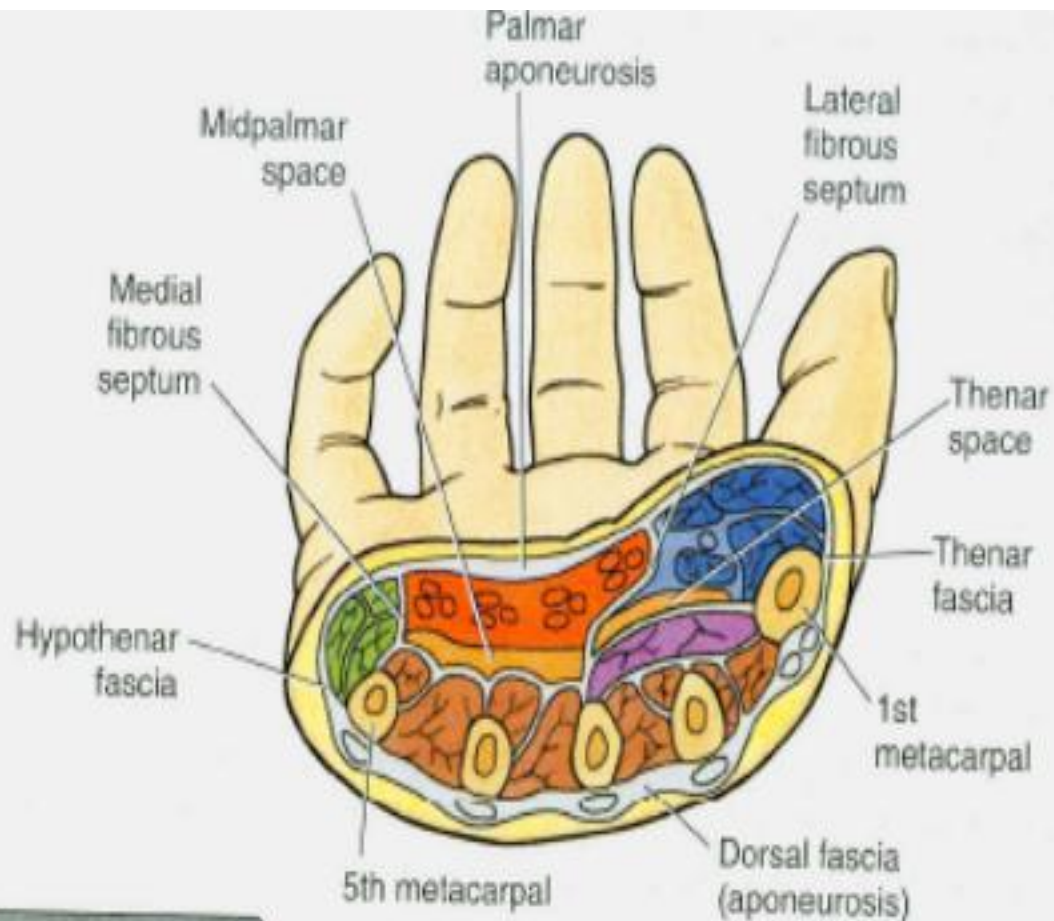
- Cruciate part (ligament)
- Anular part (ligament)



- **Medial fibrous septum**
 - Medial to this septum is the medial or hypothenar compartment containing the hypothenar muscles
- **Lateral fibrous septum**
 - Lateral to the septum is the lateral or thenar compartment containing the thenar muscles
- **Central compartment**
 - Between the hypothenar and the thenar compartments
 - Contains flexor tendons and their sheaths, the lumbrical muscles, the superficial palmar arterial arch, and the digital vessels and nerves



A Anterior view



B Transverse section

Key for A and B

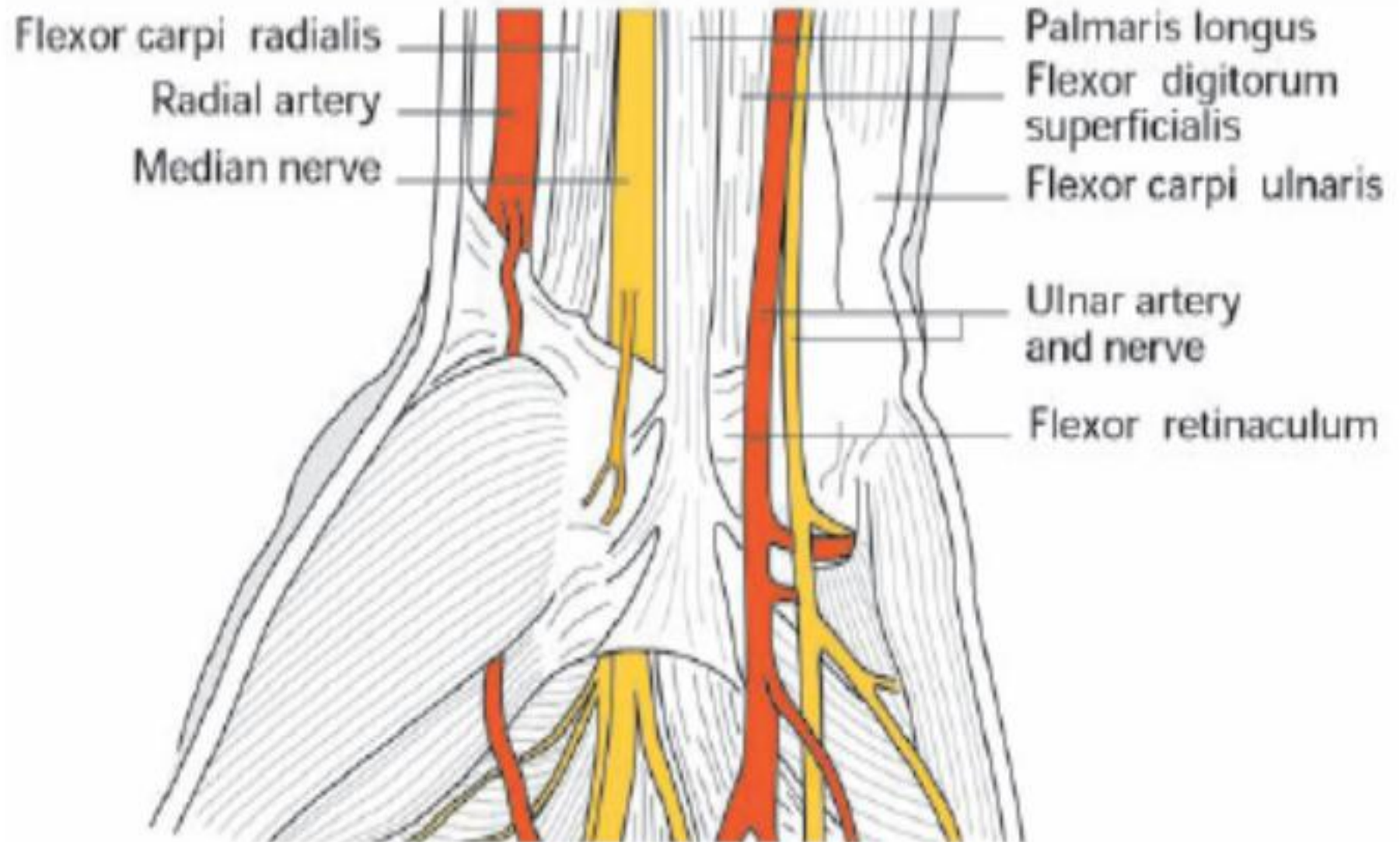
5 Compartments:

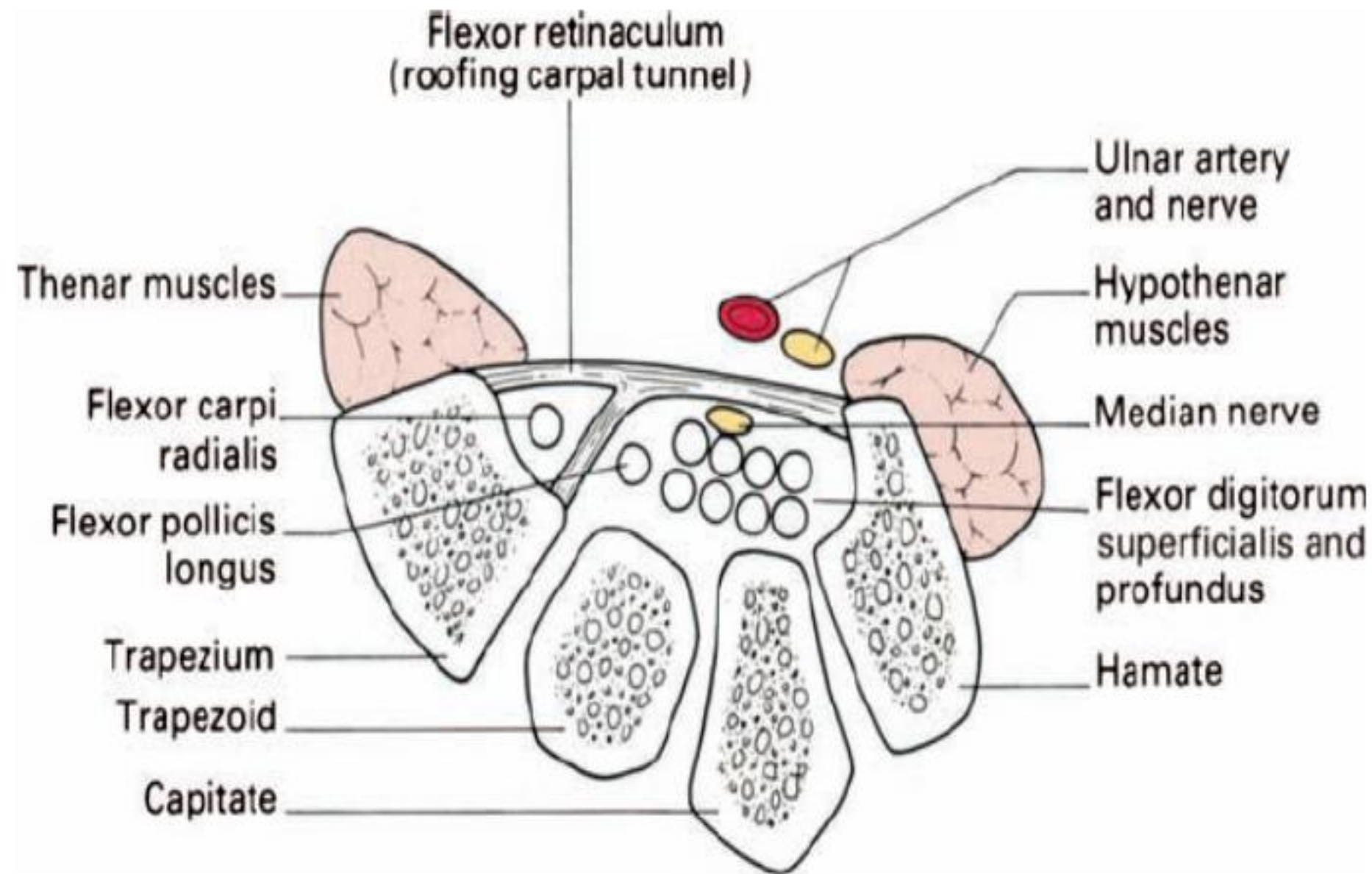
- 1. Hypothenar
- 2. Thenar
- 3. Central
- 4. Adductor
- 5. Interosseous

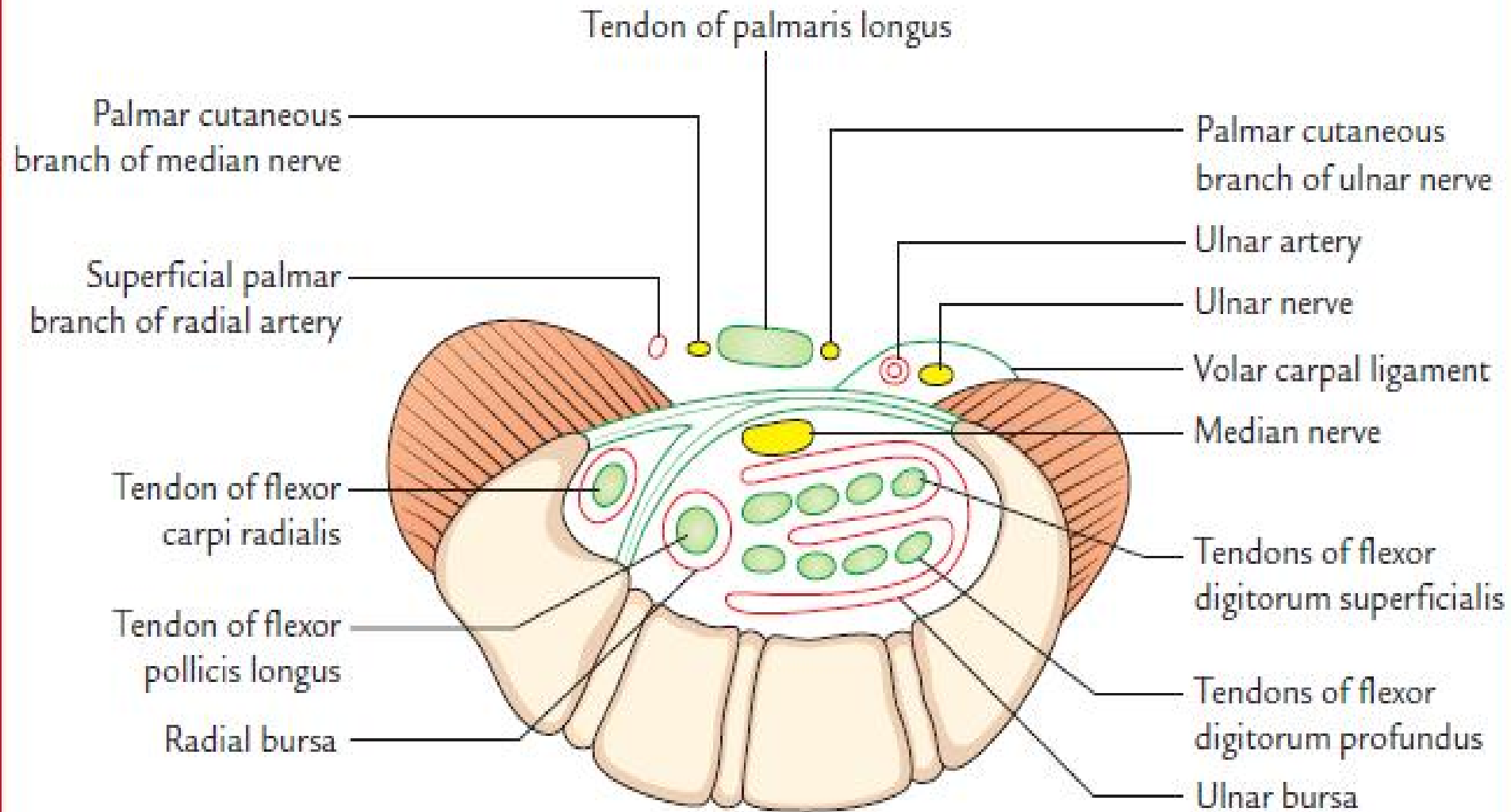
- Fibrous septa
- Midpalmar and thenar spaces

Flexor retinaculum

- Fibrous band which bridges the anterior concavity of carpus and converts it into a tunnel, carpal flexor canal
 - **Structures passing superficial to the retinaculum:**
 - Tendon of palmaris longus
 - Palmar cutaneous branches of median and ulnar nerves
 - Ulnar vessels and ulnar nerve
 - **Structures passing deep to the retinaculum:**
 - Median nerve
 - Tendons of flexor digitorum superficialis, profundus, flexor pollicis longus







Carpal Tunnel Syndrome

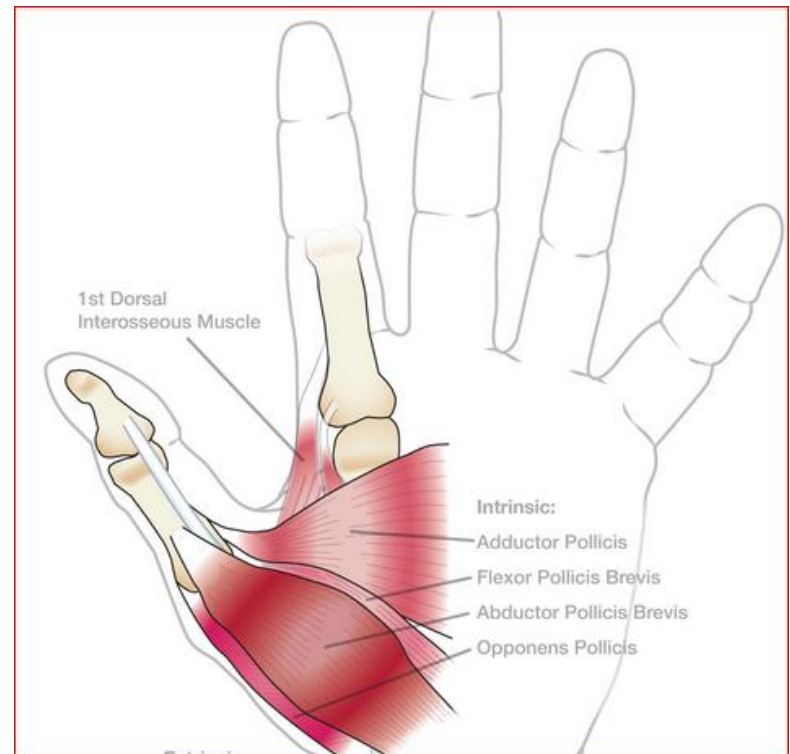
- Results from any lesion that significantly reduces the size of the carpal tunnel or, more commonly, increases the size of some of the structures that pass through it
- **The median nerve is the most sensitive structure in the carpal tunnel and, therefore, is the most affected**
- The median nerve has two terminal sensory branches that supply the skin of the hand; hence anesthesia may occur in the lateral three and a half digits
- The palmar cutaneous branch of the median nerve arises proximal to and does not pass through the carpal tunnel; thus sensation in the central palm remains unaffected

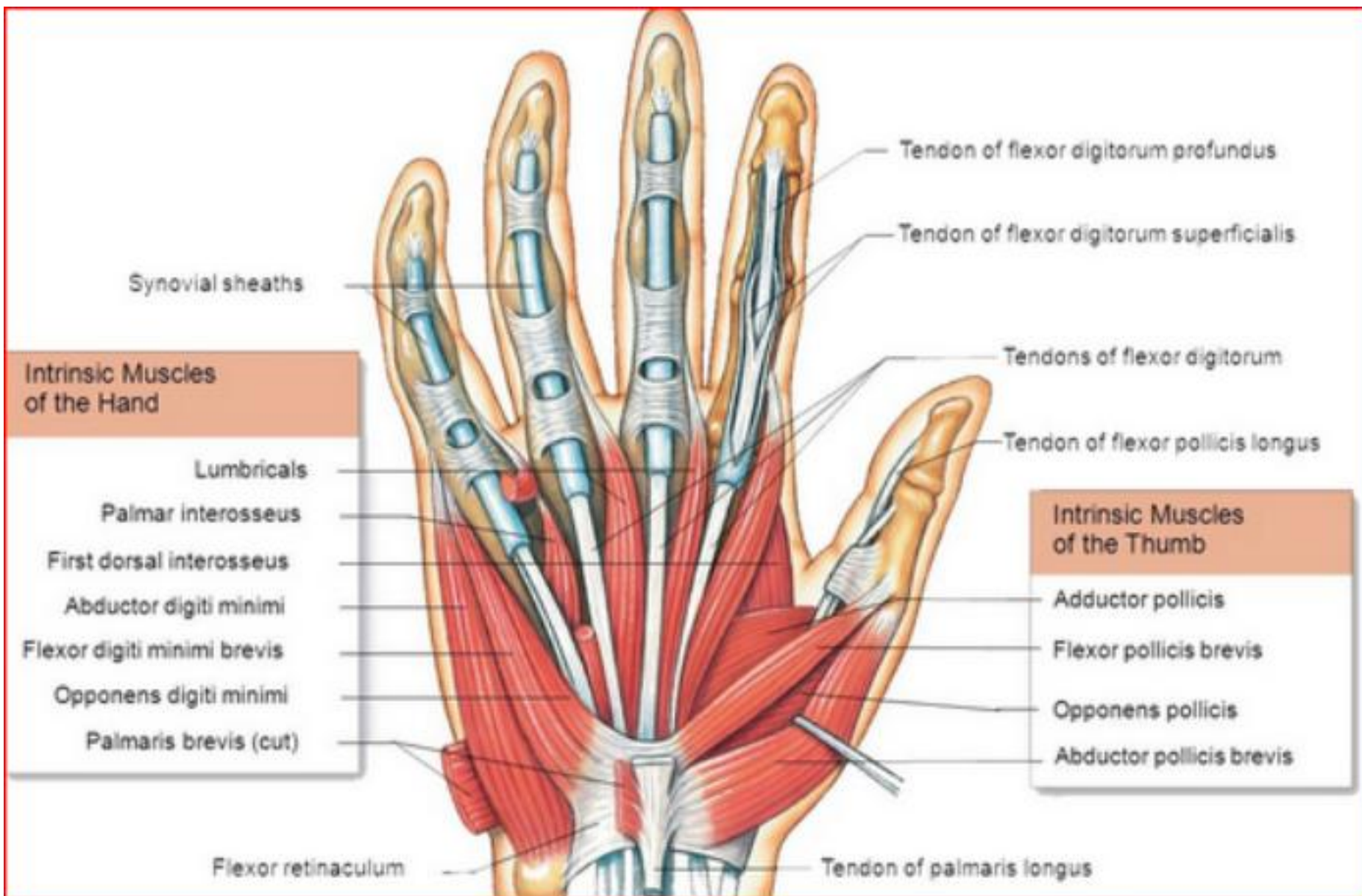
Muscles of the hand

- The intrinsic muscles of the hand are located in five compartments
 - ◆ **Thenar muscles in thenar compartment:** abductor pollicis brevis, flexor pollicis brevis, and opponens pollicis
 - ◆ **Hypothenar muscles in hypothenar compartment:** abductor digiti minimi, flexor digiti minimi brevis and opponens digiti minimi
 - ◆ **Adductor pollicis** in adductor compartment
 - ◆ **Lumbricals** in central compartment
 - ◆ **Interossei** in interosseous compartment

Thenar muscles

- Form thenar eminence on lateral surface of palm
- Responsible for opposition of thumb
- **Four muscles**
 - Opponens pollicis
 - Abductor pollicis brevis
 - Flexor pollicis brevis
 - **Adductor pollicis**





Opponens pollicis

- Innervation: recurrent branch of median nerve (C8 & T1)
- Action: opposition

Abductor pollicis brevis

- Innervation: recurrent branch of median nerve (C8 & T1)
- Action: abducts thumb; helps to oppose it

Flexor pollicis brevis

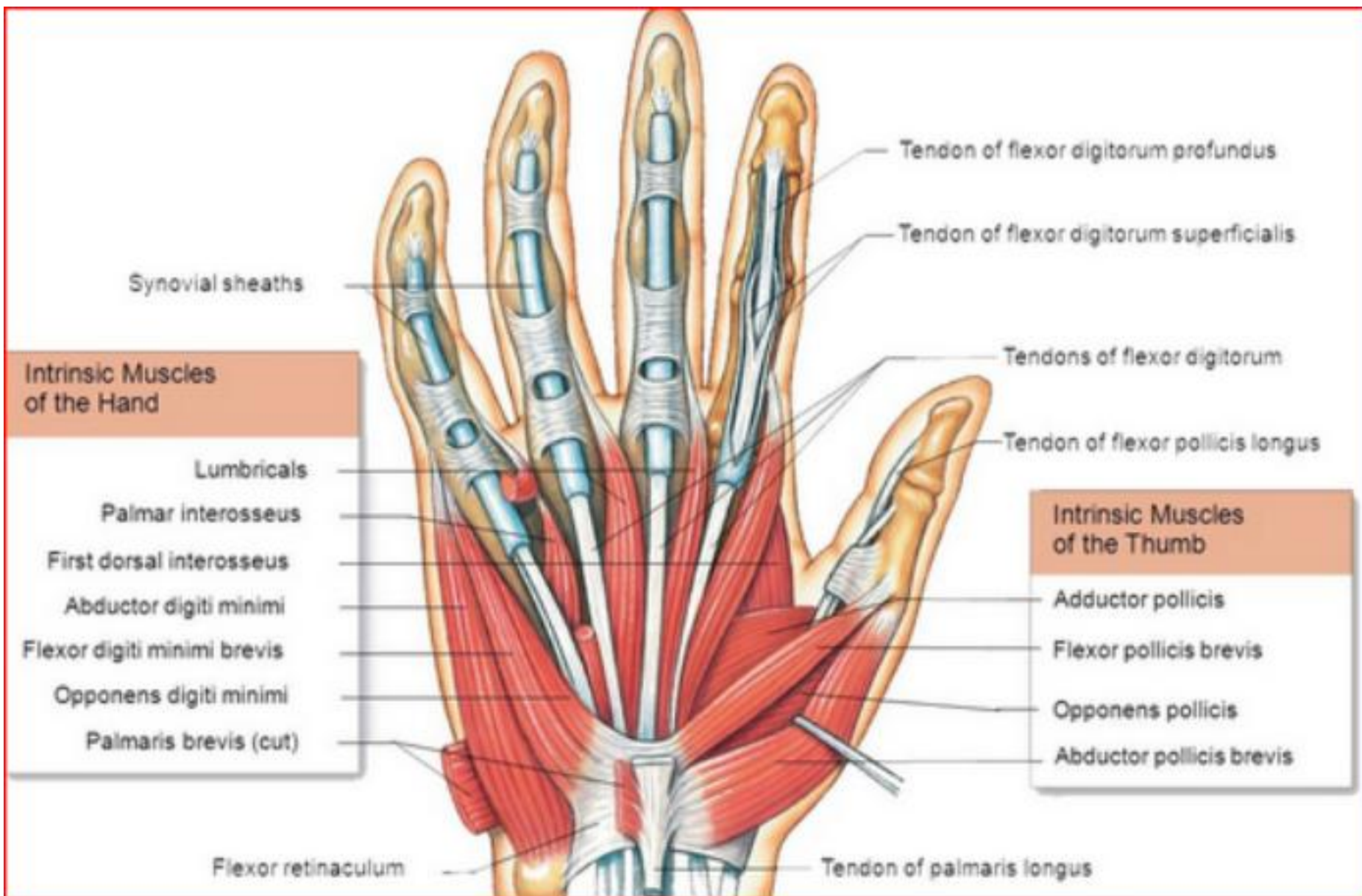
- Innervation:
 - Superficial head: recurrent branch of median nerve (C8 & T1)
 - Deep head: deep branch of ulnar nerve (C8 & T1)
- Action: flexes thumb

Adductor pollicis

- Innervation: deep branch of ulnar nerve (C8 & T1)
 - Action: adducts thumb

Hypothenar muscles

- In hypothenar compartment
- Produce hypothenar eminence
- Three muscles
 - Abductor digiti minimi
 - Flexor digiti minimi brevis
 - Opponens digiti minimi
- **Palmaris brevis**
 - In subcutaneous tissue of hypothenar eminence; not in hypothenar compartment
 - Covers and protects ulnar nerve and artery
 - All innervated by deep branch of **ulnar nerve(C8,T1)**

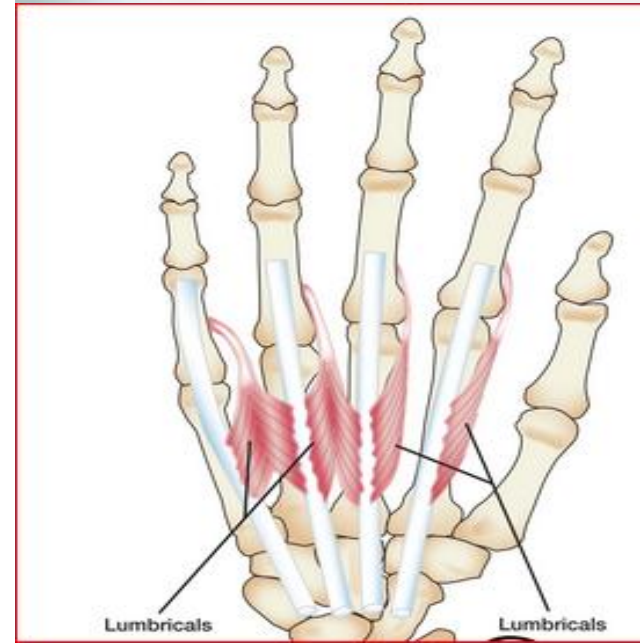
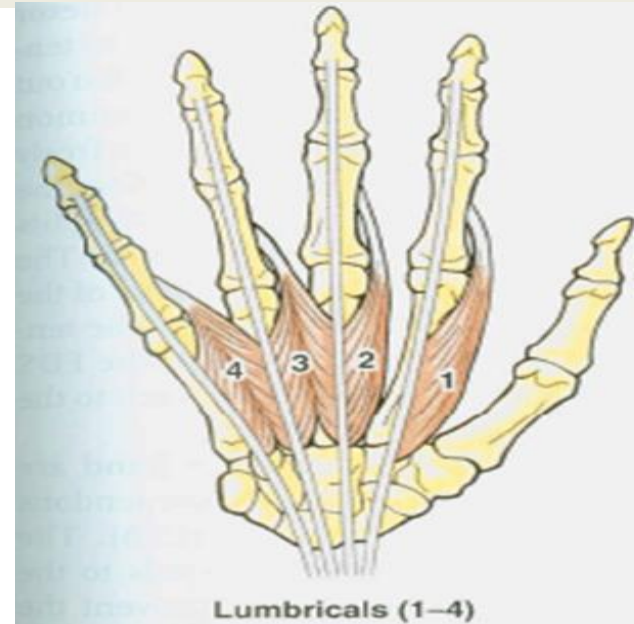


Short muscles

There are four lumbricals muscles in the palmar surface of the hand

Lumbricals, 1st to 4th

- **Innervation:**
 - The lateral two (1st and 2nd):
Median nerve (C8, T1)
 - The medial two (3rd and 4th):
Deep branch of ulnar nerve (C8, T1)
- **Action:** Flex metacarpophalangeal joints, and extend interphalangeal joints of 2nd - 5th fingers

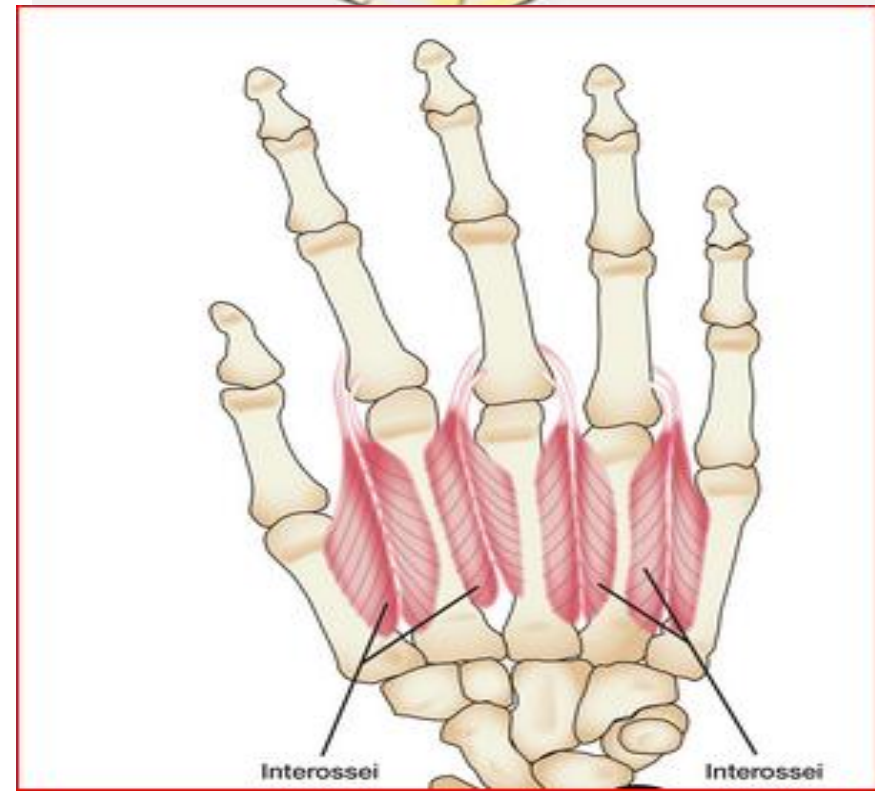
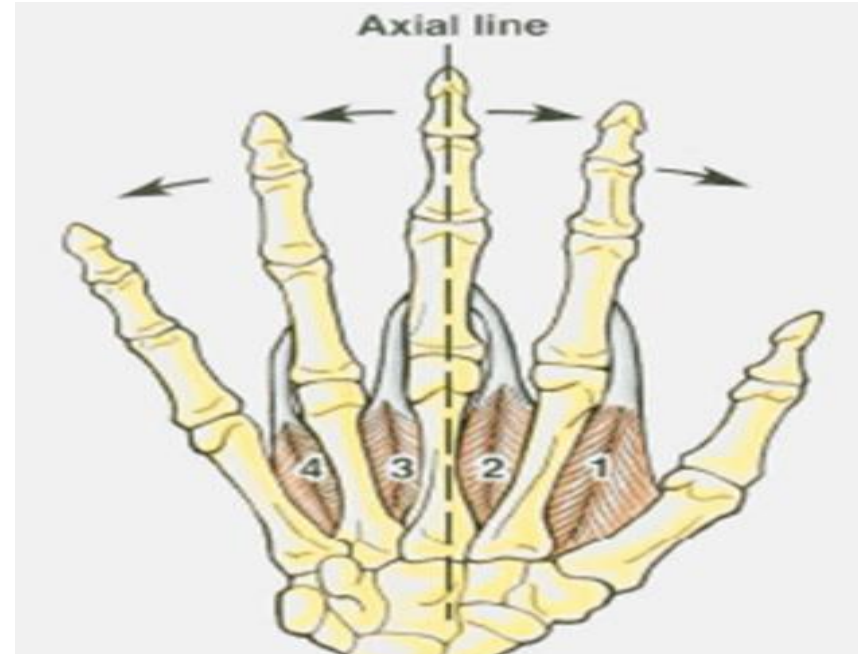


Interossei muscles

- ✚ There are 7 interossei muscles
- ✚ Four in the dorsal surface
- ✚ Three in the palmar surface
- ✚ All innervated by deep branch of ulnar nerve

Dorsal interossei, 1st to 4th

- **Innervation:** Deep branch of ulnar nerve (C8, T1)
- **Action:** Abduct 2nd to 4th fingers from axial line; act with lumbricals in flexing metacarpophalangeal joints and extending interphalangeal joints



Palmar interossei, 1st to 3rd

- **Innervation:** Deep branch of ulnar nerve (C8, T1)
- **Action:** Adduct 2nd , 4th , and 5th fingers toward axial line; assist lumbricals in flexing metacarpophalangeal joints and extending interphalangeal joints



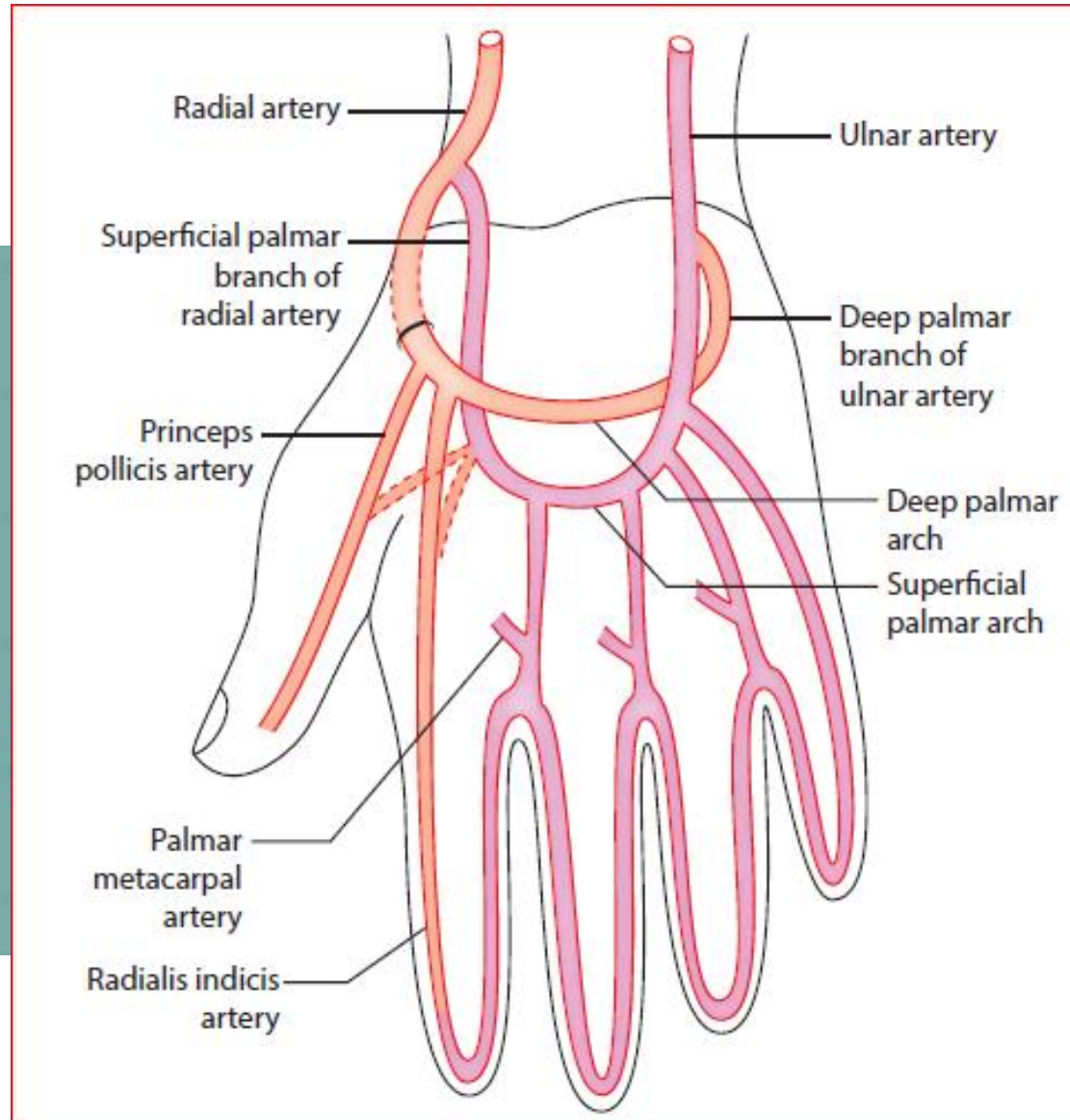
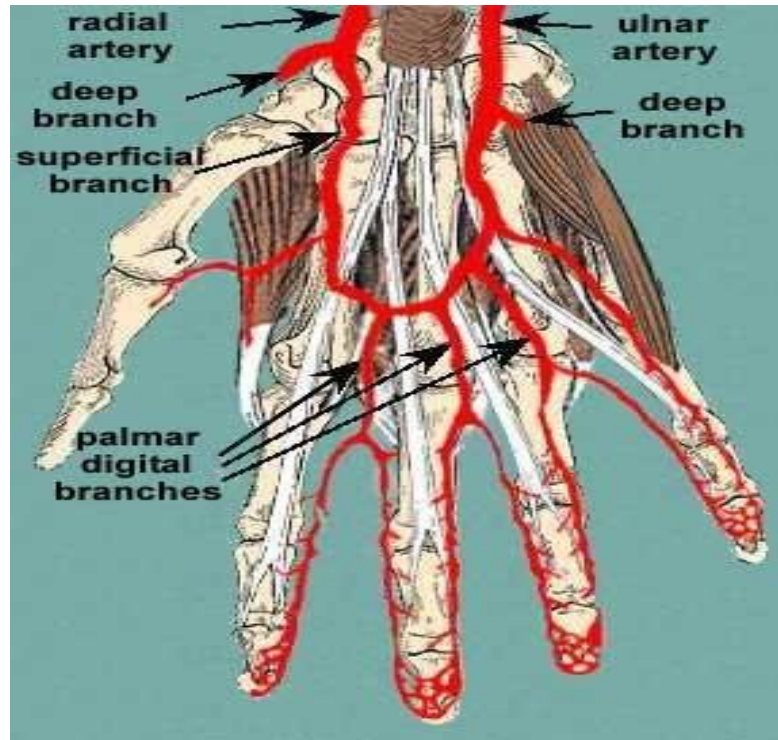
Arteries of Hand

Ulnar artery

- ✦ Enters the hand **anterior to the flexor retinaculum** between the pisiform and the hook of hamate via the ulnar canal (Guyon canal)
- ✦ Lies lateral to the ulnar nerve
- ✦ Divides in to two terminal branches
 - ▶ **Superficial palmar arch**
 - ▶ **Deep palmar branch**

■ Superficial palmar arch

- ✦ Formed by main termination of the **ulnar artery**
- ✦ It gives rise to three common palmar digital arteries that anastomose with palmar metacarpal arteries from the deep palmar arch
- ✦ Each common palmar digital artery divides into a pair of proper palmar digital arteries that run along the adjacent side of the 2nd - 4th fingers

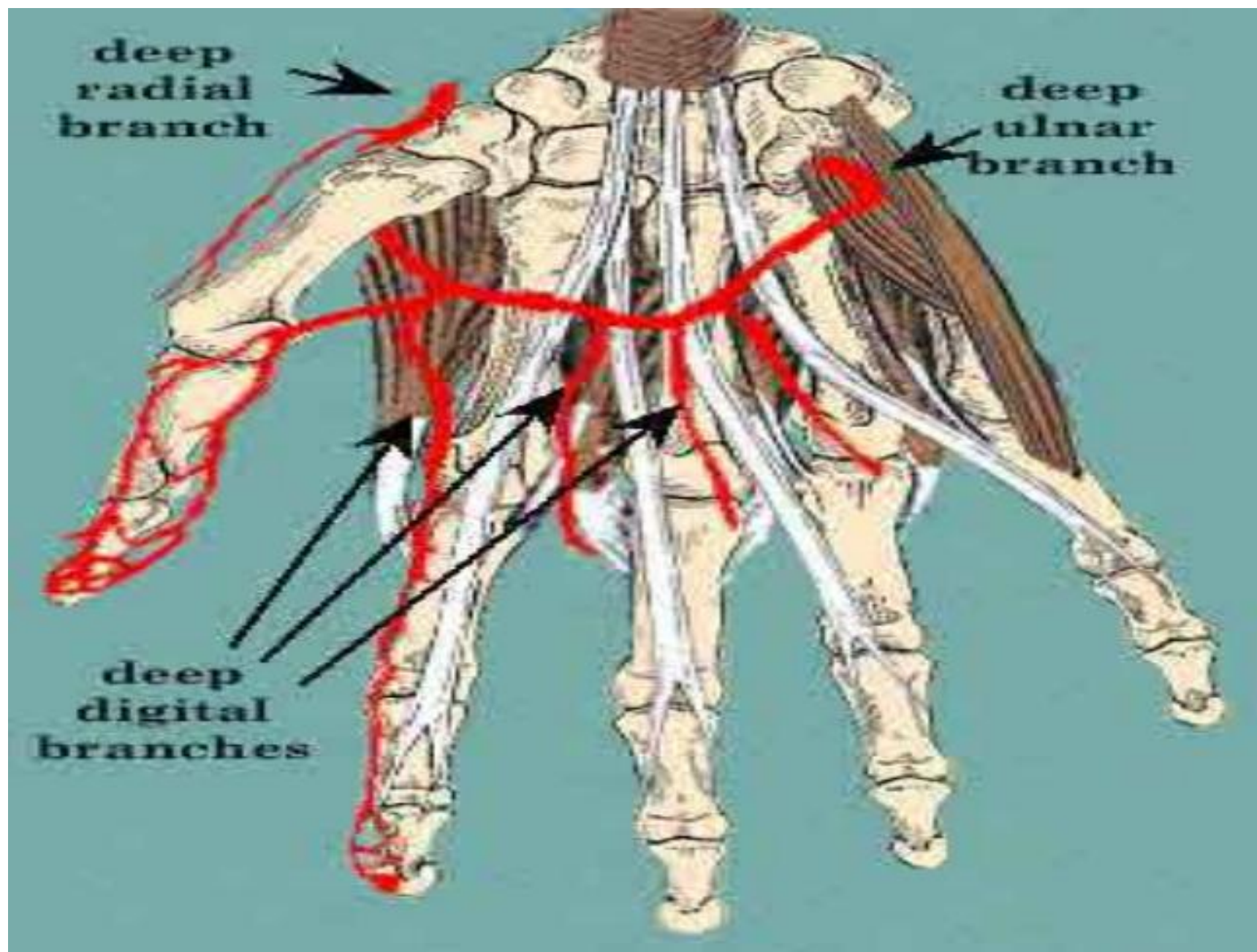


Radial artery

- ◆ Curves dorsally around the scaphoid and trapezium in the floor of anatomical snuff box
- ◆ Enters the palm by passing between the heads of 1st dorsal interosseous muscle
- ◆ Turns medially and passes between the heads of the adductor pollicis
- ◆ Ends by anastomosing with the deep branch of the ulnar artery to form the deep palmar arch

Deep palmar arch

- ✚ formed mainly by **radial artery**
- ✚ lies across the metacarpals just distal to their bases
- ✚ gives rise to three palmar metacarpal arteries, princeps pollicis artery and radialis indicis artery

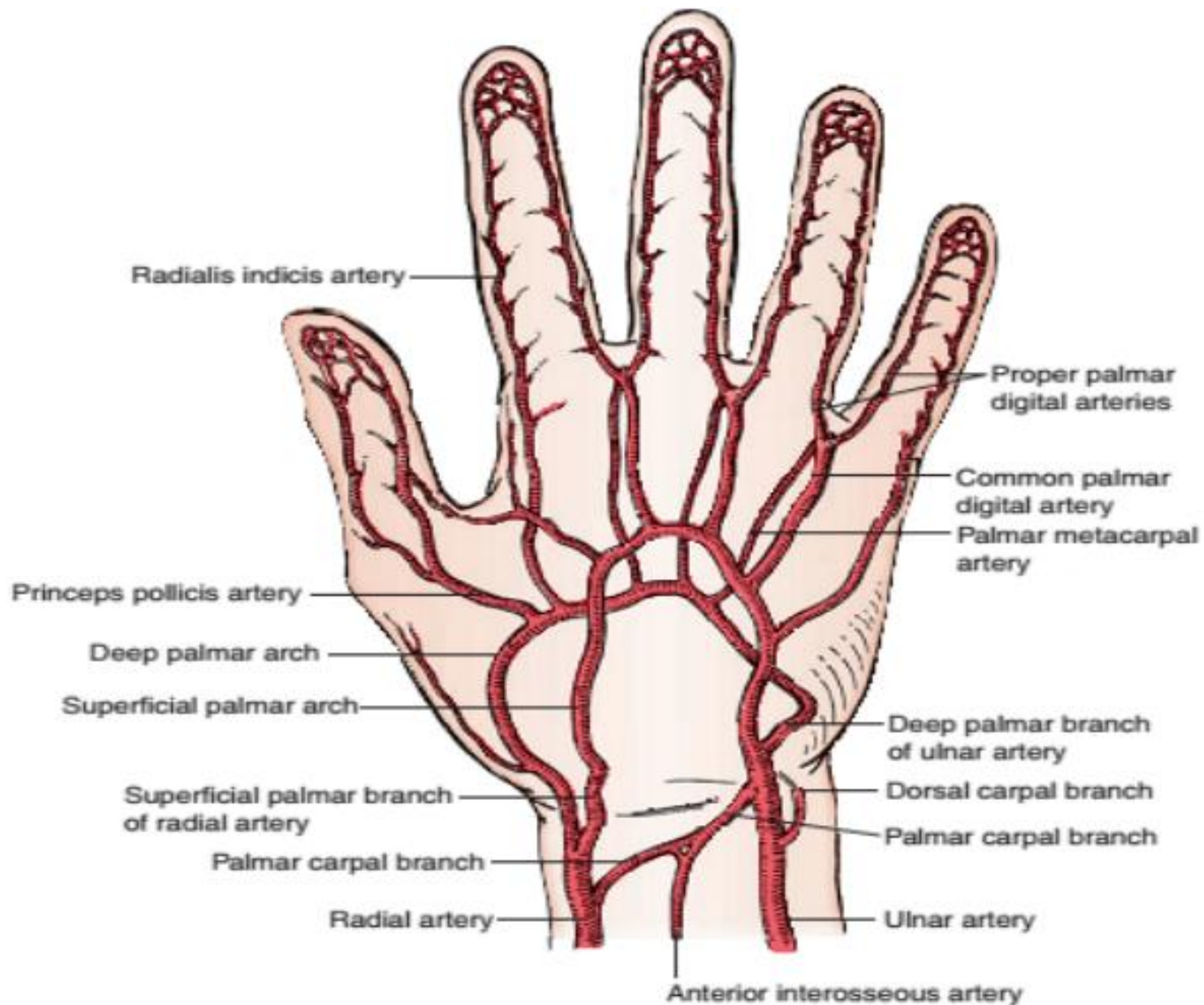


Veins of hand

- Superficial and deep palmar venous arches
 - ✚ associated with superficial and deep palmar arterial arches
 - ✚ drain into the deep veins of the forearm

Dorsal digital veins

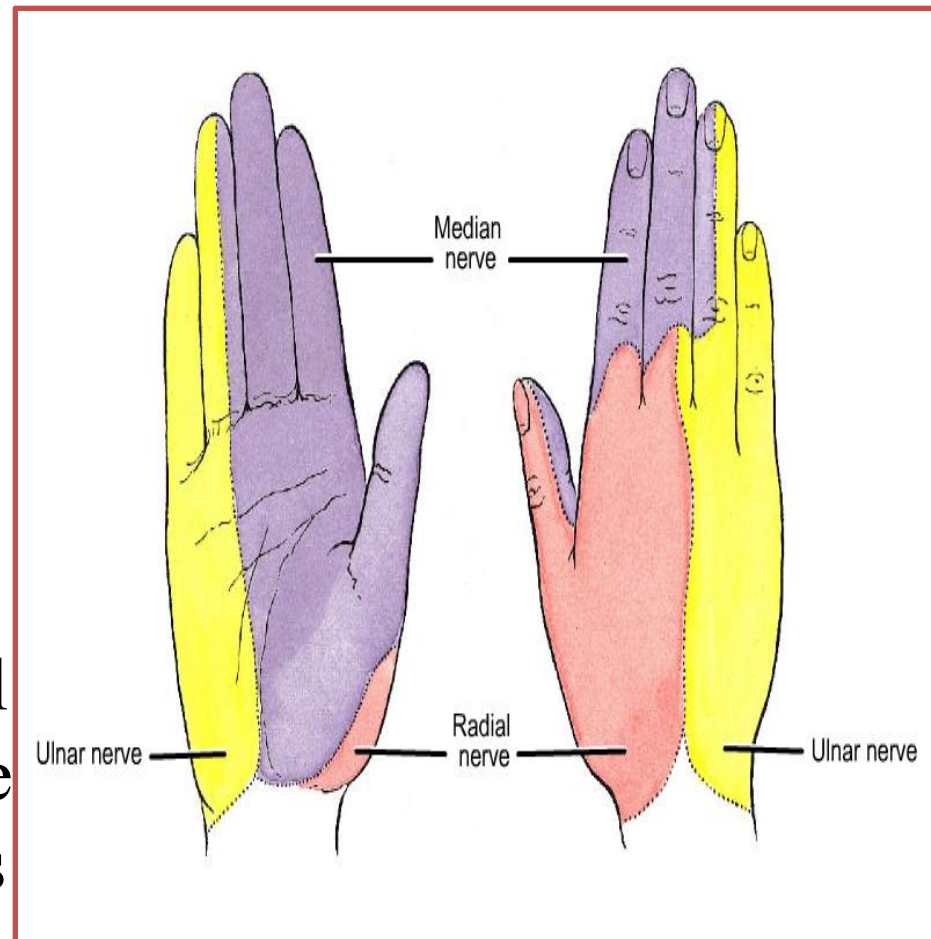
- ✚ drain into three dorsal metacarpal veins, which unite to form the dorsal venous network
- ✚ this network is prolonged proximally on the lateral side as **cephalic vein** and on the medial side as **basilic vein**



Nerves of hand

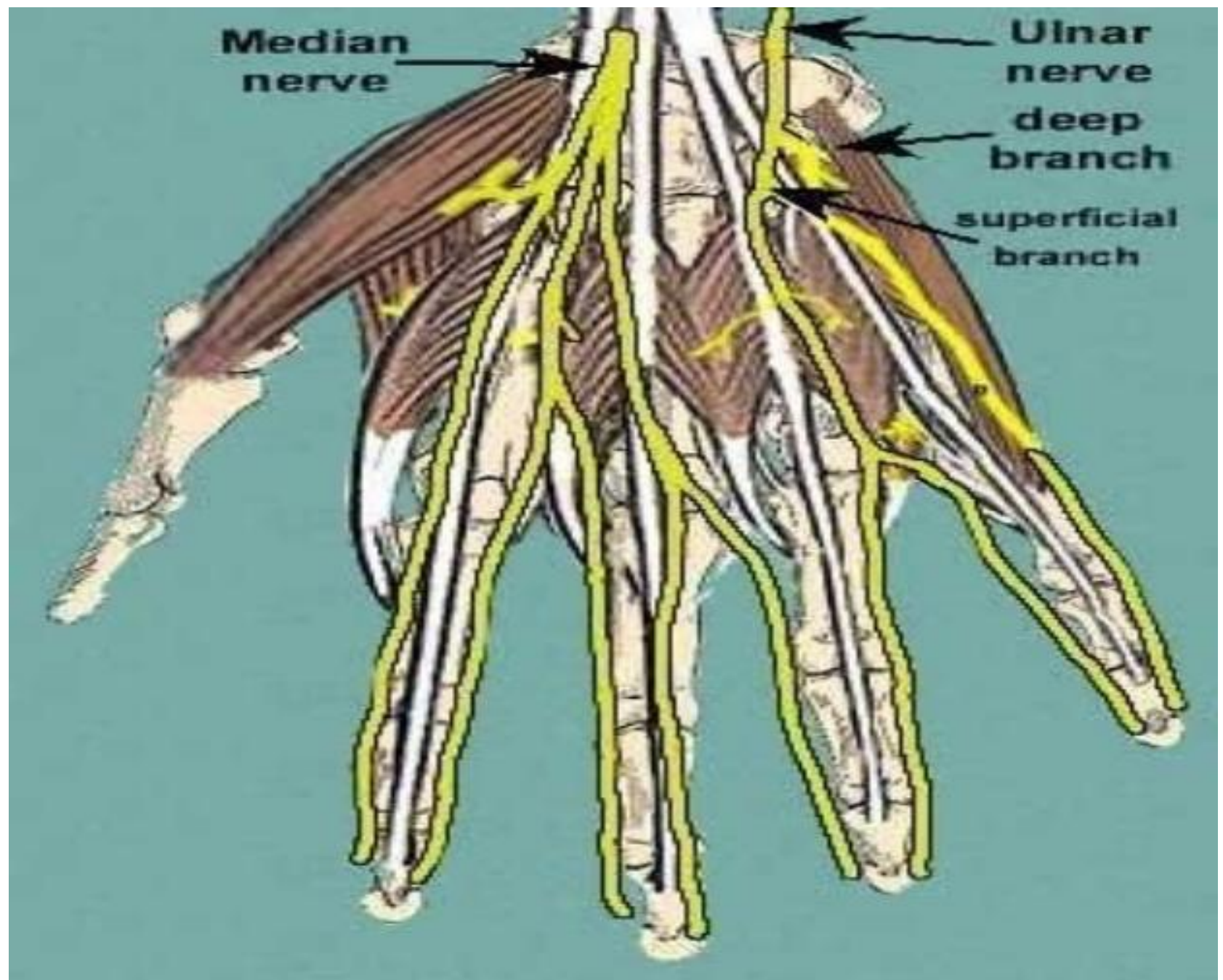
Median nerve

- Enters the hand through carpal tunnel
- Supplies
 - ✦ two and a half thenar muscles and the 1st and 2nd lumbricals
 - ✦ skin on the lateral palmar surface, the sides of the first three digits, the lateral half of the 4th digit, and the dorsum of the distal halves of these digits



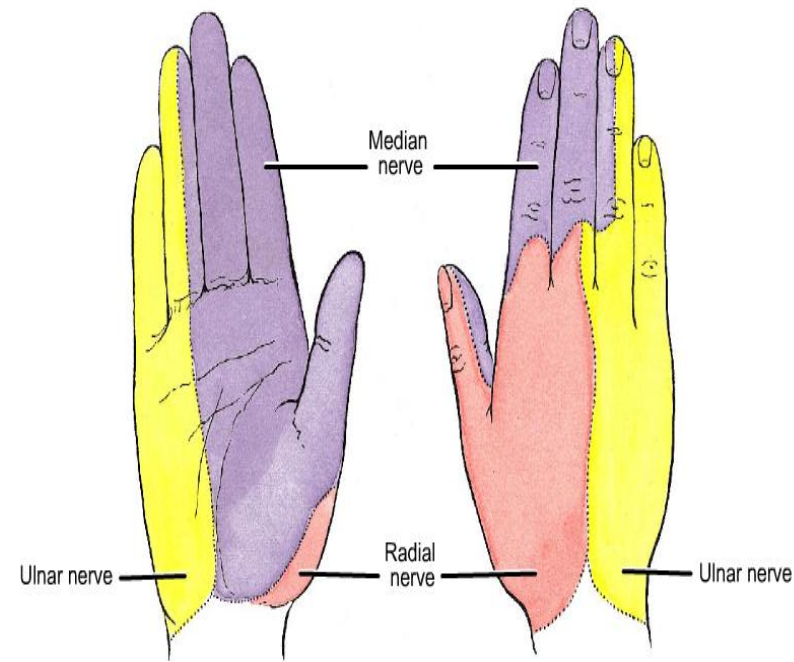
Ulnar nerve

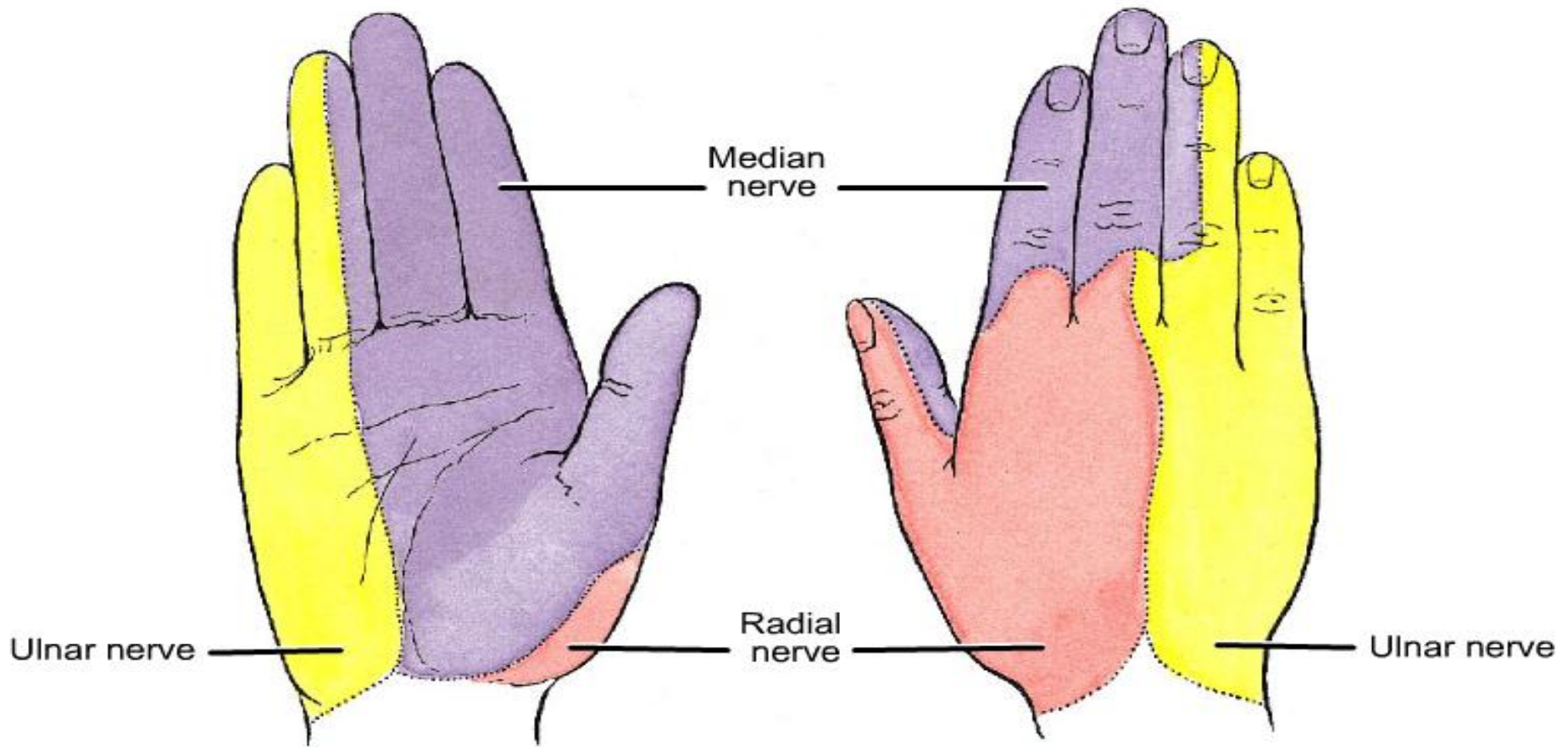
- It pass deep to the tendon of the flexor carpi ulnaris
- Continues distally to the wrist via the ulnar canal
- The ulnar nerve gives off
 - **palmar cutaneous branch**
 - supplies skin on the medial side of the palm
 - **dorsal cutaneous branch**
 - supplies the medial half of dorsum of the hand, the 5th finger, and the medial half of the 4th finger
- Ends at the flexor retinaculum by dividing into 2 branches
 - **superficial branch**: supplies anterior surface of medial one and a half fingers
 - **deep branch**: supplies hypothenar muscles, medial two lumbricals, adductor pollicis, deep head of FPB and all the interossei



Radial nerve

- Supplies no hand muscles
- Supply the skin and fascia over the lateral two thirds of the dorsum of the hand, the thumb, and the proximal parts of the lateral two and a half digits





Thank you!!

Upper Limb: Clinical Conditions



Nerve injury of the upper limb

- Many deformities of the upper limb, particularly those resulting from nerve injuries, are readily interpreted anatomically.

Upper Lesions of Brachial Plexus (Erb-Duchenne Palsy):

- Upper lesions of brachial plexus are usually the result of tearing the 5th & 6th roots of brachial plexus.
- This may occur in infants during a difficult delivery or in adults following a violent fall on the shoulder or a blow to the shoulder.

Nerve injury of Upper Limb

Upper Lesions of the Brachial Plexus:

- The major nerves involved are:
- *suprascapular nerve (1)*
- *musculocutaneous nerve (2)*
- *Axillary (3)*
- ➔ lateral rotation of humerus is lost, due to *suprascapular nerve lesion*: ➔ **humerus is medially rotated:**
 - Supraspinatus
 - Infraspinatus
 - Teres Minor

Nerve injury of the upper limb

Upper Lesions of the Brachial Plexus:

➔ Forearm pronated:

- flexion of forearm, supination of the forearm, weak flexion of shoulder

- biceps brachii
- brachialis

➔ adduction of shoulder: paralysis of deltoid

- Lateral rotation: - infraspinatus; teres minor

- Another name for this lesion is 'porters tip' or "waiter's tip"

Upper Lesions of the Brachial Plexus:



Lower Brachial Plexus Lesions (Klumpke Palsy)

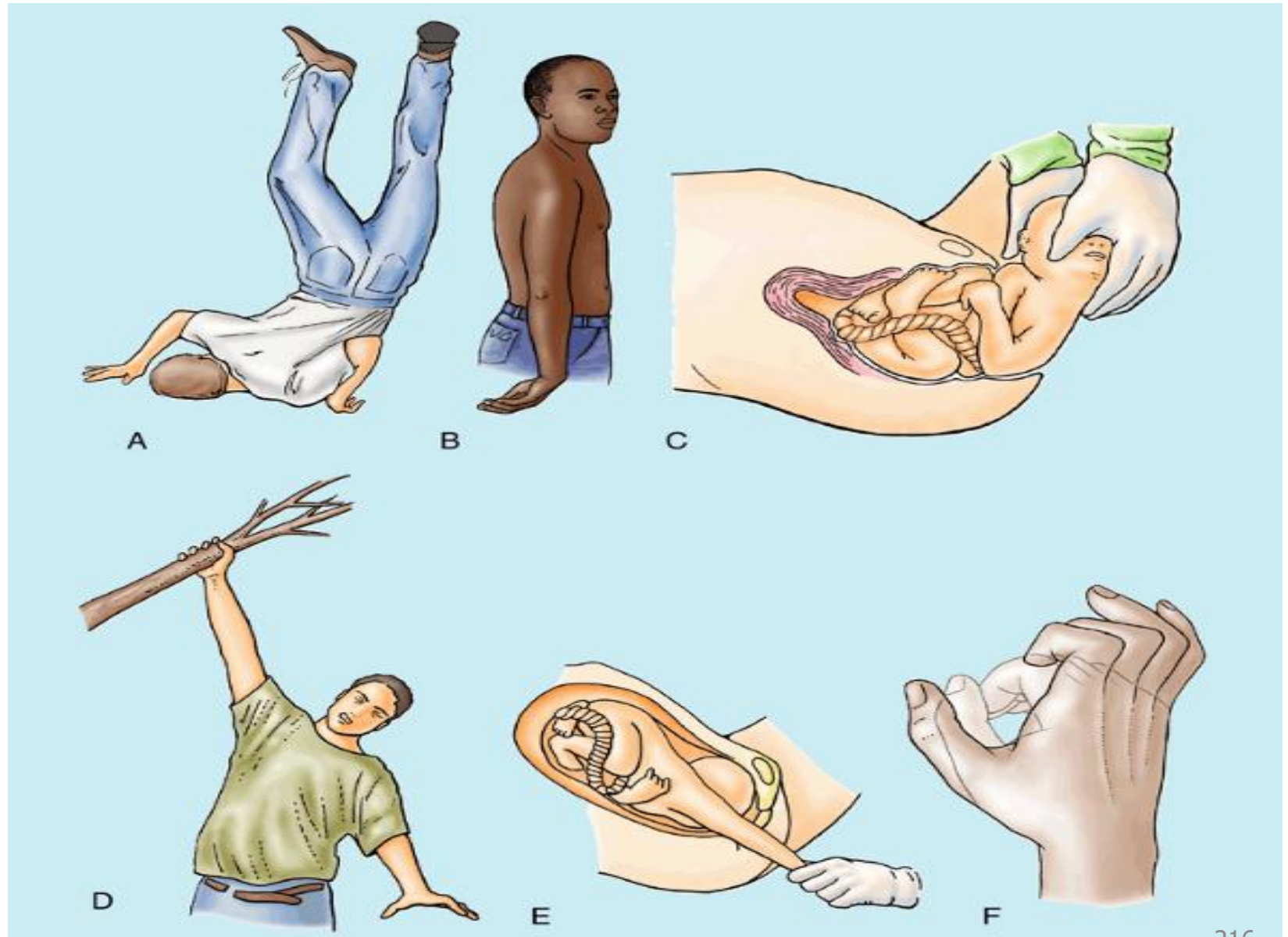
Lower brachial plexus lesions:

- injuries caused by excessive abduction of arm: as a result of someone clutching for an object when falling from a height.
- **1st thoracic nerve (T1)** is usually torn
 - ➔ fibers from T1 help form ***ulnar & median nn.***
 - ➔ small muscles of hand (***interossei & lumbricals***) affected.
- hand has a ***clawed appearance*** due to ***hyperextension of metacarpophalangeal joints & flexion of interphalangeal joints.***

Lower Brachial Plexus Lesions

- ➔ ***extensor digitorum*** is unopposed by lumbricals & interossei & extends the metacarpophalangeal joints.
- Because ***flexor digitorum superficialis & profundus*** are unopposed by lumbricals & interossei, ***middle & terminal phalanges*** are flexed.
- There is also ***sensory loss*** along ***medial side of forearm, hand & medial 2 fingers***
- Lower brachial plexus lesions may also result from malignant metastases from lungs in lower deep cervical lymph nodes & an aberrant ***cervical rib***.

Upper & Lower Brachial Plexus Lesions

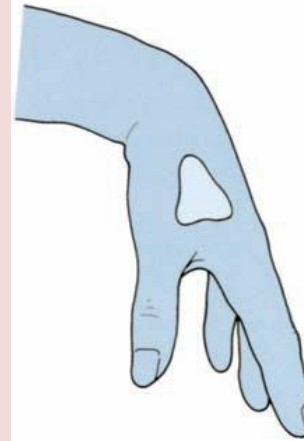


Nerve injury of the upper limb

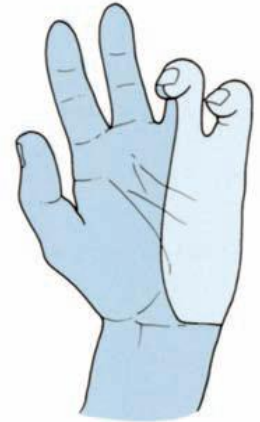
Deformities of the hand:

- (a) Radial n palsy—wrist drop.
- (b) Ulnar nerve palsy—‘main engriffe’ or claw hand.
- (c) Median nerve palsy—‘monkey’s hand’.
- (d) Volkmann’s contracture—another claw hand deformity.

•The pale blue areas represent the usual distribution of anaesthesia.



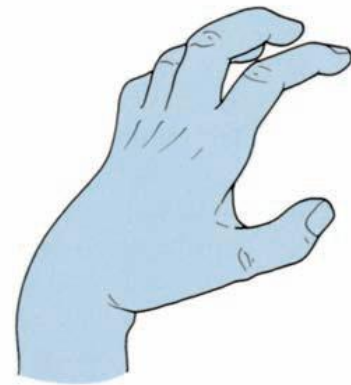
(a)



(b)



(c)



(d)

Nerve injuries of the upper limb

a. Radial Nerve Injury:

- most frequently injured nerve (e.g. fractures of shaft of humerus).

1. Sensory loss:

- skin on lat. 2/3 of dorsum of the hand & lateral 2&1/2 fingers
- skin on back of arm & forearm.

2. Motor loss:

- inability to extend wrist (**wrist drop**) due to paralysis of long extensors;
- inability to extend **metacarpophalangeal joints** due to paralysis of **long extensors**;
- difficulty in **lateral & medial movements** of hand due to paralysis of **radial & ulnar extensors**;
- inability to **extend** or **abduct thumb** due to paralysis of **long abductor & extensors** of the **thumb**.

Nerve injuries of upper limb

Injury to Radial Nerve in the Arm:

1. Injury superior to origin of its branches to triceps brachii:

- ➔ results in paralysis of triceps, brachioradials, supinator, anconeus and extensor muscles of wrist & fingers.
- ➔ Also loss of sensation in areas of skin supplied by this nerve.

2. Injured in radial groove:

- ➔ triceps may not be completely paralyzed but only weakened because only the medial head is affected;
- ➔ muscles in posterior compartment of forearm are paralyzed.
- characteristic clinical sign of radial nerve injury is **wrist drop** (inability to extend wrist & fingers at metacarpophalangeal joints.
- ➔ the relaxed wrist assumes a partly flexed position due to unopposed action of flexor muscles & gravity.

Nerve injuries of the upper limb

b. Median Nerve Injury:

- lesions may be very disabling & crippling
 1. Sensory loss:
 - skin of **lateral 2/3** of the palm;
 - skin of **thumb & lateral 2 ½ fingers** of palm of the hand;
 - skin of **lateral 2 ½ distal phalanges** of dorsum of hand.
 2. Motor loss: (assuming nerve damage is near the elbow):
 - **Weakness in flexion of wrist** → **flexor carpi radialis** paralysis.
 - **inability to flex thumb, index & middle fingers;**
 - **weak pronation;**
 - **paralysis & atrophy of thenar muscles**, resulting in **loss of opposition** of the **thumb**: → resulting in **flat hand** → often referred to as a “**Simian**” or “**ape-like**” hand.
- **Median nerve lesions** are **crippling** because of **loss of control of thumb &** to some extent, **loss of control of index & middle fingers.**

Nerve injuries of the upper limb

Median Nerve Injury in the elbow region:

- flexion of proximal interphalangeal joints (IPJ) of 1st, 2nd & 3rd digits is lost & flexion of 4th & 5th fingers is weakened.
- Flexion of the distal IPJ of the 2nd & 3rd digits is also lost.
- Flexion of the distal IPJ of the 4th & 5th digits is not affected because medial part of FDP is not affected (ulnar nerve).
- ability to flex metacarpophalangeal joints (MPJ) of 2nd & 3rd digits is affected because median nerve supply 1st & 2nd lumbricals:
➔ Thus, when the person attempts to make a first, 2nd & 3rd digits remain partially extended ("**hand of benediction**").
- Thenar muscle function is also lost, as in carpal tunnel syndrome.

Nerve injuries of the upper limb

c. Ulnar nerve lesions:

1. Sensory loss:

- skin on **medial 1/3** of the palm & dorsum of the hand;
- skin on **medial 1 ½ fingers** of the palm & dorsum of the hand.

2. Motor deficits:

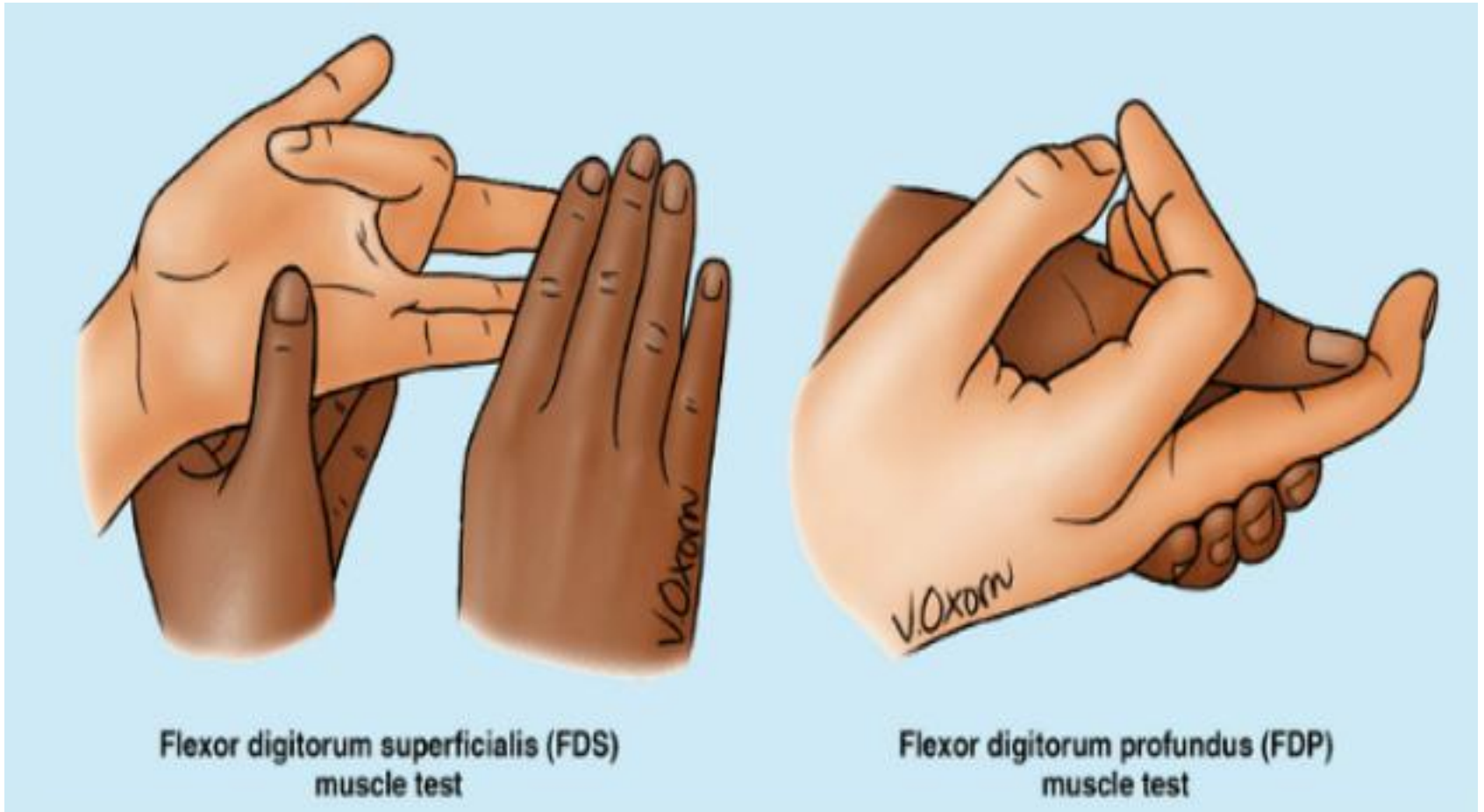
- result from paralysis of small muscles of the hand, supplied by **ulnar nerve**.
- results in “**claw hand**”: → due to paralysis of the **interossei** & the **medial 2 lumbricals**:
- In the normal hand, **interossei & lumbricals** hold balance b/n **long flexors** & **long extensors** of fingers:
 - When these small muscles are paralysed, the **balance** is **lost**, & **long flexors flex** the **IPJ** & **long extensors extend** the **MPJ**.
- **inability** to **oppose thumb & fingers** due to **paralysis of adductor pollicis**.

Nerve injuries of the upper limb

Ulnar Nerve Injury:

- More than 27% of nerve lesions of upper limb affect ulnar nerve.
- Ulnar nerve injuries usually occur in four places: **(1) posterior to medial epicondyle of humerus. (2) In cubital tunnel (formed by tendinous arch connecting humeral & ulnar heads of FCU), (3) at the wrist, (4) in the hand.**
- the person has difficulty making a fist because, in the absence of opposition, the metacarpophalangeal joints become hypertextended, and he or she cannot flex the 4th and 5th fingers at the distal interphalangeal joints when trying to make a fist.
- In addition, the person cannot extend the interphalangeal joints when trying to straighten the fingers. This characteristic appearance of the hand, resulting from a distal lesion of the ulnar nerve, is known as **claw hand (*main en griffe*)**.
- ➔ The deformity results from atrophy of the interosseous muscles supplied by ulnar nerve & the claw is produced by unopposed action of extensors & FDP.

Nerve injuries of the upper limb



Nerve injuries of the upper limb

Injury to the Axillary Nerve (C5, C6):

- usually injured during fracture of surgical neck of humerus.
 - ➔ It may also be damaged during inferior dislocation of glenohumeral joint & by compression from incorrect use of crutches.
 - ➔ deltoid atrophies.
 - ➔ As deltoid atrophies, the rounded contour of shoulder disappears.
 - ➔ This gives the shoulder a flattened appearance & produces a slight hollow inferior to the acromion.
 - ➔ a loss of sensation over lateral side of proximal (upper) part of arm: ➔ area supplied by the superior lateral cutaneous nerve of the arm.
- *The deltoid is a common site for intramuscular injection of drugs:*
 - ➔ *axillary nerve runs transversely under cover of deltoid at level of surgical neck of humerus.*
 - ➔ *Awareness of its location avoids injury during surgical approaches to the shoulder.*

Nerve injuries of the upper limb

Paralysis of the Serratus Anterior:

- When serratus anterior is paralyzed due to injury to **long thoracic nerve**, medial border & inferior angle of scapula moves laterally & posteriorly away from thoracic wall,
- ➔ giving scapula the appearance of a wing, (**Winged Scapula**) especially when the person leans on a hand or presses upper limb against a wall or When the arm is raised.



Nerve injuries of the upper limb

Paralysis of the Serratus Anterior:

- In addition, upper limb cannot be abducted above horizontal level → serratus anterior is unable to rotate glenoid cavity superiorly to allow complete abduction of the limb.
 - long thoracic n. is protected when limbs are on the sides
- ➔ But because it courses on superficial aspect of serratus anterior, it is especially vulnerable when the limbs are elevated, as in a knife fight.

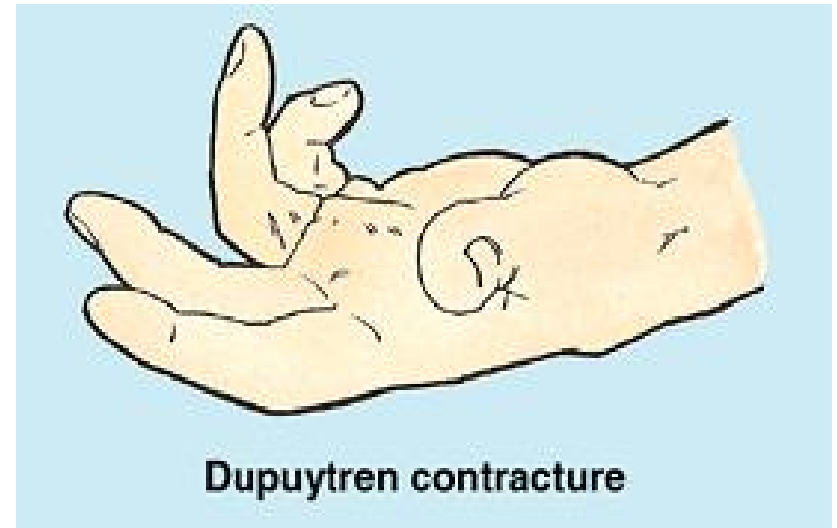
Injury to the Dorsal Scapular Nerve:

- ➔ *Injury to dorsal scapular nerve, nerve to the rhomboids, on one side, causes the scapula on the affected side to be located farther from the midline.*

Upper Limb: Clinical Conditions

Dupuytren Contracture:

- progressive shortening, thickening, fibrosis of palmar aponeurosis on medial side of the hand:
- pulls the **4th & 5th fingers** into ***partial flexion at the MPJ & proximal IPJ.***



Mallet finger:

- Avulsion by long flexor tendon.
- Results in hyperflexion of distal IPJ.



Upper Limb: Clinical Conditions

Carpal Tunnel Syndrome:

- results from any lesion that significantly reduces size of carpal tunnel or increases in size of structures that pass thru it .
- ➔ Fluid retention, infection, & excessive exercise of the fingers may cause swelling of tendons or their synovial sheaths.
- ➔ **median nerve** is the most affected.
- ➔ paresthesia (tingling), hypoesthesia (diminished sensation), or anesthesia (absence of sensation may occur in the skin of lateral three & a half digits.

Upper Limb: Clinical Conditions

Rotator Cuff Injuries:

- Injury or disease may damage the musculotendinous rotator cuff, producing instability of the glenohumeral joint
- Trauma may tear or rupture one or more of the tendons of the **SITS** muscles
- **supraspinatus muscle** is most commonly involved;
- Degenerative tendonitis of the rotator cuff is common, especially in older people.

Upper Limb: Clinical Conditions

Fractures of the Humerus:

- The following parts of the humerus are in direct contact with the indicated nerves:-
- ***Surgical neck: axillary nerve & posterior humeral circumflex artery*** as they pass thru quadrangular space.
- ***Radial groove: radial nerve & profunda brachii artery*** in the spiral groove.
- ***Distal end (supra condylar part) of humerus: median nerve & brachial artery***
- ***Medial epicondyle: ulnar nerve*** which may be compressed in the groove behind medial **epicondyle ('funny bone').'**

Upper Limb: Clinical Conditions

Fracture of the Clavicle:

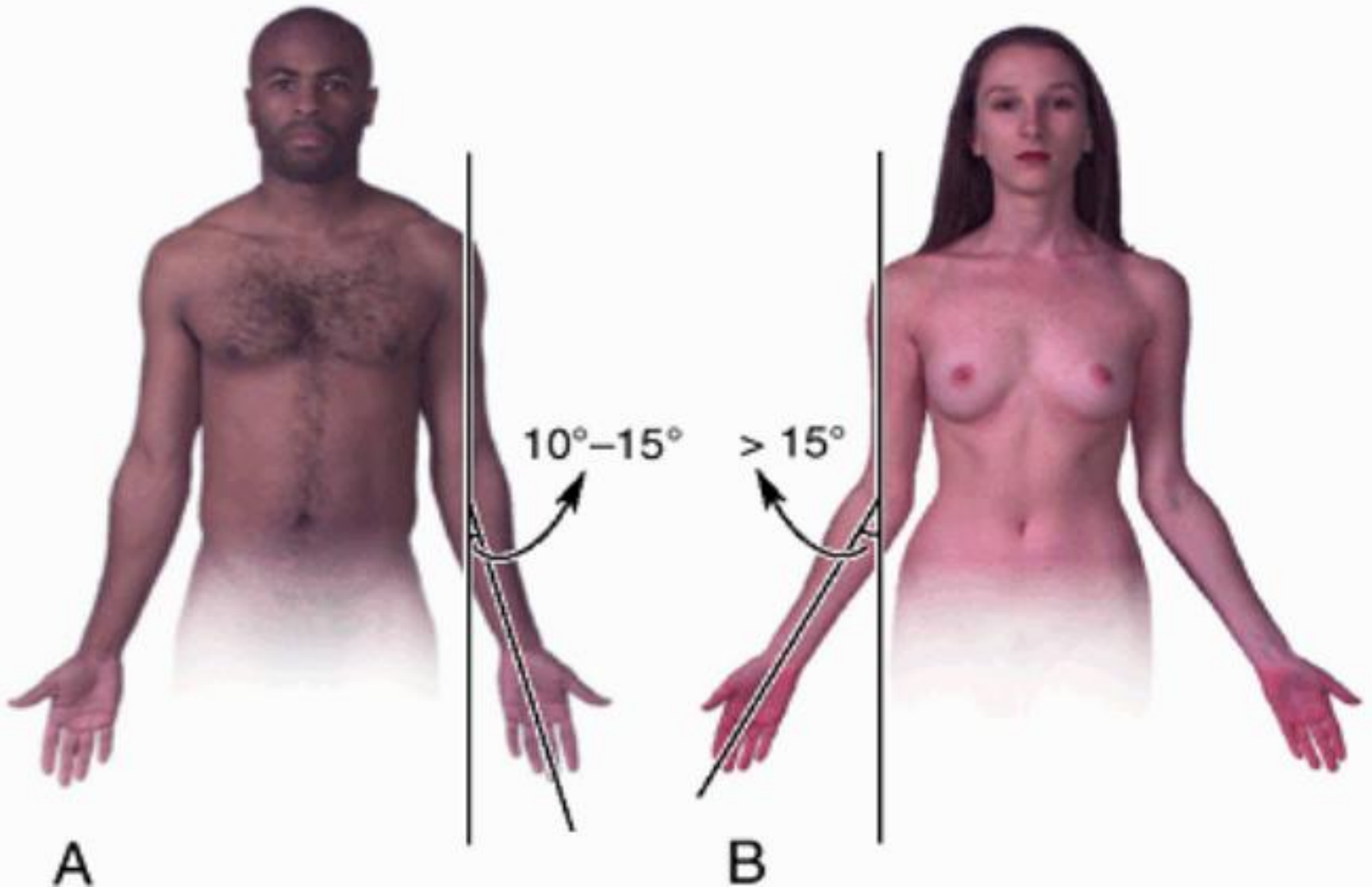
- Occurs at junction of middle/lateral thirds.
- Upward displacement of proximal segment (by sternocleidomastoid muscle) & downward displacement of distal segment (by deltoid muscle & gravity).
- Could result in injury to lower brachial plexus & hemorrhage of subclavian vein.

Upper Limb: Clinical Conditions

Fracture of Humeral Tubercles:

- Fracture of **greater tubercle** occurs by direct trauma
→ affects supraspinatus, infraspinatus, & teres minor muscles, all which attach there by tendons.
- Fracture of **lesser tubercle** accompanies posterior dislocation of shoulder joint,
- the bone fragment has insertion of subscapularis tendon.

Carrying Angle of Elbow Joint



Upper Limb: Clinical Conditions

Colle's Fracture of the Wrist:-

- Is a fracture of lower end of radius
→ distal fragment is displaced posteriorly, producing the ***dinner fork deformity***.
- If distal fragment is displaced anteriorly: → the ***reverse Colle's fracture*** known as ***Smith's fracture***.

Fracture of the Scaphoid: -

- Occurs with a fall on the outstretched hand & may damage the radial artery.

Fracture of the Hamate:-

- May injure ulnar nerve & ulnar artery.
- ulnar nerve & artery may be compressed in Guyon's canal (pisiform-hamate tunnel), formed b/n hook of hamate & pisiform.

Upper Limb: Clinical Conditions

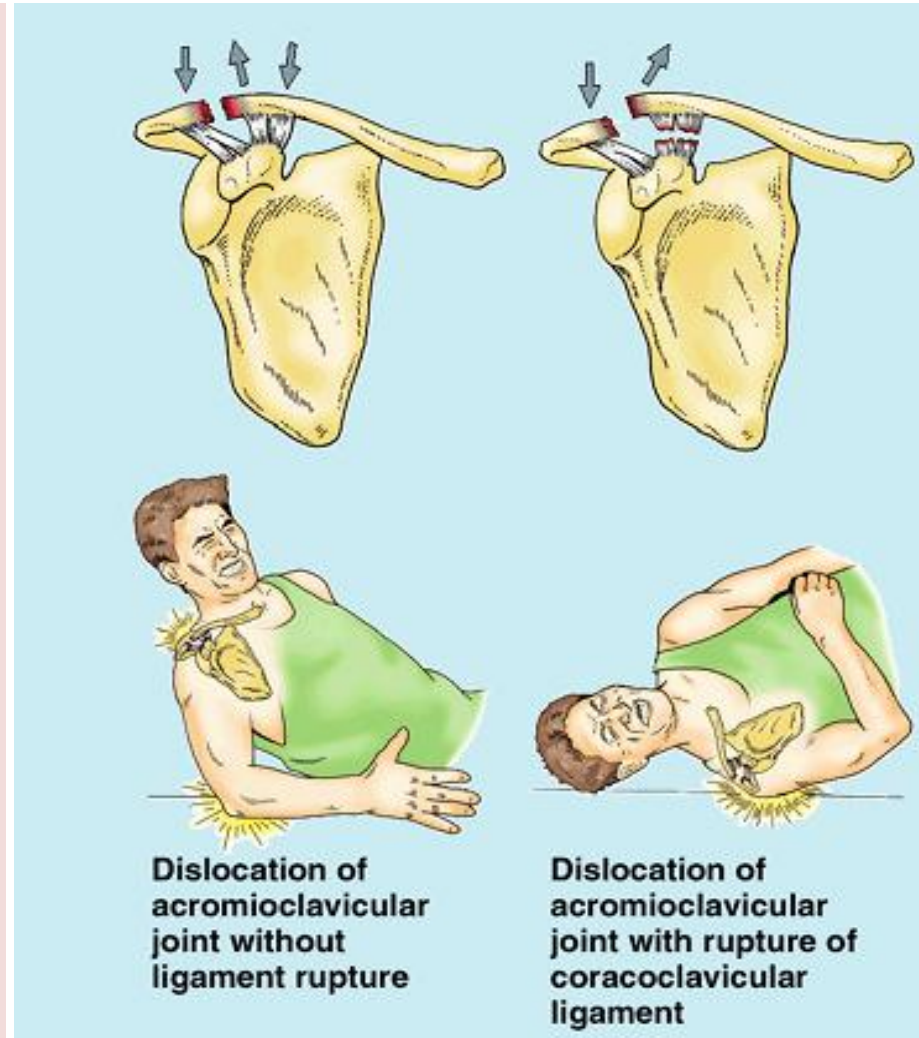
Ulnar Nerve Injury, Entrapment, & Claw Hand:

- *Clinical Features:*
 - ➔ Entrapment of deep branch of ulnar nerve produces a motor loss exhibited by a weak pinch, weak finger spread against resistance, deficit in actively flexing MPJ & interosseous atrophy.
 - ➔ Compression of the superficial branch features burning sensations in the 4th & 5th digits.
- Deep palpation of pisiform-hamate tunnel initiates or aggravates the pain.

Upper Limb: Clinical Conditions

Dislocation of Acromioclavicular Joint:

- Results from a fall on the shoulder with the impact taken by the acromion.
- It is called **shoulder separation** because the shoulder is separated from the clavicle with possible rupture of the coracoclavicular ligament.



Upper Limb: Clinical Conditions

Golfer's Wrist:

- If a full golf swing hits the ground or a hard object other than the ball, an isolated fracture of the wrist may result.
- Results from violent contraction of the flexor carpi ulnaris insertion thru the pisiform-hamate ligament.
- may show a fracture of the hamate.

Boxer's Wrist (Lunate Injury):

- Lunate dislocation and/or fracture may be seen in a fall on the outstretched hand,
- most common in boxers whose hands are carelessly wrapped.
- Damage to the median nerve is a complication.

Mammary gland

Lymphatic drainage:

- is of great clinical significance owing to its role in the spread of malignant tumours.
- may be divided into ***cutaneous & parenchymal drainage***.

a. Cutaneous Drainage:

1. ***lateral*** – to the anterior pectoral nodes
2. ***medial*** – to the ***internal mammary nodes & cutaneous lymphatics*** of the opposite side
3. ***superior*** – to the ***medial (posterior) pectoral nodes & apical (infraclavicular) nodes***.
4. ***inferior*** – to the ***cutaneous lymphatics*** of ***upper abdominal wall & the liver***.

b. **Parenchymal Drainage:** receives channels from the breast proper & skin of the nipple & areola → drain into a ***subareolar plexus***.

→ Efferents from ***subareolar plexus*** drain into the ***anterior axillary nodes***.

- the ***subareolar plexus*** communicates with the ***lymphatics of the deep fascia*** → anatomical basis for removal of a wide area in radical mastectomy

Lymphatic drainage & breast cancer

Prognosis in Cancer based on lymphatic drainage:

- Upper-outer Quadrant → most favourable
- Lower-outer quadrant → less favourable
- Upper-inner Quadrant → dangerous because of its proximity to the mediastinum
- Lower-inner Quadrant → most dangerous because of its proximity to the peritoneal cavity.

Venous spread of cancer of the breast:

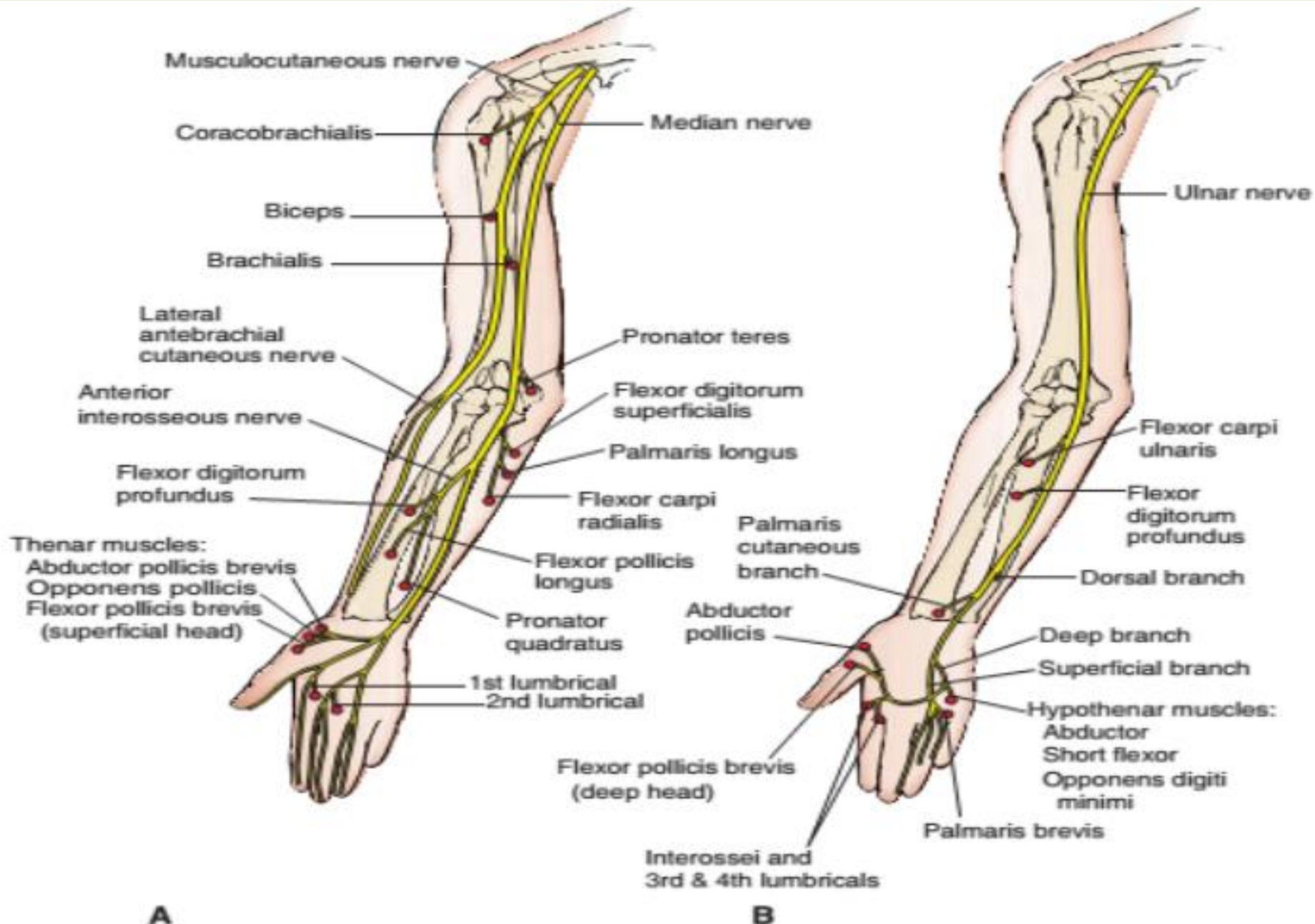
- As the breast is drained by ***axillary, internal thoracic & intercostal veins***, cancer spreads by venous channels to:
 - a) ***lungs*** through ***axillary vein*** and ***superior vena cava***
 - b) ***liver*** via the ***azygous veins***
 - c) the ***bones of axial skeleton*** via the ***intercostal veins*** which communicates with the ***vertebral venous plexus***.

Lymphatic drainage & breast cancer

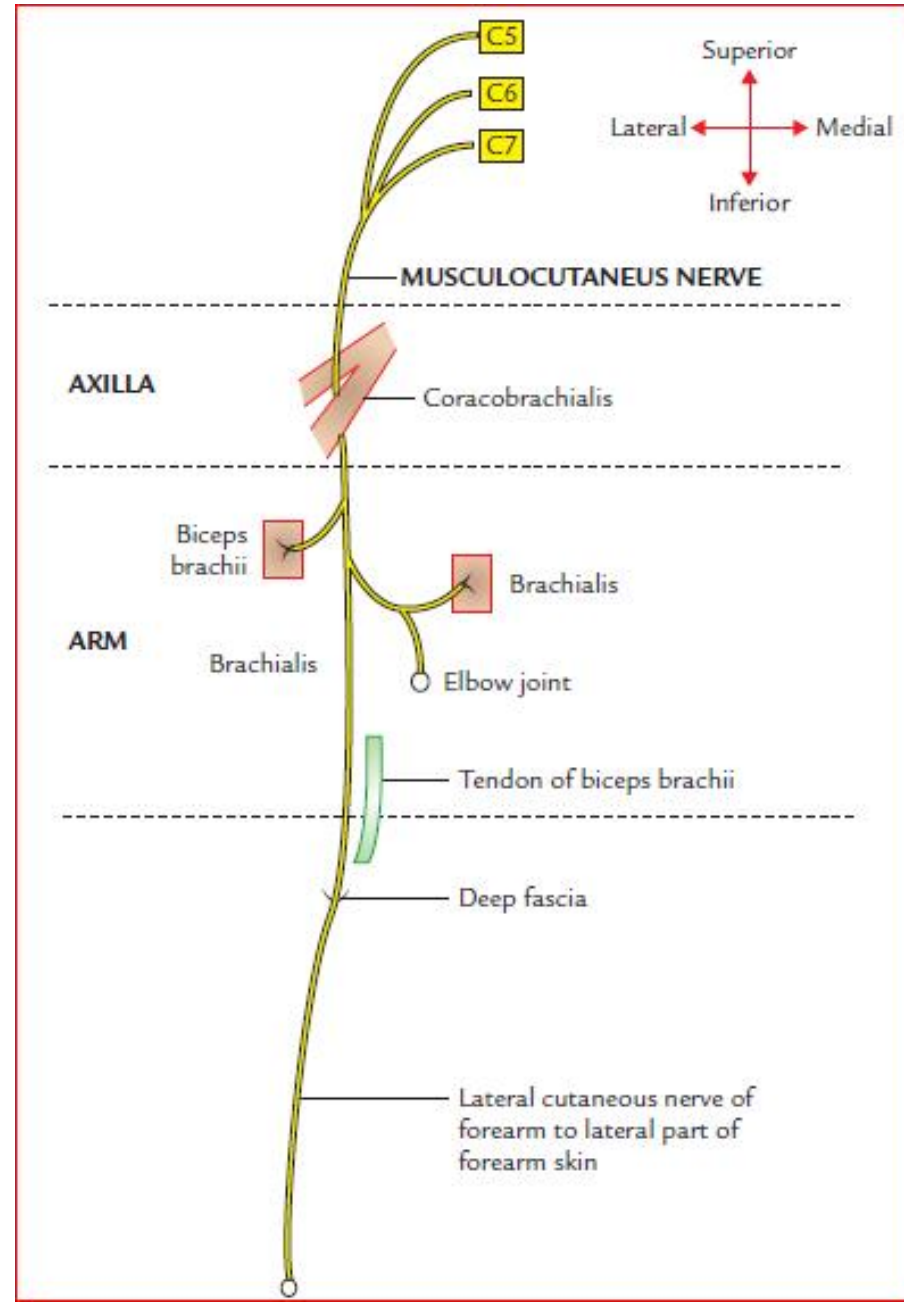
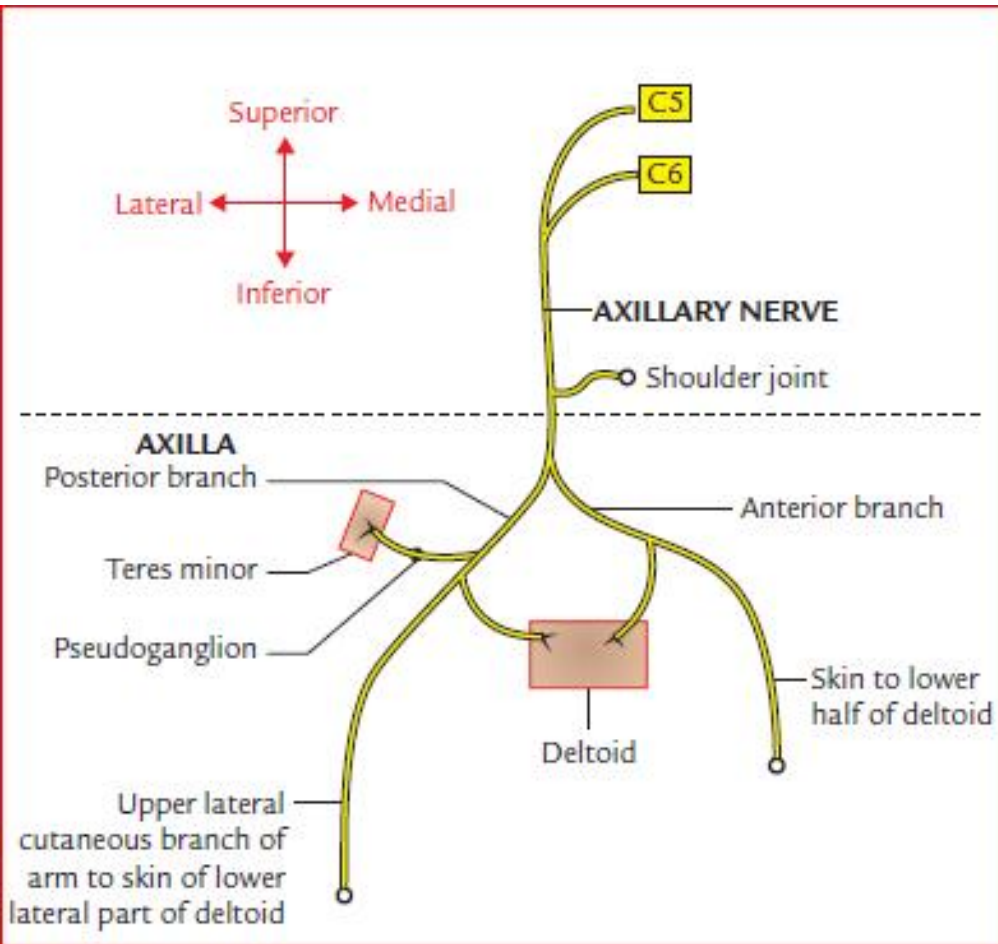
Anatomic basis of physical signs in carcinoma of the breast:

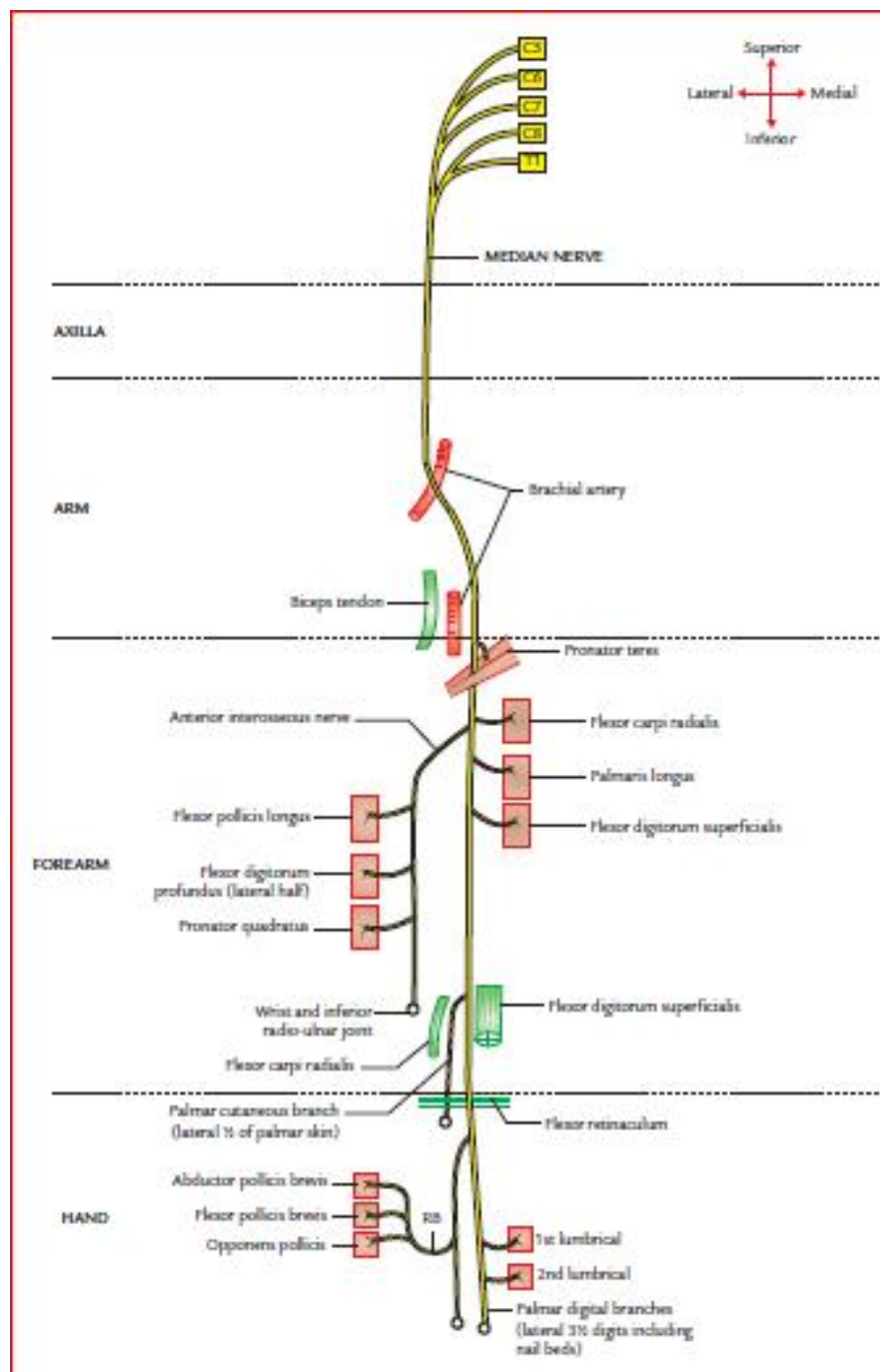
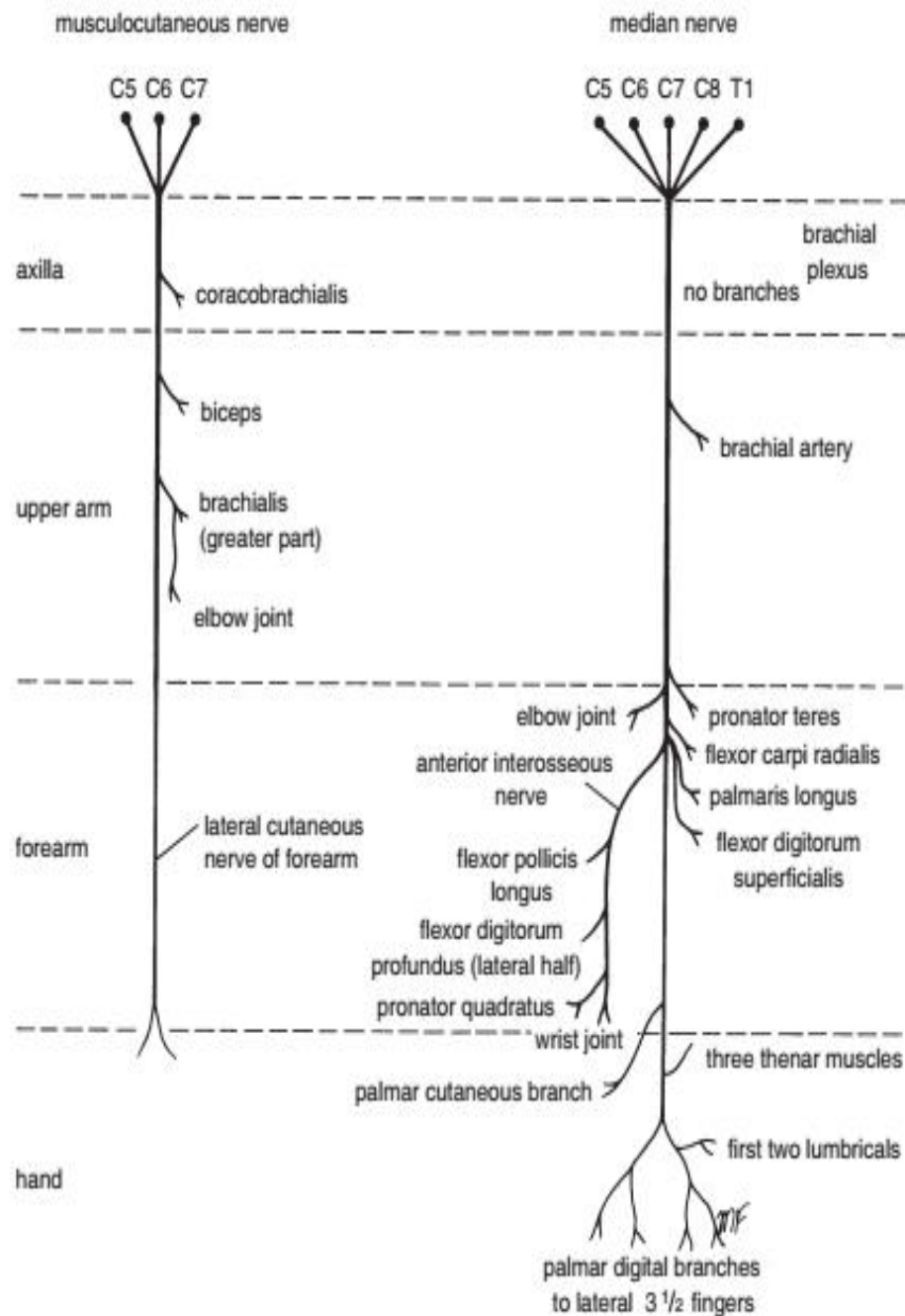
- 1. ***Fixation or depression of overlying skin***: is due to invasion of ***fibrous septa*** which extend from the skin to the lobes of the gland by cancer cells producing shortening of the septa resulting in fixation or depression of the skin.
- 2. ***Fixation, retraction or inversion of the nipple***: is due to neoplastic involvement of the lactiferous ducts.
- 3. ***Fixation of the breast to the underlying muscle***: is seen in advanced cancer when the cancer has invaded the deep fascia of ***pectoralis major muscle***.
- knowledge of anatomy of the breast is of v. important as carcinoma of the breast is twice as common as any other carcinoma in women.
- earliest sign of serious pathology is a lump in the breast.
- The above three signs are fairly advanced signs.

Summary muscular branch

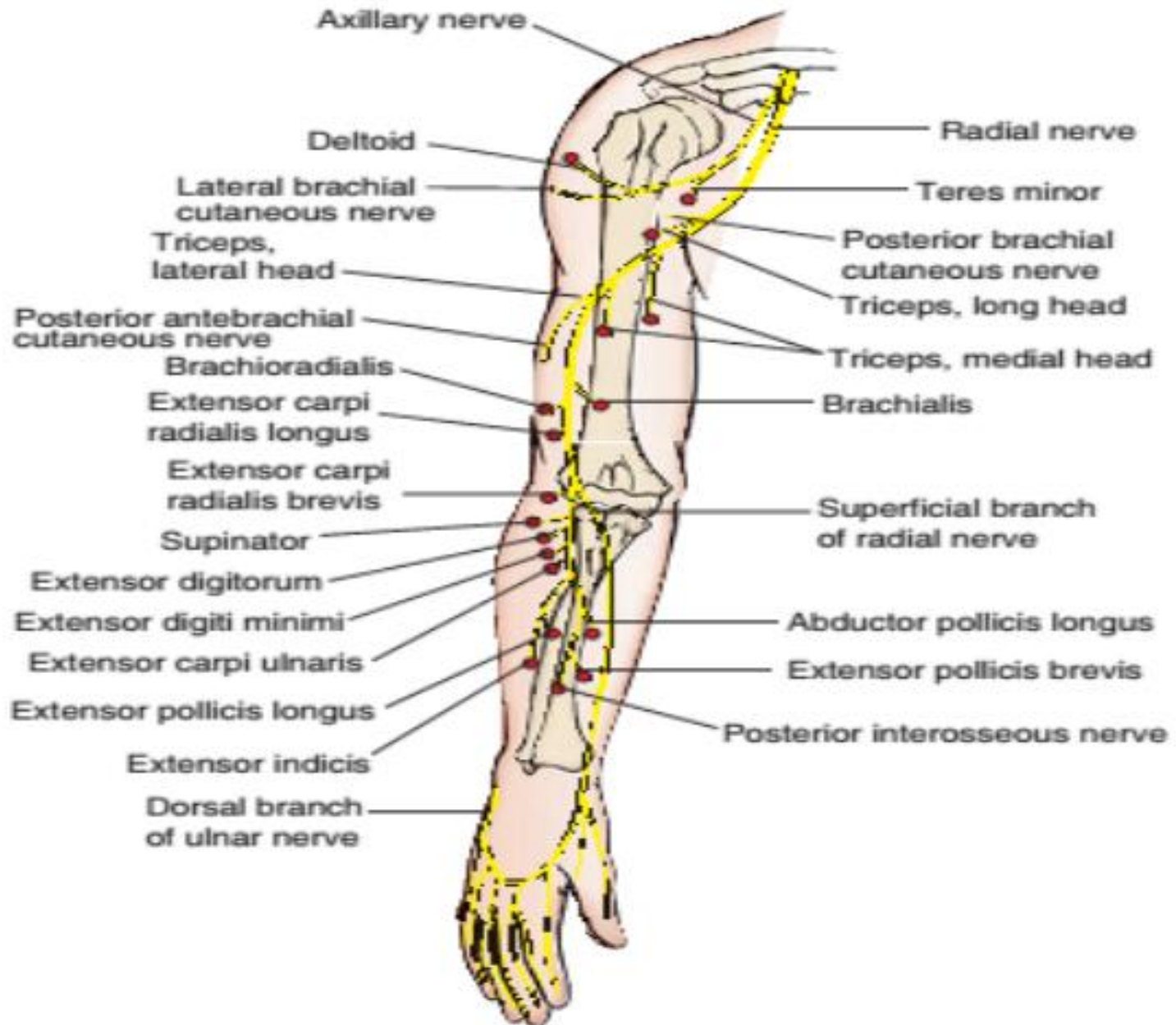


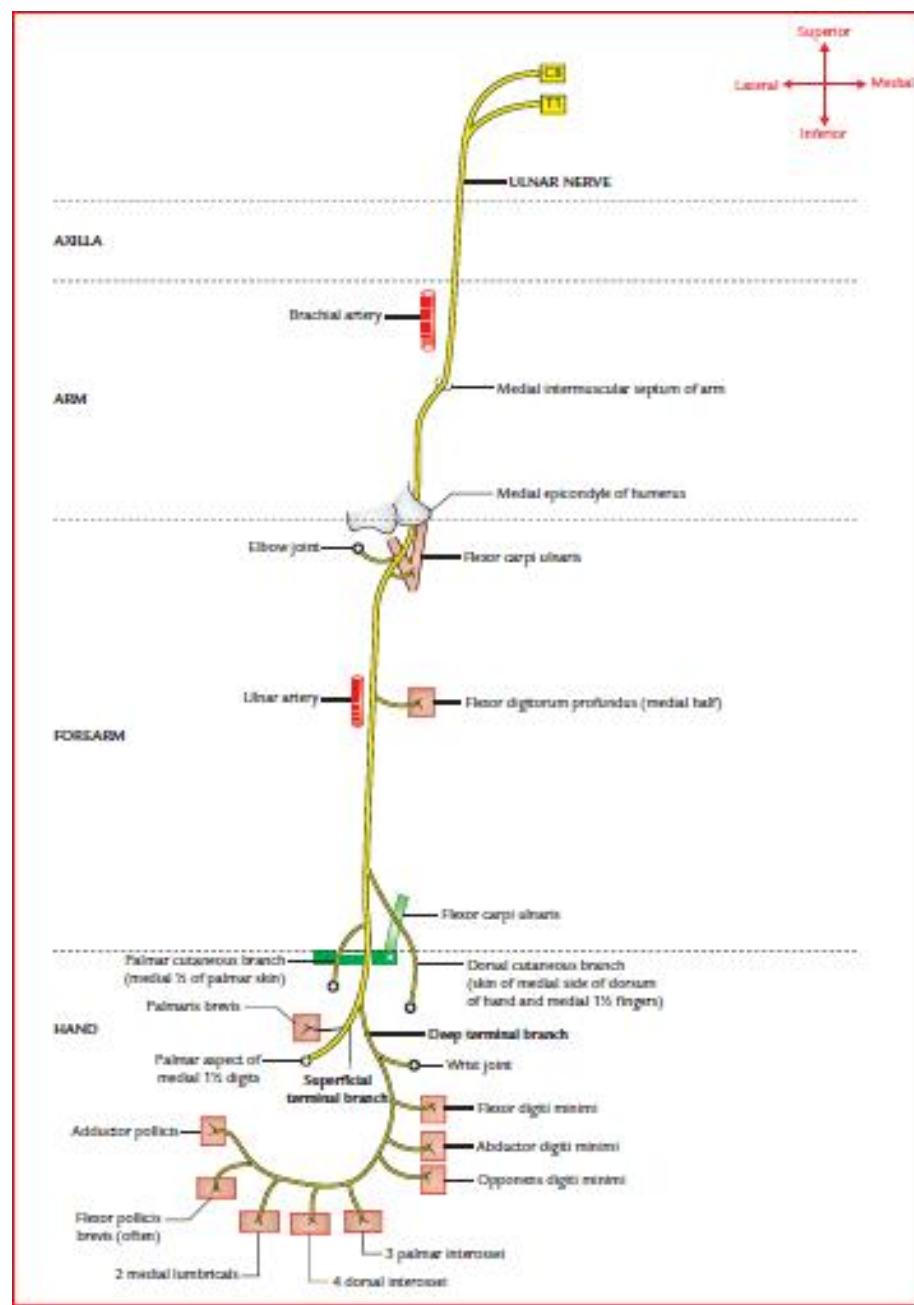
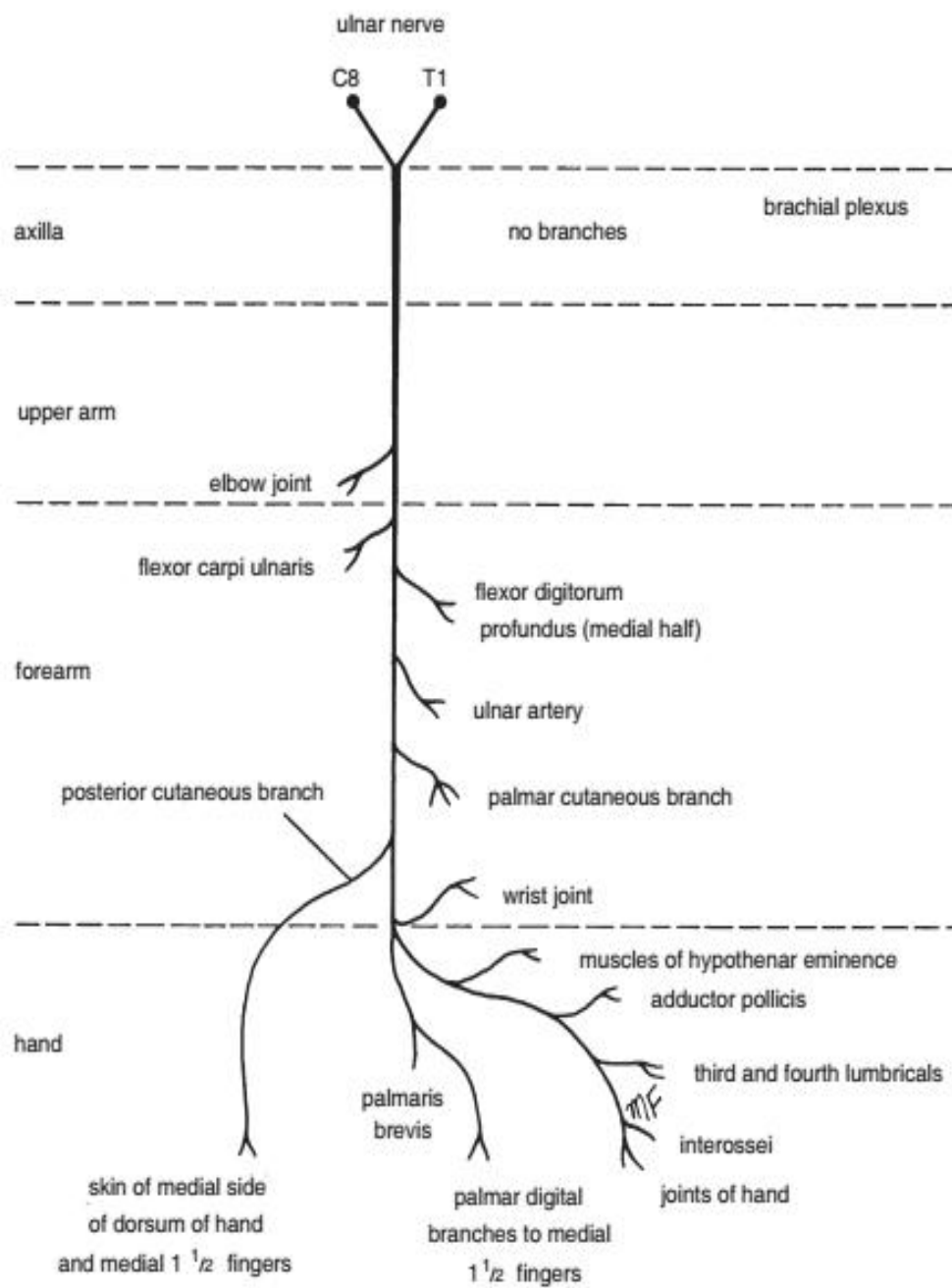
Major nerve in upper limb





Summary muscular branch





axillary nerve

C5

C6

axilla

brachial plexus

shoulder joint

anterior branch

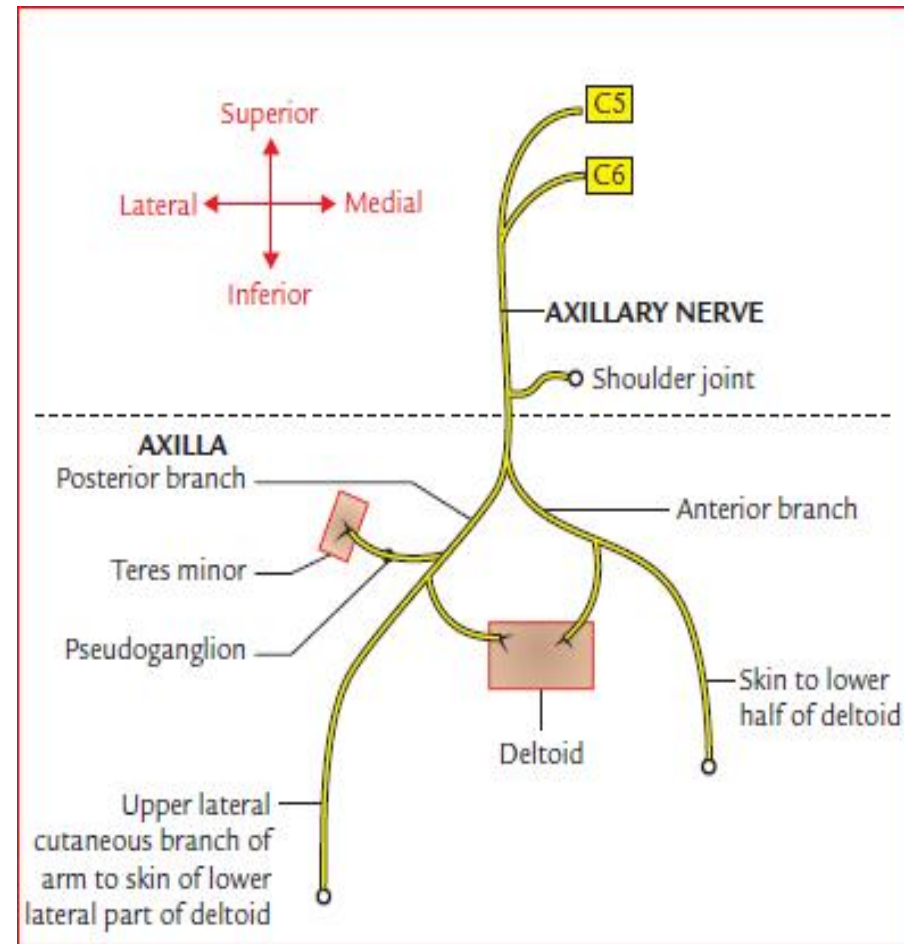
posterior branch

scapular
region

teres minor

deltoid

upper lateral cutaneous
nerve of arm to skin
over lower half of deltoid



skin over lower
half of deltoid

